liblava

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1 Hierarchical Index 1
1.1 Class Hierarchy . . . 1

2 Class Index 5
2.1 Class List . . . 5

3 File Index 11
3.1 File List . . . 11

4 Class Documentation 15
4.1 lava::allocator Struct Reference . . . 15
4.1.1 Detailed Description . . . 16
4.1.2 Constructor & Destructor Documentation . . . 16
4.1.2.1 allocator() . . . 16
4.1.3 Member Function Documentation . . . 16
4.1.3.1 create() . . . 16
4.1.3.2 get() . . . 16
4.1.3.3 make() . . . 17
4.1.3.4 valid() . . . 17
4.2 lava::app Struct Reference . . . 17
4.2.1 Detailed Description . . . 19
4.2.2 Constructor & Destructor Documentation . . . 19
4.2.2.1 app() [1/2] . . . 19
4.2.2.2 app() [2/2] . . . 20
4.2.3 Member Function Documentation . . . 20
4.2.3.1 block_cmd() . . . 20
4.2.3.2 draw_about() . . . 20
4.2.3.3 get_frame_counter() . . . 21
4.2.3.4 parse_config() . . . 21
4.2.3.5 screenshot() . . . 21
4.2.3.6 setup() . . . 21
4.2.3.7 v_sync() . . . 22
4.3 lava::app_config Struct Reference . . . 22
4.3.1 Detailed Description . . . 23
4.3.2 Member Function Documentation . . . 23
4.3.2.1 get_config() . . . 23
4.3.2.2 set_config() . . . 23
4.4 lava::attachment Struct Reference . . . 24
4.4.1 Detailed Description . . . 25
4.4.2 Constructor & Destructor Documentation . . . 25
4.4.2.1 attachment() . . . 25
4.4.3 Member Function Documentation . . . 25
4.4.3.1 get_description() . . . 25

4.4.3.2 make() . . . 25
4.4.3.3 set_final_layout() . . . 26
4.4.3.4 set_format() . . . 26
4.4.3.5 set_initial_layout() . . . 26
4.4.3.6 set_layouts() . . . 27
4.4.3.7 set_load_op() . . . 27
4.4.3.8 set_op() . . . 27
4.4.3.9 set_samples() . . . 27
4.4.3.10 set_stencil_load_op() . . . 28
4.4.3.11 set_stencil_op() . . . 28
4.4.3.12 set_stencil_store_op() . . . 28
4.4.3.13 set_store_op() . . . 29
4.5 lava::benchmark_data Struct Reference . . . 29
4.5.1 Detailed Description . . . 30
4.6 lava::descriptor::binding Struct Reference . . . 30
4.6.1 Detailed Description . . . 30
4.6.2 Member Function Documentation . . . 31
4.6.2.1 get() . . . 31
4.6.2.2 make() . . . 31
4.6.2.3 set() . . . 31
4.6.2.4 set_count() . . . 32
4.6.2.5 set_samplers() . . . 32
4.6.2.6 set_stage_flags() . . . 32
4.6.2.7 set_type() . . . 32
4.7 lava::block Struct Reference . . . 33
4.7.1 Detailed Description . . . 34
4.7.2 Member Function Documentation . . . 34
4.7.2.1 activated() . . . 34
4.7.2.2 add_cmd() . . . 35
4.7.2.3 add_command() . . . 35
4.7.2.4 create() . . . 35
4.7.2.5 get_buffers() . . . 36
4.7.2.6 get_cmd_order() . . . 36
4.7.2.7 get_command_buffer() [1/2] . . . 36
4.7.2.8 get_command_buffer() [2/2] . . . 37
4.7.2.9 get_commands() . . . 37
4.7.2.10 get_current_frame() . . . 37
4.7.2.11 get_device() . . . 38
4.7.2.12 get_frame_count() . . . 38
4.7.2.13 make() . . . 38
4.7.2.14 process() . . . 38
4.7.2.15 remove_cmd() . . . 39

4.7.2.16 remove_command() . . . 39
4.7.2.17 set_active() . . . 39
4.8 lava::buffer Struct Reference . . . 40
4.8.1 Detailed Description . . . 41
4.8.2 Member Function Documentation . . . 41
4.8.2.1 create() . . . 41
4.8.2.2 create_mapped() . . . 42
4.8.2.3 flush() . . . 42
4.8.2.4 get() . . . 42
4.8.2.5 get_address() . . . 43
4.8.2.6 get_allocation() . . . 43
4.8.2.7 get_allocation_info() . . . 43
4.8.2.8 get_descriptor_info() . . . 43
4.8.2.9 get_device() . . . 44
4.8.2.10 get_device_memory() . . . 44
4.8.2.11 get_mapped_data() . . . 44
4.8.2.12 get_size() . . . 44
4.8.2.13 make() . . . 45
4.8.2.14 valid() . . . 45
4.9 lava::json_file::callback Struct Reference . . . 45
4.9.1 Detailed Description . . . 46
4.10 lava::swapchain::callback Struct Reference . . . 46
4.10.1 Detailed Description . . . 46
4.11 lava::camera Struct Reference . . . 47
4.11.1 Detailed Description . . . 48
4.11.2 Member Function Documentation . . . 49
4.11.2.1 activated() . . . 49
4.11.2.2 create() . . . 49
4.11.2.3 get_descriptor_info() . . . 49
4.11.2.4 get_projection() . . . 50
4.11.2.5 get_view() . . . 50
4.11.2.6 get_view_projection() . . . 50
4.11.2.7 handle() [1/3] . . . 50
4.11.2.8 handle() [2/3] . . . 51
4.11.2.9 handle() [3/3] . . . 51
4.11.2.10 moving() . . . 51
4.11.2.11 set_active() . . . 52
4.11.2.12 set_movement_keys() . . . 52
4.11.2.13 update_view() [1/2] . . . 52
4.11.2.14 update_view() [2/2] . . . 53
4.11.2.15 valid() . . . 53
4.12 lava::cdata Struct Reference . . . 53

4.12.1 Detailed Description . . . 54
4.12.2 Constructor & Destructor Documentation . . . 54
4.12.2.1 cdata() [1/2] . . . 54
4.12.2.2 cdata() [2/2] . . . 54
4.13 lava::command Struct Reference . . . 55
4.13.1 Detailed Description . . . 56
4.13.2 Member Function Documentation . . . 56
4.13.2.1 create() . . . 56
4.13.2.2 destroy() . . . 56
4.13.2.3 make() . . . 56
4.14 lava::compute_pipeline Struct Reference . . . 57
4.14.1 Detailed Description . . . 58
4.14.2 Member Function Documentation . . . 58
4.14.2.1 bind() . . . 58
4.14.2.2 copy_from() . . . 58
4.14.2.3 copy_to() . . . 58
4.14.2.4 get_shader_stage() . . . 59
4.14.2.5 make() . . . 59
4.14.2.6 set() . . . 59
4.14.2.7 set_shader_stage() . . . 60
4.15 lava::imgui::config Struct Reference . . . 60
4.15.1 Detailed Description . . . 61
4.16 lava::configurable Struct Reference . . . 61
4.16.1 Detailed Description . . . 61
4.16.2 Member Function Documentation . . . 61
4.16.2.1 get_config() . . . 61
4.16.2.2 set_config() . . . 62
4.17 lava::render_pipeline::create_info Struct Reference . . . 62
4.17.1 Detailed Description . . . 62
4.18 lava::device::create_param Struct Reference . . . 63
4.18.1 Detailed Description . . . 64
4.18.2 Member Function Documentation . . . 64
4.18.2.1 add_dedicated_queues() . . . 64
4.18.2.2 add_queue() . . . 64
4.18.2.3 add_queues() . . . 64
4.18.2.4 verify_queues() . . . 65
4.19 lava::instance::create_param Struct Reference . . . 65
4.19.1 Detailed Description . . . 66
4.20 lava::data Struct Reference . . . 66
4.20.1 Detailed Description . . . 67
4.20.2 Constructor & Destructor Documentation . . . 67
4.20.2.1 data() . . . 67

4.20.3 Member Function Documentation . . . 67
4.20.3.1 allocate() . . . 67
4.20.3.2 end() . . . 68
4.20.3.3 set() . . . 68
4.21 lava::data_provider Struct Reference . . . 68
4.21.1 Detailed Description . . . 69
4.22 lava::instance::debug_config Struct Reference . . . 69
4.22.1 Detailed Description . . . 69
4.23 lava::descriptor Struct Reference . . . 70
4.23.1 Detailed Description . . . 71
4.23.2 Member Function Documentation . . . 71
4.23.2.1 add() . . . 71
4.23.2.2 add_binding() . . . 71
4.23.2.3 allocate() [1/2] . . . 73
4.23.2.4 allocate() [2/2] . . . 73
4.23.2.5 allocate_set() . . . 73
4.23.2.6 allocate_sets() . . . 74
4.23.2.7 create() . . . 74
4.23.2.8 free() [1/2] . . . 74
4.23.2.9 free() [2/2] . . . 75
4.23.2.10 free_set() . . . 75
4.23.2.11 free_sets() . . . 75
4.23.2.12 get() . . . 76
4.23.2.13 get_binding_count() . . . 76
4.23.2.14 get_bindings() . . . 76
4.23.2.15 get_device() . . . 76
4.23.2.16 make() . . . 77
4.24 lava::device Struct Reference . . . 77
4.24.1 Detailed Description . . . 79
4.24.2 Member Function Documentation . . . 79
4.24.2.1 alloc() . . . 79
4.24.2.2 call() . . . 79
4.24.2.3 compute_queue() . . . 79
4.24.2.4 compute_queues() . . . 80
4.24.2.5 create() . . . 80
4.24.2.6 get() . . . 80
4.24.2.7 get_allocator() . . . 80
4.24.2.8 get_compute_queue() . . . 80
4.24.2.9 get_compute_queues() . . . 81
4.24.2.10 get_features() . . . 81
4.24.2.11 get_graphics_queue() . . . 81
4.24.2.12 get_graphics_queues() . . . 82

4.24.2.13 get_physical_device() . . . 82
4.24.2.14 get_properties() . . . 82
4.24.2.15 get_queues() . . . 82
4.24.2.16 get_transfer_queue() . . . 82
4.24.2.17 get_transfer_queues() . . . 83
4.24.2.18 get_vk_physical_device() . . . 83
4.24.2.19 graphics_queue() . . . 83
4.24.2.20 graphics_queues() . . . 84
4.24.2.21 make() . . . 84
4.24.2.22 queues() . . . 84
4.24.2.23 set_allocator() . . . 84
4.24.2.24 surface_supported() . . . 84
4.24.2.25 transfer_queue() . . . 85
4.24.2.26 transfer_queues() . . . 85
4.24.2.27 wait_for_idle() . . . 85
4.25 lava::device_table Struct Reference . . . 86
4.25.1 Detailed Description . . . 87
4.25.2 Member Function Documentation . . . 87
4.25.2.1 vkAcquireNextImageKHR() . . . 88
4.25.2.2 vkAllocateCommandBuffers() [1/2] . . . 88
4.25.2.3 vkAllocateCommandBuffers() [2/2] . . . 88
4.25.2.4 vkCreateCommandPool() [1/3] . . . 88
4.25.2.5 vkCreateCommandPool() [2/3] . . . 89
4.25.2.6 vkCreateCommandPool() [3/3] . . . 89
4.25.2.7 vkCreateFence() [1/2] . . . 89
4.25.2.8 vkCreateFence() [2/2] . . . 89
4.25.2.9 vkCreateImageView() [1/2] . . . 90
4.25.2.10 vkCreateImageView() [2/2] . . . 90
4.25.2.11 vkCreateSampler() [1/2] . . . 90
4.25.2.12 vkCreateSampler() [2/2] . . . 90
4.25.2.13 vkCreateSemaphore() [1/2] . . . 91
4.25.2.14 vkCreateSemaphore() [2/2] . . . 91
4.25.2.15 vkCreateShaderModule() [1/2] . . . 91
4.25.2.16 vkCreateShaderModule() [2/2] . . . 91
4.25.2.17 vkCreateSwapchainKHR() [1/2] . . . 92
4.25.2.18 vkCreateSwapchainKHR() [2/2] . . . 92
4.25.2.19 vkDestroyCommandPool() . . . 92
4.25.2.20 vkDestroyFence() . . . 92
4.25.2.21 vkDestroyImageView() . . . 93
4.25.2.22 vkDestroySampler() . . . 93
4.25.2.23 vkDestroySemaphore() . . . 93
4.25.2.24 vkDestroySwapchainKHR() . . . 93

4.25.2.25 vkFreeCommandBuffers() . . . 94
4.25.2.26 vkGetSwapchainImagesKHR() . . . 94
4.25.2.27 vkQueuePresentKHR() . . . 94
4.25.2.28 vkQueueSubmit() . . . 94
4.25.2.29 vkResetFences() . . . 95
4.25.2.30 vkUpdateDescriptorSets() [1/7] . . . 95
4.25.2.31 vkUpdateDescriptorSets() [2/7] . . . 95
4.25.2.32 vkUpdateDescriptorSets() [3/7] . . . 95
4.25.2.33 vkUpdateDescriptorSets() [4/7] . . . 96
4.25.2.34 vkUpdateDescriptorSets() [5/7] . . . 96
4.25.2.35 vkUpdateDescriptorSets() [6/7] . . . 96
4.25.2.36 vkUpdateDescriptorSets() [7/7] . . . 96
4.25.2.37 vkWaitForFences() . . . 97
4.26 lava::dispatcher Struct Reference . . . 97
4.26.1 Detailed Description . . . 97
4.26.2 Member Function Documentation . . . 98
4.26.2.1 add_message() . . . 98
4.26.2.2 setup() . . . 98
4.26.2.3 update() . . . 98
4.27 lava::driver Struct Reference . . . 99
4.27.1 Detailed Description . . . 99
4.27.2 Member Function Documentation . . . 100
4.27.2.1 add_stage() . . . 100
4.27.2.2 get_stages() . . . 101
4.27.2.3 instance() . . . 101
4.27.2.4 run() . . . 101
4.28 lava::engine Struct Reference . . . 102
4.28.1 Detailed Description . . . 102
4.28.2 Member Function Documentation . . . 102
4.28.2.1 setup() . . . 103
4.29 lava::entity Struct Reference . . . 103
4.29.1 Detailed Description . . . 104
4.29.2 Member Function Documentation . . . 104
4.29.2.1 get_id() . . . 104
4.30 lava::file Struct Reference . . . 104
4.30.1 Detailed Description . . . 105
4.30.2 Constructor & Destructor Documentation . . . 105
4.30.2.1 file() . . . 105
4.30.3 Member Function Documentation . . . 106
4.30.3.1 get_path() . . . 106
4.30.3.2 get_size() . . . 106
4.30.3.3 get_type() . . . 106

4.30.3.4 open() . . . 106
4.30.3.5 opened() . . . 107
4.30.3.6 read() [1/2] . . . 107
4.30.3.7 read() [2/2] . . . 107
4.30.3.8 seek() . . . 108
4.30.3.9 tell() . . . 108
4.30.3.10 writable() . . . 108
4.30.3.11 write() . . . 109
4.31 lava::file_data Struct Reference . . . 109
4.31.1 Detailed Description . . . 110
4.31.2 Constructor & Destructor Documentation . . . 110
4.31.2.1 file_data() . . . 110
4.32 lava::file_delete Struct Reference . . . 110
4.32.1 Detailed Description . . . 111
4.32.2 Constructor & Destructor Documentation . . . 111
4.32.2.1 file_delete() . . . 111
4.33 lava::file_format Struct Reference . . . 111
4.33.1 Detailed Description . . . 112
4.34 lava::file_system Struct Reference . . . 112
4.34.1 Detailed Description . . . 113
4.34.2 Member Function Documentation . . . 113
4.34.2.1 create_folder() . . . 113
4.34.2.2 enumerate_files() . . . 113
4.34.2.3 exists() . . . 114
4.34.2.4 get_app() . . . 114
4.34.2.5 get_base_dir() . . . 114
4.34.2.6 get_ext() . . . 115
4.34.2.7 get_full_base_dir() . . . 115
4.34.2.8 get_org() . . . 115
4.34.2.9 get_pref_dir() . . . 115
4.34.2.10 get_real_dir() . . . 116
4.34.2.11 get_res_dir() . . . 116
4.34.2.12 get_version() . . . 116
4.34.2.13 initialize() . . . 116
4.34.2.14 mount() . . . 117
4.34.2.15 mount_base() . . . 117
4.34.2.16 mount_res() . . . 118
4.34.2.17 ready() . . . 118
4.35 lava::imgui::font Struct Reference . . . 118
4.35.1 Detailed Description . . . 119
4.36 lava::forward_shading Struct Reference . . . 119
4.36.1 Detailed Description . . . 119

4.36.2 Member Function Documentation . . . 119
4.36.2.1 create() . . . 119
4.36.2.2 get_depth_stencil() . . . 120
4.36.2.3 get_pass() . . . 120
4.36.2.4 get_vk_pass() . . . 120
4.37 lava::frame Struct Reference . . . 121
4.37.1 Detailed Description . . . 122
4.37.2 Constructor & Destructor Documentation . . . 122
4.37.2.1 frame() [1/2] . . . 122
4.37.2.2 frame() [2/2] . . . 123
4.37.3 Member Function Documentation . . . 123
4.37.3.1 add_run() . . . 123
4.37.3.2 add_run_end() . . . 123
4.37.3.3 add_run_once() . . . 124
4.37.3.4 get_cmd_line() . . . 124
4.37.3.5 get_env() . . . 124
4.37.3.6 get_name() . . . 125
4.37.3.7 get_running_time() . . . 125
4.37.3.8 get_running_time_sec() . . . 125
4.37.3.9 ready() . . . 125
4.37.3.10 remove() . . . 125
4.37.3.11 run() . . . 126
4.37.3.12 set_wait_for_events() . . . 126
4.37.3.13 shut_down() . . . 126
4.37.3.14 waiting_for_events() . . . 127
4.38 lava::frame_env Struct Reference . . . 127
4.38.1 Detailed Description . . . 128
4.38.2 Constructor & Destructor Documentation . . . 128
4.38.2.1 frame_env() . . . 128
4.39 lava::gamepad Struct Reference . . . 128
4.39.1 Detailed Description . . . 129
4.39.2 Constructor & Destructor Documentation . . . 129
4.39.2.1 gamepad() . . . 129
4.39.3 Member Function Documentation . . . 129
4.39.3.1 get_id() . . . 129
4.39.3.2 get_name() . . . 130
4.39.3.3 get_pad_id() . . . 130
4.39.3.4 pressed() . . . 130
4.39.3.5 ready() . . . 130
4.39.3.6 update() . . . 131
4.39.3.7 value() . . . 131
4.40 lava::gamepad_manager Struct Reference . . . 131

4.40.1 Detailed Description . . . 132
4.40.2 Member Function Documentation . . . 132
4.40.2.1 add() . . . 132
4.40.2.2 remove() . . . 132
4.40.2.3 singleton() . . . 132
4.41 lava::global_logger Struct Reference . . . 133
4.41.1 Detailed Description . . . 133
4.41.2 Member Function Documentation . . . 133
4.41.2.1 get() . . . 133
4.41.2.2 set() . . . 133
4.41.2.3 singleton() . . . 134
4.42 lava::hex_cell Struct Reference . . . 134
4.42.1 Detailed Description . . . 135
4.42.2 Member Function Documentation . . . 135
4.42.2.1 add() . . . 135
4.42.2.2 scale() . . . 135
4.42.2.3 substract() . . . 135
4.42.2.4 to_pair() . . . 136
4.43 lava::hex_fractional_cell Struct Reference . . . 136
4.43.1 Detailed Description . . . 136
4.44 lava::hex_grid Struct Reference . . . 137
4.44.1 Detailed Description . . . 137
4.44.2 Constructor & Destructor Documentation . . . 137
4.44.2.1 hex_grid() . . . 137
4.44.3 Member Function Documentation . . . 138
4.44.3.1 find() . . . 138
4.44.3.2 to_pixel() . . . 138
4.44.3.3 update() . . . 138
4.45 lava::hex_layout Struct Reference . . . 139
4.45.1 Detailed Description . . . 139
4.46 lava::hex_offset_coord Struct Reference . . . 139
4.46.1 Detailed Description . . . 140
4.47 lava::hex_orientation Struct Reference . . . 140
4.47.1 Detailed Description . . . 140
4.48 lava::hex_point Struct Reference . . . 140
4.48.1 Detailed Description . . . 141
4.49 lava::imgui::icon_font Struct Reference . . . 141
4.49.1 Detailed Description . . . 141
4.50 lava::id Struct Reference . . . 142
4.50.1 Detailed Description . . . 142
4.50.2 Member Function Documentation . . . 142
4.50.2.1 to_string() . . . 143

4.50.2.2 valid() . . . 143
4.51 lava::id_listeners< T > Struct Template Reference . . . 143
4.51.1 Detailed Description . . . 143
4.51.2 Member Function Documentation . . . 144
4.51.2.1 add() . . . 144
4.51.2.2 get_list() . . . 144
4.51.2.3 remove() . . . 144
4.52 lava::id_registry< T, Meta > Struct Template Reference . . . 145
4.52.1 Detailed Description . . . 145
4.52.2 Member Function Documentation . . . 146
4.52.2.1 add() . . . 146
4.52.2.2 create() . . . 146
4.52.2.3 exists() . . . 146
4.52.2.4 get() . . . 147
4.52.2.5 get_all() . . . 147
4.52.2.6 get_all_meta() . . . 148
4.52.2.7 get_meta() . . . 148
4.52.2.8 remove() . . . 148
4.52.2.9 update() . . . 149
4.53 lava::ids Struct Reference . . . 149
4.53.1 Detailed Description . . . 149
4.53.2 Member Function Documentation . . . 149
4.53.2.1 instance() . . . 150
4.53.2.2 next() . . . 150
4.54 lava::image Struct Reference . . . 150
4.54.1 Detailed Description . . . 152
4.54.2 Constructor & Destructor Documentation . . . 152
4.54.2.1 image() . . . 152
4.54.3 Member Function Documentation . . . 152
4.54.3.1 create() . . . 152
4.54.3.2 destroy() . . . 153
4.54.3.3 get() . . . 153
4.54.3.4 get_allocation() . . . 153
4.54.3.5 get_depth() . . . 153
4.54.3.6 get_device() . . . 154
4.54.3.7 get_format() . . . 154
4.54.3.8 get_info() . . . 154
4.54.3.9 get_size() . . . 154
4.54.3.10 get_subresource_range() . . . 155
4.54.3.11 get_view() . . . 155
4.54.3.12 get_view_info() . . . 155
4.54.3.13 make() . . . 155

4.54.3.14 set_aspect_mask() . . . 156
4.54.3.15 set_component() . . . 156
4.54.3.16 set_flags() . . . 156
4.54.3.17 set_layer_count() . . . 156
4.54.3.18 set_layout() . . . 157
4.54.3.19 set_level_count() . . . 157
4.54.3.20 set_tiling() . . . 157
4.54.3.21 set_usage() . . . 158
4.54.3.22 set_view_type() . . . 158
4.55 lava::image_loader Struct Reference . . . 158
4.55.1 Detailed Description . . . 159
4.55.2 Constructor & Destructor Documentation . . . 159
4.55.2.1 image_loader() [1/2] . . . 159
4.55.2.2 image_loader() [2/2] . . . 159
4.55.3 Member Function Documentation . . . 159
4.55.3.1 get() . . . 160
4.55.3.2 get_channels() . . . 160
4.55.3.3 get_dimensions() . . . 160
4.55.3.4 ready() . . . 160
4.55.3.5 size() . . . 161
4.56 lava::imgui Struct Reference . . . 161
4.56.1 Detailed Description . . . 162
4.56.2 Constructor & Destructor Documentation . . . 162
4.56.2.1 imgui() . . . 162
4.56.3 Member Function Documentation . . . 163
4.56.3.1 activated() . . . 163
4.56.3.2 capture_keyboard() . . . 163
4.56.3.3 capture_mouse() . . . 163
4.56.3.4 create() [1/3] . . . 163
4.56.3.5 create() [2/3] . . . 164
4.56.3.6 create() [3/3] . . . 164
4.56.3.7 get_ini_file() . . . 165
4.56.3.8 get_input_callback() . . . 165
4.56.3.9 get_pipeline() . . . 165
4.56.3.10 ready() . . . 165
4.56.3.11 set_active() . . . 165
4.56.3.12 set_ini_file() . . . 166
4.56.3.13 setup() [1/2] . . . 166
4.56.3.14 setup() [2/2] . . . 166
4.56.3.15 upload_fonts() . . . 167
4.57 lava::input Struct Reference . . . 167
4.57.1 Detailed Description . . . 168

4.57.2 Member Function Documentation . . . 168
4.57.2.1 add() . . . 168
4.57.2.2 get_mouse_position() . . . 168
4.57.2.3 remove() . . . 168
4.57.2.4 set_mouse_position() . . . 169
4.58 lava::input_callback Struct Reference . . . 169
4.58.1 Detailed Description . . . 170
4.58.2 Member Typedef Documentation . . . 170
4.58.2.1 func . . . 170
4.59 lava::input_events< T > Struct Template Reference . . . 170
4.59.1 Detailed Description . . . 171
4.59.2 Member Function Documentation . . . 171
4.59.2.1 add() . . . 171
4.60 lava::instance Struct Reference . . . 171
4.60.1 Detailed Description . . . 172
4.60.2 Member Function Documentation . . . 172
4.60.2.1 create() . . . 172
4.60.2.2 get() . . . 173
4.60.2.3 get_debug_config() . . . 173
4.60.2.4 get_first_physical_device() . . . 173
4.60.2.5 get_info() . . . 174
4.60.2.6 get_physical_devices() . . . 174
4.60.2.7 singleton() . . . 174
4.61 lava::instance_info Struct Reference . . . 174
4.61.1 Detailed Description . . . 175
4.62 lava::interface Struct Reference . . . 175
4.62.1 Detailed Description . . . 176
4.63 lava::internal_version Struct Reference . . . 176
4.63.1 Detailed Description . . . 176
4.64 lava::json_file Struct Reference . . . 176
4.64.1 Detailed Description . . . 177
4.64.2 Constructor & Destructor Documentation . . . 177
4.64.2.1 json_file() . . . 177
4.64.3 Member Function Documentation . . . 177
4.64.3.1 add() . . . 178
4.64.3.2 get() . . . 178
4.64.3.3 load() . . . 178
4.64.3.4 remove() . . . 178
4.64.3.5 save() . . . 179
4.64.3.6 set() . . . 179
4.65 lava::key_event Struct Reference . . . 179
4.65.1 Detailed Description . . . 180

4.65.2 Member Function Documentation . . . 180
4.65.2.1 active() . . . 180
4.65.2.2 pressed() [1/2] . . . 180
4.65.2.3 pressed() [2/2] . . . 181
4.65.2.4 released() . . . 181
4.65.2.5 repeated() . . . 182
4.66 lava::texture::layer Struct Reference . . . 182
4.66.1 Detailed Description . . . 182
4.67 lava::log_config Struct Reference . . . 183
4.67.1 Detailed Description . . . 183
4.68 lava::memory Struct Reference . . . 183
4.68.1 Detailed Description . . . 184
4.68.2 Member Function Documentation . . . 184
4.68.2.1 alloc() . . . 184
4.68.2.2 instance() . . . 184
4.68.2.3 set_callbacks() . . . 184
4.68.2.4 set_use_custom_cpu_callbacks() . . . 185
4.69 lava::mesh_meta Struct Reference . . . 185
4.69.1 Detailed Description . . . 185
4.70 lava::mesh_template< T > Struct Template Reference . . . 186
4.70.1 Detailed Description . . . 187
4.70.2 Member Function Documentation . . . 187
4.70.2.1 add_data() . . . 187
4.70.2.2 bind() . . . 188
4.70.2.3 bind_draw() . . . 188
4.70.2.4 create() . . . 188
4.70.2.5 draw() . . . 189
4.70.2.6 empty() . . . 189
4.70.2.7 get_data() . . . 189
4.70.2.8 get_index_buffer() . . . 190
4.70.2.9 get_indices() [1/2] . . . 190
4.70.2.10 get_indices() [2/2] . . . 190
4.70.2.11 get_indices_count() . . . 190
4.70.2.12 get_vertex_buffer() . . . 191
4.70.2.13 get_vertices() [1/2] . . . 191
4.70.2.14 get_vertices() [2/2] . . . 191
4.70.2.15 get_vertices_count() . . . 191
4.70.2.16 make() . . . 192
4.70.2.17 reload() . . . 192
4.70.2.18 set_data() . . . 192
4.71 lava::mesh_template_data< T > Struct Template Reference . . . 192
4.71.1 Detailed Description . . . 193

4.71.2 Member Function Documentation . . . 193
4.71.2.1 move() . . . 193
4.71.2.2 scale() . . . 194
4.71.2.3 scale_vector() . . . 194
4.72 lava::texture::mip_level Struct Reference . . . 194
4.72.1 Detailed Description . . . 195
4.73 lava::mouse_active_event Struct Reference . . . 195
4.73.1 Detailed Description . . . 196
4.74 lava::mouse_button_event Struct Reference . . . 196
4.74.1 Detailed Description . . . 196
4.74.2 Member Function Documentation . . . 197
4.74.2.1 pressed() . . . 197
4.74.2.2 released() . . . 197
4.75 lava::mouse_move_event Struct Reference . . . 197
4.75.1 Detailed Description . . . 198
4.76 lava::mouse_position Struct Reference . . . 198
4.76.1 Detailed Description . . . 198
4.77 lava::no_copy_no_move Struct Reference . . . 199
4.77.1 Detailed Description . . . 199
4.78 lava::pair_hash Struct Reference . . . 200
4.78.1 Detailed Description . . . 200
4.78.2 Member Function Documentation . . . 200
4.78.2.1 operator()() . . . 200
4.79 lava::path_drop_event Struct Reference . . . 201
4.79.1 Detailed Description . . . 201
4.80 lava::physical_device Struct Reference . . . 201
4.80.1 Detailed Description . . . 203
4.80.2 Constructor & Destructor Documentation . . . 203
4.80.2.1 physical_device() . . . 203
4.80.3 Member Function Documentation . . . 203
4.80.3.1 create_default_device_param() . . . 203
4.80.3.2 get() . . . 203
4.80.3.3 get_device_name() . . . 204
4.80.3.4 get_device_type_string() . . . 204
4.80.3.5 get_driver_version() . . . 204
4.80.3.6 get_extension_properties() . . . 204
4.80.3.7 get_features() . . . 205
4.80.3.8 get_memory_properties() . . . 205
4.80.3.9 get_properties() . . . 205
4.80.3.10 get_queue_family() . . . 205
4.80.3.11 get_queue_family_properties() . . . 206
4.80.3.12 initialize() . . . 206

4.80.3.13 make() . . . 206
4.80.3.14 supported() . . . 207
4.80.3.15 surface_supported() . . . 207
4.80.3.16 swapchain_supported() . . . 207
4.81 lava::pipeline Struct Reference . . . 208
4.81.1 Detailed Description . . . 209
4.81.2 Constructor & Destructor Documentation . . . 209
4.81.2.1 pipeline() . . . 209
4.81.3 Member Function Documentation . . . 210
4.81.3.1 activated() . . . 210
4.81.3.2 auto_bind() . . . 210
4.81.3.3 bind() . . . 210
4.81.3.4 create() . . . 211
4.81.3.5 get() . . . 211
4.81.3.6 get_device() . . . 211
4.81.3.7 get_layout() . . . 211
4.81.3.8 ready() . . . 212
4.81.3.9 set_active() . . . 212
4.81.3.10 set_auto_bind() . . . 213
4.81.3.11 set_layout() . . . 213
4.81.3.12 setup() . . . 213
4.82 lava::pipeline_layout Struct Reference . . . 214
4.82.1 Detailed Description . . . 215
4.82.2 Member Function Documentation . . . 215
4.82.2.1 add() [1/2] . . . 215
4.82.2.2 add() [2/2] . . . 215
4.82.2.3 add_descriptor() . . . 215
4.82.2.4 add_push_constant_range() . . . 216
4.82.2.5 bind() . . . 216
4.82.2.6 bind_descriptor_set() . . . 216
4.82.2.7 create() . . . 217
4.82.2.8 get() . . . 217
4.82.2.9 get_descriptors() . . . 217
4.82.2.10 get_device() . . . 218
4.82.2.11 get_push_constant_ranges() . . . 218
4.82.2.12 make() . . . 218
4.83 lava::platform Struct Reference . . . 218
4.83.1 Detailed Description . . . 219
4.83.2 Member Function Documentation . . . 219
4.83.2.1 create() [1/2] . . . 219
4.83.2.2 create() [2/2] . . . 220
4.83.2.3 create_device() . . . 220

4.83.2.4 get_devices() . . . 220
4.83.2.5 remove() . . . 220
4.84 lava::descriptor::pool Struct Reference . . . 221
4.84.1 Detailed Description . . . 222
4.84.2 Member Function Documentation . . . 222
4.84.2.1 create() . . . 222
4.84.2.2 get() . . . 222
4.84.2.3 get_device() . . . 223
4.84.2.4 get_max() . . . 223
4.84.2.5 get_sizes() . . . 223
4.84.2.6 make() . . . 223
4.85 lava::producer Struct Reference . . . 224
4.85.1 Detailed Description . . . 225
4.85.2 Member Function Documentation . . . 225
4.85.2.1 add_mesh() . . . 225
4.85.2.2 add_texture() . . . 225
4.85.2.3 compile_shader() . . . 225
4.85.2.4 create_mesh() . . . 226
4.85.2.5 create_texture() . . . 226
4.85.2.6 get_mesh() . . . 227
4.85.2.7 get_shader() . . . 227
4.85.2.8 get_texture() . . . 227
4.85.2.9 reload_shader() . . . 228
4.86 lava::property::prop Struct Reference . . . 228
4.86.1 Detailed Description . . . 229
4.86.2 Constructor & Destructor Documentation . . . 229
4.86.2.1 prop() . . . 229
4.87 lava::property Struct Reference . . . 229
4.87.1 Detailed Description . . . 230
4.87.2 Member Function Documentation . . . 230
4.87.2.1 add() . . . 231
4.87.2.2 check() . . . 232
4.87.2.3 empty() . . . 232
4.87.2.4 exists() . . . 232
4.87.2.5 get_all() . . . 233
4.87.2.6 get_config() . . . 233
4.87.2.7 get_filename() . . . 233
4.87.2.8 load() . . . 234
4.87.2.9 load_all() . . . 234
4.87.2.10 operator()() . . . 234
4.87.2.11 parse() . . . 235
4.87.2.12 set_config() . . . 235

4.87.2.13 set_filename() . . . 235
4.87.2.14 unload() . . . 235
4.88 lava::pseudo_random_generator Struct Reference . . . 236
4.88.1 Detailed Description . . . 236
4.88.2 Constructor & Destructor Documentation . . . 236
4.88.2.1 pseudo_random_generator() . . . 236
4.88.3 Member Function Documentation . . . 237
4.88.3.1 get() . . . 237
4.88.3.2 set_seed() . . . 237
4.89 lava::queue Struct Reference . . . 237
4.89.1 Detailed Description . . . 238
4.89.2 Member Function Documentation . . . 238
4.89.2.1 operator<() . . . 238
4.89.2.2 valid() . . . 238
4.90 lava::queue_family_info Struct Reference . . . 239
4.90.1 Detailed Description . . . 239
4.90.2 Member Function Documentation . . . 239
4.90.2.1 add() . . . 239
4.90.2.2 count() . . . 240
4.91 lava::queue_info Struct Reference . . . 240
4.91.1 Detailed Description . . . 240
4.92 lava::random_generator Struct Reference . . . 241
4.92.1 Detailed Description . . . 241
4.92.2 Member Function Documentation . . . 241
4.92.2.1 get() [1/2] . . . 241
4.92.2.2 get() [2/2] . . . 241
4.93 lava::rect Struct Reference . . . 242
4.93.1 Detailed Description . . . 243
4.93.2 Constructor & Destructor Documentation . . . 243
4.93.2.1 rect() [1/3] . . . 243
4.93.2.2 rect() [2/3] . . . 243
4.93.2.3 rect() [3/3] . . . 243
4.93.3 Member Function Documentation . . . 244
4.93.3.1 contains() . . . 244
4.93.3.2 get_end_point() . . . 244
4.93.3.3 get_origin() . . . 245
4.93.3.4 get_size() . . . 245
4.93.3.5 move() . . . 245
4.93.3.6 set_size() . . . 245
4.94 lava::render_pass Struct Reference . . . 246
4.94.1 Detailed Description . . . 247
4.94.2 Constructor & Destructor Documentation . . . 247

4.94.2.1 render_pass() . . . 247
4.94.3 Member Function Documentation . . . 248
4.94.3.1 add() [1/4] . . . 248
4.94.3.2 add() [2/4] . . . 248
4.94.3.3 add() [3/4] . . . 248
4.94.3.4 add() [4/4] . . . 248
4.94.3.5 add_front() . . . 249
4.94.3.6 create() . . . 249
4.94.3.7 exists_subpass() . . . 249
4.94.3.8 get() . . . 250
4.94.3.9 get_clear_color() . . . 250
4.94.3.10 get_clear_values() . . . 250
4.94.3.11 get_device() . . . 250
4.94.3.12 get_subpass() . . . 250
4.94.3.13 get_subpass_count() . . . 251
4.94.3.14 get_subpasses() . . . 251
4.94.3.15 get_target_callback() . . . 251
4.94.3.16 make() . . . 251
4.94.3.17 process() . . . 252
4.94.3.18 remove() . . . 252
4.94.3.19 set_clear_color() . . . 252
4.94.3.20 set_clear_values() . . . 253
4.95 lava::render_pipeline Struct Reference . . . 253
4.95.1 Detailed Description . . . 256
4.95.2 Constructor & Destructor Documentation . . . 256
4.95.2.1 render_pipeline() . . . 256
4.95.3 Member Function Documentation . . . 256
4.95.3.1 add() . . . 257
4.95.3.2 add_color_blend_attachment() . . . 257
4.95.3.3 add_dynamic_state() . . . 257
4.95.3.4 add_shader() . . . 257
4.95.3.5 add_shader_stage() . . . 258
4.95.3.6 auto_line_width() . . . 258
4.95.3.7 auto_sizing() . . . 258
4.95.3.8 bind() . . . 259
4.95.3.9 copy_from() . . . 259
4.95.3.10 copy_to() . . . 259
4.95.3.11 create() . . . 259
4.95.3.12 get_line_width() . . . 260
4.95.3.13 get_render_pass() . . . 260
4.95.3.14 get_scissor() . . . 260
4.95.3.15 get_shader_stages() . . . 261

4.95.3.16 get_sizing() . . . 261
4.95.3.17 get_subpass() . . . 261
4.95.3.18 get_viewport() . . . 261
4.95.3.19 make() . . . 261
4.95.3.20 set() . . . 262
4.95.3.21 set_auto_line_width() . . . 262
4.95.3.22 set_auto_size() . . . 262
4.95.3.23 set_depth_compare_op() . . . 263
4.95.3.24 set_depth_test_and_write() . . . 263
4.95.3.25 set_dynamic_states() . . . 263
4.95.3.26 set_input_topology() . . . 263
4.95.3.27 set_line_width() [1/2] . . . 264
4.95.3.28 set_line_width() [2/2] . . . 264
4.95.3.29 set_rasterization_cull_mode() . . . 264
4.95.3.30 set_rasterization_front_face() . . . 265
4.95.3.31 set_rasterization_polygon_mode() . . . 265
4.95.3.32 set_render_pass() . . . 265
4.95.3.33 set_scissor() . . . 265
4.95.3.34 set_sizing() . . . 266
4.95.3.35 set_subpass() . . . 266
4.95.3.36 set_vertex_input_attribute() . . . 266
4.95.3.37 set_vertex_input_attributes() . . . 267
4.95.3.38 set_vertex_input_binding() . . . 267
4.95.3.39 set_vertex_input_bindings() . . . 267
4.95.3.40 set_viewport() . . . 267
4.95.3.41 set_viewport_and_scissor() . . . 268
4.96 lava::render_target Struct Reference . . . 268
4.96.1 Detailed Description . . . 270
4.96.2 Member Function Documentation . . . 270
4.96.2.1 add_callback() . . . 270
4.96.2.2 create() . . . 270
4.96.2.3 get_backbuffer() . . . 271
4.96.2.4 get_backbuffer_image() . . . 271
4.96.2.5 get_backbuffers() . . . 271
4.96.2.6 get_device() . . . 272
4.96.2.7 get_format() . . . 272
4.96.2.8 get_frame_count() . . . 272
4.96.2.9 get_image() . . . 272
4.96.2.10 get_size() . . . 273
4.96.2.11 get_swapchain() . . . 273
4.96.2.12 make() . . . 273
4.96.2.13 reload_request() . . . 273

4.96.2.14 remove_callback() . . . 273
4.96.2.15 resize() . . . 274
4.97 lava::renderer Struct Reference . . . 274
4.97.1 Detailed Description . . . 275
4.97.2 Member Function Documentation . . . 275
4.97.2.1 begin_frame() . . . 275
4.97.2.2 create() . . . 275
4.97.2.3 end_frame() . . . 276
4.97.2.4 frame() . . . 276
4.97.2.5 get_device() . . . 276
4.97.2.6 get_frame() . . . 277
4.98 lava::driver::result Struct Reference . . . 277
4.98.1 Detailed Description . . . 277
4.99 lava::reversion_wrapper< T > Struct Template Reference . . . 277
4.99.1 Detailed Description . . . 277
4.100 lava::run_time Struct Reference . . . 278
4.100.1 Detailed Description . . . 278
4.101 lava::scoped_label< T > Struct Template Reference . . . 278
4.101.1 Detailed Description . . . 279
4.101.2 Constructor & Destructor Documentation . . . 279
4.101.2.1 scoped_label() . . . 279
4.102 lava::scroll_event Struct Reference . . . 279
4.102.1 Detailed Description . . . 280
4.103 lava::scroll_offset Struct Reference . . . 280
4.103.1 Detailed Description . . . 280
4.104 lava::pipeline::shader_stage Struct Reference . . . 281
4.104.1 Detailed Description . . . 281
4.104.2 Member Function Documentation . . . 281
4.104.2.1 add_specialization_entry() . . . 281
4.104.2.2 create() . . . 282
4.104.2.3 get_create_info() . . . 282
4.104.2.4 make() . . . 282
4.104.2.5 set_stage() . . . 283
4.105 lava::stage Struct Reference . . . 283
4.105.1 Detailed Description . . . 284
4.105.2 Constructor & Destructor Documentation . . . 284
4.105.2.1 stage() . . . 284
4.106 lava::staging Struct Reference . . . 284
4.106.1 Detailed Description . . . 285
4.106.2 Member Function Documentation . . . 285
4.106.2.1 add() . . . 285
4.106.2.2 busy() . . . 285

4.106.2.3 stage() . . . 285
4.107 lava::window::state Struct Reference . . . 286
4.107.1 Detailed Description . . . 287
4.108 lava::subpass Struct Reference . . . 287
4.108.1 Detailed Description . . . 288
4.108.2 Member Function Documentation . . . 288
4.108.2.1 activated() . . . 289
4.108.2.2 add() . . . 289
4.108.2.3 add_front() . . . 289
4.108.2.4 add_preserve_attachment() . . . 289
4.108.2.5 get_description() . . . 290
4.108.2.6 make() . . . 290
4.108.2.7 process() . . . 290
4.108.2.8 remove() . . . 291
4.108.2.9 set() . . . 291
4.108.2.10 set_active() . . . 291
4.108.2.11 set_color_attachment() [1/2] . . . 291
4.108.2.12 set_color_attachment() [2/2] . . . 292
4.108.2.13 set_color_attachments() . . . 292
4.108.2.14 set_depth_stencil_attachment() [1/2] . . . 292
4.108.2.15 set_depth_stencil_attachment() [2/2] . . . 293
4.108.2.16 set_input_attachment() [1/2] . . . 293
4.108.2.17 set_input_attachment() [2/2] . . . 293
4.108.2.18 set_input_attachments() . . . 293
4.108.2.19 set_preserve_attachments() . . . 294
4.108.2.20 set_resolve_attachment() [1/2] . . . 294
4.108.2.21 set_resolve_attachment() [2/2] . . . 294
4.108.2.22 set_resolve_attachments() . . . 295
4.109 lava::subpass_dependency Struct Reference . . . 295
4.109.1 Detailed Description . . . 296
4.109.2 Member Function Documentation . . . 296
4.109.2.1 get_dependency() . . . 296
4.109.2.2 make() . . . 296
4.109.2.3 set_access_mask() . . . 297
4.109.2.4 set_dependency_flags() . . . 297
4.109.2.5 set_dst_access_mask() . . . 297
4.109.2.6 set_dst_stage_mask() . . . 297
4.109.2.7 set_dst_subpass() . . . 298
4.109.2.8 set_src_access_mask() . . . 298
4.109.2.9 set_src_stage_mask() . . . 298
4.109.2.10 set_src_subpass() . . . 299
4.109.2.11 set_stage_mask() . . . 299

4.109.2.12 set_subpass() . . . 299
4.110 lava::surface_format_request Struct Reference . . . 299
4.110.1 Detailed Description . . . 300
4.110.2 Member Data Documentation . . . 300
4.110.2.1 formats . . . 300
4.111 lava::swapchain Struct Reference . . . 300
4.111.1 Detailed Description . . . 301
4.111.2 Member Function Documentation . . . 301
4.111.2.1 add_callback() . . . 302
4.111.2.2 create() . . . 303
4.111.2.3 get() . . . 303
4.111.2.4 get_backbuffer_count() . . . 303
4.111.2.5 get_backbuffers() . . . 304
4.111.2.6 get_color_space() . . . 304
4.111.2.7 get_device() . . . 304
4.111.2.8 get_format() . . . 304
4.111.2.9 get_size() . . . 305
4.111.2.10 reload_request() . . . 305
4.111.2.11 remove_callback() . . . 305
4.111.2.12 resize() . . . 305
4.111.2.13 surface_supported() . . . 306
4.111.2.14 v_sync() . . . 306
4.112 lava::target_callback Struct Reference . . . 306
4.112.1 Detailed Description . . . 307
4.113 lava::telegram Struct Reference . . . 307
4.113.1 Detailed Description . . . 308
4.113.2 Constructor & Destructor Documentation . . . 308
4.113.2.1 telegram() . . . 308
4.113.3 Member Function Documentation . . . 308
4.113.3.1 operator<() . . . 309
4.113.3.2 operator==() . . . 309
4.114 lava::texture Struct Reference . . . 309
4.114.1 Detailed Description . . . 311
4.114.2 Member Function Documentation . . . 311
4.114.2.1 create() . . . 311
4.114.2.2 get_descriptor_info() . . . 311
4.114.2.3 get_format() . . . 312
4.114.2.4 get_image() . . . 312
4.114.2.5 get_size() . . . 312
4.114.2.6 get_type() . . . 312
4.114.2.7 make() . . . 313
4.114.2.8 stage() . . . 313

4.114.2.9 upload() . . . 314
4.115 lava::thread_pool Struct Reference . . . 314
4.115.1 Detailed Description . . . 315
4.115.2 Member Function Documentation . . . 315
4.115.2.1 enqueue() . . . 315
4.115.2.2 setup() . . . 315
4.116 lava::timer Struct Reference . . . 315
4.116.1 Detailed Description . . . 316
4.116.2 Member Function Documentation . . . 316
4.116.2.1 elapsed() . . . 316
4.117 lava::tooltip Struct Reference . . . 316
4.117.1 Detailed Description . . . 317
4.117.2 Constructor & Destructor Documentation . . . 317
4.117.2.1 tooltip() . . . 317
4.118 lava::tooltip_list Struct Reference . . . 317
4.118.1 Detailed Description . . . 318
4.118.2 Member Function Documentation . . . 318
4.118.2.1 add() . . . 318
4.118.2.2 format_string() . . . 318
4.118.2.3 get_list() . . . 319
4.118.2.4 set() . . . 319
4.119 lava::unique_data Struct Reference . . . 319
4.119.1 Detailed Description . . . 320
4.119.2 Constructor & Destructor Documentation . . . 320
4.119.2.1 unique_data() [1/2] . . . 320
4.119.2.2 unique_data() [2/2] . . . 320
4.120 lava::version Struct Reference . . . 321
4.120.1 Detailed Description . . . 321
4.121 lava::vertex Struct Reference . . . 321
4.121.1 Detailed Description . . . 322
4.121.2 Member Function Documentation . . . 322
4.121.2.1 operator==() . . . 322
4.122 lava::vk_result Struct Reference . . . 322
4.122.1 Detailed Description . . . 323
4.122.2 Member Function Documentation . . . 323
4.122.2.1 operator bool() . . . 323
4.123 lava::window Struct Reference . . . 323
4.123.1 Detailed Description . . . 326
4.123.2 Constructor & Destructor Documentation . . . 326
4.123.2.1 window() . . . 326
4.123.3 Member Function Documentation . . . 327
4.123.3.1 assign() . . . 327

4.123.3.2 close_request() . . . 327
4.123.3.3 create() . . . 327
4.123.3.4 create_surface() . . . 328
4.123.3.5 decorated() . . . 328
4.123.3.6 floating() . . . 328
4.123.3.7 focused() . . . 328
4.123.3.8 fullscreen() . . . 329
4.123.3.9 get() . . . 329
4.123.3.10 get_aspect_ratio() . . . 329
4.123.3.11 get_framebuffer_size() [1/2] . . . 329
4.123.3.12 get_framebuffer_size() [2/2] . . . 329
4.123.3.13 get_monitor() . . . 330
4.123.3.14 get_mouse_position() [1/2] . . . 330
4.123.3.15 get_mouse_position() [2/2] . . . 330
4.123.3.16 get_position() . . . 331
4.123.3.17 get_save_name() . . . 331
4.123.3.18 get_size() [1/2] . . . 331
4.123.3.19 get_size() [2/2] . . . 331
4.123.3.20 get_state() . . . 332
4.123.3.21 get_title() . . . 332
4.123.3.22 handle_resize() . . . 332
4.123.3.23 hovered() . . . 333
4.123.3.24 iconified() . . . 333
4.123.3.25 maximized() . . . 333
4.123.3.26 resizable() . . . 333
4.123.3.27 resize_request() . . . 334
4.123.3.28 save_title() . . . 334
4.123.3.29 set_decorated() . . . 334
4.123.3.30 set_floating() . . . 334
4.123.3.31 set_fullscreen() . . . 335
4.123.3.32 set_icon() . . . 335
4.123.3.33 set_mouse_position() . . . 335
4.123.3.34 set_position() . . . 335
4.123.3.35 set_resizable() . . . 336
4.123.3.36 set_save_name() . . . 336
4.123.3.37 set_size() . . . 336
4.123.3.38 set_title() . . . 337
4.123.3.39 show_save_title() . . . 337
4.123.3.40 switch_mode() . . . 337
4.123.3.41 switch_mode_request() . . . 337
4.123.3.42 visible() . . . 338

5 File Documentation 339
5.1 liblava/app.hpp File Reference . . . 339
5.1.1 Detailed Description . . . 339
5.2 app.hpp . . . 339
5.3 liblava/app/app.hpp File Reference . . . 340
5.3.1 Detailed Description . . . 340
5.4 app.hpp . . . 341
5.5 liblava/app/benchmark.hpp File Reference . . . 342
5.5.1 Detailed Description . . . 343
5.5.2 Function Documentation . . . 343
5.5.2.1 benchmark() . . . 343
5.5.2.2 parse() . . . 344
5.5.2.3 write_frames_json() . . . 344
5.6 benchmark.hpp . . . 344
5.7 liblava/app/camera.hpp File Reference . . . 345
5.7.1 Detailed Description . . . 345
5.8 camera.hpp . . . 346
5.9 liblava/app/config.hpp File Reference . . . 347
5.9.1 Detailed Description . . . 348
5.9.2 Function Documentation . . . 348
5.9.2.1 set_window_icon() . . . 348
5.10 config.hpp . . . 348
5.11 liblava/app/forward_shading.hpp File Reference . . . 349
5.11.1 Detailed Description . . . 349
5.12 forward_shading.hpp . . . 350
5.13 liblava/app/icon.hpp File Reference . . . 350
5.13.1 Detailed Description . . . 350
5.14 icon.hpp . . . 351
5.15 liblava/app/imgui.hpp File Reference . . . 364
5.15.1 Detailed Description . . . 365
5.15.2 Function Documentation . . . 365
5.15.2.1 imgui_left_spacing() . . . 365
5.15.2.2 setup_imgui_font() . . . 365
5.15.2.3 setup_imgui_font_icons() . . . 366
5.16 imgui.hpp . . . 366
5.17 liblava/asset.hpp File Reference . . . 369
5.17.1 Detailed Description . . . 369
5.18 asset.hpp . . . 369
5.19 liblava/asset/image_loader.hpp File Reference . . . 369
5.19.1 Detailed Description . . . 370
5.20 image_loader.hpp . . . 370
5.21 liblava/asset/load_mesh.hpp File Reference . . . 371

5.21.1 Detailed Description . . . 371
5.21.2 Function Documentation . . . 371
5.21.2.1 load_mesh() . . . 371
5.22 load_mesh.hpp . . . 372
5.23 liblava/asset/load_texture.hpp File Reference . . . 372
5.23.1 Detailed Description . . . 372
5.23.2 Function Documentation . . . 372
5.23.2.1 create_default_texture() . . . 372
5.23.2.2 load_texture() [1/2] . . . 373
5.23.2.3 load_texture() [2/2] . . . 373
5.24 load_texture.hpp . . . 374
5.25 liblava/asset/write_image.hpp File Reference . . . 374
5.25.1 Detailed Description . . . 374
5.25.2 Function Documentation . . . 375
5.25.2.1 write_image_png() . . . 375
5.26 write_image.hpp . . . 375
5.27 liblava/base.hpp File Reference . . . 375
5.27.1 Detailed Description . . . 376
5.28 base.hpp . . . 376
5.29 liblava/base/base.hpp File Reference . . . 376
5.29.1 Detailed Description . . . 379
5.29.2 Function Documentation . . . 379
5.29.2.1 check() . . . 379
5.29.2.2 failed() . . . 379
5.29.2.3 to_api_version() . . . 380
5.29.2.4 to_string() . . . 380
5.29.2.5 to_version() [1/2] . . . 380
5.29.2.6 to_version() [2/2] . . . 381
5.29.2.7 version_to_string() . . . 381
5.30 base.hpp . . . 381
5.31 liblava/base/debug_utils.hpp File Reference . . . 383
5.31.1 Detailed Description . . . 386
5.31.2 Function Documentation . . . 386
5.31.2.1 begin_label() [1/2] . . . 386
5.31.2.2 begin_label() [2/2] . . . 387
5.31.2.3 end_label() [1/2] . . . 387
5.31.2.4 end_label() [2/2] . . . 387
5.31.2.5 insert_label() [1/2] . . . 387
5.31.2.6 insert_label() [2/2] . . . 388
5.31.2.7 set_acceleration_structure_name() . . . 388
5.31.2.8 set_acceleration_structure_nv_name() . . . 388
5.31.2.9 set_acceleration_structure_nv_tag() . . . 388

5.31.2.10 set_acceleration_structure_tag() . . . 389
5.31.2.11 set_buffer_name() . . . 389
5.31.2.12 set_buffer_tag() . . . 389
5.31.2.13 set_buffer_view_name() . . . 389
5.31.2.14 set_buffer_view_tag() . . . 390
5.31.2.15 set_command_buffer_name() . . . 390
5.31.2.16 set_command_buffer_tag() . . . 390
5.31.2.17 set_command_pool_name() . . . 390
5.31.2.18 set_command_pool_tag() . . . 391
5.31.2.19 set_debug_report_callback_name() . . . 391
5.31.2.20 set_debug_report_callback_tag() . . . 391
5.31.2.21 set_debug_utils_messenger_name() . . . 391
5.31.2.22 set_debug_utils_messenger_tag() . . . 392
5.31.2.23 set_deferred_operation_name() . . . 392
5.31.2.24 set_deferred_operation_tag() . . . 392
5.31.2.25 set_descriptor_pool_name() . . . 392
5.31.2.26 set_descriptor_pool_tag() . . . 393
5.31.2.27 set_descriptor_set_layout_name() . . . 393
5.31.2.28 set_descriptor_set_layout_tag() . . . 393
5.31.2.29 set_descriptor_set_name() . . . 393
5.31.2.30 set_descriptor_set_tag() . . . 394
5.31.2.31 set_descriptor_update_template_name() . . . 394
5.31.2.32 set_descriptor_update_template_tag() . . . 394
5.31.2.33 set_device_memory_name() . . . 394
5.31.2.34 set_device_memory_tag() . . . 395
5.31.2.35 set_device_name() . . . 395
5.31.2.36 set_device_tag() . . . 395
5.31.2.37 set_display_mode_name() . . . 395
5.31.2.38 set_display_mode_tag() . . . 396
5.31.2.39 set_display_name() . . . 396
5.31.2.40 set_display_tag() . . . 396
5.31.2.41 set_event_name() . . . 396
5.31.2.42 set_event_tag() . . . 397
5.31.2.43 set_fence_name() . . . 397
5.31.2.44 set_fence_tag() . . . 397
5.31.2.45 set_framebuffer_name() . . . 397
5.31.2.46 set_framebuffer_tag() . . . 398
5.31.2.47 set_image_name() . . . 398
5.31.2.48 set_image_tag() . . . 398
5.31.2.49 set_image_view_name() . . . 398
5.31.2.50 set_image_view_tag() . . . 399
5.31.2.51 set_indirect_commands_layout_name() . . . 399

5.31.2.52 set_indirect_commands_layout_tag() . . . 399
5.31.2.53 set_instance_name() . . . 399
5.31.2.54 set_instance_tag() . . . 400
5.31.2.55 set_name() . . . 400
5.31.2.56 set_object_name() . . . 400
5.31.2.57 set_object_tag() . . . 401
5.31.2.58 set_performance_configuration_name() . . . 401
5.31.2.59 set_performance_configuration_tag() . . . 401
5.31.2.60 set_physical_device_name() . . . 401
5.31.2.61 set_physical_device_tag() . . . 402
5.31.2.62 set_pipeline_cache_name() . . . 402
5.31.2.63 set_pipeline_cache_tag() . . . 402
5.31.2.64 set_pipeline_layout_name() . . . 402
5.31.2.65 set_pipeline_layout_tag() . . . 403
5.31.2.66 set_pipeline_name() . . . 403
5.31.2.67 set_pipeline_tag() . . . 403
5.31.2.68 set_private_data_slot_name() . . . 403
5.31.2.69 set_private_data_slot_tag() . . . 404
5.31.2.70 set_query_pool_name() . . . 404
5.31.2.71 set_query_pool_tag() . . . 404
5.31.2.72 set_queue_name() . . . 404
5.31.2.73 set_queue_tag() . . . 405
5.31.2.74 set_render_pass_name() . . . 405
5.31.2.75 set_render_pass_tag() . . . 405
5.31.2.76 set_sampler_name() . . . 405
5.31.2.77 set_sampler_tag() . . . 406
5.31.2.78 set_sampler_ycbcr_conversion_name() . . . 406
5.31.2.79 set_sampler_ycbcr_conversion_tag() . . . 406
5.31.2.80 set_semaphore_name() . . . 406
5.31.2.81 set_semaphore_tag() . . . 407
5.31.2.82 set_shader_module_name() . . . 407
5.31.2.83 set_shader_module_tag() . . . 407
5.31.2.84 set_surface_name() . . . 407
5.31.2.85 set_surface_tag() . . . 408
5.31.2.86 set_swapchain_name() . . . 408
5.31.2.87 set_swapchain_tag() . . . 408
5.31.2.88 set_tag() . . . 408
5.31.2.89 set_validation_cache_name() . . . 409
5.31.2.90 set_validation_cache_tag() . . . 409
5.32 debug_utils.hpp . . . 409
5.33 liblava/base/device.hpp File Reference . . . 422
5.33.1 Detailed Description . . . 423

5.33.2 Function Documentation . . . 423
5.33.2.1 create_shader_module() . . . 423
5.33.2.2 one_time_submit() . . . 423
5.33.2.3 one_time_submit_pool() . . . 424
5.34 device.hpp . . . 424
5.35 liblava/base/device_table.hpp File Reference . . . 427
5.35.1 Detailed Description . . . 427
5.36 device_table.hpp . . . 427
5.37 liblava/base/instance.hpp File Reference . . . 431
5.37.1 Detailed Description . . . 432
5.37.2 Function Documentation . . . 432
5.37.2.1 check() . . . 432
5.37.2.2 enumerate_extension_properties() . . . 432
5.37.2.3 enumerate_layer_properties() . . . 434
5.37.2.4 get_instance_version() . . . 434
5.38 instance.hpp . . . 434
5.39 liblava/base/memory.hpp File Reference . . . 436
5.39.1 Detailed Description . . . 436
5.39.2 Function Documentation . . . 436
5.39.2.1 create_allocator() . . . 436
5.39.2.2 find_memory_type() . . . 437
5.39.2.3 find_memory_type_with_properties() . . . 437
5.40 memory.hpp . . . 438
5.41 liblava/base/physical_device.hpp File Reference . . . 439
5.41.1 Detailed Description . . . 439
5.42 physical_device.hpp . . . 439
5.43 liblava/base/platform.hpp File Reference . . . 440
5.43.1 Detailed Description . . . 441
5.44 platform.hpp . . . 441
5.45 liblava/base/queue.hpp File Reference . . . 441
5.45.1 Detailed Description . . . 442
5.45.2 Function Documentation . . . 443
5.45.2.1 add_dedicated_queues() . . . 443
5.45.2.2 add_queues() . . . 443
5.45.2.3 set_all_queues() . . . 444
5.45.2.4 set_default_queues() . . . 444
5.45.2.5 verify_queues() . . . 444
5.45.3 Variable Documentation . . . 444
5.45.3.1 default_queue_flags . . . 445
5.46 queue.hpp . . . 445
5.47 liblava/block/attachment.hpp File Reference . . . 446
5.47.1 Detailed Description . . . 446

5.48 attachment.hpp . . . 447
5.49 liblava/block.hpp File Reference . . . 448
5.49.1 Detailed Description . . . 448
5.50 block.hpp . . . 448
5.51 liblava/block/block.hpp File Reference . . . 449
5.51.1 Detailed Description . . . 449
5.52 block.hpp . . . 449
5.53 liblava/block/compute_pipeline.hpp File Reference . . . 451
5.53.1 Detailed Description . . . 451
5.54 compute_pipeline.hpp . . . 452
5.55 liblava/block/descriptor.hpp File Reference . . . 452
5.55.1 Detailed Description . . . 453
5.56 descriptor.hpp . . . 453
5.57 liblava/block/pipeline.hpp File Reference . . . 455
5.57.1 Detailed Description . . . 456
5.57.2 Function Documentation . . . 456
5.57.2.1 create_pipeline_shader_stage() . . . 456
5.58 pipeline.hpp . . . 456
5.59 liblava/block/pipeline_layout.hpp File Reference . . . 458
5.59.1 Detailed Description . . . 458
5.60 pipeline_layout.hpp . . . 459
5.61 liblava/block/render_pass.hpp File Reference . . . 460
5.61.1 Detailed Description . . . 460
5.62 render_pass.hpp . . . 460
5.63 liblava/block/render_pipeline.hpp File Reference . . . 462
5.63.1 Detailed Description . . . 462
5.63.2 Function Documentation . . . 463
5.63.2.1 create_pipeline_color_blend_attachment() . . . 463
5.64 render_pipeline.hpp . . . 463
5.65 liblava/block/subpass.hpp File Reference . . . 466
5.65.1 Detailed Description . . . 466
5.66 subpass.hpp . . . 466
5.67 liblava/core.hpp File Reference . . . 468
5.67.1 Detailed Description . . . 469
5.68 core.hpp . . . 469
5.69 liblava/core/data.hpp File Reference . . . 469
5.69.1 Detailed Description . . . 470
5.69.2 Function Documentation . . . 470
5.69.2.1 align() [1/2] . . . 471
5.69.2.2 align() [2/2] . . . 472
5.69.2.3 align_up() . . . 472
5.69.2.4 alloc_data() . . . 473

5.69.2.5 as_ptr() . . . 473
5.69.2.6 free_data() . . . 473
5.69.2.7 human_readable() . . . 474
5.69.2.8 next_pow_2() . . . 474
5.69.2.9 realloc_data() . . . 474
5.70 data.hpp . . . 475
5.71 liblava/core/id.hpp File Reference . . . 477
5.71.1 Detailed Description . . . 478
5.71.2 Function Documentation . . . 478
5.71.2.1 add_id_map() . . . 478
5.71.2.2 remove_id_map() . . . 479
5.71.2.3 to_id() . . . 479
5.72 id.hpp . . . 480
5.73 liblava/core/misc.hpp File Reference . . . 482
5.73.1 Detailed Description . . . 483
5.73.2 Function Documentation . . . 484
5.73.2.1 append() . . . 484
5.73.2.2 begin() . . . 484
5.73.2.3 contains() . . . 485
5.73.2.4 end() . . . 485
5.73.2.5 exists() . . . 485
5.73.2.6 ltrim() . . . 486
5.73.2.7 ltrim_copy() . . . 486
5.73.2.8 remove() . . . 486
5.73.2.9 remove_chars() . . . 488
5.73.2.10 remove_chars_copy() . . . 488
5.73.2.11 remove_chars_if_not() . . . 489
5.73.2.12 remove_chars_if_not_copy() . . . 489
5.73.2.13 remove_nondigit() . . . 489
5.73.2.14 remove_nondigit_copy() . . . 490
5.73.2.15 remove_punctuation_marks() . . . 490
5.73.2.16 reverse() . . . 490
5.73.2.17 rtrim() . . . 491
5.73.2.18 rtrim_copy() . . . 491
5.73.2.19 trim() . . . 491
5.73.2.20 trim_copy() . . . 492
5.74 misc.hpp . . . 492
5.75 liblava/core/time.hpp File Reference . . . 494
5.75.1 Detailed Description . . . 495
5.75.2 Function Documentation . . . 495
5.75.2.1 get_current_time() . . . 495
5.75.2.2 get_current_timestamp() . . . 496

5.75.2.3 get_current_timestamp_ms() . . . 496
5.75.2.4 timestamp() . . . 496
5.75.2.5 to_delta() . . . 497
5.75.2.6 to_dt() . . . 497
5.75.2.7 to_ms() [1/2] . . . 497
5.75.2.8 to_ms() [2/2] . . . 497
5.75.2.9 to_sec() . . . 499
5.76 time.hpp . . . 499
5.77 liblava/core/types.hpp File Reference . . . 501
5.77.1 Detailed Description . . . 504
5.77.2 Macro Definition Documentation . . . 504
5.77.2.1 ENUM_FLAG_OPERATORS . . . 504
5.77.3 Typedef Documentation . . . 505
5.77.3.1 c16 . . . 505
5.77.3.2 c32 . . . 505
5.77.3.3 c8 . . . 505
5.77.3.4 i16 . . . 505
5.77.3.5 i32 . . . 506
5.77.3.6 i64 . . . 506
5.77.3.7 i8 . . . 506
5.77.3.8 uc16 . . . 506
5.77.3.9 uc32 . . . 506
5.77.3.10 uc8 . . . 507
5.77.3.11 ui16 . . . 507
5.77.3.12 ui32 . . . 507
5.77.3.13 ui64 . . . 507
5.77.3.14 ui8 . . . 507
5.77.4 Function Documentation . . . 508
5.77.4.1 hash_combine() . . . 508
5.77.4.2 hash_val() [1/3] . . . 508
5.77.4.3 hash_val() [2/3] . . . 509
5.77.4.4 hash_val() [3/3] . . . 509
5.77.4.5 str() . . . 509
5.77.4.6 to_char() . . . 509
5.77.4.7 to_i32() . . . 510
5.77.4.8 to_i64() . . . 510
5.77.4.9 to_index() . . . 510
5.77.4.10 to_r32() . . . 511
5.77.4.11 to_r64() . . . 511
5.77.4.12 to_size_t() . . . 511
5.77.4.13 to_ui32() . . . 512
5.77.4.14 to_ui64() . . . 512

5.78 types.hpp . . . 512
5.79 liblava/core/version.hpp File Reference . . . 515
5.79.1 Detailed Description . . . 516
5.79.2 Function Documentation . . . 516
5.79.2.1 to_version() . . . 516
5.80 version.hpp . . . 517
5.81 liblava/engine.hpp File Reference . . . 517
5.81.1 Detailed Description . . . 518
5.82 engine.hpp . . . 518
5.83 liblava/engine/engine.hpp File Reference . . . 518
5.83.1 Detailed Description . . . 518
5.84 engine.hpp . . . 519
5.85 liblava/engine/producer.hpp File Reference . . . 519
5.85.1 Detailed Description . . . 520
5.86 producer.hpp . . . 520
5.87 liblava/engine/property.hpp File Reference . . . 521
5.87.1 Detailed Description . . . 521
5.88 property.hpp . . . 522
5.89 liblava/file.hpp File Reference . . . 523
5.89.1 Detailed Description . . . 523
5.90 file.hpp . . . 523
5.91 liblava/file/file.hpp File Reference . . . 523
5.91.1 Detailed Description . . . 524
5.91.2 Function Documentation . . . 524
5.91.2.1 file_error() . . . 524
5.92 file.hpp . . . 525
5.93 liblava/file/file_system.hpp File Reference . . . 526
5.93.1 Detailed Description . . . 526
5.94 file_system.hpp . . . 526
5.95 liblava/file/file_utils.hpp File Reference . . . 527
5.95.1 Detailed Description . . . 528
5.95.2 Function Documentation . . . 528
5.95.2.1 extension() [1/2] . . . 528
5.95.2.2 extension() [2/2] . . . 529
5.95.2.3 get_filename_from() . . . 529
5.95.2.4 load_file_data() . . . 529
5.95.2.5 read_file() . . . 530
5.95.2.6 remove_existing_path() . . . 530
5.95.2.7 write_file() . . . 531
5.96 file_utils.hpp . . . 531
5.97 json.hpp . . . 532
5.98 liblava/file/json_file.hpp File Reference . . . 532

5.98.1 Detailed Description . . . 532
5.99 json_file.hpp . . . 533
5.100 liblava/frame.hpp File Reference . . . 533
5.100.1 Detailed Description . . . 534
5.101 frame.hpp . . . 534
5.102 liblava/frame/frame.hpp File Reference . . . 534
5.102.1 Detailed Description . . . 535
5.102.2 Function Documentation . . . 535
5.102.2.1 handle_events() [1/3] . . . 535
5.102.2.2 handle_events() [2/3] . . . 536
5.102.2.3 handle_events() [3/3] . . . 536
5.102.2.4 now() . . . 536
5.103 frame.hpp . . . 536
5.104 liblava/frame/argh.hpp File Reference . . . 538
5.104.1 Detailed Description . . . 539
5.104.2 Function Documentation . . . 539
5.104.2.1 log_command_line() . . . 539
5.105 argh.hpp . . . 539
5.106 liblava/frame/driver.hpp File Reference . . . 540
5.106.1 Detailed Description . . . 541
5.106.2 Macro Definition Documentation . . . 541
5.106.2.1 LAVA_STAGE . . . 541
5.107 driver.hpp . . . 541
5.108 liblava/frame/gamepad.hpp File Reference . . . 542
5.108.1 Detailed Description . . . 543
5.108.2 Function Documentation . . . 543
5.108.2.1 gamepads() . . . 543
5.109 gamepad.hpp . . . 544
5.110 liblava/frame/input.hpp File Reference . . . 545
5.110.1 Detailed Description . . . 548
5.110.2 Function Documentation . . . 548
5.110.2.1 check_mod() . . . 548
5.110.2.2 get_scancode() . . . 549
5.110.2.3 to_string() [1/2] . . . 549
5.110.2.4 to_string() [2/2] . . . 549
5.111 input.hpp . . . 550
5.112 liblava/frame/render_target.hpp File Reference . . . 555
5.112.1 Detailed Description . . . 556
5.112.2 Function Documentation . . . 556
5.112.2.1 create_target() . . . 556
5.112.2.2 create_target_v_sync() . . . 557
5.113 render_target.hpp . . . 557

5.114 liblava/frame/renderer.hpp File Reference . . . 559
5.114.1 Detailed Description . . . 559
5.115 renderer.hpp . . . 560
5.116 liblava/frame/swapchain.hpp File Reference . . . 560
5.116.1 Detailed Description . . . 561
5.117 swapchain.hpp . . . 561
5.118 liblava/frame/window.hpp File Reference . . . 562
5.118.1 Detailed Description . . . 563
5.118.2 Function Documentation . . . 563
5.118.2.1 create_surface() . . . 563
5.118.2.2 get_window() . . . 564
5.119 window.hpp . . . 564
5.120 liblava/lava.hpp File Reference . . . 567
5.120.1 Detailed Description . . . 567
5.121 lava.hpp . . . 567
5.122 liblava/resource.hpp File Reference . . . 568
5.122.1 Detailed Description . . . 568
5.123 resource.hpp . . . 568
5.124 liblava/resource/buffer.hpp File Reference . . . 568
5.124.1 Detailed Description . . . 569
5.124.2 Function Documentation . . . 569
5.124.2.1 buffer_usage_to_possible_access() . . . 569
5.124.2.2 buffer_usage_to_possible_stages() . . . 569
5.125 buffer.hpp . . . 570
5.126 liblava/resource/format.hpp File Reference . . . 571
5.126.1 Detailed Description . . . 572
5.126.2 Function Documentation . . . 573
5.126.2.1 format_align_dim() . . . 573
5.126.2.2 format_aspect_mask() . . . 573
5.126.2.3 format_bgr() . . . 573
5.126.2.4 format_block_dim() . . . 574
5.126.2.5 format_block_size() . . . 574
5.126.2.6 format_depth() . . . 574
5.126.2.7 format_depth_stencil() . . . 575
5.126.2.8 format_num_blocks() . . . 575
5.126.2.9 format_srgb() . . . 575
5.126.2.10 format_stencil() . . . 576
5.126.2.11 get_supported_depth_format() . . . 576
5.126.2.12 get_supported_format() . . . 577
5.126.2.13 get_surface_format() . . . 577
5.126.2.14 image_memory_barrier() . . . 577
5.126.2.15 insert_image_memory_barrier() . . . 578

5.126.2.16 set_image_layout() [1/2] . . . 578
5.126.2.17 set_image_layout() [2/2] . . . 579
5.126.2.18 support_blit() . . . 579
5.126.2.19 support_vertex_buffer_format() . . . 580
5.127 format.hpp . . . 580
5.128 liblava/resource/image.hpp File Reference . . . 582
5.128.1 Detailed Description . . . 582
5.128.2 Function Documentation . . . 582
5.128.2.1 create_image() . . . 582
5.128.2.2 grab_image() . . . 583
5.129 image.hpp . . . 583
5.130 liblava/resource/mesh.hpp File Reference . . . 585
5.130.1 Detailed Description . . . 586
5.130.2 Function Documentation . . . 586
5.130.2.1 create_mesh() . . . 586
5.130.2.2 create_mesh_data() . . . 587
5.130.2.3 make_primitive_indices_cube() . . . 588
5.130.2.4 make_primitive_normals_cube() . . . 588
5.130.2.5 make_primitive_positions_cube() . . . 588
5.130.2.6 make_primitive_uvs_cube() . . . 589
5.131 mesh.hpp . . . 589
5.132 liblava/resource/primitive.hpp File Reference . . . 596
5.132.1 Detailed Description . . . 597
5.133 primitive.hpp . . . 597
5.134 liblava/resource/texture.hpp File Reference . . . 598
5.134.1 Detailed Description . . . 598
5.135 texture.hpp . . . 599
5.136 liblava/test.hpp File Reference . . . 600
5.136.1 Detailed Description . . . 600
5.137 test.hpp . . . 601
5.138 liblava/util.hpp File Reference . . . 601
5.138.1 Detailed Description . . . 601
5.139 util.hpp . . . 601
5.140 liblava/util/hex.hpp File Reference . . . 601
5.140.1 Detailed Description . . . 604
5.140.2 Function Documentation . . . 604
5.140.2.1 hex_calculate_inner_radius() . . . 604
5.140.2.2 hex_cell_from_pair() . . . 605
5.140.2.3 hex_corner_offset() . . . 605
5.140.2.4 hex_diagonal() . . . 605
5.140.2.5 hex_diagonal_neighbor() . . . 606
5.140.2.6 hex_direction() . . . 606

5.140.2.7 hex_distance() . . . 607
5.140.2.8 hex_get() . . . 607
5.140.2.9 hex_get_corner() . . . 607
5.140.2.10 hex_get_s() . . . 608
5.140.2.11 hex_is_valid() . . . 608
5.140.2.12 hex_length() . . . 608
5.140.2.13 hex_lerp() . . . 609
5.140.2.14 hex_line() . . . 609
5.140.2.15 hex_neighbor() . . . 610
5.140.2.16 hex_opposite() . . . 610
5.140.2.17 hex_pixel_to_cell() . . . 610
5.140.2.18 hex_polygon_corners() . . . 611
5.140.2.19 hex_q_doubled_from_cube() . . . 611
5.140.2.20 hex_q_doubled_to_cube() . . . 611
5.140.2.21 hex_q_offset_from_cube() . . . 612
5.140.2.22 hex_q_offset_to_cube() . . . 612
5.140.2.23 hex_r_doubled_from_cube() . . . 613
5.140.2.24 hex_r_doubled_to_cube() . . . 613
5.140.2.25 hex_r_offset_from_cube() . . . 613
5.140.2.26 hex_r_offset_to_cube() . . . 614
5.140.2.27 hex_round() . . . 614
5.140.2.28 hex_to_pixel() . . . 614
5.140.2.29 to_string() . . . 615
5.140.3 Variable Documentation . . . 615
5.140.3.1 hex_cardinal_directions . . . 615
5.140.3.2 hex_diagonals . . . 616
5.140.3.3 hex_directions . . . 616
5.140.3.4 hex_layout_flat . . . 616
5.140.3.5 hex_layout_point_y . . . 617
5.141 hex.hpp . . . 617
5.142 liblava/util/log.hpp File Reference . . . 622
5.142.1 Detailed Description . . . 623
5.142.2 Function Documentation . . . 623
5.142.2.1 int_version_string() . . . 624
5.142.2.2 internal_version_string() . . . 624
5.142.2.3 log() . . . 624
5.142.2.4 setup_log() . . . 624
5.142.2.5 teardown_log() . . . 625
5.142.2.6 to_string() [1/4] . . . 625
5.142.2.7 to_string() [2/4] . . . 625
5.142.2.8 to_string() [3/4] . . . 626
5.142.2.9 to_string() [4/4] . . . 626

5.142.2.10 version_string() . . . 626
5.143 log.hpp . . . 627
5.144 liblava/util/math.hpp File Reference . . . 628
5.144.1 Detailed Description . . . 629
5.144.2 Function Documentation . . . 629
5.144.2.1 ceil_div() . . . 629
5.144.2.2 hash256() . . . 630
5.144.2.3 perspective_matrix() . . . 630
5.144.3 Variable Documentation . . . 631
5.144.3.1 default_color . . . 631
5.145 math.hpp . . . 631
5.146 liblava/util/random.hpp File Reference . . . 632
5.146.1 Detailed Description . . . 633
5.146.2 Function Documentation . . . 633
5.146.2.1 random() [1/2] . . . 633
5.146.2.2 random() [2/2] . . . 634
5.147 random.hpp . . . 634
5.148 liblava/util/telegram.hpp File Reference . . . 635
5.148.1 Detailed Description . . . 635
5.149 telegram.hpp . . . 636
5.150 liblava/util/thread.hpp File Reference . . . 637
5.150.1 Detailed Description . . . 638
5.150.2 Function Documentation . . . 638
5.150.2.1 sleep() [1/2] . . . 638
5.150.2.2 sleep() [2/2] . . . 638
5.151 thread.hpp . . . 638

Index **641**

Chapter 1

Hierarchical Index

1.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

lava::allocator 15
lava::attachment 24
lava::benchmark_data 29
lava::descriptor::binding 30
lava::json_file::callback 45
lava::swapchain::callback 46
lava::cdata 53
lava::imgui::config 60
lava::render_pipeline::create_info 62
lava::device::create_param 63
lava::instance::create_param 65
lava::data 66
  lava::unique_data 319
    lava::file_data 109
lava::data_provider 68
lava::instance::debug_config 69
lava::device_table 86
  lava::device 77
lava::dispatcher 97
lava::driver 99
lava::file_format 111
lava::imgui::font 118
lava::forward_shading 119
lava::frame_env 127
lava::gamepad 128
lava::gamepad_manager 131
lava::global_logger 133
lava::hex_cell 134
lava::hex_fractional_cell 136
lava::hex_grid 137
lava::hex_layout 139
lava::hex_offset_coord 139
lava::hex_orientation 140
lava::hex_point 140
lava::imgui::icon_font 141

lava::id . . . 142
lava::id_listeners< T > . . . 143
lava::id_listeners< key_event > . . . 143
lava::id_listeners< mouse_active_event > . . . 143
lava::id_listeners< mouse_button_event > . . . 143
lava::id_listeners< mouse_move_event > . . . 143
lava::id_listeners< path_drop_event > . . . 143
lava::id_listeners< scroll_event > . . . 143
lava::id_registry< T, Meta > . . . 145
lava::id_registry< lava::mesh_template, string > . . . 145
lava::id_registry< lava::texture, string > . . . 145
lava::ids . . . 149
lava::image_loader . . . 158
lava::imgui . . . 161
lava::input . . . 167
lava::input_callback . . . 169
lava::instance_info . . . 174
lava::interface . . . 175
- lava::configurable . . . 61
  - lava::app_config . . . 22
  - lava::property . . . 229
- lava::entity . . . 103
  - lava::block . . . 33
  - lava::buffer . . . 40
  - lava::camera . . . 47
  - lava::command . . . 55
  - lava::descriptor . . . 70
  - lava::descriptor::pool . . . 221
  - lava::device . . . 77
  - lava::image . . . 150
  - lava::mesh_template< T > . . . 186
  - lava::physical_device . . . 201
  - lava::pipeline . . . 208
    - lava::compute_pipeline . . . 57
    - lava::render_pipeline . . . 253
  - lava::pipeline_layout . . . 214
  - lava::render_pass . . . 246
  - lava::render_target . . . 268
  - lava::renderer . . . 274
  - lava::subpass . . . 287
  - lava::swapchain . . . 300
  - lava::texture . . . 309
  - lava::window . . . 323
- lava::frame . . . 121
  - lava::app . . . 17
    - lava::engine . . . 102

lava::internal_version . . . 176
lava::json_file . . . 176
lava::key_event . . . 179
lava::texture::layer . . . 182
T::list
- lava::input_events< key_event > . . . 170
- lava::input_events< scroll_event > . . . 170
- lava::input_events< mouse_move_event > . . . 170
- lava::input_events< mouse_button_event > . . . 170
- lava::input_events< mouse_active_event > . . . 170
- lava::input_events< path_drop_event > . . . 170
- lava::input_events< T > . . . 170

lava::log_config . . . 183
lava::mesh_meta . . . 185
lava::mesh_template_data< T > . . . 192
lava::mesh_template_data< vertex > . . . 192
lava::texture::mip_level . . . 194
lava::mouse_active_event . . . 195
lava::mouse_button_event . . . 196
lava::mouse_move_event . . . 197
lava::mouse_position . . . 198
lava::no_copy_no_move . . . 199
- lava::entity . . . 103
- lava::file . . . 104
- lava::file_delete . . . 110
- lava::file_system . . . 112
- lava::frame . . . 121
- lava::instance . . . 171
- lava::memory . . . 183

lava::pair_hash . . . 200
lava::path_drop_event . . . 201
lava::platform . . . 218
lava::producer . . . 224
lava::property::prop . . . 228
lava::pseudo_random_generator . . . 236
lava::queue . . . 237
lava::queue_family_info . . . 239
lava::queue_info . . . 240
lava::random_generator . . . 241
lava::rect . . . 242
lava::driver::result . . . 277
lava::reversion_wrapper< T > . . . 277
lava::run_time . . . 278
lava::scoped_label< T > . . . 278
lava::scroll_event . . . 279
lava::scroll_offset . . . 280
lava::pipeline::shader_stage . . . 281
lava::stage . . . 283
lava::staging . . . 284
lava::window::state . . . 286
lava::subpass_dependency . . . 295
lava::surface_format_request . . . 299
lava::target_callback . . . 306
lava::telegram . . . 307
lava::thread_pool . . . 314
lava::timer . . . 315
lava::tooltip . . . 316
lava::tooltip_list . . . 317
lava::version . . . 321
lava::vertex . . . 321
lava::vk_result . . . 322

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

lava::allocator
Vulkan allocator 15
lava::app
Application with basic functionality 17
lava::app_config
Application configuration 22
lava::attachment
Attachment description 24
lava::benchmark_data
Benchmark data 29
lava::descriptor::binding
Descriptor binding 30
lava::block
Block of commands 33
lava::buffer
Buffer 40
lava::json_file::callback
Json file callback 45
lava::swapchain::callback
Swapchain callback 46
lava::camera
First Person / Look At camera 47
lava::cdata
Const data wrapper 53
lava::command
Block command 55
lava::compute_pipeline
Compute pipeline 57
lava::imgui::config
ImGui configuration 60
lava::configurable
Configurable interface 61
lava::render_pipeline::create_info
Render pipeline create information 62
lava::device::create_param
Device create parameters 63

lava::instance::create_param
Instance create parameters . . . 65
lava::data
Data wrapper . . . 66
lava::data_provider
Data provider . . . 68
lava::instance::debug_config
Debug configuration . . . 69
lava::descriptor
Descriptor . . . 70
lava::device
Vulkan device . . . 77
lava::device_table
Device functions . . . 86
lava::dispatcher
Telegram dispatcher . . . 97
lava::driver
Stage driver . . . 99
lava::engine
Engine . . . 102
lava::entity
Entity . . . 103
lava::file
File . . . 104
lava::file_data
File data . . . 109
lava::file_delete
File delete guard . . . 110
lava::file_format
File format . . . 111
lava::file_system
File system . . . 112
lava::imgui::font
ImGui font settings . . . 118
lava::forward_shading
Forward shading . . . 119
lava::frame
Framework . . . 121
lava::frame_env
Framework environment . . . 127
lava::gamepad
Gamepad . . . 128
lava::gamepad_manager
Gamepad manager . . . 131
lava::global_logger
Global logger . . . 133
lava::hex_cell
Hex cell . . . 134
lava::hex_fractional_cell
Hex fractional cell . . . 136
lava::hex_grid
Hex grid . . . 137
lava::hex_layout
Hex layout . . . 139
lava::hex_offset_coord
Hex offset coordinates . . . 139
lava::hex_orientation
Hex orientation . . . 140

lava::hex_point
Hex point 140
lava::imgui::icon_font
ImGui icon font settings 141
lava::id
Identification 142
lava::id_listeners< T >
Id listeners 143
lava::id_registry< T, Meta >
Id registry 145
lava::ids
Id factory 149
lava::image
Image 150
lava::image_loader
Load image data from file and memory 158
lava::imgui
ImGui integration 161
lava::input
Input handling 167
lava::input_callback
Input callback 169
lava::input_events< T >
List of input events 170
lava::instance
Vulkan instance 171
lava::instance_info
Vulkan instance information 174
lava::interface
Interface 175
lava::internal_version
Internal version 176
lava::json_file
Json file 176
lava::key_event
Key event 179
lava::texture::layer
Texture layer 182
lava::log_config
Log configuration 183
lava::memory
Vulkan memory 183
lava::mesh_meta
Mesh meta 185
lava::mesh_template< T >
Temporary templated mesh 186
lava::mesh_template_data< T >
Templated mesh data 192
lava::texture::mip_level
Texture mip level 194
lava::mouse_active_event
Mouse active event 195
lava::mouse_button_event
Mouse button event 196
lava::mouse_move_event
Mouse move event 197
lava::mouse_position
Input mouse position 198

lava::no_copy_no_move
No copy and no move object . . . 199
lava::pair_hash
Pair hash . . . 200
lava::path_drop_event
Path drop event . . . 201
lava::physical_device
Vulkan physical device . . . 201
lava::pipeline
Pipeline . . . 208
lava::pipeline_layout
Pipeline layout . . . 214
lava::platform
Stage platform . . . 218
lava::descriptor::pool
Descriptor pool . . . 221
lava::producer
Producer . . . 224
lava::property::prop
Prop . . . 228
lava::property
Props master . . . 229
lava::pseudo_random_generator
Pseudo random generator . . . 236
lava::queue
Device queue . . . 237
lava::queue_family_info
Queue family information . . . 239
lava::queue_info
Queue information . . . 240
lava::random_generator
Random generator . . . 241
lava::rect
Rectangle . . . 242
lava::render_pass
Render pass . . . 246
lava::render_pipeline
Render pipeline (Graphics) . . . 253
lava::render_target
Render target . . . 268
lava::renderer
Plain renderer . . . 274
lava::driver::result
Driver result . . . 277
lava::reversion_wrapper< T >
Reversion Wrapper . . . 277
lava::run_time
Run time . . . 278
lava::scoped_label< T >
Scoped debug util label . . . 278
lava::scroll_event
Scroll event . . . 279
lava::scroll_offset
Input scroll offset . . . 280
lava::pipeline::shader_stage
Shader stage . . . 281
lava::stage
Stage . . . 283

lava::staging
Texture staging 284
lava::window::state
Window state 286
lava::subpass
Subpass 287
lava::subpass_dependency
Subpass dependency 295
lava::surface_format_request
Surface format request 299
lava::swapchain
Swaphchain 300
lava::target_callback
Target callback 306
lava::telegram
Telegram 307
lava::texture
Texture 309
lava::thread_pool
Thread pool 314
lava::timer
Timer 315
lava::tooltip
Tooltip 316
lava::tooltip_list
Tooltip list 317
lava::unique_data
Unique data wrapper 319
lava::version
Version 321
lava::vertex
Vertex 321
lava::vk_result
Vulkan result 322
lava::window
Window 323

Chapter 3

File Index

3.1 File List

Here is a list of all documented files with brief descriptions:

liblava/app.hpp
App module . . . 339
liblava/asset.hpp
Asset module . . . 369
liblava/base.hpp
Base module . . . 375
liblava/block.hpp
Block module . . . 448
liblava/core.hpp
Core module . . . 468
liblava/engine.hpp
Engine module . . . 517
liblava/file.hpp
File module . . . 523
liblava/frame.hpp
Frame module . . . 533
liblava/lava.hpp
All lava modules . . . 567
liblava/resource.hpp
Resource module . . . 568
liblava/test.hpp
Unit tests . . . 600
liblava/util.hpp
Util module . . . 601
liblava/app/app.hpp
Application with basic functionality . . . 340
liblava/app/benchmark.hpp
Benchmark . . . 342
liblava/app/camera.hpp
First Person / Look At camera . . . 345
liblava/app/config.hpp
Application configuration . . . 347
liblava/app/forward_shading.hpp
Forward shading . . . 349
liblava/app/icon.hpp
App default icon data . . . 350

liblava/app/imgui.hpp
ImGui integration . . . 364
liblava/asset/image_loader.hpp
Load image data from file and memory . . . 369
liblava/asset/load_mesh.hpp
Load mesh from file . . . 371
liblava/asset/load_texture.hpp
Load texture from file . . . 372
liblava/asset/write_image.hpp
Write image data to file . . . 374
liblava/base/base.hpp
Vulkan base types . . . 376
liblava/base/debug_utils.hpp
Debug utilities . . . 383
liblava/base/device.hpp
Vulkan device . . . 422
liblava/base/device_table.hpp
Device functions . . . 427
liblava/base/instance.hpp
Vulkan instance . . . 431
liblava/base/memory.hpp
Vulkan allocator . . . 436
liblava/base/physical_device.hpp
Vulkan physical device . . . 439
liblava/base/platform.hpp
Stage platform . . . 440
liblava/base/queue.hpp
Device queue . . . 441
liblava/block/attachment.hpp
Attachment description . . . 446
liblava/block/block.hpp
Command buffer model . . . 449
liblava/block/compute_pipeline.hpp
Compute pipeline . . . 451
liblava/block/descriptor.hpp
Descriptor definition . . . 452
liblava/block/pipeline.hpp
Pipeline . . . 455
liblava/block/pipeline_layout.hpp
Pipeline layout . . . 458
liblava/block/render_pass.hpp
Render pass . . . 460
liblava/block/render_pipeline.hpp
Render pipeline (Graphics) . . . 462
liblava/block/subpass.hpp
Subpass . . . 466
liblava/core/data.hpp
Data wrapper . . . 469
liblava/core/id.hpp
Object Identification . . . 477
liblava/core/misc.hpp
Miscellaneous helpers . . . 482
liblava/core/time.hpp
Run time . . . 494
liblava/core/types.hpp
Basic types . . . 501
liblava/core/version.hpp
Version information . . . 515

liblava/engine/engine.hpp
Engine . . . 518
liblava/engine/producer.hpp
Producer . . . 519
liblava/engine/property.hpp
Props master . . . 521
liblava/file/file.hpp
File access . . . 523
liblava/file/file_system.hpp
File system . . . 526
liblava/file/file_utils.hpp
File utilities . . . 527
liblava/file/json.hpp . . . 532
liblava/file/json_file.hpp
Json file . . . 532
liblava/frame/argh.hpp
Json . . . 538
liblava/frame/driver.hpp
Stage driver . . . 540
liblava/frame/frame.hpp
Framework . . . 534
liblava/frame/gamepad.hpp
Gamepad manager . . . 542
liblava/frame/input.hpp
Input handling . . . 545
liblava/frame/render_target.hpp
Render target . . . 555
liblava/frame/renderer.hpp
Plain renderer . . . 559
liblava/frame/swapchain.hpp
Swapchain . . . 560
liblava/frame/window.hpp
Window . . . 562
liblava/resource/buffer.hpp
Vulkan buffer . . . 568
liblava/resource/format.hpp
Vulkan format . . . 571
liblava/resource/image.hpp
Vulkan image . . . 582
liblava/resource/mesh.hpp
Vulkan mesh . . . 585
liblava/resource/primitive.hpp
Vulkan primitives . . . 596
liblava/resource/texture.hpp
Vulkan texture . . . 598
liblava/util/hex.hpp
Hex point, cell and grid . . . 601
liblava/util/log.hpp
Logging . . . 622
liblava/util/math.hpp
Math helpers . . . 628
liblava/util/random.hpp
Random generator . . . 632
liblava/util/telegram.hpp
Telegram dispatcher . . . 635
liblava/util/thread.hpp
Thread pool . . . 637

Chapter 4

Class Documentation

4.1 lava::allocator Struct Reference

Vulkan allocator.

```
#include <memory.hpp>
```

Public Types

- using **ptr** = std::shared_ptr< allocator >

  Shared pointer to a allocator.

Public Member Functions

- **allocator** ()=default

  Construct a new allocator.
- allocator (VmaAllocator allocator)

  Construct a new allocator.
- bool create (device_cptr device, VmaAllocatorCreateFlags flags=0)

  Create a new allocator.
- void **destroy** ()

  Destroy the allocator.
- bool valid () const

  Check if allocator is valid.
- VmaAllocator get () const

  Get the VMA allocator.

Static Public Member Functions

- static ptr make ()

  Make a new allocator.

4.1.1 Detailed Description

Vulkan allocator.

4.1.2 Constructor & Destructor Documentation

4.1.2.1 allocator()

```
lava::allocator::allocator (
            VmaAllocator allocator )  [inline], [explicit]
```

Construct a new allocator.

Parameters

allocator	VMA allocator

4.1.3 Member Function Documentation

4.1.3.1 create()

```
bool lava::allocator::create (
            device_cptr device,
            VmaAllocatorCreateFlags flags = 0 )
```

Create a new allocator.

Parameters

device	Vulkan device
flags	VMA allocator create flags

Returns

Create was successful or failed

4.1.3.2 get()

```
VmaAllocator lava::allocator::get ( ) const  [inline]
```

Get the VMA allocator.

Returns

VmaAllocator VMA allocator

4.1.3.3 make()

```
static ptr lava::allocator::make ( ) [inline], [static]
```

Make a new allocator.

Returns

ptr Shared pointer to allocator

4.1.3.4 valid()

```
bool lava::allocator::valid ( ) const [inline]
```

Check if allocator is valid.

Returns

Allocator is valid or not

The documentation for this struct was generated from the following file:

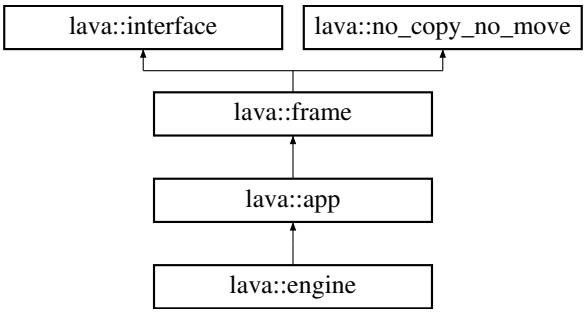
- liblava/base/memory.hpp

4.2 lava::app Struct Reference

Application with basic functionality.

```
#include <app.hpp>
```

Inheritance diagram for lava::app:



Public Types

- using **update_func** = std::function< bool(delta)>

  Update function.
- using **create_func** = std::function< bool()>

  Create function.
- using **destroy_func** = std::function< void()>

  Destroy function.
- using **process_func** = std::function< void(VkCommandBuffer, index)>

  Process function.
- using **setup_func** = std::function< bool()>

  Set up function.

Public Member Functions

- app (frame_env::ref env)

  Construct a new app.
- app (name name, argh::parser cmd_line={})

  Construct a new app.
- virtual bool setup ()

  Set up the application.
- bool v_sync () const

  V-Sync setting.
- ui32 get_frame_counter () const

  Get the frame counter.
- void draw_about (bool separator=draw_separator, bool fps=draw_fps, bool spacing=draw_spacing) const

  Draw about information.
- id::ref block_cmd () const

  Get id of the block command.
- string screenshot ()

  Take screenshot and save it to file.
- void parse_config (argh::parser cmd_line)

  Parse config.

Public Attributes

- window_t **window**

  Main window.
- input_t **input**

  Window input.
- imgui_t **imgui**

  ImGui handling.
- imgui::config **imgui_config**

  ImGui configuration.
- tooltip_list **tooltips**

  Tooltip list.
- device_p **device** = nullptr

  Vulkan device.
- camera_t **camera**

Main camera.

- gamepad **pad**

  Gamepad.
- staging_t **staging**

  Texture staging.
- block_t **block**

  Basic block.
- renderer_t **renderer**

  Plain renderer.
- forward_shading **shading**

  Forward shading.
- render_target::ptr **target**

  Render target.
- run_time_t **run_time**

  Run time.
- file_system **fs**

  File system.
- VkPipelineCache **pipeline_cache** = nullptr

  Pipeline cache.
- update_func **on_update**

  Function called on application update.
- create_func **on_create**

  Function called on application create.
- destroy_func **on_destroy**

  Function called on application destroy.
- app_config **config**

  Application configuration.
- json_file **config_file**

  Configuration file.
- process_func **on_process**

  Function called on application process.
- setup_func **on_setup**

  Function called on application setup.

4.2.1 Detailed Description

Application with basic functionality.

4.2.2 Constructor & Destructor Documentation

4.2.2.1 app() [1/2]

```
lava::app::app (
            frame_env::ref env )  [explicit]
```

Construct a new app.

Parameters

env	Frame environment

4.2.2.2 app() [2/2]

```
lava::app::app (
            name name,
            argh::parser cmd_line = {} )  [explicit]
```

Construct a new app.

Parameters

name	Application name
cmd_line	Command line arguments

4.2.3 Member Function Documentation

4.2.3.1 block_cmd()

```
id::ref lava::app::block_cmd ( ) const  [inline]
```

Get id of the block command.

Returns

id::ref Id to access the command

4.2.3.2 draw_about()

```
void lava::app::draw_about (
            bool separator = draw_separator,
            bool fps = draw_fps,
            bool spacing = draw_spacing ) const
```

Draw about information.

Parameters

separator	Prepend separator
fps	FPS counter
spacing	Spacing

4.2.3.3 get_frame_counter()

```
ui32 lava::app::get_frame_counter ( ) const  [inline]
```

Get the frame counter.

Returns

ui32 Number of rendered frames

4.2.3.4 parse_config()

```
void lava::app::parse_config (
            argh::parser cmd_line )
```

Parse config.

Parameters

cmd_line	Command line arguments

4.2.3.5 screenshot()

```
string lava::app::screenshot ( )
```

Take screenshot and save it to file.

Returns

string Screenshot file path (empty: failed)

4.2.3.6 setup()

```
virtual bool lava::app::setup ( )  [virtual]
```

Set up the application.

Returns

Setup was successful or failed

Reimplemented in lava::engine.

4.2.3.7 v_sync()

```
bool lava::app::v_sync ( ) const  [inline]
```

V-Sync setting.

Returns

V-Sync is active or not

The documentation for this struct was generated from the following file:

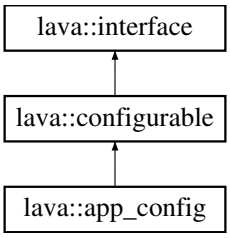
- liblava/app/app.hpp

4.3 lava::app_config Struct Reference

Application configuration.

```
#include <config.hpp>
```

Inheritance diagram for lava::app_config:



Public Member Functions

- void set_config (json_ref j) override

  Set config.
- json get_config () const override

  Get config.
- void **update_window_state** ()

  Update window state.

Public Attributes

- app * **context** = nullptr

  Application.
- name **org** = _liblava_

  Organization name.
- name **ext** = "zip"

  Preferred compression file format.
- bool **save_window** = true

  Save window state.
- bool **handle_key_events** = true

  Handle key events.
- bool **v_sync** = false

  Activate V-Sync.
- surface_format_request **surface**

  Request surface formats.
- index **physical_device** = 0

  Physical device index.
- imgui::font **imgui_font**

  ImGui font settings.
- string **id** = _default_

  Identification.
- window::state::optional **window_state**

  Window state if available.

4.3.1 Detailed Description

Application configuration.

4.3.2 Member Function Documentation

4.3.2.1 get_config()

```
json lava::app_config::get_config ( ) const  [override], [virtual]
```

Get config.

Returns

json Json file

Implements lava::configurable.

4.3.2.2 set_config()

```
void lava::app_config::set_config (
            json_ref j )  [override], [virtual]
```

Set config.

Parameters

j	Json file

Implements lava::configurable.

The documentation for this struct was generated from the following file:

- liblava/app/config.hpp

4.4 lava::attachment Struct Reference

Attachment description.

```
#include <attachment.hpp>
```

Public Types

- using **ptr** = std::shared_ptr< attachment >

  Shared pointer to attachment.
- using **list** = std::vector< ptr >

  List of attachments.

Public Member Functions

- attachment (VkFormat format=VK_FORMAT_UNDEFINED, VkSampleCountFlagBits samples=VK_↩ SAMPLE_COUNT_1_BIT)

  Construct a new attachment.
- VkAttachmentDescription const & get_description () const

  Get the description.
- void set_format (VkFormat format)

  Set the format.
- void set_samples (VkSampleCountFlagBits samples)

  Set the samples.
- void set_op (VkAttachmentLoadOp load_op, VkAttachmentStoreOp store_op)

  Set the op.
- void set_load_op (VkAttachmentLoadOp load_op)

  Set the load op.
- void set_store_op (VkAttachmentStoreOp store_op)

  Set the store op.
- void set_stencil_op (VkAttachmentLoadOp load_op, VkAttachmentStoreOp store_op)

  Set the stencil op.
- void set_stencil_load_op (VkAttachmentLoadOp load_op)

  Set the stencil load op.
- void set_stencil_store_op (VkAttachmentStoreOp store_op)

  Set the stencil store op.
- void set_layouts (VkImageLayout initial, VkImageLayout final)

  Set the layouts.
- void set_initial_layout (VkImageLayout layout)

  Set the initial layout.
- void set_final_layout (VkImageLayout layout)

  Set the final layout.

Static Public Member Functions

- static ptr make (VkFormat format=VK_FORMAT_UNDEFINED, VkSampleCountFlagBits samples=VK_↩ SAMPLE_COUNT_1_BIT)

  Make a new attachment.

4.4.1 Detailed Description

Attachment description.

4.4.2 Constructor & Destructor Documentation

4.4.2.1 attachment()

```
lava::attachment::attachment (
            VkFormat format = VK_FORMAT_UNDEFINED,
            VkSampleCountFlagBits samples = VK_SAMPLE_COUNT_1_BIT )  [inline], [explicit]
```

Construct a new attachment.

Parameters

format	Attachment format
samples	Sample count flag bits

4.4.3 Member Function Documentation

4.4.3.1 get_description()

```
VkAttachmentDescription const & lava::attachment::get_description ( ) const  [inline]
```

Get the description.

Returns

VkAttachmentDescription const& Attachment description

4.4.3.2 make()

```
static ptr lava::attachment::make (
            VkFormat format = VK_FORMAT_UNDEFINED,
            VkSampleCountFlagBits samples = VK_SAMPLE_COUNT_1_BIT )  [inline], [static]
```

Make a new attachment.

Parameters

format	Attachment format
samples	Sample count flag bits

Returns

ptr Shared pointer to attachment

4.4.3.3 set_final_layout()

```
void lava::attachment::set_final_layout (
            VkImageLayout layout )  [inline]
```

Set the final layout.

Parameters

layout	Image layout

4.4.3.4 set_format()

```
void lava::attachment::set_format (
            VkFormat format )  [inline]
```

Set the format.

Parameters

format	Attachment format

4.4.3.5 set_initial_layout()

```
void lava::attachment::set_initial_layout (
            VkImageLayout layout )  [inline]
```

Set the initial layout.

Parameters

layout	Image layout

4.4.3.6 set_layouts()

```
void lava::attachment::set_layouts (
            VkImageLayout initial,
            VkImageLayout final )  [inline]
```

Set the layouts.

Parameters

initial	Initial image layout
final	Final image layout

4.4.3.7 set_load_op()

```
void lava::attachment::set_load_op (
            VkAttachmentLoadOp load_op )  [inline]
```

Set the load op.

Parameters

load_op	Attachment load op

4.4.3.8 set_op()

```
void lava::attachment::set_op (
            VkAttachmentLoadOp load_op,
            VkAttachmentStoreOp store_op )  [inline]
```

Set the op.

Parameters

load_op	Attachment load op
store_op	Attachment store op

4.4.3.9 set_samples()

```
void lava::attachment::set_samples (
```

```
            VkSampleCountFlagBits samples )  [inline]
```

Set the samples.

Parameters

samples	Attachment sample count flag bits

4.4.3.10 set_stencil_load_op()

```
void lava::attachment::set_stencil_load_op (
            VkAttachmentLoadOp load_op )  [inline]
```

Set the stencil load op.

Parameters

load_op	Attachment load op

4.4.3.11 set_stencil_op()

```
void lava::attachment::set_stencil_op (
            VkAttachmentLoadOp load_op,
            VkAttachmentStoreOp store_op )  [inline]
```

Set the stencil op.

Parameters

load_op	Attachment load op
store_op	Attachment store op

4.4.3.12 set_stencil_store_op()

```
void lava::attachment::set_stencil_store_op (
            VkAttachmentStoreOp store_op )  [inline]
```

Set the stencil store op.

Parameters

store_op	Attachment store op

4.4.3.13 set_store_op()

```
void lava::attachment::set_store_op (
            VkAttachmentStoreOp store_op ) [inline]
```

Set the store op.

Parameters

store_op	Attachment store op

The documentation for this struct was generated from the following file:

- liblava/block/attachment.hpp

4.5 lava::benchmark_data Struct Reference

Benchmark data.

```
#include <benchmark.hpp>
```

Public Types

- using **list** = std::vector< ui32 >

  List of frame times.

Public Attributes

- ms **time** = ms{ 10000 }

  Benchmark duration.
- ms **offset** = ms{ 5000 }

  Warm up time.
- string **file** = _benchmark_json_

  Output file.
- string **path**

  Output path (empty: pref_dir)
- bool **exit** = true

  Close app after benchmark.
- ui32 **buffer_size** = 100000

  Pre-allocated buffer size for results.
- list **values**

  Benchmark results.
- index **current** = 0

  Current frame index.
- ms **start_timestamp** = ms{ 0 }

  Benchmark start timestamp.

4.5.1 Detailed Description

Benchmark data.

The documentation for this struct was generated from the following file:

- liblava/app/benchmark.hpp

4.6 lava::descriptor::binding Struct Reference

Descriptor binding.

```
#include <descriptor.hpp>
```

Public Types

- using **ptr** = std::shared_ptr< binding >

  Shared pointer to binding.
- using **list** = std::vector< ptr >

  List of bindings.

Public Member Functions

- **binding** ()

  Construct a new binding.
- VkDescriptorSetLayoutBinding const & get () const

  Get the Vulkan descriptor set layout binding.
- void set (index index)

  Det the binding index.
- void set_type (VkDescriptorType descriptor_type)

  Set the type.
- void set_count (ui32 descriptor_count)

  Set the count.
- void set_stage_flags (VkShaderStageFlags stage_flags)

  Set the stage flags.
- void set_samplers (VkSampler const *immutable_samplers)

  Set the samplers.

Static Public Member Functions

- static ptr make (index index)

  Make a new descriptor binding.

4.6.1 Detailed Description

Descriptor binding.

4.6.2 Member Function Documentation

4.6.2.1 get()

```
VkDescriptorSetLayoutBinding const & lava::descriptor::binding::get ( ) const  [inline]
```

Get the Vulkan descriptor set layout binding.

Returns

VkDescriptorSetLayoutBinding const& Vulkan binding

4.6.2.2 make()

```
static ptr lava::descriptor::binding::make (
            index index )  [inline], [static]
```

Make a new descriptor binding.

Parameters

index	Binding index

Returns

ptr Shared pointer to descriptor binding

4.6.2.3 set()

```
void lava::descriptor::binding::set (
            index index )  [inline]
```

Det the binding index.

Parameters

index	Binding index

4.6.2.4 set_count()

```
void lava::descriptor::binding::set_count (
            ui32 descriptor_count ) [inline]
```

Set the count.

Parameters

descriptor_count	Descriptor count

4.6.2.5 set_samplers()

```
void lava::descriptor::binding::set_samplers (
            VkSampler const * immutable_samplers ) [inline]
```

Set the samplers.

Parameters

immutable_samplers	Pointer to immutable samplers

4.6.2.6 set_stage_flags()

```
void lava::descriptor::binding::set_stage_flags (
            VkShaderStageFlags stage_flags ) [inline]
```

Set the stage flags.

Parameters

stage_flags	Shader stage flags

4.6.2.7 set_type()

```
void lava::descriptor::binding::set_type (
            VkDescriptorType descriptor_type ) [inline]
```

Set the type.

Parameters

descriptor_type	Descriptor type

The documentation for this struct was generated from the following file:

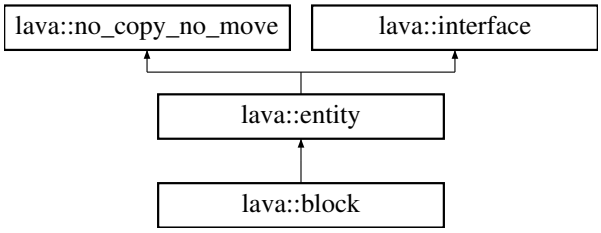
- liblava/block/descriptor.hpp

4.7 lava::block Struct Reference

Block of commands.

```
#include <block.hpp>
```

Inheritance diagram for lava::block:



Public Types

- using **ptr** = std::shared_ptr< block >

  Shared pointer to block.
- using **cptr** = command const ∗

  Const pointer to block.
- using **map** = std::map< id, ptr >

  Map of blocks.
- using **clist** = std::vector< cptr >

  List of blocks.

Public Member Functions

- ∼**block** ()

  Destroy the block.
- bool create (device_p device, index frame_count, index queue_family)

  Create a new block.
- void **destroy** ()

  Destroy the block.
- index get_frame_count () const

  Get the frame count.
- id add_cmd (command::process_func func, bool active=true)
- id add_command (command::process_func func, bool active=true)

Add a command.

- void remove_cmd (id::ref cmd_id)
- void remove_command (id::ref cmd_id)

  Remove the command.
- bool process (index frame)

  Process the block.
- index get_current_frame () const

  Get the current frame.
- VkCommandBuffer get_command_buffer (id::ref cmd_id) const

  Get the command buffer.
- VkCommandBuffer get_command_buffer (id::ref cmd_id, index frame) const

  Get the command buffer.
- VkCommandBuffers get_buffers ()

  Get the buffers.
- command::map const & get_commands () const

  Get the commands.
- command::clist const & get_cmd_order () const

  Get the cmd order.
- bool activated (id::ref cmd_id)

  Check if command is activated.
- bool set_active (id::ref cmd_id, bool active=true)

  Set the command active.
- device_p get_device ()

  Get the device.

Static Public Member Functions

- static ptr make ()

  Make a new block.

4.7.1 Detailed Description

Block of commands.

4.7.2 Member Function Documentation

4.7.2.1 activated()

```
bool lava::block::activated (
            id::ref cmd_id )
```

Check if command is activated.

Parameters

cmd↩ _id	Command id

Returns

Command is active or not

4.7.2.2 add_cmd()

```
id lava::block::add_cmd (
            command::process_func func,
            bool active = true )
```

See also

add_command

4.7.2.3 add_command()

```
id lava::block::add_command (
            command::process_func func,
            bool active = true )  [inline]
```

Add a command.

Parameters

func	Command function
active	Active state

Returns

id Command id

4.7.2.4 create()

```
bool lava::block::create (
            device_p device,
            index frame_count,
            index queue_family )
```

Create a new block.

Parameters

device	Vulkan device
frame_count	Number of frames
queue_family	Queue family index

Returns

Create was successful or failed

4.7.2.5 get_buffers()

```
VkCommandBuffers lava::block::get_buffers ( ) [inline]
```

Get the buffers.

Returns

VkCommandBuffers List of Vulkan command buffers

4.7.2.6 get_cmd_order()

```
command::clist const & lava::block::get_cmd_order ( ) const [inline]
```

Get the cmd order.

Returns

command::list const& List of commands

4.7.2.7 get_command_buffer() [1/2]

```
VkCommandBuffer lava::block::get_command_buffer (
            id::ref cmd_id ) const [inline]
```

Get the command buffer.

Parameters

cmd↩ _id	Command id

Returns

VkCommandBuffer Vulkan command buffer

4.7.2.8 get_command_buffer() [2/2]

```
VkCommandBuffer lava::block::get_command_buffer (
            id::ref cmd_id,
            index frame ) const  [inline]
```

Get the command buffer.

Parameters

cmd↩ _id	Command id
frame	Frame index

Returns

VkCommandBuffer Vulkan command buffer

4.7.2.9 get_commands()

```
command::map const & lava::block::get_commands ( ) const  [inline]
```

Get the commands.

Returns

command::map const& Map of commands

4.7.2.10 get_current_frame()

```
index lava::block::get_current_frame ( ) const  [inline]
```

Get the current frame.

Returns

index Current frame

4.7.2.11 get_device()

```
device_p lava::block::get_device ( ) [inline]
```

Get the device.

Returns

device_p Vulkan device

4.7.2.12 get_frame_count()

```
index lava::block::get_frame_count ( ) const [inline]
```

Get the frame count.

Returns

index Number of frames

4.7.2.13 make()

```
static ptr lava::block::make ( ) [inline], [static]
```

Make a new block.

Returns

ptr Shared pointer to block

4.7.2.14 process()

```
bool lava::block::process (
            index frame )
```

Process the block.

Parameters

frame	Frame index

Returns

Process was successful or aborted

4.7.2.15 remove_cmd()

```
void lava::block::remove_cmd (
            id::ref cmd_id )
```

See also

remove_command

4.7.2.16 remove_command()

```
void lava::block::remove_command (
            id::ref cmd_id )  [inline]
```

Remove the command.

Parameters

cmd↩ _id	Command id

4.7.2.17 set_active()

```
bool lava::block::set_active (
            id::ref cmd_id,
            bool active = true )
```

Set the command active.

Parameters

cmd↩ _id	Command id
active	Active state

Returns

Set was successful or failed

The documentation for this struct was generated from the following file:

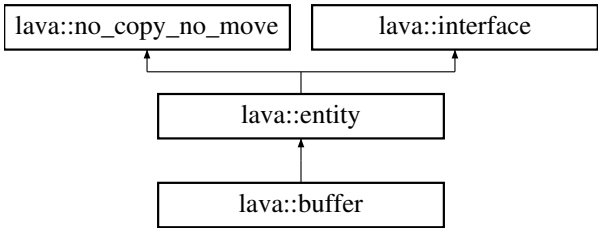
- liblava/block/block.hpp

4.8 lava::buffer Struct Reference

Buffer.

```
#include <buffer.hpp>
```

Inheritance diagram for lava::buffer:



Public Types

- using **ptr** = std::shared_ptr< buffer >

  Shared pointer to buffer.
- using **list** = std::vector< ptr >

  List of buffers.

Public Member Functions

- ~**buffer** ()

  Destroy the buffer.
- bool create (device_p device, void const *data, size_t size, VkBufferUsageFlags usage, bool mapped=false, VmaMemoryUsage memory_usage=VMA_MEMORY_USAGE_GPU_ONLY)

  Create a new buffer.
- bool create_mapped (device_p device, void const *data, size_t size, VkBufferUsageFlags usage, Vma↩MemoryUsage memory_usage=VMA_MEMORY_USAGE_CPU_TO_GPU)

  Create a new mapped buffer.
- void **destroy** ()

  Destroy the buffer.
- device_p get_device ()

  Get the device.
- bool valid () const

  Check if the buffer is valid.
- VkBuffer get () const

  Get the buffer.
- VkDescriptorBufferInfo const * get_descriptor_info () const

  Get the descriptor information.
- VkDeviceAddress get_address () const

Get the address of the buffer.

- VkDeviceSize get_size () const

  Get the size of the buffer.
- void * get_mapped_data () const

  Get the mapped data.
- VkDeviceMemory get_device_memory () const

  Get the device memory of the buffer.
- void flush (VkDeviceSize offset=0, VkDeviceSize size=VK_WHOLE_SIZE)

  Flush the buffer data.
- VmaAllocation const & get_allocation () const

  Get the allocation.
- VmaAllocationInfo const & get_allocation_info () const

  Get the allocation information.

Static Public Member Functions

- static ptr make ()

  Make a new buffer.

4.8.1 Detailed Description

Buffer.

4.8.2 Member Function Documentation

4.8.2.1 create()

```
bool lava::buffer::create (
            device_p device,
            void const * data,
            size_t size,
            VkBufferUsageFlags usage,
            bool mapped = false,
            VmaMemoryUsage memory_usage = VMA_MEMORY_USAGE_GPU_ONLY )
```

Create a new buffer.

Parameters

device	Vulkan device
data	Buffer data
size	Data size
usage	Buffer usage flags
mapped	Map the buffer
memory_usage	Memory usage

Returns

Create was successful or failed

4.8.2.2 create_mapped()

```
bool lava::buffer::create_mapped (
            device_p device,
            void const * data,
            size_t size,
            VkBufferUsageFlags usage,
            VmaMemoryUsage memory_usage = VMA_MEMORY_USAGE_CPU_TO_GPU )
```

Create a new mapped buffer.

Parameters

device	Vulkan device
data	Buffer data
size	Data size
usage	Buffer usage flags
memory_usage	Memory usage

Returns

Create was successful or failed

4.8.2.3 flush()

```
void lava::buffer::flush (
            VkDeviceSize offset = 0,
            VkDeviceSize size = VK_WHOLE_SIZE )
```

Flush the buffer data.

Parameters

offset	Offset device size
size	Data device size

4.8.2.4 get()

```
VkBuffer lava::buffer::get ( ) const  [inline]
```

Get the buffer.

Returns

VkBuffer Vulkan buffer

4.8.2.5 get_address()

```
VkDeviceAddress lava::buffer::get_address ( ) const
```

Get the address of the buffer.

Returns

VkDeviceAddress Device address

4.8.2.6 get_allocation()

```
VmaAllocation const & lava::buffer::get_allocation ( ) const  [inline]
```

Get the allocation.

Returns

VmaAllocation const& Allocation

4.8.2.7 get_allocation_info()

```
VmaAllocationInfo const & lava::buffer::get_allocation_info ( ) const  [inline]
```

Get the allocation information.

Returns

VmaAllocationInfo const& Allocation information

4.8.2.8 get_descriptor_info()

```
VkDescriptorBufferInfo const * lava::buffer::get_descriptor_info ( ) const  [inline]
```

Get the descriptor information.

Returns

VkDescriptorBufferInfo const* Descriptor buffer information

4.8.2.9 get_device()

```
device_p lava::buffer::get_device ( ) [inline]
```

Get the device.

Returns

device_p Vulkan device

4.8.2.10 get_device_memory()

```
VkDeviceMemory lava::buffer::get_device_memory ( ) const [inline]
```

Get the device memory of the buffer.

Returns

VkDeviceMemory Device memory

4.8.2.11 get_mapped_data()

```
void * lava::buffer::get_mapped_data ( ) const [inline]
```

Get the mapped data.

Returns

void∗ Pointer to data

4.8.2.12 get_size()

```
VkDeviceSize lava::buffer::get_size ( ) const [inline]
```

Get the size of the buffer.

Returns

VkDeviceSize Device size

4.8.2.13 make()

```
static ptr lava::buffer::make ( ) [inline], [static]
```

Make a new buffer.

Returns

ptr Shared pointer to buffer

4.8.2.14 valid()

```
bool lava::buffer::valid ( ) const [inline]
```

Check if the buffer is valid.

Returns

Buffer is valid or not

The documentation for this struct was generated from the following file:

- liblava/resource/buffer.hpp

4.9 lava::json_file::callback Struct Reference

Json file callback.

```
#include <json_file.hpp>
```

Public Types

- using **list** = std::vector< callback * >

  List of callbacks.
- using **load_func** = std::function< void(json_ref)>

  Load function.
- using **save_func** = std::function< json()>

  Save function.

Public Attributes

- load_func **on_load**

  Called on load.
- save_func **on_save**

  Called on save.

4.9.1 Detailed Description

Json file callback.

The documentation for this struct was generated from the following file:

- liblava/file/json_file.hpp

4.10 lava::swapchain::callback Struct Reference

Swapchain callback.

```
#include <swapchain.hpp>
```

Public Types

- using **list** = std::vector< callback * >

  List of callbacks.
- using **created_func** = std::function< bool() >

  Created function.
- using **destroyed_func** = std::function< void() >

  Destroyed function.

Public Attributes

- created_func **on_created**

  Called on swapchain created.
- destroyed_func **on_destroyed**

  Called on swapchain destroyed.

4.10.1 Detailed Description

Swapchain callback.

The documentation for this struct was generated from the following file:

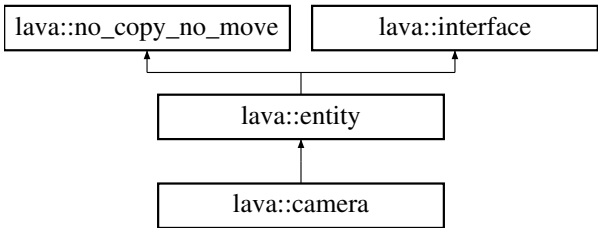
- liblava/frame/swapchain.hpp

4.11 lava::camera Struct Reference

First Person / Look At camera.

```
#include <camera.hpp>
```

Inheritance diagram for lava::camera:



Public Types

- using **ptr** = std::shared_ptr< camera >

  Shared pointer to camera.
- using **map** = std::map< id, ptr >

  Map of cameras.
- using **list** = std::vector< ptr >

  List of cameras.

Public Member Functions

- bool create (device_p device)

  Create a camera.
- void **destroy** ()

  Destroy the camera.
- void **update_projection** ()

  Update the projection.
- void update_view (delta dt, mouse_position mouse_pos)

  Update the view with mouse position.
- void update_view (delta dt, gamepad::ref pad)

  Update the view with gamepad.
- mat4 get_view () const

  Get the camera's 4x4 view matrix.
- mat4 get_projection () const

  Get the camera's 4x4 projection matrix.
- mat4 get_view_projection () const

  Get the camera's combined 4x4 view/projection matrix.
- bool handle (key_event::ref event)

  Handle key event.
- bool handle (mouse_button_event::ref event, mouse_position mouse_pos)

  Handle mouse button event.
- bool handle (scroll_event::ref event)

  Handle scroll event.

- bool valid () const

  Check if camera is valid.
- VkDescriptorBufferInfo const * get_descriptor_info () const

  Get the descriptor buffer info.
- void **upload** ()

  Upload camera state.
- void **stop** ()

  Stop camera movement.
- void **reset** ()

  Reset camera.
- void set_active (bool value=true)

  Set camera active.
- bool activated () const

  Check if camera is activated.
- bool moving () const

  Check if camera is moving.
- void set_movement_keys (keys_ref up, keys_ref down, keys_ref left, keys_ref right)

  Set keys for moving this camera.

Public Attributes

- v3 **position** = v3(0.f)

  Camera position.
- v3 **rotation** = v3(0.f)

  Camera rotation.
- r32 **rotation_speed** = 20.f

  Camera rotation speed.
- r32 **movement_speed** = 1.f

  Camera movement speed.
- r32 **zoom_speed** = 20.f

  Camera zoom speed.
- r32 **fov** = 60.f

  Field of view.
- r32 **z_near** = 0.1f

  Distance to near clipping plane along the -Z axis.
- r32 **z_far** = 256.f

  Distance to far clipping plane along the -Z axis.
- r32 **aspect_ratio** = 1.77f

  Camera aspect ratio.
- camera_mode **mode** = camera_mode::first_person

  Camera mode.
- bool **lock_z** = false

  Lock Z axis movement.
- bool **lock_rotation** = false

  Lock camera rotation.

4.11.1 Detailed Description

First Person / Look At camera.

4.11.2 Member Function Documentation

4.11.2.1 activated()

```
bool lava::camera::activated ( ) const  [inline]
```

Check if camera is activated.

Returns

Camera is active or not

4.11.2.2 create()

```
bool lava::camera::create (
            device_p device )
```

Create a camera.

Parameters

device	Vulkan device

Returns

Create was successful or failed

4.11.2.3 get_descriptor_info()

```
VkDescriptorBufferInfo const * lava::camera::get_descriptor_info ( ) const  [inline]
```

Get the descriptor buffer info.

Returns

VkDescriptorBufferInfo const* Descriptor buffer info

4.11.2.4 get_projection()

```
mat4 lava::camera::get_projection ( ) const
```

Get the camera's 4x4 projection matrix.

Returns

mat4 Projection matrix

4.11.2.5 get_view()

```
mat4 lava::camera::get_view ( ) const
```

Get the camera's 4x4 view matrix.

Returns

mat4 View matrix

4.11.2.6 get_view_projection()

```
mat4 lava::camera::get_view_projection ( ) const
```

Get the camera's combined 4x4 view/projection matrix.

Returns

mat4 Combined view/projection matrix

4.11.2.7 handle() [1/3]

```
bool lava::camera::handle (
            key_event::ref event )
```

Handle key event.

Parameters

event	Key event

Returns

Event was handled or ignored

4.11.2.8 handle() [2/3]

```
bool lava::camera::handle (
            mouse_button_event::ref event,
            mouse_position mouse_pos )
```

Handle mouse button event.

Parameters

event	Mouse button event
mouse_pos	Mouse position

Returns

Event was handled or ignored

4.11.2.9 handle() [3/3]

```
bool lava::camera::handle (
            scroll_event::ref event )
```

Handle scroll event.

Parameters

event	Scroll event

Returns

Event was handled or ignored

4.11.2.10 moving()

```
bool lava::camera::moving ( ) const  [inline]
```

Check if camera is moving.

Returns

Camera is moving or does not move

4.11.2.11 set_active()

```
void lava::camera::set_active (
            bool value = true )  [inline]
```

Set camera active.

Parameters

value	Active state

4.11.2.12 set_movement_keys()

```
void lava::camera::set_movement_keys (
            keys_ref up,
            keys_ref down,
            keys_ref left,
            keys_ref right )  [inline]
```

Set keys for moving this camera.

Parameters

up	Up inputs
down	Down inputs
left	Left inputs
right	Right inputs

4.11.2.13 update_view() [1/2]

```
void lava::camera::update_view (
            delta dt,
            gamepad::ref pad )
```

Update the view with gamepad.

Parameters

dt	Delta time
pad	Gamepad

4.11.2.14 update_view() [2/2]

```
void lava::camera::update_view (
            delta dt,
            mouse_position mouse_pos )
```

Update the view with mouse position.

Parameters

dt	Delta time
mouse_pos	Mouse position

4.11.2.15 valid()

```
bool lava::camera::valid ( ) const  [inline]
```

Check if camera is valid.

Returns

Camera is valid or not

The documentation for this struct was generated from the following file:

- liblava/app/camera.hpp

4.12 lava::cdata Struct Reference

Const data wrapper.

```
#include <data.hpp>
```

Public Types

- using **ref** = cdata const &

  Reference to const data wrapper.

Public Member Functions

- **cdata** ()=default

  Construct a new const data.
- cdata (void const *ptr, size_t length)

  Construct a new const data.
- cdata (data::ref data)

  Construct a new const data from other data.

Public Attributes

- data_cptr **ptr** = nullptr

  Const data pointer.
- size_t **size** = 0

  Size of data.

4.12.1 Detailed Description

Const data wrapper.

4.12.2 Constructor & Destructor Documentation

4.12.2.1 cdata() [1/2]

```
lava::cdata::cdata (
            void const * ptr,
            size_t length )  [inline]
```

Construct a new const data.

Parameters

ptr	Pointer to data
length	Length of data

4.12.2.2 cdata() [2/2]

```
lava::cdata::cdata (
            data::ref data )  [inline]
```

Construct a new const data from other data.

Parameters

data	Source data

The documentation for this struct was generated from the following file:

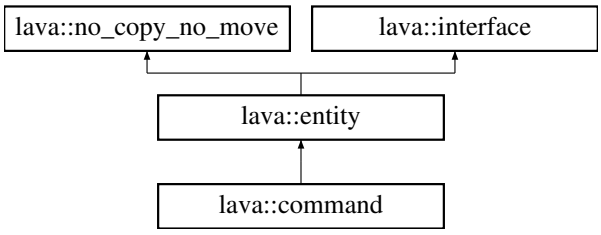
- liblava/core/data.hpp

4.13 lava::command Struct Reference

Block command.

```
#include <block.hpp>
```

Inheritance diagram for lava::command:



Public Types

- using **ptr** = std::shared_ptr< command >

  Shared pointer to command.
- using **cptr** = command const *

  Const pointer to command.
- using **map** = std::map< id, ptr >

  Map of commands.
- using **clist** = std::vector< cptr >

  List of commands.
- using **process_func** = std::function< void(VkCommandBuffer)>

  Command process function.

Public Member Functions

- bool create (device_p device, index frame_count, VkCommandPools command_pools)

  Create a new command.
- void destroy (device_p device, VkCommandPools command_pools)

  Destroy the command.

Static Public Member Functions

- static ptr make ()

  Make a new command.

Public Attributes

- VkCommandBuffers **buffers** = {}

  List of command buffers.
- process_func **on_process**

  Called on command process.
- bool **active** = true

  Active state.

4.13.1 Detailed Description

Block command.

4.13.2 Member Function Documentation

4.13.2.1 create()

```
bool lava::command::create (
            device_p device,
            index frame_count,
            VkCommandPools command_pools )
```

Create a new command.

Parameters

device	Vulkan device
frame_count	Number of frames
command_pools	List of command pools

Returns

Create was successful or failed

4.13.2.2 destroy()

```
void lava::command::destroy (
            device_p device,
            VkCommandPools command_pools )
```

Destroy the command.

Parameters

device	Vulkan device
command_pools	List of command pools

4.13.2.3 make()

```
static ptr lava::command::make ( )  [inline], [static]
```

Make a new command.

Returns

ptr Shared pointer to command

The documentation for this struct was generated from the following file:

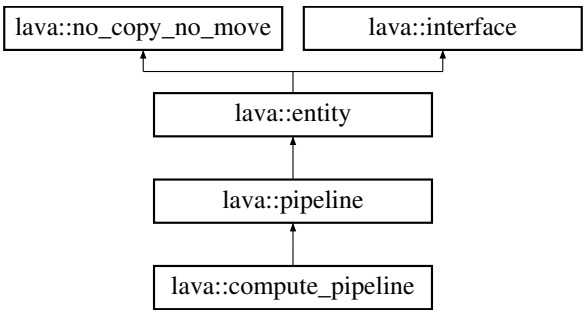
- liblava/block/block.hpp

4.14 lava::compute_pipeline Struct Reference

Compute pipeline.

```
#include <compute_pipeline.hpp>
```

Inheritance diagram for lava::compute_pipeline:



Public Types

- using **ptr** = std::shared_ptr< compute_pipeline >

  Shared pointer to compute pipeline.
- using **map** = std::map< id, ptr >

  Map of compute pipelines.
- using **list** = std::vector< ptr >

  List of compute pipelines.

Public Member Functions

- void bind (VkCommandBuffer cmdBuffer) override

  Bind the pipeline.
- bool set_shader_stage (cdata::ref data, VkShaderStageFlagBits stage)

  Set shader stage.
- void set (shader_stage::ptr const &stage)

  Set shader stage.
- shader_stage::ptr const & get_shader_stage () const

  Get the shader stage.
- void copy_to (compute_pipeline *target) const

  Copy configuration to target pipeline.
- void copy_from (ptr const &source)

  Copy configuration from source.
- **pipeline** (device_p device, VkPipelineCache pipeline_cache=0)

  Pipeline constructors.

Static Public Member Functions

- static ptr make (device_p device, VkPipelineCache pipeline_cache=0)

  Make a new compute pipeline.

Additional Inherited Members

4.14.1 Detailed Description

Compute pipeline.

4.14.2 Member Function Documentation

4.14.2.1 bind()

```
void lava::compute_pipeline::bind (
            VkCommandBuffer cmdBuffer )  [override], [virtual]
```

Bind the pipeline.

Parameters

cmdBuffer	Command buffer

Implements lava::pipeline.

4.14.2.2 copy_from()

```
void lava::compute_pipeline::copy_from (
            ptr const & source )  [inline]
```

Copy configuration from source.

Parameters

source	Compute pipeline

4.14.2.3 copy_to()

```
void lava::compute_pipeline::copy_to (
```

```
            compute_pipeline * target ) const
```

Copy configuration to target pipeline.

Parameters

target	Compute pipeline

4.14.2.4 get_shader_stage()

```
shader_stage::ptr const & lava::compute_pipeline::get_shader_stage ( ) const  [inline]
```

Get the shader stage.

Returns

shader_stage::ptr const& Shader state

4.14.2.5 make()

```
static ptr lava::compute_pipeline::make (
            device_p device,
            VkPipelineCache pipeline_cache = 0 )  [inline], [static]
```

Make a new compute pipeline.

Parameters

device	Vulkan device
pipeline_cache	Pipeline cache

Returns

ptr Shared pointer to compute pipeline

4.14.2.6 set()

```
void lava::compute_pipeline::set (
            shader_stage::ptr const & stage )  [inline]
```

Set shader stage.

Parameters

stage	Shader state

4.14.2.7 set_shader_stage()

```
bool lava::compute_pipeline::set_shader_stage (
            cdata::ref data,
            VkShaderStageFlagBits stage )
```

Set shader stage.

Parameters

data	Shader data
stage	Shader stage flag bits

Returns

Set was successful or failed

The documentation for this struct was generated from the following file:

- liblava/block/compute_pipeline.hpp

4.15 lava::imgui::config Struct Reference

ImGui configuration.

```
#include <imgui.hpp>
```

Public Attributes

- data **font_data**

  Font data.
- r32 **font_size** = default_imgui_font_size

  Font size.
- std::shared_ptr< ImGuiStyle > **style**

  Font style.
- icon_font **icon**

  Font icon settings.
- std::filesystem::path **ini_file_dir**

  ImGui state file path.

4.15.1 Detailed Description

ImGui configuration.

The documentation for this struct was generated from the following file:

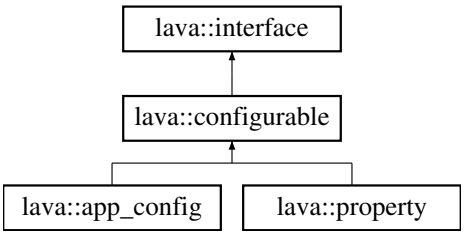
- liblava/app/imgui.hpp

4.16 lava::configurable Struct Reference

Configurable interface.

`#include <json.hpp>`

Inheritance diagram for lava::configurable:



Public Member Functions

- virtual void set_config (json_ref j)=0

  Set config.
- virtual json get_config () const =0

  Get config.

4.16.1 Detailed Description

Configurable interface.

4.16.2 Member Function Documentation

4.16.2.1 get_config()

`virtual json lava::configurable::get_config ( ) const [pure virtual]`

Get config.

Returns

json Json file

Implemented in lava::app_config, and lava::property.

4.16.2.2 set_config()

```
virtual void lava::configurable::set_config (
            json_ref j )  [pure virtual]
```

Set config.

Parameters

j	Json file

Implemented in lava::app_config, and lava::property.

The documentation for this struct was generated from the following file:

- liblava/file/json.hpp

4.17 lava::render_pipeline::create_info Struct Reference

Render pipeline create information.

```
#include <render_pipeline.hpp>
```

Public Attributes

- VkPipelineVertexInputStateCreateInfo **vertex_input_state**
  Vertex input stage.
- VkPipelineInputAssemblyStateCreateInfo **input_assembly_state**
  Input assembly state.
- VkPipelineViewportStateCreateInfo **viewport_state**
  Viewport state.
- VkPipelineMultisampleStateCreateInfo **multisample_state**
  Multisample state.
- VkPipelineDepthStencilStateCreateInfo **depth_stencil_state**
  Depth stencil state.
- VkPipelineRasterizationStateCreateInfo **rasterization_state**
  Rasterization state.

4.17.1 Detailed Description

Render pipeline create information.

The documentation for this struct was generated from the following file:

- liblava/block/render_pipeline.hpp

4.18 lava::device::create_param Struct Reference

Device create parameters.

```
#include <device.hpp>
```

Public Types

- using **ref** = create_param const &

  Reference to device create parameters.

Public Member Functions

- void **add_swapchain_extension** ()

  Add swapchain extension.
- void **add_portability_subset_extension** ()

  Add portability subset extension.
- void **set_default_queues** ()

  Set the default queues.
- void **set_all_queues** ()

  Set the all queues.
- bool add_queue (VkQueueFlags flags, r32 priority=1.f)

  Add queue.
- bool add_queues (VkQueueFlags flags, ui32 count, r32 priority=1.f)

  Add queues.
- bool add_dedicated_queues (r32 priority=1.f)

  Add all dedicated queues.
- verify_queues_result verify_queues () const

  Verify queues.

Public Attributes

- physical_device_cp **physical_device** = nullptr

  Physical device.
- VmaAllocatorCreateFlags **vma_flags** = 0

  VMA flags.
- names **extensions**

  List of extensions to enable.
- VkPhysicalDeviceFeatures **features** {}

  List of physical device features to enable.
- bool **has_features_2** = false

  Must be true if .next points to a VkPhysicalDevice2 instance.
- void const ∗ **next** = nullptr

  Create parameter next pointer (pNext)
- queue_family_info::list **queue_family_infos**

  List of queue famiy infos.

4.18.1 Detailed Description

Device create parameters.

4.18.2 Member Function Documentation

4.18.2.1 add_dedicated_queues()

```
bool lava::device::create_param::add_dedicated_queues (
            r32 priority = 1.f )
```

Add all dedicated queues.

Parameters

priority	Priority for queues

Returns

Add was successful or failed

4.18.2.2 add_queue()

```
bool lava::device::create_param::add_queue (
            VkQueueFlags flags,
            r32 priority = 1.f )  [inline]
```

Add queue.

Parameters

flags	Queue flags
priority	Priority for queue

Returns

Add was successful or failed

4.18.2.3 add_queues()

```
bool lava::device::create_param::add_queues (
            VkQueueFlags flags,
```

```
ui32 count,
r32 priority = 1.f )
```

Add queues.

Parameters

flags	Queue flags
count	Number of queues
priority	Priority for queues

Returns

Add was successful or failed

4.18.2.4 verify_queues()

```
verify_queues_result lava::device::create_param::verify_queues ( ) const
```

Verify queues.

Returns

verify_queues_result Verification result

The documentation for this struct was generated from the following file:

- liblava/base/device.hpp

4.19 lava::instance::create_param Struct Reference

Instance create parameters.

```
#include <instance.hpp>
```

Public Types

- using **ref** = create_param const &

  Reference to instance create parameters.

Public Attributes

- names **layers** {}

  List of layers to enable.
- names **extensions** {}

  List of extensions to enable.

4.19.1 Detailed Description

Instance create parameters.

The documentation for this struct was generated from the following file:

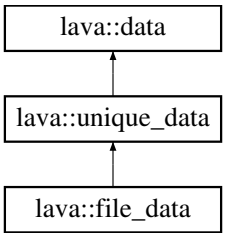
- liblava/base/instance.hpp

4.20 lava::data Struct Reference

Data wrapper.

```
#include <data.hpp>
```

Inheritance diagram for lava::data:



Public Types

- using **ref** = data const &

  Reference to data wrapper.

Public Member Functions

- **data** ()=default

  Construct a new data.
- data (auto *ptr, size_t size)

  Construct a new data.
- void set (size_t length, data_mode mode=data_mode::alloc)

  Set and allocate data by length.
- bool allocate ()

  Allocate data.
- void **free** ()

  Free data.
- data_ptr end () const

  Pointer to end of data.

Public Attributes

- data_ptr **ptr** = nullptr

  Pointer to data.
- size_t **size** = 0

  Size of data.
- size_t **alignment** = 0

  Data alignment.

4.20.1 Detailed Description

Data wrapper.

4.20.2 Constructor & Destructor Documentation

4.20.2.1 data()

```
lava::data::data (
            auto * ptr,
            size_t size ) [inline]
```

Construct a new data.

Parameters

ptr	Data pointer
size	Size of data

4.20.3 Member Function Documentation

4.20.3.1 allocate()

```
bool lava::data::allocate ( ) [inline]
```

Allocate data.

Returns

Allocate was successful or failed

4.20.3.2 end()

```
data_ptr lava::data::end ( ) const  [inline]
```

Pointer to end of data.

Returns

data_ptr Pointer to end

4.20.3.3 set()

```
void lava::data::set (
            size_t length,
            data_mode mode = data_mode::alloc )  [inline]
```

Set and allocate data by length.

Parameters

length	Length of data
mode	Data mode

The documentation for this struct was generated from the following file:

- liblava/core/data.hpp

4.21 lava::data_provider Struct Reference

Data provider.

```
#include <data.hpp>
```

Public Types

- using **alloc_func** = std::function< data_ptr(size_t, size_t)>

  Allocation function.
- using **free_func** = std::function< void()>

  Free function.
- using **realloc_func** = std::function< data_ptr(data_ptr, size_t, size_t)>

  Reallocation function.

Public Attributes

- alloc_func **on_alloc**

  Called on allocation.
- free_func **on_free**

  Called on free.
- realloc_func **on_realloc**

  Called on reallocation.

4.21.1 Detailed Description

Data provider.

The documentation for this struct was generated from the following file:

- liblava/core/data.hpp

4.22 lava::instance::debug_config Struct Reference

Debug configuration.

```
#include <instance.hpp>
```

Public Types

- using **ref** = debug_config const &

  Reference to debug configuration.

Public Attributes

- bool **validation** = false

  Validation.
- bool **render_doc** = false

  Renderdoc.
- bool **verbose** = false

  Verbose logging.
- bool **utils** = false

  Debug utils.

4.22.1 Detailed Description

Debug configuration.

The documentation for this struct was generated from the following file:

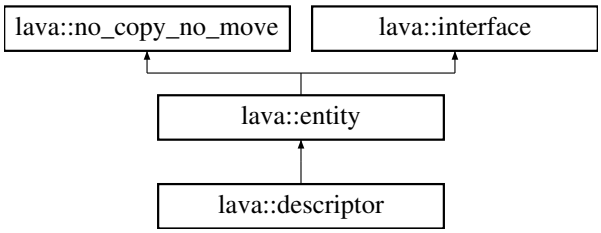
- liblava/base/instance.hpp

4.23 lava::descriptor Struct Reference

Descriptor.

```
#include <descriptor.hpp>
```

Inheritance diagram for lava::descriptor:



Classes

- struct binding

  Descriptor binding.
- struct pool

  Descriptor pool.

Public Types

- using **ptr** = std::shared_ptr< descriptor >

  Shared pointer to descriptor.
- using **list** = std::vector< ptr >

  List of descriptors.

Public Member Functions

- void add_binding (index binding, VkDescriptorType descriptor_type, VkShaderStageFlags stage_flags)

  Add binding.
- void **clear_bindings** ()

  Clear bindings.
- void add (binding::ptr const &binding)

  Add binding.
- bool create (device_p device)

  Create a new descriptor.
- void **destroy** ()

  Destroy the descriptor.
- ui32 get_binding_count () const

  Get the binding count.
- binding::list const & get_bindings ()

  Get the bindings.
- VkDescriptorSetLayout get () const

  Get descriptor set layout.
- device_p get_device ()

Get the device.

- VkDescriptorSet allocate_set (VkDescriptorPool pool)

  Allocate descriptor set.
- VkDescriptorSet allocate (VkDescriptorPool pool)
- bool free_set (VkDescriptorSet &descriptor_set, VkDescriptorPool pool)

  Free descriptor set.
- bool free (VkDescriptorSet &descriptor_set, VkDescriptorPool pool)
- VkDescriptorSets allocate_sets (ui32 size, VkDescriptorPool pool)

  Allocate descriptor sets.
- VkDescriptorSets allocate (ui32 size, VkDescriptorPool pool)
- bool free_sets (VkDescriptorSets &descriptor_sets, VkDescriptorPool pool)

  Free descriptor sets.
- bool free (VkDescriptorSets &descriptor_sets, VkDescriptorPool pool)

Static Public Member Functions

- static ptr make ()

  Make a new descriptor.

4.23.1 Detailed Description

Descriptor.

4.23.2 Member Function Documentation

4.23.2.1 add()

```
void lava::descriptor::add (
            binding::ptr const & binding )  [inline]
```

Add binding.

Parameters

binding	Descriptor binding

4.23.2.2 add_binding()

```
void lava::descriptor::add_binding (
            index binding,
            VkDescriptorType descriptor_type,
            VkShaderStageFlags stage_flags )
```

Add binding.

Parameters

binding	Index of binding
descriptor_type	Descriptor type
stage_flags	Shader stage flags

4.23.2.3 allocate() [1/2]

```
VkDescriptorSets lava::descriptor::allocate (
            ui32 size,
            VkDescriptorPool pool )  [inline]
```

See also

allocate_sets

4.23.2.4 allocate() [2/2]

```
VkDescriptorSet lava::descriptor::allocate (
            VkDescriptorPool pool )  [inline]
```

See also

allocate_set

4.23.2.5 allocate_set()

```
VkDescriptorSet lava::descriptor::allocate_set (
            VkDescriptorPool pool )
```

Allocate descriptor set.

Parameters

pool	Descriptor pool

Returns

VkDescriptorSet Descriptor set

4.23.2.6 allocate_sets()

```
VkDescriptorSets lava::descriptor::allocate_sets (
            ui32 size,
            VkDescriptorPool pool )
```

Allocate descriptor sets.

Parameters

size	Number of sets
pool	Descriptor pool

Returns

VkDescriptorSets List of descriptor sets

4.23.2.7 create()

```
bool lava::descriptor::create (
            device_p device )
```

Create a new descriptor.

Parameters

device	Vulkan device

Returns

Create was successful or failed

4.23.2.8 free() [1/2]

```
bool lava::descriptor::free (
            VkDescriptorSet & descriptor_set,
            VkDescriptorPool pool )  [inline]
```

See also

free_set

4.23.2.9 free() [2/2]

```
bool lava::descriptor::free (
            VkDescriptorSets & descriptor_sets,
            VkDescriptorPool pool )  [inline]
```

See also

free_sets

4.23.2.10 free_set()

```
bool lava::descriptor::free_set (
            VkDescriptorSet & descriptor_set,
            VkDescriptorPool pool )
```

Free descriptor set.

Parameters

descriptor_set	Descriptor set
pool	Descriptor pool

Returns

Free was successful or failed

4.23.2.11 free_sets()

```
bool lava::descriptor::free_sets (
            VkDescriptorSets & descriptor_sets,
            VkDescriptorPool pool )
```

Free descriptor sets.

Parameters

descriptor_sets	List of descriptor sets
pool	Descriptor pool

Returns

Free was successful or failed

4.23.2.12 get()

```
VkDescriptorSetLayout lava::descriptor::get ( ) const  [inline]
```

Get descriptor set layout.

Returns

VkDescriptorSetLayout Vulkan descriptor set layout

4.23.2.13 get_binding_count()

```
ui32 lava::descriptor::get_binding_count ( ) const  [inline]
```

Get the binding count.

Returns

ui32 Number of bindings

4.23.2.14 get_bindings()

```
binding::list const & lava::descriptor::get_bindings ( )  [inline]
```

Get the bindings.

Returns

binding::list const& List of bindings

4.23.2.15 get_device()

```
device_p lava::descriptor::get_device ( )  [inline]
```

Get the device.

Returns

device_p Vulkan device

4.23.2.16 make()

```
static ptr lava::descriptor::make ( )  [inline], [static]
```

Make a new descriptor.

Returns

ptr Shared pointer to descriptor

The documentation for this struct was generated from the following file:

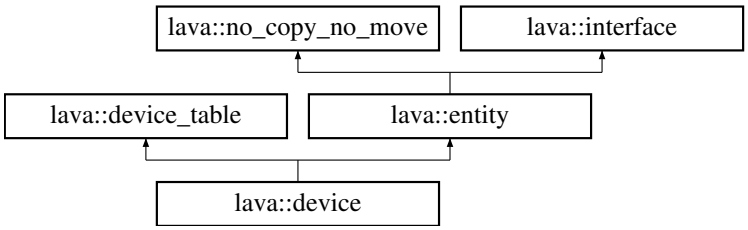
- liblava/block/descriptor.hpp

4.24 lava::device Struct Reference

Vulkan device.

```
#include <device.hpp>
```

Inheritance diagram for lava::device:



Classes

- struct create_param

  Device create parameters.

Public Types

- using **ptr** = std::shared_ptr< device >

  Shared pointer to a device.
- using **list** = std::vector< ptr >

  List of devices.

Public Member Functions

- ~**device** ()

  Destroy the device.
- bool create (create_param::ref param)

  Create a new device.
- void **destroy** ()

  Destroy the device.
- queue::ref get_graphics_queue (index index=0) const

  Get a graphics queue by index.
- queue::ref graphics_queue (index index=0) const
- queue::ref get_compute_queue (index index=0) const

  Get a compute queue by index.
- queue::ref compute_queue (index index=0) const
- queue::ref get_transfer_queue (index index=0) const

  Get a transfer queue by index.
- queue::ref transfer_queue (index index=0) const
- queue::list const & get_graphics_queues () const

  Get the list of graphics queues.
- queue::list const & graphics_queues () const
- queue::list const & get_compute_queues () const

  Get the list of compute queues.
- queue::list const & compute_queues () const
- queue::list const & get_transfer_queues () const

  Get the list of transfer queues.
- queue::list const & transfer_queues () const
- queue::list const & get_queues () const

  Get all queues.
- queue::list const & queues () const
- VkDevice get () const

  Get the Vulkan device.
- VolkDeviceTable const & call () const

  Get the Volk device table.
- bool wait_for_idle () const

  Wait for idle.
- physical_device_cp get_physical_device () const

  Get the physical device.
- VkPhysicalDevice get_vk_physical_device () const

  Get the Vulkan physical device.
- VkPhysicalDeviceFeatures const & get_features () const

  Get the physical device features.
- VkPhysicalDeviceProperties const & get_properties () const

  Get the physical device properties.
- bool surface_supported (VkSurfaceKHR surface) const

  Check if surface is supported by this device.
- void set_allocator (allocator::ptr value)

  Set the allocator for this device.
- allocator::ptr get_allocator ()

  Get the allocator of this device.
- VmaAllocator alloc () const

  Get the VMA allocator.

Static Public Member Functions

- static ptr make ()

  Make a new device.

Additional Inherited Members

4.24.1 Detailed Description

Vulkan device.

4.24.2 Member Function Documentation

4.24.2.1 alloc()

```
VmaAllocator lava::device::alloc ( ) const  [inline]
```

Get the VMA allocator.

Returns

VmaAllocator VMA allocator

4.24.2.2 call()

```
VolkDeviceTable const & lava::device::call ( ) const  [inline]
```

Get the Volk device table.

Returns

VolkDeviceTable const& Volk device table

4.24.2.3 compute_queue()

```
queue::ref lava::device::compute_queue (
            index index = 0 ) const  [inline]
```

See also

get_compute_queue

4.24.2.4 compute_queues()

```
queue::list const & lava::device::compute_queues ( ) const  [inline]
```

See also

get_compute_queues

4.24.2.5 create()

```
bool lava::device::create (
            create_param::ref param )
```

Create a new device.

Parameters

param	Create parameters

Returns

Create was successful or failed

4.24.2.6 get()

```
VkDevice lava::device::get ( ) const  [inline]
```

Get the Vulkan device.

Returns

VkDevice Vulkan device

4.24.2.7 get_allocator()

```
allocator::ptr lava::device::get_allocator ( )  [inline]
```

Get the allocator of this device.

Returns

allocator::ptr Allocator

4.24.2.8 get_compute_queue()

```
queue::ref lava::device::get_compute_queue (
            index index = 0 ) const  [inline]
```

Get a compute queue by index.

Parameters

index	Index of queue

Returns

queue::ref Compute queue

4.24.2.9 get_compute_queues()

```
queue::list const & lava::device::get_compute_queues ( ) const  [inline]
```

Get the list of compute queues.

Returns

queue::list const& Compute queues

4.24.2.10 get_features()

```
VkPhysicalDeviceFeatures const & lava::device::get_features ( ) const
```

Get the physical device features.

Returns

VkPhysicalDeviceFeatures const& Features

4.24.2.11 get_graphics_queue()

```
queue::ref lava::device::get_graphics_queue (
            index index = 0 ) const  [inline]
```

Get a graphics queue by index.

Parameters

index	Index of queue

Returns

queue::ref Graphics queue

4.24.2.12 get_graphics_queues()

```
queue::list const & lava::device::get_graphics_queues ( ) const  [inline]
```

Get the list of graphics queues.

Returns

queue::list const& Graphics queues

4.24.2.13 get_physical_device()

```
physical_device_cp lava::device::get_physical_device ( ) const  [inline]
```

Get the physical device.

Returns

physical_device_cp Physical device

4.24.2.14 get_properties()

```
VkPhysicalDeviceProperties const & lava::device::get_properties ( ) const
```

Get the physical device properties.

Returns

VkPhysicalDeviceProperties const& Properties

4.24.2.15 get_queues()

```
queue::list const & lava::device::get_queues ( ) const  [inline]
```

Get all queues.

Returns

queue::list const& List of all queues

4.24.2.16 get_transfer_queue()

```
queue::ref lava::device::get_transfer_queue (
            index index = 0 ) const  [inline]
```

Get a transfer queue by index.

Parameters

index	Index of queue

Returns

queue::ref Transfer queue

4.24.2.17 get_transfer_queues()

```
queue::list const & lava::device::get_transfer_queues ( ) const  [inline]
```

Get the list of transfer queues.

Returns

queue::list const& Transfer queues

4.24.2.18 get_vk_physical_device()

```
VkPhysicalDevice lava::device::get_vk_physical_device ( ) const
```

Get the Vulkan physical device.

Returns

VkPhysicalDevice Vulkan physical device

4.24.2.19 graphics_queue()

```
queue::ref lava::device::graphics_queue (
            index index = 0 ) const  [inline]
```

See also

get_graphics_queue

4.24.2.20 graphics_queues()

```
queue::list const & lava::device::graphics_queues ( ) const  [inline]
```

See also

get_graphics_queues

4.24.2.21 make()

```
static ptr lava::device::make ( )  [inline], [static]
```

Make a new device.

Returns

ptr Shared pointer to device

4.24.2.22 queues()

```
queue::list const & lava::device::queues ( ) const  [inline]
```

See also

get_queues

4.24.2.23 set_allocator()

```
void lava::device::set_allocator (
            allocator::ptr value )  [inline]
```

Set the allocator for this device.

Parameters

value	Allocator

4.24.2.24 surface_supported()

```
bool lava::device::surface_supported (
```

```
            VkSurfaceKHR surface ) const
```

Check if surface is supported by this device.

Parameters

surface	Surface to check

Returns

Surface is supported or not

4.24.2.25 transfer_queue()

```
queue::ref lava::device::transfer_queue (
            index index = 0 ) const  [inline]
```

See also

get_transfer_queue

4.24.2.26 transfer_queues()

```
queue::list const & lava::device::transfer_queues ( ) const  [inline]
```

See also

get_transfer_queues

4.24.2.27 wait_for_idle()

```
bool lava::device::wait_for_idle ( ) const  [inline]
```

Wait for idle.

Returns

Wait was successful or failed

The documentation for this struct was generated from the following file:

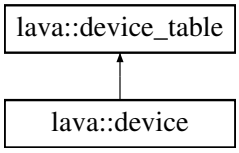
- liblava/base/device.hpp

4.25 lava::device_table Struct Reference

Device functions.

```
#include <device_table.hpp>
```

Inheritance diagram for lava::device_table:



Public Member Functions

- void **load_table** ()

  Load device table.
- vk_result vkCreateImageView (const VkImageViewCreateInfo *pCreateInfo, const VkAllocationCallbacks *pAllocator, VkImageView *pView)
- vk_result vkCreateImageView (const VkImageViewCreateInfo *pCreateInfo, VkImageView *pView)
- vk_result vkCreateSampler (const VkSamplerCreateInfo *pCreateInfo, const VkAllocationCallbacks *p↩Allocator, VkSampler *pSampler)
- vk_result vkCreateSampler (const VkSamplerCreateInfo *pCreateInfo, VkSampler *pSampler)
- vk_result vkCreateShaderModule (const VkShaderModuleCreateInfo *pCreateInfo, const VkAllocation↩Callbacks *pAllocator, VkShaderModule *pShaderModule)
- vk_result vkCreateShaderModule (const VkShaderModuleCreateInfo *pCreateInfo, VkShaderModule *p↩ShaderModule)
- vk_result vkCreateFence (const VkFenceCreateInfo *pCreateInfo, const VkAllocationCallbacks *pAllocator, VkFence *pFence)
- vk_result vkCreateFence (const VkFenceCreateInfo *pCreateInfo, VkFence *pFence)
- vk_result vkCreateSemaphore (const VkSemaphoreCreateInfo *pCreateInfo, const VkAllocationCallbacks *pAllocator, VkSemaphore *pSemaphore)
- vk_result vkCreateSemaphore (const VkSemaphoreCreateInfo *pCreateInfo, VkSemaphore *pSemaphore)
- vk_result vkWaitForFences (uint32_t fenceCount, const VkFence *pFences, VkBool32 waitAll, uint64_t timeout)
- vk_result vkResetFences (uint32_t fenceCount, const VkFence *pFences)
- vk_result vkQueueSubmit (VkQueue queue, uint32_t submitCount, const VkSubmitInfo *pSubmits, VkFence fence)
- vk_result vkAcquireNextImageKHR (VkSwapchainKHR swapchain, uint64_t timeout, VkSemaphore semaphore, VkFence fence, uint32_t *pImageIndex)
- vk_result vkQueuePresentKHR (VkQueue queue, const VkPresentInfoKHR *pPresentInfo)
- vk_result vkCreateSwapchainKHR (const VkSwapchainCreateInfoKHR *pCreateInfo, const VkAllocation↩Callbacks *pAllocator, VkSwapchainKHR *pSwapchain)
- vk_result vkCreateSwapchainKHR (const VkSwapchainCreateInfoKHR *pCreateInfo, VkSwapchainKHR *pSwapchain)
- void vkDestroySwapchainKHR (VkSwapchainKHR swapchain, const VkAllocationCallbacks *p↩Allocator=memory::instance().alloc())
- vk_result vkGetSwapchainImagesKHR (VkSwapchainKHR swapchain, uint32_t *pSwapchainImageCount, VkImage *pSwapchainImages)
- vk_result vkCreateCommandPool (const VkCommandPoolCreateInfo *pCreateInfo, const VkAllocation↩Callbacks *pAllocator, VkCommandPool *pCommandPool)
- vk_result vkCreateCommandPool (const VkCommandPoolCreateInfo *pCreateInfo, VkCommandPool *p↩CommandPool)

- vk_result vkCreateCommandPool (index queue_family, VkCommandPool *pCommandPool)
- vk_result vkAllocateCommandBuffers (const VkCommandBufferAllocateInfo *pAllocateInfo, VkCommandBuffer *pCommandBuffers)
- vk_result vkAllocateCommandBuffers (VkCommandPool commandPool, uint32_t commandBufferCount, VkCommandBuffer *pCommandBuffers, VkCommandBufferLevel level=VK_COMMAND_BUFFER_LEVEL_PRIMARY)
- void vkDestroyImageView (VkImageView imageView, const VkAllocationCallbacks *pAllocator=memory::instance().alloc())
- void vkDestroyFence (VkFence fence, const VkAllocationCallbacks *pAllocator=memory::instance().alloc())
- void vkDestroySemaphore (VkSemaphore semaphore, const VkAllocationCallbacks *pAllocator=memory::instance().alloc())
- void vkFreeCommandBuffers (VkCommandPool commandPool, uint32_t commandBufferCount, const VkCommandBuffer *pCommandBuffers)
- void vkDestroyCommandPool (VkCommandPool commandPool, const VkAllocationCallbacks *pAllocator=memory::instance().alloc())
- void vkDestroySampler (VkSampler sampler, const VkAllocationCallbacks *pAllocator=memory::instance().alloc())
- void vkUpdateDescriptorSets (uint32_t descriptorWriteCount, const VkWriteDescriptorSet *pDescriptorWrites, uint32_t descriptorCopyCount=0, const VkCopyDescriptorSet *pDescriptorCopies=nullptr)
- template<std::size_t SIZE>
  void vkUpdateDescriptorSets (std::array< VkWriteDescriptorSet, SIZE > const &descriptor_writes)
- template<std::size_t SIZE>
  void vkUpdateDescriptorSets (std::array< VkCopyDescriptorSet, SIZE > const &descriptor_copies)
- template<std::size_t WRITE_SIZE, std::size_t COPY_SIZE>
  void vkUpdateDescriptorSets (std::array< VkWriteDescriptorSet, WRITE_SIZE > const &descriptor_writes, std::array< VkCopyDescriptorSet, COPY_SIZE > const &descriptor_copies)
- void vkUpdateDescriptorSets (std::initializer_list< VkWriteDescriptorSet > descriptor_writes)
- void vkUpdateDescriptorSets (std::initializer_list< VkCopyDescriptorSet > descriptor_copies)
- void vkUpdateDescriptorSets (std::initializer_list< VkWriteDescriptorSet > descriptor_writes, std::initializer_list< VkCopyDescriptorSet > descriptor_copies)

Public Attributes

- VkDevice **vk_device** = nullptr

  Vulkan device.
- VolkDeviceTable **table** = {}

  Volk device table.

4.25.1 Detailed Description

Device functions.

4.25.2 Member Function Documentation

4.25.2.1 vkAcquireNextImageKHR()

```
vk_result lava::device_table::vkAcquireNextImageKHR (
            VkSwapchainKHR swapchain,
            uint64_t timeout,
            VkSemaphore semaphore,
            VkFence fence,
            uint32_t * pImageIndex )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3-extensions/man/html/vk↩AcquireNextImageKHR.html

4.25.2.2 vkAllocateCommandBuffers() [1/2]

```
vk_result lava::device_table::vkAllocateCommandBuffers (
            const VkCommandBufferAllocateInfo * pAllocateInfo,
            VkCommandBuffer * pCommandBuffers )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩AllocateCommandBuffers

4.25.2.3 vkAllocateCommandBuffers() [2/2]

```
vk_result lava::device_table::vkAllocateCommandBuffers (
            VkCommandPool commandPool,
            uint32_t commandBufferCount,
            VkCommandBuffer * pCommandBuffers,
            VkCommandBufferLevel level = VK_COMMAND_BUFFER_LEVEL_PRIMARY )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩AllocateCommandBuffers

4.25.2.4 vkCreateCommandPool() [1/3]

```
vk_result lava::device_table::vkCreateCommandPool (
            const VkCommandPoolCreateInfo * pCreateInfo,
            const VkAllocationCallbacks * pAllocator,
            VkCommandPool * pCommandPool )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩CreateCommandPool

4.25.2.5 vkCreateCommandPool() [2/3]

```
vk_result lava::device_table::vkCreateCommandPool (
            const VkCommandPoolCreateInfo * pCreateInfo,
            VkCommandPool * pCommandPool )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩CreateCommandPool

4.25.2.6 vkCreateCommandPool() [3/3]

```
vk_result lava::device_table::vkCreateCommandPool (
            index queue_family,
            VkCommandPool * pCommandPool )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩CreateCommandPool

4.25.2.7 vkCreateFence() [1/2]

```
vk_result lava::device_table::vkCreateFence (
            const VkFenceCreateInfo * pCreateInfo,
            const VkAllocationCallbacks * pAllocator,
            VkFence * pFence )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩CreateFence

4.25.2.8 vkCreateFence() [2/2]

```
vk_result lava::device_table::vkCreateFence (
            const VkFenceCreateInfo * pCreateInfo,
            VkFence * pFence )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩CreateFence

4.25.2.9 vkCreateImageView() [1/2]

```
vk_result lava::device_table::vkCreateImageView (
            const VkImageViewCreateInfo * pCreateInfo,
            const VkAllocationCallbacks * pAllocator,
            VkImageView * pView )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩CreateImageView

4.25.2.10 vkCreateImageView() [2/2]

```
vk_result lava::device_table::vkCreateImageView (
            const VkImageViewCreateInfo * pCreateInfo,
            VkImageView * pView )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩CreateImageView

4.25.2.11 vkCreateSampler() [1/2]

```
vk_result lava::device_table::vkCreateSampler (
            const VkSamplerCreateInfo * pCreateInfo,
            const VkAllocationCallbacks * pAllocator,
            VkSampler * pSampler )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩CreateSampler

4.25.2.12 vkCreateSampler() [2/2]

```
vk_result lava::device_table::vkCreateSampler (
            const VkSamplerCreateInfo * pCreateInfo,
            VkSampler * pSampler )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩CreateSampler

4.25.2.13 vkCreateSemaphore() [1/2]

```
vk_result lava::device_table::vkCreateSemaphore (
            const VkSemaphoreCreateInfo * pCreateInfo,
            const VkAllocationCallbacks * pAllocator,
            VkSemaphore * pSemaphore )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vkCreateSemaphore

4.25.2.14 vkCreateSemaphore() [2/2]

```
vk_result lava::device_table::vkCreateSemaphore (
            const VkSemaphoreCreateInfo * pCreateInfo,
            VkSemaphore * pSemaphore )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vkCreateSemaphore

4.25.2.15 vkCreateShaderModule() [1/2]

```
vk_result lava::device_table::vkCreateShaderModule (
            const VkShaderModuleCreateInfo * pCreateInfo,
            const VkAllocationCallbacks * pAllocator,
            VkShaderModule * pShaderModule )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vkCreateShaderModule

4.25.2.16 vkCreateShaderModule() [2/2]

```
vk_result lava::device_table::vkCreateShaderModule (
            const VkShaderModuleCreateInfo * pCreateInfo,
            VkShaderModule * pShaderModule )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vkCreateShaderModule

4.25.2.17 vkCreateSwapchainKHR() [1/2]

```
vk_result lava::device_table::vkCreateSwapchainKHR (
            const VkSwapchainCreateInfoKHR * pCreateInfo,
            const VkAllocationCallbacks * pAllocator,
            VkSwapchainKHR * pSwapchain )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3-extensions/man/html/vkCreateSwapchainKHR.html

4.25.2.18 vkCreateSwapchainKHR() [2/2]

```
vk_result lava::device_table::vkCreateSwapchainKHR (
            const VkSwapchainCreateInfoKHR * pCreateInfo,
            VkSwapchainKHR * pSwapchain )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3-extensions/man/html/vkCreateSwapchainKHR.html

4.25.2.19 vkDestroyCommandPool()

```
void lava::device_table::vkDestroyCommandPool (
            VkCommandPool commandPool,
            const VkAllocationCallbacks * pAllocator = memory::instance().alloc() )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vkDestroyCommandPool

4.25.2.20 vkDestroyFence()

```
void lava::device_table::vkDestroyFence (
            VkFence fence,
            const VkAllocationCallbacks * pAllocator = memory::instance().alloc() )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vkDestroyFence

4.25.2.21 vkDestroyImageView()

```
void lava::device_table::vkDestroyImageView (
            VkImageView imageView,
            const VkAllocationCallbacks * pAllocator = memory::instance().alloc() )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩DestroyImageView

4.25.2.22 vkDestroySampler()

```
void lava::device_table::vkDestroySampler (
            VkSampler sampler,
            const VkAllocationCallbacks * pAllocator = memory::instance().alloc() )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩DestroySampler

4.25.2.23 vkDestroySemaphore()

```
void lava::device_table::vkDestroySemaphore (
            VkSemaphore semaphore,
            const VkAllocationCallbacks * pAllocator = memory::instance().alloc() )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩DestroySemaphore

4.25.2.24 vkDestroySwapchainKHR()

```
void lava::device_table::vkDestroySwapchainKHR (
            VkSwapchainKHR swapchain,
            const VkAllocationCallbacks * pAllocator = memory::instance().alloc() )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3-extensions/man/html/vk↩DestroySwapchainKHR.html

4.25.2.25 vkFreeCommandBuffers()

```
void lava::device_table::vkFreeCommandBuffers (
            VkCommandPool commandPool,
            uint32_t commandBufferCount,
            const VkCommandBuffer * pCommandBuffers )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩FreeCommandBuffers

4.25.2.26 vkGetSwapchainImagesKHR()

```
vk_result lava::device_table::vkGetSwapchainImagesKHR (
            VkSwapchainKHR swapchain,
            uint32_t * pSwapchainImageCount,
            VkImage * pSwapchainImages )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3-extensions/man/html/vk↩GetSwapchainImagesKHR.html

4.25.2.27 vkQueuePresentKHR()

```
vk_result lava::device_table::vkQueuePresentKHR (
            VkQueue queue,
            const VkPresentInfoKHR * pPresentInfo )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3-extensions/man/html/vk↩QueuePresentKHR.html

4.25.2.28 vkQueueSubmit()

```
vk_result lava::device_table::vkQueueSubmit (
            VkQueue queue,
            uint32_t submitCount,
            const VkSubmitInfo * pSubmits,
            VkFence fence )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩QueueSubmit

4.25.2.29 vkResetFences()

```
vk_result lava::device_table::vkResetFences (
            uint32_t fenceCount,
            const VkFence * pFences )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vkResetFences

4.25.2.30 vkUpdateDescriptorSets() [1/7]

```
template<std::size_t SIZE>
void lava::device_table::vkUpdateDescriptorSets (
            std::array< VkCopyDescriptorSet, SIZE > const & descriptor_copies )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vkUpdateDescriptorSets

4.25.2.31 vkUpdateDescriptorSets() [2/7]

```
template<std::size_t SIZE>
void lava::device_table::vkUpdateDescriptorSets (
            std::array< VkWriteDescriptorSet, SIZE > const & descriptor_writes )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vkUpdateDescriptorSets

4.25.2.32 vkUpdateDescriptorSets() [3/7]

```
template<std::size_t WRITE_SIZE, std::size_t COPY_SIZE>
void lava::device_table::vkUpdateDescriptorSets (
            std::array< VkWriteDescriptorSet, WRITE_SIZE > const & descriptor_writes,
            std::array< VkCopyDescriptorSet, COPY_SIZE > const & descriptor_copies )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vkUpdateDescriptorSets

4.25.2.33 vkUpdateDescriptorSets() [4/7]

```
void lava::device_table::vkUpdateDescriptorSets (
            std::initializer_list< VkCopyDescriptorSet > descriptor_copies )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩UpdateDescriptorSets

4.25.2.34 vkUpdateDescriptorSets() [5/7]

```
void lava::device_table::vkUpdateDescriptorSets (
            std::initializer_list< VkWriteDescriptorSet > descriptor_writes )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩UpdateDescriptorSets

4.25.2.35 vkUpdateDescriptorSets() [6/7]

```
void lava::device_table::vkUpdateDescriptorSets (
            std::initializer_list< VkWriteDescriptorSet > descriptor_writes,
            std::initializer_list< VkCopyDescriptorSet > descriptor_copies )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩UpdateDescriptorSets

4.25.2.36 vkUpdateDescriptorSets() [7/7]

```
void lava::device_table::vkUpdateDescriptorSets (
            uint32_t descriptorWriteCount,
            const VkWriteDescriptorSet * pDescriptorWrites,
            uint32_t descriptorCopyCount = 0,
            const VkCopyDescriptorSet * pDescriptorCopies = nullptr )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩UpdateDescriptorSets

4.25.2.37 vkWaitForFences()

```
vk_result lava::device_table::vkWaitForFences (
            uint32_t fenceCount,
            const VkFence * pFences,
            VkBool32 waitAll,
            uint64_t timeout )  [inline]
```

See also

https://www.khronos.org/registry/vulkan/specs/1.3/html/vkspec.html#vk↩WaitForFences

The documentation for this struct was generated from the following file:

- liblava/base/device_table.hpp

4.26 lava::dispatcher Struct Reference

Telegram dispatcher.

```
#include <telegram.hpp>
```

Public Types

- using **message_func** = std::function< void(telegram::ref, id::ref)>

  Message function.

Public Member Functions

- void setup (ui32 thread_count)

  Set up the dispatcher.
- void **teardown** ()

  Tear down the dispatcher.
- void update (ms current)

  Update the dispatcher.
- void add_message (id::ref receiver, id::ref sender, id::ref message, ms delay={}, any const &info={})

  Add message to dispatcher.

Public Attributes

- message_func **on_message**

  Called on message handling.

4.26.1 Detailed Description

Telegram dispatcher.

4.26.2 Member Function Documentation

4.26.2.1 add_message()

```
void lava::dispatcher::add_message (
            id::ref receiver,
            id::ref sender,
            id::ref message,
            ms delay = {},
            any const & info = {} )  [inline]
```

Add message to dispatcher.

Parameters

receiver	Receiver id
sender	Sender id
message	Telegram message
delay	Delay time
info	Telegram information

4.26.2.2 setup()

```
void lava::dispatcher::setup (
            ui32 thread_count )  [inline]
```

Set up the dispatcher.

Parameters

thread_count	Number of threads

4.26.2.3 update()

```
void lava::dispatcher::update (
            ms current )  [inline]
```

Update the dispatcher.

Parameters

current	Time in milliseconds

The documentation for this struct was generated from the following file:

- liblava/util/telegram.hpp

4.27 lava::driver Struct Reference

Stage driver.

```
#include <driver.hpp>
```

Classes

- struct result

  Driver result.

Public Types

- enum error { **stages_empty** = -1 , **stage_not_found** = -2 , **undef_run** = -3 }

  Driver error codes.
- using **run_func** = std::function< result(argh::parser)>

  Run function.

Public Member Functions

- void add_stage (stage *stage)

  Add a stage.
- stage::map const & get_stages () const

  Get all stages.
- i32 run (argh::parser cmd_line={})

  Run the driver.

Static Public Member Functions

- static driver & instance ()

  Get driver instance.

Public Attributes

- run_func **on_run**

  Called if no stage has been selected.

4.27.1 Detailed Description

Stage driver.

4.27.2 Member Function Documentation

4.27.2.1 add_stage()

```
void lava::driver::add_stage (
            stage * stage )  [inline]
```

Add a stage.

Parameters

stage	Stage to add

4.27.2.2 get_stages()

```
stage::map const & lava::driver::get_stages ( ) const  [inline]
```

Get all stages.

Returns

stage::map const& Map of stages

4.27.2.3 instance()

```
static driver & lava::driver::instance ( )  [inline], [static]
```

Get driver instance.

Returns

driver& Stage driver

4.27.2.4 run()

```
i32 lava::driver::run (
            argh::parser cmd_line = {} )
```

Run the driver.

Parameters

cmd_line	Command line arguments

Returns

i32 Result code

The documentation for this struct was generated from the following file:

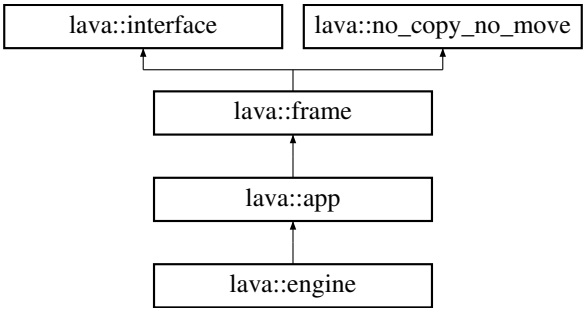
- liblava/frame/driver.hpp

4.28 lava::engine Struct Reference

Engine.

```
#include <engine.hpp>
```

Inheritance diagram for lava::engine:



Public Member Functions

- bool setup () override

  Set up the engine.
- **app** (frame_env::ref env)

  App constructors.
- **app** (name name, argh::parser cmd_line={})

  App constructors.

Public Attributes

- property **props**

  Props master.
- producer_t **producer**

  Producer.

Additional Inherited Members

4.28.1 Detailed Description

Engine.

4.28.2 Member Function Documentation

4.28.2.1 setup()

```
bool lava::engine::setup ( ) [override], [virtual]
```

Set up the engine.

Returns

Setup was successful or failed

Reimplemented from lava::app.

The documentation for this struct was generated from the following file:

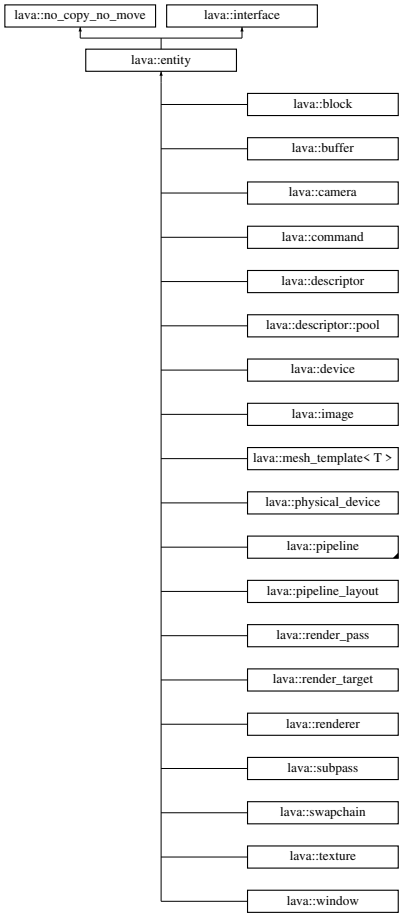
- liblava/engine/engine.hpp

4.29 lava::entity Struct Reference

Entity.

```
#include <id.hpp>
```

Inheritance diagram for lava::entity:



Public Member Functions

- **entity** ()

  Construct a new entity.
- id::ref get_id () const

  Get the id of entity.

4.29.1 Detailed Description

Entity.

4.29.2 Member Function Documentation

4.29.2.1 get_id()

```
id::ref lava::entity::get_id ( ) const  [inline]
```

Get the id of entity.

Returns

id::ref Entity id

The documentation for this struct was generated from the following file:

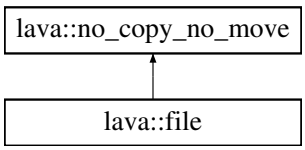
- liblava/core/id.hpp

4.30 lava::file Struct Reference

File.

```
#include <file.hpp>
```

Inheritance diagram for lava::file:



Public Types

- using **ref** = file const &

  Reference to file.

Public Member Functions

- file (string_ref path="", file_mode mode=file_mode::read)

  Construct a new file.
- ~**file** ()

  Destroy the file.
- bool open (string_ref path, file_mode mode=file_mode::read)

  Open the file.
- void **close** ()

  Close the file.
- bool opened () const

  Check if the file is opened.
- i64 get_size () const

  Get the size of the file.
- i64 read (data_ptr data)

  Read data from file.
- i64 read (data_ptr data, ui64 size)

  Read data from file (limited size)
- i64 write (data_cptr data, ui64 size)

  Write data to file.
- i64 seek (ui64 position)

  Seek to position in the file.
- i64 tell () const

  Get the current position in the file.
- bool writable () const

  Check if the file is in write mode.
- file_type get_type () const

  Get the file type.
- string_ref get_path () const

  Get the path of the file.

4.30.1 Detailed Description

File.

4.30.2 Constructor & Destructor Documentation

4.30.2.1 file()

```
lava::file::file (
            string_ref path = "",
            file_mode mode = file_mode::read )  [explicit]
```

Construct a new file.

Parameters

path	Name of file
mode	File mode

4.30.3 Member Function Documentation

4.30.3.1 get_path()

```
string_ref lava::file::get_path ( ) const  [inline]
```

Get the path of the file.

Returns

name File path

4.30.3.2 get_size()

```
i64 lava::file::get_size ( ) const
```

Get the size of the file.

Returns

i64 File size

4.30.3.3 get_type()

```
file_type lava::file::get_type ( ) const  [inline]
```

Get the file type.

Returns

file_type Type of file

4.30.3.4 open()

```
bool lava::file::open (
            string_ref path,
            file_mode mode = file_mode::read )
```

Open the file.

Parameters

path	Name of file
mode	File mode

Returns

Open was successful or failed

4.30.3.5 opened()

```
bool lava::file::opened ( ) const
```

Check if the file is opened.

Returns

File is opened or not

4.30.3.6 read() [1/2]

```
i64 lava::file::read (
            data_ptr data )  [inline]
```

Read data from file.

Parameters

data	Data to read

Returns

i64 File size

4.30.3.7 read() [2/2]

```
i64 lava::file::read (
            data_ptr data,
            ui64 size )
```

Read data from file (limited size)

Parameters

data	Data to read
size	File size

Returns

i64 File size

4.30.3.8 seek()

```
i64 lava::file::seek (
            ui64 position )
```

Seek to position in the file.

Parameters

position	Position to seek to

Returns

i64 Current position

4.30.3.9 tell()

```
i64 lava::file::tell ( ) const
```

Get the current position in the file.

Returns

i64 Current position

4.30.3.10 writable()

```
bool lava::file::writable ( ) const  [inline]
```

Check if the file is in write mode.

Returns

File is writable or only readable

4.30.3.11 write()

```
i64 lava::file::write (
            data_cptr data,
            ui64 size )
```

Write data to file.

Parameters

data	Data to write
size	File size

Returns

i64 File size

The documentation for this struct was generated from the following file:

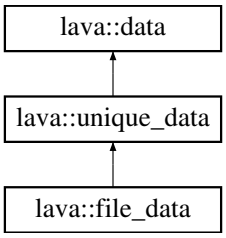
- liblava/file/file.hpp

4.31 lava::file_data Struct Reference

File data.

```
#include <file_utils.hpp>
```

Inheritance diagram for lava::file_data:



Public Types

- using **ref** = file_data const &

  Reference to file data.

Public Member Functions

- file_data (string_ref filename)

  Construct a new file data.
- **unique_data** (size_t length=0, data_mode mode=data_mode::alloc)

  Unique data constructors.
- **unique_data** (data::ref data)

  Unique data constructors.

Public Attributes

- string **filename**

  Name of file.

4.31.1 Detailed Description

File data.

4.31.2 Constructor & Destructor Documentation

4.31.2.1 file_data()

```
lava::file_data::file_data (
            string_ref filename )  [inline], [explicit]
```

Construct a new file data.

Parameters

filename	Name of file

The documentation for this struct was generated from the following file:

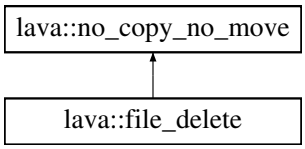
- liblava/file/file_utils.hpp

4.32 lava::file_delete Struct Reference

File delete guard.

`#include <file_utils.hpp>`

Inheritance diagram for lava::file_delete:



Public Member Functions

- file_delete (string filename="")

  Construct a new file delete.
- ~**file_delete** ()

  Destroy the file delete.

Public Attributes

- string **filename**

  Name of file.
- bool **active** = true

  Active state.

4.32.1 Detailed Description

File delete guard.

4.32.2 Constructor & Destructor Documentation

4.32.2.1 file_delete()

```
lava::file_delete::file_delete (
            string filename = "" )  [inline], [explicit]
```

Construct a new file delete.

Parameters

filename	Name of file

The documentation for this struct was generated from the following file:

- liblava/file/file_utils.hpp

4.33 lava::file_format Struct Reference

File format.

```
#include <texture.hpp>
```

Public Types

- using **list** = std::vector< file_format >

  List of file formats.

Public Attributes

- string **path**

  File path.
- VkFormat **format** = VK_FORMAT_UNDEFINED

  File format.

4.33.1 Detailed Description

File format.

The documentation for this struct was generated from the following file:

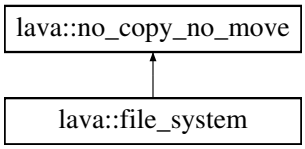
- liblava/resource/texture.hpp

4.34 lava::file_system Struct Reference

File system.

```
#include <file_system.hpp>
```

Inheritance diagram for lava::file_system:



Public Member Functions

- int_version get_version ()

  Get the version.
- string get_base_dir ()

  Get the base directory.
- string get_full_base_dir (string_ref path)

  Get the path relative to base directory.
- string get_pref_dir ()

  Get the preferences directory.
- string get_res_dir ()

  Get the resource directory.
- bool mount (string_ref path)

  Mount path.
- bool mount_base (string_ref base_dir_path)

  Mount base directory path.
- bool exists (string_ref file)

  Check if file exists.
- string get_real_dir (string_ref file)

  Get the real directory of file.
- string_list enumerate_files (string_ref path)

  Enumerate files in directory.
- bool initialize (string_ref argv_0, string_ref org, string_ref app, string_ref ext)

  Initialize the file system.
- void **terminate** ()

  Terminate the file system.
- string_list mount_res ()

Mount resource directories and files.

- bool create_folder (string_ref name="data")

  Create a folder in the preferences directory.
- void **clean_pref_dir** ()

  Clean preferences directory.
- string_ref get_org () const

  Get the organization name.
- string_ref get_app () const

  Get the application name.
- string_ref get_ext () const

  Get the extension name.
- bool ready () const

  Check if file system is ready.

4.34.1 Detailed Description

File system.

4.34.2 Member Function Documentation

4.34.2.1 create_folder()

```
bool lava::file_system::create_folder (
            string_ref name = "data" )
```

Create a folder in the preferences directory.

Parameters

name	Name of folder (default: data)

Returns

Folder created or not

4.34.2.2 enumerate_files()

```
string_list lava::file_system::enumerate_files (
            string_ref path )
```

Enumerate files in directory.

Parameters

path	Target directory

Returns

string_list List of files

4.34.2.3 exists()

```
bool lava::file_system::exists (
            string_ref file )
```

Check if file exists.

Parameters

file	File to check

Returns

File exists or not found

4.34.2.4 get_app()

```
string_ref lava::file_system::get_app ( ) const [inline]
```

Get the application name.

Returns

string_ref Name of application

4.34.2.5 get_base_dir()

```
string lava::file_system::get_base_dir ( )
```

Get the base directory.

Returns

string Base directory

4.34.2.6 get_ext()

```
string_ref lava::file_system::get_ext ( ) const [inline]
```

Get the extension name.

Returns

string_ref Name of extension

4.34.2.7 get_full_base_dir()

```
string lava::file_system::get_full_base_dir (
            string_ref path )
```

Get the path relative to base directory.

Parameters

path	Path to add to base directory

Returns

string Relative base directory path

4.34.2.8 get_org()

```
string_ref lava::file_system::get_org ( ) const [inline]
```

Get the organization name.

Returns

string_ref Name of organization

4.34.2.9 get_pref_dir()

```
string lava::file_system::get_pref_dir ( )
```

Get the preferences directory.

Returns

string Preferences directory

4.34.2.10 get_real_dir()

```
string lava::file_system::get_real_dir (
            string_ref file )
```

Get the real directory of file.

Parameters

file	Target file

Returns

string Real directory of file

4.34.2.11 get_res_dir()

```
string lava::file_system::get_res_dir ( )
```

Get the resource directory.

Returns

string Resource directory

4.34.2.12 get_version()

```
int_version lava::file_system::get_version ( )
```

Get the version.

Returns

int_version Internal version

4.34.2.13 initialize()

```
bool lava::file_system::initialize (
            string_ref argv_0,
            string_ref org,
            string_ref app,
            string_ref ext )
```

Initialize the file system.

Parameters

argv↩ _0	First command line argument
org	Organization name
app	Application name
ext	Extension name

Returns

Initialize was successful or failed

4.34.2.14 mount()

```
bool lava::file_system::mount (
            string_ref path )
```

Mount path.

Parameters

path	Path to mount

Returns

Mount was successful or failed

4.34.2.15 mount_base()

```
bool lava::file_system::mount_base (
            string_ref base_dir_path )
```

Mount base directory path.

Parameters

base_dir_path	Base directory path

Returns

Mount was successful or failed

4.34.2.16 mount_res()

```
string_list lava::file_system::mount_res ( )
```

Mount resource directories and files.

Returns

string_list List of mounted resources

4.34.2.17 ready()

```
bool lava::file_system::ready ( ) const  [inline]
```

Check if file system is ready.

Returns

File system is ready or not

The documentation for this struct was generated from the following file:

- liblava/file/file_system.hpp

4.35 lava::imgui::font Struct Reference

ImGui font settings.

```
#include <imgui.hpp>
```

Public Types

- using **ref** = font const &

  Const font reference.

Public Attributes

- string **file**

  Font file.
- r32 **size** = 21.f

  Font size.
- string **icon_file**

  Font icon file.
- r32 **icon_size** = 21.f

  Font icon size.
- ui16 **icon_range_begin** = 0

  Font range begin.
- ui16 **icon_range_end** = 0

  Font range end.

4.35.1 Detailed Description

ImGui font settings.

The documentation for this struct was generated from the following file:

- liblava/app/imgui.hpp

4.36 lava::forward_shading Struct Reference

Forward shading.

```
#include <forward_shading.hpp>
```

Public Member Functions

- **forward_shading** ()=default

  Construct a new forward shading.
- ~**forward_shading** ()

  Destroy the forward shading.
- bool create (render_target::ptr target)

  Create a forward shading for a render target.
- void **destroy** ()

  Destroy the forward shading.
- render_pass::ptr get_pass () const

  Get the render pass.
- VkRenderPass get_vk_pass () const

  Get the Vulkan render pass.
- image::ptr get_depth_stencil () const

  Get the depth stencil image.

4.36.1 Detailed Description

Forward shading.

4.36.2 Member Function Documentation

4.36.2.1 create()

```
bool lava::forward_shading::create (
            render_target::ptr target )
```

Create a forward shading for a render target.

Parameters

target	Render target

Returns

Create was successful or failed

4.36.2.2 get_depth_stencil()

```
image::ptr lava::forward_shading::get_depth_stencil ( ) const  [inline]
```

Get the depth stencil image.

Returns

image::ptr Depth stencil Image

4.36.2.3 get_pass()

```
render_pass::ptr lava::forward_shading::get_pass ( ) const  [inline]
```

Get the render pass.

Returns

render_pass::ptr Render pass

4.36.2.4 get_vk_pass()

```
VkRenderPass lava::forward_shading::get_vk_pass ( ) const  [inline]
```

Get the Vulkan render pass.

Returns

VkRenderPass Vulkan Render pass

The documentation for this struct was generated from the following file:

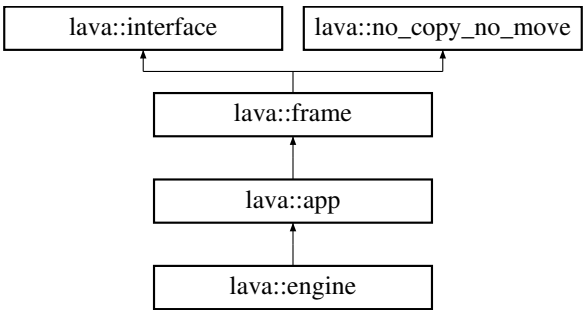
- liblava/app/forward_shading.hpp

4.37 lava::frame Struct Reference

Framework.

```
#include <frame.hpp>
```

Inheritance diagram for lava::frame:



Public Types

- using **ptr** = std::shared_ptr< frame >

  Shared pointer to framework.
- using **result** = i32

  Framework result.
- using **run_func** = std::function< bool(id::ref)>

  Run function.
- using **run_func_ref** = run_func const &

  Reference to run function.
- using **run_end_func** = std::function< void()>

  Run end function.
- using **run_end_func_ref** = run_end_func const &

  Reference to run end function.
- using **run_once_func** = std::function< bool()>

  Run once function.
- using **run_once_func_ref** = run_once_func const &

  Reference to run once function.

Public Member Functions

- frame (argh::parser cmd_line)

  Construct a new framework.
- frame (frame_env env)

  Construct a new framework.
- ~**frame** () override

  Destroy the framework.
- bool ready () const

  Check if framework is ready.
- result run ()

  Run the framework.

- bool shut_down ()

  Shut down the framework.
- id add_run (run_func_ref func)

  Add run to framework.
- id add_run_end (run_end_func_ref func)

  Add run end to framework.
- void add_run_once (run_once_func_ref func)

  Add run once to framework.
- bool remove (id::ref func_id)

  Remove a function from framework.
- ms get_running_time () const

  Get the running time.
- r64 get_running_time_sec () const

  Get the running time in seconds.
- cmd_line get_cmd_line () const

  Get the command line arguments.
- frame_env::ref get_env () const

  Get the framework environment.
- name get_name () const

  Get the name of application.
- bool waiting_for_events () const

  Check if framework is waiting for events.
- void set_wait_for_events (bool value=true)

  Set wait for events in framework.

Public Attributes

- platform_t **platform**

  Stage platform.

4.37.1 Detailed Description

Framework.

4.37.2 Constructor & Destructor Documentation

4.37.2.1 frame() [1/2]

```
lava::frame::frame (
            argh::parser cmd_line )  [explicit]
```

Construct a new framework.

Parameters

cmd_line	Command line arguments

4.37.2.2 frame() [2/2]

```
lava::frame::frame (
            frame_env env )  [explicit]
```

Construct a new framework.

Parameters

env	Framework environment

4.37.3 Member Function Documentation

4.37.3.1 add_run()

```
id lava::frame::add_run (
            run_func_ref func )
```

Add run to framework.

Parameters

func	Run function

Returns

id Id of function

4.37.3.2 add_run_end()

```
id lava::frame::add_run_end (
            run_end_func_ref func )
```

Add run end to framework.

Parameters

func	Run end function

Returns

id Id of function

4.37.3.3 add_run_once()

```
void lava::frame::add_run_once (
            run_once_func_ref func ) [inline]
```

Add run once to framework.

Parameters

func	Run once function

4.37.3.4 get_cmd_line()

```
cmd_line lava::frame::get_cmd_line ( ) const [inline]
```

Get the command line arguments.

Returns

cmd_line Command line arguments

4.37.3.5 get_env()

```
frame_env::ref lava::frame::get_env ( ) const [inline]
```

Get the framework environment.

Returns

frame_env::ref Framework environment

4.37.3.6 get_name()

```
name lava::frame::get_name ( ) const [inline]
```

Get the name of application.

Returns

name Name of application

4.37.3.7 get_running_time()

```
ms lava::frame::get_running_time ( ) const [inline]
```

Get the running time.

Returns

ms Time since start of framework

4.37.3.8 get_running_time_sec()

```
r64 lava::frame::get_running_time_sec ( ) const [inline]
```

Get the running time in seconds.

Returns

r64 Time since start of framework

4.37.3.9 ready()

```
bool lava::frame::ready ( ) const [inline]
```

Check if framework is ready.

Returns

Framework is ready or not

4.37.3.10 remove()

```
bool lava::frame::remove (
            id::ref func_id )
```

Remove a function from framework.

Parameters

func↩ _id	Id of function

Returns

Remove was successful or failed

4.37.3.11 run()

```
result lava::frame::run ( )
```

Run the framework.

Returns

result Run result

4.37.3.12 set_wait_for_events()

```
void lava::frame::set_wait_for_events (
            bool value = true )  [inline]
```

Set wait for events in framework.

Parameters

value	Wait for events state

4.37.3.13 shut_down()

```
bool lava::frame::shut_down ( )
```

Shut down the framework.

Returns

Shut down was successful or failed

4.37.3.14 waiting_for_events()

```
bool lava::frame::waiting_for_events ( ) const  [inline]
```

Check if framework is waiting for events.

Returns

Framework waits for events or not

The documentation for this struct was generated from the following file:

- liblava/frame/frame.hpp

4.38 lava::frame_env Struct Reference

Framework environment.

```
#include <frame.hpp>
```

Public Types

- using **ref** = frame_env const &

  Reference to frame environment.

Public Member Functions

- **frame_env** ()

  Construct a new frame environment.
- frame_env (name app_name, argh::parser cmd_line)

  Construct a new frame environment.
- void **set_default** ()

  Set default settings.

Public Attributes

- argh::parser **cmd_line**

  Command line arguments.
- log_config **log**

  Logging configuration.
- instance_info **info**

  Instance information.
- instance::create_param **param**

  Instance create parameters.
- instance::debug_config **debug**

  Intance debug configuration.

4.38.1 Detailed Description

Framework environment.

4.38.2 Constructor & Destructor Documentation

4.38.2.1 frame_env()

```
lava::frame_env::frame_env (
            name app_name,
            argh::parser cmd_line )  [inline], [explicit]
```

Construct a new frame environment.

Parameters

app_name	Name of application
cmd_line	Command line arguments

The documentation for this struct was generated from the following file:

- liblava/frame/frame.hpp

4.39 lava::gamepad Struct Reference

Gamepad.

```
#include <gamepad.hpp>
```

Public Types

- using **list** = std::vector< gamepad >

  List of gamepads.
- using **ref** = gamepad const &

  Reference to gamepad.

Public Member Functions

- gamepad (gamepad_id_ref pad_id=gamepad_id::_1)

  Construct a new gamepad.
- bool ready () const

  Check if gamepad is active.
- bool update ()

Update gamepad.

- bool pressed (gamepad_button_ref button) const

  Check if gamepad button is pressed.
- r32 value (gamepad_axis_ref axis) const

  Get value of axis.
- gamepad_id_ref get_pad_id () const

  Get the gamepad id.
- ui32 get_id () const

  Get the gamepad id as integer.
- name get_name () const

  Get the name.

4.39.1 Detailed Description

Gamepad.

4.39.2 Constructor & Destructor Documentation

4.39.2.1 gamepad()

```
lava::gamepad::gamepad (
            gamepad_id_ref pad_id = gamepad_id::_1 )  [explicit]
```

Construct a new gamepad.

Parameters

pad↩ _id	Gamepad id

4.39.3 Member Function Documentation

4.39.3.1 get_id()

```
ui32 lava::gamepad::get_id ( ) const  [inline]
```

Get the gamepad id as integer.

Returns

ui32 Integer gamepad id

4.39.3.2 get_name()

```
name lava::gamepad::get_name ( ) const
```

Get the name.

Returns

name Name of gamepad

4.39.3.3 get_pad_id()

```
gamepad_id_ref lava::gamepad::get_pad_id ( ) const [inline]
```

Get the gamepad id.

Returns

gamepad_id_ref Gamepad id

4.39.3.4 pressed()

```
bool lava::gamepad::pressed (
            gamepad_button_ref button ) const [inline]
```

Check if gamepad button is pressed.

Parameters

button	Gamepad button to check

Returns

Button is pressed or not

4.39.3.5 ready()

```
bool lava::gamepad::ready ( ) const
```

Check if gamepad is active.

Returns

Gamepad is active or not

4.39.3.6 update()

```
bool lava::gamepad::update ( )
```

Update gamepad.

Returns

Update was successful or failed

4.39.3.7 value()

```
r32 lava::gamepad::value (
            gamepad_axis_ref axis ) const  [inline]
```

Get value of axis.

Parameters

axis	Target axis

Returns

r32 Axis value

The documentation for this struct was generated from the following file:

- liblava/frame/gamepad.hpp

4.40 lava::gamepad_manager Struct Reference

Gamepad manager.

```
#include <gamepad.hpp>
```

Public Types

- using **listener_func** = std::function< bool(gamepad, bool)>

  Gamepad listener function.

Public Member Functions

- id add (listener_func listener)

  Add listener.
- void remove (id::ref func_id)

  Remove listener.

Static Public Member Functions

- static gamepad_manager & singleton ()

  Get gamepad manager singleton.

4.40.1 Detailed Description

Gamepad manager.

4.40.2 Member Function Documentation

4.40.2.1 add()

```
id lava::gamepad_manager::add (
            listener_func listener )
```

Add listener.

Parameters

listener	Gamepad listener function

Returns

id Id of function

4.40.2.2 remove()

```
void lava::gamepad_manager::remove (
            id::ref func_id )
```

Remove listener.

Parameters

func↩ _id	Id of function

4.40.2.3 singleton()

```
static gamepad_manager & lava::gamepad_manager::singleton ( )  [inline], [static]
```

Get gamepad manager singleton.

Returns

gamepad_manager& Gamepad manager

The documentation for this struct was generated from the following file:

- liblava/frame/gamepad.hpp

4.41 lava::global_logger Struct Reference

Global logger.

`#include <log.hpp>`

Public Member Functions

- logger get ()

  Get logger.
- void set (lava::logger l)

  Set logger.
- void **reset** ()

  Reset logger.

Static Public Member Functions

- static global_logger & singleton ()

  Get global logger singleton.

4.41.1 Detailed Description

Global logger.

4.41.2 Member Function Documentation

4.41.2.1 get()

```
logger lava::global_logger::get ( ) [inline]
```

Get logger.

Returns

logger Logger

4.41.2.2 set()

```
void lava::global_logger::set (
            lava::logger l ) [inline]
```

Set logger.

Parameters

l	Logger

4.41.2.3 singleton()

```
static global_logger & lava::global_logger::singleton ( ) [inline], [static]
```

Get global logger singleton.

Returns

log_global& Global logger

The documentation for this struct was generated from the following file:

- liblava/util/log.hpp

4.42 lava::hex_cell Struct Reference

Hex cell.

```
#include <hex.hpp>
```

Public Types

- using **list** = std::vector< hex_cell >

  List of hex cells.
- using **pair** = std::pair< i32, i32 >

  Hex pair (Q and R)
- using **map** = std::unordered_map< pair, index, pair_hash >

  Map of hex cells.

Public Member Functions

- auto **operator<=>** (hex_cell const &) const =default

  Compare operator.
- pair to_pair () const

  Get the pair.
- void add (hex_cell const &cell)

  Add hex cell.
- void substract (hex_cell const &cell)

  Substract hex cell.
- void scale (i32 factor)

  Scale the hex cell.
- void **rotate_left** ()

  Rotate to left.
- void **rotate_right** ()

  Rotate to right.

Public Attributes

- i32 **q** {}

  Q axis.
- i32 **r** {}

  R axis.
- i32 **s** {}

  S axis.

4.42.1 Detailed Description

Hex cell.

4.42.2 Member Function Documentation

4.42.2.1 add()

```
void lava::hex_cell::add (
            hex_cell const & cell )  [inline]
```

Add hex cell.

Parameters

cell	Another hex cell

4.42.2.2 scale()

```
void lava::hex_cell::scale (
            i32 factor )  [inline]
```

Scale the hex cell.

Parameters

factor	Scaling factor

4.42.2.3 substract()

```
void lava::hex_cell::substract (
            hex_cell const & cell )  [inline]
```

Substract hex cell.

Parameters

cell	Another hex cell

4.42.2.4 to_pair()

```
pair lava::hex_cell::to_pair ( ) const [inline]
```

Get the pair.

Returns

pair Hex pair

The documentation for this struct was generated from the following file:

- liblava/util/hex.hpp

4.43 lava::hex_fractional_cell Struct Reference

Hex fractional cell.

```
#include <hex.hpp>
```

Public Attributes

- r32 **q** {}
  Q axis.
- r32 **r** {}
  R axis.
- r32 **s** {}
  S axis.

4.43.1 Detailed Description

Hex fractional cell.

The documentation for this struct was generated from the following file:

- liblava/util/hex.hpp

4.44 lava::hex_grid Struct Reference

Hex grid.

```
#include <hex.hpp>
```

Public Member Functions

- hex_grid (r32 radius=hex_default_outer_radius)

  Construct a new hex grid.
- void update (hex_orientation orientation=hex_layout_point_y)

  Update the hex grid.
- hex_cell find (r32 x, r32 y) const

  Find the hex cell from X and Y coordinates.
- hex_point to_pixel (hex_cell const &cell) const

  Get the hex point from hex cell.

Public Attributes

- r32 **inner_radius** = 0.f

  Hex inner radius.
- r32 **outer_radius** = hex_default_outer_radius

  Hex outer radius.
- hex_layout **layout**

  Hex layout.

4.44.1 Detailed Description

Hex grid.

4.44.2 Constructor & Destructor Documentation

4.44.2.1 hex_grid()

```
lava::hex_grid::hex_grid (
            r32 radius = hex_default_outer_radius )  [inline]
```

Construct a new hex grid.

Parameters

radius	Hex outer radius

4.44.3 Member Function Documentation

4.44.3.1 find()

```
hex_cell lava::hex_grid::find (
            r32 x,
            r32 y ) const  [inline]
```

Find the hex cell from X and Y coordinates.

Parameters

x	X coordinate
y	Y coordinate

Returns

hex_cell Hex cell

4.44.3.2 to_pixel()

```
hex_point lava::hex_grid::to_pixel (
            hex_cell const & cell ) const  [inline]
```

Get the hex point from hex cell.

Parameters

cell	Hex cell

Returns

hex_point Hex point

4.44.3.3 update()

```
void lava::hex_grid::update (
            hex_orientation orientation = hex_layout_point_y )  [inline]
```

Update the hex grid.

Parameters

orientation	Hex orientation

The documentation for this struct was generated from the following file:

- liblava/util/hex.hpp

4.45 lava::hex_layout Struct Reference

Hex layout.

```
#include <hex.hpp>
```

Public Attributes

- hex_orientation **orientation**

  Hex orientation.
- hex_point **origin**

  Hex origin.
- hex_point **size**

  Hex size.

4.45.1 Detailed Description

Hex layout.

The documentation for this struct was generated from the following file:

- liblava/util/hex.hpp

4.46 lava::hex_offset_coord Struct Reference

Hex offset coordinates.

```
#include <hex.hpp>
```

Public Attributes

- i32 **col** {}

  Column coordinate.
- i32 **row** {}

  Row coordinate.

4.46.1 Detailed Description

Hex offset coordinates.

The documentation for this struct was generated from the following file:

- liblava/util/hex.hpp

4.47 lava::hex_orientation Struct Reference

Hex orientation.

```
#include <hex.hpp>
```

Public Attributes

- r32 **f0** {}

  F0 value.
- r32 **f1** {}

  F1 value.
- r32 **f2** {}

  F2 value.
- r32 **f3** {}

  F3 value.
- r32 **b0** {}

  B0 value.
- r32 **b1** {}

  B1 value.
- r32 **b2** {}

  B2 value.
- r32 **b3** {}

  B3 value.
- r32 **start_angle** {}

  Start angle.

4.47.1 Detailed Description

Hex orientation.

The documentation for this struct was generated from the following file:

- liblava/util/hex.hpp

4.48 lava::hex_point Struct Reference

Hex point.

```
#include <hex.hpp>
```

Public Types

- using **list** = std::vector< hex_point >

  List of hex points.

Public Attributes

- r32 **x** {}

  X coordinate.
- r32 **y** {}

  Y coordinate.

4.48.1 Detailed Description

Hex point.

The documentation for this struct was generated from the following file:

- liblava/util/hex.hpp

4.49 lava::imgui::icon_font Struct Reference

ImGui icon font settings.

```
#include <imgui.hpp>
```

Public Attributes

- data **font_data**

  Icon font data.
- ui16 **range_begin** = 0

  Icon range begin.
- ui16 **range_end** = 0

  Icon range end.
- r32 **size** = default_imgui_font_size

  Default icon font size.

4.49.1 Detailed Description

ImGui icon font settings.

The documentation for this struct was generated from the following file:

- liblava/app/imgui.hpp

4.50 lava::id Struct Reference

Identification.

```
#include <id.hpp>
```

Public Types

- using **ref** = id const &

  Reference to id.
- using **set** = std::set< id >

  Set of ids.
- using **set_ref** = set const &

  Reference to set of ids.
- using **list** = std::vector< id >

  List if ids.
- using **map** = std::map< id, id >

  Map of ids.
- using **index_map** = std::map< id, index >

  Index map by ids.
- using **string_map** = std::map< id, string >

  String map by ids.

Public Member Functions

- bool valid () const

  Check if the id is valid.
- string to_string () const

  Convert the id to string.
- void **invalidate** ()

  Invalidate id.
- auto **operator<=>** (id const &) const =default

  Compare operator.

Public Attributes

- index **value** = no_index

  Value.

4.50.1 Detailed Description

Identification.

4.50.2 Member Function Documentation

4.50.2.1 to_string()

```
string lava::id::to_string ( ) const  [inline]
```

Convert the id to string.

Returns

string String representation of id

4.50.2.2 valid()

```
bool lava::id::valid ( ) const  [inline]
```

Check if the id is valid.

Returns

Id is valid or not

The documentation for this struct was generated from the following file:

- liblava/core/id.hpp

4.51 lava::id_listeners< T > Struct Template Reference

Id listeners.

```
#include <id.hpp>
```

Public Member Functions

- id add (typename T::func const &listener)

  Add listener to map.
- void remove (id &id)

  Remove listener from map by id.
- T::listeners const & get_list () const

  Get the list.

4.51.1 Detailed Description

template<typename T>
struct lava::id_listeners< T >

Id listeners.

Template Parameters

T	Listener

4.51.2 Member Function Documentation

4.51.2.1 add()

```
template<typename T >
id lava::id_listeners< T >::add (
            typename T::func const & listener )  [inline]
```

Add listener to map.

Parameters

listener	Target listener

Returns

id Id of listener

4.51.2.2 get_list()

```
template<typename T >
T::listeners const & lava::id_listeners< T >::get_list ( ) const  [inline]
```

Get the list.

Returns

T::listeners const& List of listeners

4.51.2.3 remove()

```
template<typename T >
void lava::id_listeners< T >::remove (
            id & id )  [inline]
```

Remove listener from map by id.

Parameters

id	Id of listener

The documentation for this struct was generated from the following file:

- liblava/core/id.hpp

4.52 lava::id_registry< T, Meta > Struct Template Reference

Id registry.

`#include <id.hpp>`

Public Types

- using **ptr** = std::shared_ptr< T >

  Shared pointer to id registry.
- using **map** = std::map< id, ptr >

  Map of id registries.
- using **meta_map** = std::map< id, Meta >

  Map of ids with meta.

Public Member Functions

- id create (Meta info={})

  Create a new object in registry.
- void add (ptr object, Meta info={})

  Add a object with meta to registry.
- bool exists (id::ref object_id) const

  Check if object exists in registry.
- ptr get (id::ref object_id) const

  Get the object by id.
- Meta const & get_meta (id::ref object_id) const

  Get the meta by id.
- map const & get_all () const

  Get all objects.
- meta_map const & get_all_meta () const

  Get all meta objects.
- bool update (id::ref object_id, Meta const &meta)

  Update meta of object.
- void remove (id::ref object_id)

  Remove object from registry.
- void **clear** ()

  Clear the registry.

4.52.1 Detailed Description

template<typename T, typename Meta>
struct lava::id_registry< T, Meta >

Id registry.

Template Parameters

T	Type of objects hold in registry
Meta	Meta type for object

4.52.2 Member Function Documentation

4.52.2.1 add()

```
template<typename T , typename Meta >
void lava::id_registry< T, Meta >::add (
            ptr object,
            Meta info = {} )  [inline]
```

Add a object with meta to registry.

Parameters

object	Object to add
info	Meta of object

4.52.2.2 create()

```
template<typename T , typename Meta >
id lava::id_registry< T, Meta >::create (
            Meta info = {} )  [inline]
```

Create a new object in registry.

Parameters

info	Meta information

Returns

id Object id

4.52.2.3 exists()

```
template<typename T , typename Meta >
bool lava::id_registry< T, Meta >::exists (
            id::ref object_id ) const  [inline]
```

Check if object exists in registry.

Parameters

object↩ _id	Object to check

Returns

Object exists or not

4.52.2.4 get()

```
template<typename T , typename Meta >
ptr lava::id_registry< T, Meta >::get (
            id::ref object_id ) const  [inline]
```

Get the object by id.

Parameters

object↩ _id	Object id

Returns

ptr Shared pointer to object

4.52.2.5 get_all()

```
template<typename T , typename Meta >
map const & lava::id_registry< T, Meta >::get_all ( ) const  [inline]
```

Get all objects.

Returns

map const& Map with objects

4.52.2.6 get_all_meta()

```
template<typename T , typename Meta >
meta_map const & lava::id_registry< T, Meta >::get_all_meta ( ) const  [inline]
```

Get all meta objects.

Returns

meta_map const& Map with metas

4.52.2.7 get_meta()

```
template<typename T , typename Meta >
Meta const & lava::id_registry< T, Meta >::get_meta (
            id::ref object_id ) const  [inline]
```

Get the meta by id.

Parameters

object_id	Object id

Returns

Meta Meta object

4.52.2.8 remove()

```
template<typename T , typename Meta >
void lava::id_registry< T, Meta >::remove (
            id::ref object_id )  [inline]
```

Remove object from registry.

Parameters

object_id	Object id

4.52.2.9 update()

```
template<typename T , typename Meta >
bool lava::id_registry< T, Meta >::update (
            id::ref object_id,
            Meta const & meta )  [inline]
```

Update meta of object.

Parameters

object↩ _id	Object id
meta	Meta to update

Returns

Meta updated or not

The documentation for this struct was generated from the following file:

- liblava/core/id.hpp

4.53 lava::ids Struct Reference

Id factory.

```
#include <id.hpp>
```

Public Member Functions

- id next ()

  Get next id from factory.

Static Public Member Functions

- static ids & instance ()

  Get id factory instance.

4.53.1 Detailed Description

Id factory.

4.53.2 Member Function Documentation

4.53.2.1 instance()

```
static ids & lava::ids::instance ( ) [inline], [static]
```

Get id factory instance.

Returns

ids& Id factory

4.53.2.2 next()

```
id lava::ids::next ( ) [inline]
```

Get next id from factory.

Returns

id Next id

The documentation for this struct was generated from the following file:

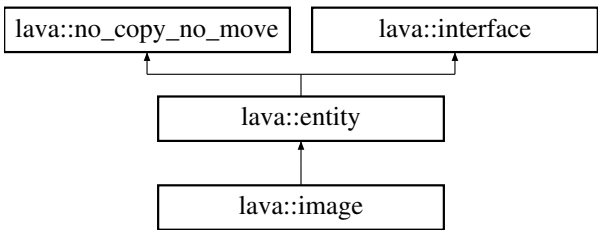
- liblava/core/id.hpp

4.54 lava::image Struct Reference

Image.

```
#include <image.hpp>
```

Inheritance diagram for lava::image:



Public Types

- using **ptr** = std::shared_ptr< image >

  Shared pointer to image.
- using **map** = std::map< id, ptr >

  Map of images.
- using **list** = std::vector< ptr >

  List of images.

Public Member Functions

- image (VkFormat format, VkImage vk_image=0)

  Construct a new image.
- bool create (device_p device, uv2 size, VmaMemoryUsage memory_usage=VMA_MEMORY_USAGE_↩ GPU_ONLY, bool mip_levels_generation=false)

  Create a new image.
- void destroy (bool view_only=false)

  Destroy the image.
- void **destroy_view** ()

  Destroy the image view.
- device_p get_device ()

  Get the device.
- VkFormat get_format () const

  Get the format of the image.
- uv2 get_size () const

  Get the size of the image.
- ui32 get_depth () const

  Get the depth of the image.
- VkImage get () const

  Get the image.
- VkImageView get_view () const

  Get the image view.
- VkImageCreateInfo const & get_info () const

  Get the image create information.
- VkImageViewCreateInfo const & get_view_info () const

  Get the image view create information.
- VkImageSubresourceRange const & get_subresource_range () const

  Get the subresource range of the image.
- void set_flags (VkImageCreateFlagBits flags)

  Set the image create flags.
- void set_tiling (VkImageTiling tiling)

  Set the image tiling.
- void set_usage (VkImageUsageFlags usage)

  Set the image usage.
- void set_layout (VkImageLayout initial)

  Set the initial layout of the image.
- void set_aspect_mask (VkImageAspectFlags aspectMask)

  Set the aspect mask of the image.
- void set_level_count (ui32 levels)

  Set the level count of the image.
- void set_layer_count (ui32 layers)

  Set the layer count of the image.
- void set_component (VkComponentMapping mapping={})

  Set the component mapping of the image.
- void set_view_type (VkImageViewType type)

  Set the view type of the image.
- VmaAllocation const & get_allocation () const

  Get the allocation of the image.

Static Public Member Functions

- static ptr make (VkFormat format, VkImage vk_image=0)

  Make a new image.

4.54.1 Detailed Description

Image.

4.54.2 Constructor & Destructor Documentation

4.54.2.1 image()

```
lava::image::image (
            VkFormat format,
            VkImage vk_image = 0 )  [explicit]
```

Construct a new image.

Parameters

format	Image format
vk_image	Vulkan image

4.54.3 Member Function Documentation

4.54.3.1 create()

```
bool lava::image::create (
            device_p device,
            uv2 size,
            VmaMemoryUsage memory_usage = VMA_MEMORY_USAGE_GPU_ONLY,
            bool mip_levels_generation = false )
```

Create a new image.

Parameters

device	Vulkan device
size	Image size
memory_usage	Memory usage
mip_levels_generation	Enable mip levels generation

Returns

Create was successful or failed

4.54.3.2 destroy()

```
void lava::image::destroy (
            bool view_only = false )
```

Destroy the image.

Parameters

view_only	Destroy only the image view

4.54.3.3 get()

```
VkImage lava::image::get ( ) const  [inline]
```

Get the image.

Returns

VkImage Vulkan image

4.54.3.4 get_allocation()

```
VmaAllocation const & lava::image::get_allocation ( ) const  [inline]
```

Get the allocation of the image.

Returns

VmaAllocation const& Image allocation

4.54.3.5 get_depth()

```
ui32 lava::image::get_depth ( ) const  [inline]
```

Get the depth of the image.

Returns

ui32 Image depth

4.54.3.6 get_device()

```
device_p lava::image::get_device ( ) [inline]
```

Get the device.

Returns

device_p Vulkan device

4.54.3.7 get_format()

```
VkFormat lava::image::get_format ( ) const [inline]
```

Get the format of the image.

Returns

VkFormat Image format

4.54.3.8 get_info()

```
VkImageCreateInfo const & lava::image::get_info ( ) const [inline]
```

Get the image create information.

Returns

VkImageCreateInfo const& Image create information

4.54.3.9 get_size()

```
uv2 lava::image::get_size ( ) const [inline]
```

Get the size of the image.

Returns

uv2 Image size

4.54.3.10 get_subresource_range()

```
VkImageSubresourceRange const & lava::image::get_subresource_range ( ) const  [inline]
```

Get the subresource range of the image.

Returns

VkImageSubresourceRange const& Image subresource range

4.54.3.11 get_view()

```
VkImageView lava::image::get_view ( ) const  [inline]
```

Get the image view.

Returns

VkImageView Vulkan image view

4.54.3.12 get_view_info()

```
VkImageViewCreateInfo const & lava::image::get_view_info ( ) const  [inline]
```

Get the image view create information.

Returns

VkImageViewCreateInfo const& Image view create information

4.54.3.13 make()

```
static ptr lava::image::make (
            VkFormat format,
            VkImage vk_image = 0 )  [inline], [static]
```

Make a new image.

Parameters

format	Image format
vk_image	Vulkan image

Returns

ptr Shared pointer to image

4.54.3.14 set_aspect_mask()

```
void lava::image::set_aspect_mask (
            VkImageAspectFlags aspectMask )  [inline]
```

Set the aspect mask of the image.

Parameters

aspectMask	Image aspect flags

4.54.3.15 set_component()

```
void lava::image::set_component (
            VkComponentMapping mapping = {} )  [inline]
```

Set the component mapping of the image.

Parameters

mapping	Component mapping

4.54.3.16 set_flags()

```
void lava::image::set_flags (
            VkImageCreateFlagBits flags )  [inline]
```

Set the image create flags.

Parameters

flags	Image create flag bits

4.54.3.17 set_layer_count()

```
void lava::image::set_layer_count (
```

```
            ui32 layers )  [inline]
```

Set the layer count of the image.

Parameters

layers	Number of layers

4.54.3.18 set_layout()

```
void lava::image::set_layout (
            VkImageLayout initial )  [inline]
```

Set the initial layout of the image.

Parameters

initial	Initial image layout

4.54.3.19 set_level_count()

```
void lava::image::set_level_count (
            ui32 levels )  [inline]
```

Set the level count of the image.

Parameters

levels	Number of levels

4.54.3.20 set_tiling()

```
void lava::image::set_tiling (
            VkImageTiling tiling )  [inline]
```

Set the image tiling.

Parameters

tiling	Image tiling

4.54.3.21 set_usage()

```
void lava::image::set_usage (
            VkImageUsageFlags usage )  [inline]
```

Set the image usage.

Parameters

usage	Image usage flags

4.54.3.22 set_view_type()

```
void lava::image::set_view_type (
            VkImageViewType type )  [inline]
```

Set the view type of the image.

Parameters

type	Image view type

The documentation for this struct was generated from the following file:

- liblava/resource/image.hpp

4.55 lava::image_loader Struct Reference

Load image data from file and memory.

```
#include <image_loader.hpp>
```

Public Member Functions

- image_loader (string_ref filename)

  Construct a new image data from file.
- image_loader (cdata::ref image)

  Construct a new image data from memory.
- ~**image_loader** ()

  Destroy the image data.
- bool ready () const

  Check if data is ready.

- data_cptr get () const

  Get image data.
- size_t size () const

  Get image data size.
- uv2 get_dimensions () const

  Get image dimensions.
- ui32 get_channels () const

  Get image channel count.

4.55.1 Detailed Description

Load image data from file and memory.

4.55.2 Constructor & Destructor Documentation

4.55.2.1 image_loader() [1/2]

```
lava::image_loader::image_loader (
            string_ref filename )  [explicit]
```

Construct a new image data from file.

Parameters

filename	File data to load

4.55.2.2 image_loader() [2/2]

```
lava::image_loader::image_loader (
            cdata::ref image )  [explicit]
```

Construct a new image data from memory.

Parameters

image	Memory data to load

4.55.3 Member Function Documentation

4.55.3.1 get()

```
data_cptr lava::image_loader::get ( ) const  [inline]
```

Get image data.

Returns

data_ptr Image data pointer

4.55.3.2 get_channels()

```
ui32 lava::image_loader::get_channels ( ) const  [inline]
```

Get image channel count.

Returns

ui32 Channel count

4.55.3.3 get_dimensions()

```
uv2 lava::image_loader::get_dimensions ( ) const  [inline]
```

Get image dimensions.

Returns

uv2 Image dimensions

4.55.3.4 ready()

```
bool lava::image_loader::ready ( ) const  [inline]
```

Check if data is ready.

Returns

Data is ready or not

4.55.3.5 size()

```
size_t lava::image_loader::size ( ) const  [inline]
```

Get image data size.

Returns

size_t Image data size

The documentation for this struct was generated from the following file:

- liblava/asset/image_loader.hpp

4.56 lava::imgui Struct Reference

ImGui integration.

```
#include <imgui.hpp>
```

Classes

- struct config

  ImGui configuration.
- struct font

  ImGui font settings.
- struct icon_font

  ImGui icon font settings.

Public Types

- using **draw_func** = std::function< void()>

  Draw function.

Public Member Functions

- **imgui** ()=default

  Construct a new ImGui.
- imgui (GLFWwindow *window)

  Construct a new ImGui.
- ~**imgui** ()

  Destroy the ImGui.
- void setup (GLFWwindow *window, config config)

  Set up ImGui with configuration.
- void setup (GLFWwindow *window)

  Set up default ImGui.
- bool create (render_pipeline::ptr pipeline, index max_frames)

Create pipeline for ImGui.

- bool create (device_p device, index max_frames, VkPipelineCache pipeline_cache)

  Create pipeline for ImGui with device.
- bool create (device_p device, index max_frames, VkRenderPass pass, VkPipelineCache pipeline_cache=0)

  Create pipeline for ImGui with device and render pass.
- bool upload_fonts (texture::ptr texture)

  Upload font texture.
- void **destroy** ()

  Destroy ImGui.
- bool ready () const

  Check if ImGui is ready.
- render_pipeline::ptr get_pipeline ()

  Get the pipeline.
- bool capture_mouse () const

  Check if mouse capture is active.
- bool capture_keyboard () const

  Check if keyboard capture is active.
- void set_active (bool value=true)

  Set ImGui active.
- bool activated () const

  Check if ImGui is activated.
- void **toggle** ()

  Togge active state.
- void set_ini_file (std::filesystem::path dir)

  Set the ini file.
- std::filesystem::path get_ini_file () const

  Get the ini file.
- void **convert_style_to_srgb** ()

  Convert style to sRGB.
- input_callback const & get_input_callback () const

  Get the input callback.

Public Attributes

- draw_func **on_draw**

  Function called on ImGui draw.

4.56.1 Detailed Description

ImGui integration.

4.56.2 Constructor & Destructor Documentation

4.56.2.1 imgui()

```
lava::imgui::imgui (
            GLFWwindow * window )  [inline], [explicit]
```

Construct a new ImGui.

Parameters

window	Window for ImGui

4.56.3 Member Function Documentation

4.56.3.1 activated()

```
bool lava::imgui::activated ( ) const  [inline]
```

Check if ImGui is activated.

Returns

ImGui is active or not

4.56.3.2 capture_keyboard()

```
bool lava::imgui::capture_keyboard ( ) const
```

Check if keyboard capture is active.

Returns

Capture is active or not

4.56.3.3 capture_mouse()

```
bool lava::imgui::capture_mouse ( ) const
```

Check if mouse capture is active.

Returns

Capture is active or not

4.56.3.4 create() [1/3]

```
bool lava::imgui::create (
            device_p device,
            index max_frames,
            VkPipelineCache pipeline_cache )  [inline]
```

Create pipeline for ImGui with device.

Parameters

device	Vulkan device
max_frames	Number of frames
pipeline_cache	Pipeline cache

Returns

Create was successful or failed

4.56.3.5 create() [2/3]

```
bool lava::imgui::create (
            device_p device,
            index max_frames,
            VkRenderPass pass,
            VkPipelineCache pipeline_cache = 0 )  [inline]
```

Create pipeline for ImGui with device and render pass.

Parameters

device	Vulkan device
max_frames	Number of frames
pass	Render pass
pipeline_cache	Pipeline cache

Returns

Create was successful or failed

4.56.3.6 create() [3/3]

```
bool lava::imgui::create (
            render_pipeline::ptr pipeline,
            index max_frames )
```

Create pipeline for ImGui.

Parameters

pipeline	Render pipeline
max_frames	Number of frames

Returns

Create was successful or failed

4.56.3.7 get_ini_file()

```
std::filesystem::path lava::imgui::get_ini_file ( ) const  [inline]
```

Get the ini file.

Returns

fs::path Path of file

4.56.3.8 get_input_callback()

```
input_callback const & lava::imgui::get_input_callback ( ) const  [inline]
```

Get the input callback.

Returns

input_callback const& Input callback

4.56.3.9 get_pipeline()

```
render_pipeline::ptr lava::imgui::get_pipeline ( )  [inline]
```

Get the pipeline.

Returns

render_pipeline::ptr Render pipeline

4.56.3.10 ready()

```
bool lava::imgui::ready ( ) const  [inline]
```

Check if ImGui is ready.

Returns

ImGui is ready or not

4.56.3.11 set_active()

```
void lava::imgui::set_active (
            bool value = true )  [inline]
```

Set ImGui active.

Parameters

value	Active state

4.56.3.12 set_ini_file()

```
void lava::imgui::set_ini_file (
            std::filesystem::path dir )
```

Set the ini file.

Parameters

dir	Path for file

4.56.3.13 setup() [1/2]

```
void lava::imgui::setup (
            GLFWwindow * window )  [inline]
```

Set up default ImGui.

Parameters

window	Target window

4.56.3.14 setup() [2/2]

```
void lava::imgui::setup (
            GLFWwindow * window,
            config config )
```

Set up ImGui with configuration.

Parameters

window	Target window
config	Configuration

4.56.3.15 upload_fonts()

```
bool lava::imgui::upload_fonts (
            texture::ptr texture )
```

Upload font texture.

Parameters

texture	Texture to upload

Returns

Upload was successful or failed

The documentation for this struct was generated from the following file:

- liblava/app/imgui.hpp

4.57 lava::input Struct Reference

Input handling.

```
#include <input.hpp>
```

Public Member Functions

- void **handle_events** ()

  Handle events.
- void add (input_callback::cptr callback)

  Add callback to the input handling.
- void remove (input_callback::cptr callback)

  Remove callback from the input handling.
- mouse_position_ref get_mouse_position () const

  Get the mouse position.
- void set_mouse_position (mouse_position_ref position)

  Set the mouse position.

Public Attributes

- input_key_events **key**

  List of key events.
- input_scroll_events **scroll**

  List of scroll events.
- input_mouse_move_events **mouse_move**

  List of mouse move events.
- input_mouse_button_events **mouse_button**

  List of mouse button events.
- input_mouse_active_events **mouse_active**

  List of mouse active events.
- input_path_drop_events **path_drop**

  List of path drop events.

4.57.1 Detailed Description

Input handling.

4.57.2 Member Function Documentation

4.57.2.1 add()

```
void lava::input::add (
            input_callback::cptr callback )  [inline]
```

Add callback to the input handling.

Parameters

callback	Callback to add

4.57.2.2 get_mouse_position()

```
mouse_position_ref lava::input::get_mouse_position ( ) const  [inline]
```

Get the mouse position.

Returns

mouse_position_ref Current mouse position

4.57.2.3 remove()

```
void lava::input::remove (
            input_callback::cptr callback )  [inline]
```

Remove callback from the input handling.

Parameters

callback	Callback to remove

4.57.2.4 set_mouse_position()

```
void lava::input::set_mouse_position (
            mouse_position_ref position ) [inline]
```

Set the mouse position.

Parameters

position	Current mouse position

The documentation for this struct was generated from the following file:

- liblava/frame/input.hpp

4.58 lava::input_callback Struct Reference

Input callback.

```
#include <input.hpp>
```

Public Types

- using **cptr** = input_callback const ∗
  Const pointer to input callback.
- using **list** = std::vector< input_callback ∗ >
  List of input callbacks.
- using **clist** = std::vector< cptr >
  List of const input callbacks.
- template<typename T >
  using func = std::function< bool(typename T::ref)>
  Input callback functions.

Public Attributes

- key_event::func **on_key_event**
  Called on key event.
- scroll_event::func **on_scroll_event**
  Called on scroll event.
- mouse_move_event::func **on_mouse_move_event**
  Called on mouse move event.
- mouse_button_event::func **on_mouse_button_event**
  Called on mouse button event.
- mouse_active_event::func **on_mouse_active_event**
  Called on mouse active event.
- path_drop_event::func **on_path_drop_event**
  Called on path drop event.

4.58.1 Detailed Description

Input callback.

4.58.2 Member Typedef Documentation

4.58.2.1 func

```
template<typename T >
using lava::input_callback::func = std::function<bool(typename T::ref)>
```

Input callback functions.

Template Parameters

T	Type of callback

The documentation for this struct was generated from the following file:

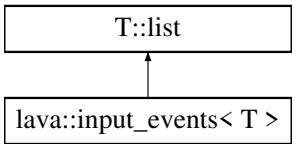
- liblava/frame/input.hpp

4.59 lava::input_events< T > Struct Template Reference

List of input events.

```
#include <input.hpp>
```

Inheritance diagram for lava::input_events< T >:



Public Member Functions

- void add (typename T::ref event)

  Add event to list.

Public Attributes

- id_listeners< T > **listeners**

  List of event listeners.

4.59.1 Detailed Description

template<typename T>
struct lava::input_events< T >

List of input events.

Template Parameters

T	Type of event

4.59.2 Member Function Documentation

4.59.2.1 add()

```
template<typename T >
void lava::input_events< T >::add (
            typename T::ref event )  [inline]
```

Add event to list.

Parameters

event	Event to add

The documentation for this struct was generated from the following file:

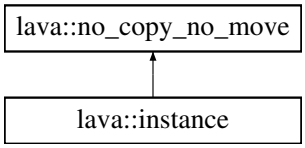
- liblava/frame/input.hpp

4.60 lava::instance Struct Reference

Vulkan instance.

```
#include <instance.hpp>
```

Inheritance diagram for lava::instance:



Classes

- struct create_param

  Instance create parameters.
- struct debug_config

  Debug configuration.

Public Member Functions

- bool create (create_param &param, debug_config::ref debug, instance_info::ref info)

  Create a new instance.
- void **destroy** ()

  Destroy the instance.
- physical_device::list const & get_physical_devices () const

  Get the physical devices.
- physical_device::ref get_first_physical_device () const

  Get the first physical device.
- VkInstance get () const

  Get the Vulkan instance.
- debug_config::ref get_debug_config () const

  Get the debug configuration.
- instance_info::ref get_info () const

  Get the instance information.

Static Public Member Functions

- static instance & singleton ()

  Instance singleton.

4.60.1 Detailed Description

Vulkan instance.

4.60.2 Member Function Documentation

4.60.2.1 create()

```
bool lava::instance::create (
            create_param & param,
            debug_config::ref debug,
            instance_info::ref info )
```

Create a new instance.

Parameters

param	Create parameters
debug	Debug configuration
info	Instance information

Returns

Create was successful or failed

4.60.2.2 get()

```
VkInstance lava::instance::get ( ) const  [inline]
```

Get the Vulkan instance.

Returns

VkInstance Vulkan instance

4.60.2.3 get_debug_config()

```
debug_config::ref lava::instance::get_debug_config ( ) const  [inline]
```

Get the debug configuration.

Returns

debug_config::ref Debug configuration

4.60.2.4 get_first_physical_device()

```
physical_device::ref lava::instance::get_first_physical_device ( ) const  [inline]
```

Get the first physical device.

Returns

physical_device::ref Physcial device

4.60.2.5 get_info()

```
instance_info::ref lava::instance::get_info ( ) const  [inline]
```

Get the instance information.

Returns

instance_info::ref Instance information

4.60.2.6 get_physical_devices()

```
physical_device::list const & lava::instance::get_physical_devices ( ) const  [inline]
```

Get the physical devices.

Returns

physical_device::list const& List of physical devices

4.60.2.7 singleton()

```
static instance & lava::instance::singleton ( )  [inline], [static]
```

Instance singleton.

Returns

instance& Instance

The documentation for this struct was generated from the following file:

- liblava/base/instance.hpp

4.61 lava::instance_info Struct Reference

Vulkan instance information.

```
#include <instance.hpp>
```

Public Types

- using **ref** = instance_info const &

  Reference to a instance.

Public Attributes

- name **app_name** = _lava_

  Name of application.
- name **engine_name** = _liblava_

  Name of engine.
- int_version **app_version**

  Version of application.
- int_version **engine_version**

  Version of engine.
- api_version **req_api_version** = api_version::v1_0

  Required Vulkan API version.

4.61.1 Detailed Description

Vulkan instance information.

The documentation for this struct was generated from the following file:

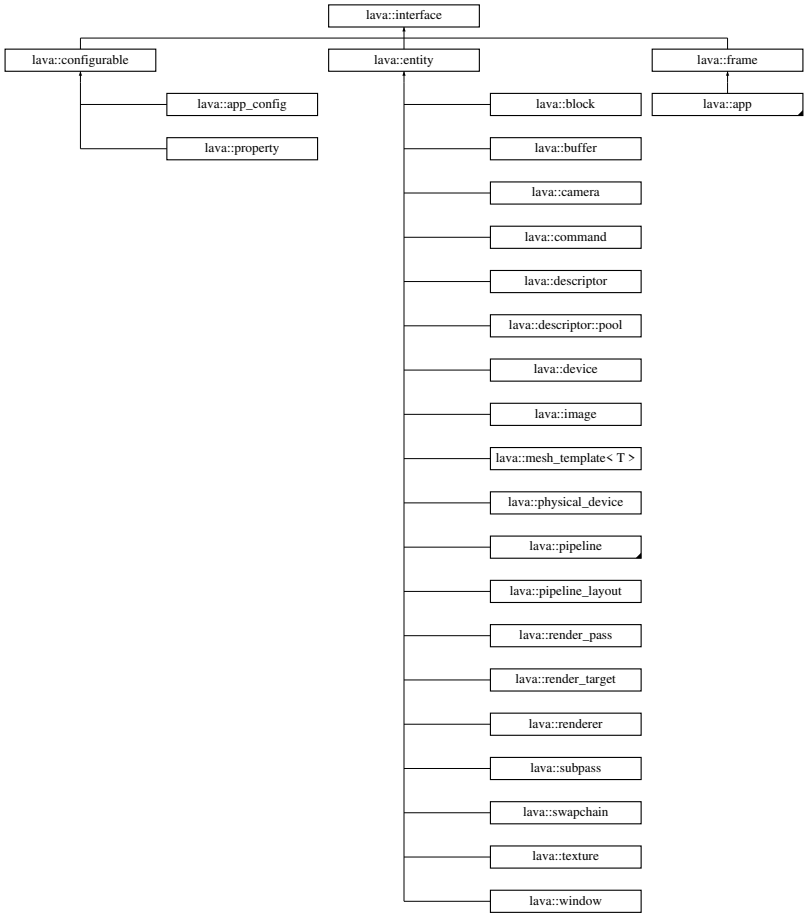
- liblava/base/instance.hpp

4.62 lava::interface Struct Reference

Interface.

```
#include <types.hpp>
```

Inheritance diagram for lava::interface:



Public Member Functions

- virtual ~**interface** ()=default

  Destroy the interface.

4.62.1 Detailed Description

Interface.

The documentation for this struct was generated from the following file:

- liblava/core/types.hpp

4.63 lava::internal_version Struct Reference

Internal version.

```
#include <version.hpp>
```

Public Member Functions

- auto **operator**<=> (internal_version const &) const =default

  Default compare operators.

Public Attributes

- i32 **major** = LAVA_VERSION_MAJOR

  Major version.
- i32 **minor** = LAVA_VERSION_MINOR

  Minor version.
- i32 **patch** = LAVA_VERSION_PATCH

  Patch version.

4.63.1 Detailed Description

Internal version.

The documentation for this struct was generated from the following file:

- liblava/core/version.hpp

4.64 lava::json_file Struct Reference

Json file.

```
#include <json_file.hpp>
```

Classes

- struct callback

  Json file callback.

Public Member Functions

- json_file (string_ref path="config.json")

  Construct a new json file.
- void add (callback *callback)

  Add callback to json file.
- void remove (callback *callback)

  Remove callback from json file.
- void **clear** ()

  Clear all callbacks.
- void set (string_ref value)

  Set path of the json file.
- string_ref get () const

  Get path of the json file.
- bool load ()

  Load the json file.
- bool save ()

  Save the json file.

4.64.1 Detailed Description

Json file.

4.64.2 Constructor & Destructor Documentation

4.64.2.1 json_file()

```
lava::json_file::json_file (
            string_ref path = "config.json" )  [explicit]
```

Construct a new json file.

Parameters

path	Name of file

4.64.3 Member Function Documentation

4.64.3.1 add()

```
void lava::json_file::add (
            callback * callback )
```

Add callback to json file.

Parameters

callback	Callback to add

4.64.3.2 get()

```
string_ref lava::json_file::get ( ) const  [inline]
```

Get path of the json file.

Returns

name Name of file

4.64.3.3 load()

```
bool lava::json_file::load ( )
```

Load the json file.

Returns

Load was successful or failed

4.64.3.4 remove()

```
void lava::json_file::remove (
            callback * callback )
```

Remove callback from json file.

Parameters

callback	Callback to remove

4.64.3.5 save()

```
bool lava::json_file::save ( )
```

Save the json file.

Returns

Save was successful or failed

4.64.3.6 set()

```
void lava::json_file::set (
            string_ref value )  [inline]
```

Set path of the json file.

Parameters

value	Name of file

The documentation for this struct was generated from the following file:

- liblava/file/json_file.hpp

4.65 lava::key_event Struct Reference

Key event.

```
#include <input.hpp>
```

Public Types

- using **ref** = key_event const &

  Reference to key event.
- using **func** = std::function< bool(ref)>

  Key event function.
- using **listeners** = std::map< id, func >

  List of key event listeners.
- using **list** = std::vector< key_event >

  List of key events.

Public Member Functions

- bool pressed (key_ref k) const

  Check if key is pressed.
- bool released (key_ref k) const

  Check if key is released.
- bool repeated (key_ref k) const

  Check if key is repeated.
- bool active () const

  Check if key is active.
- bool pressed (key_ref k, mod_ref m) const

  Check if key is pressed with mod.

Public Attributes

- id **sender**

  Sender id.
- key_t **key**

  Input key.
- action_t **action**

  Input action.
- mod_t **mod**

  Input mod.
- i32 **scancode** = 0

  Input scan code.

4.65.1 Detailed Description

Key event.

4.65.2 Member Function Documentation

4.65.2.1 active()

```
bool lava::key_event::active ( ) const  [inline]
```

Check if key is active.

Returns

Key is pressed (and repeated) or not active

4.65.2.2 pressed() [1/2]

```
bool lava::key_event::pressed (
            key_ref k ) const  [inline]
```

Check if key is pressed.

Parameters

k	Key to check

Returns

Key is pressed or not

4.65.2.3 pressed() [2/2]

```
bool lava::key_event::pressed (
            key_ref k,
            mod_ref m ) const [inline]
```

Check if key is pressed with mod.

Parameters

k	Key to check
m	Mods to check

Returns

Key is pressed with mod or not

4.65.2.4 released()

```
bool lava::key_event::released (
            key_ref k ) const [inline]
```

Check if key is released.

Parameters

k	Key to check

Returns

Key is released or not

4.65.2.5 repeated()

```
bool lava::key_event::repeated (
            key_ref k ) const [inline]
```

Check if key is repeated.

Parameters

k	Key to check

Returns

Key is repeated or not

The documentation for this struct was generated from the following file:

- liblava/frame/input.hpp

4.66 lava::texture::layer Struct Reference

Texture layer.

```
#include <texture.hpp>
```

Public Types

- using **list** = std::vector< layer >

  List of layers.

Public Attributes

- mip_level::list **levels**

  List of mip levels.

4.66.1 Detailed Description

Texture layer.

The documentation for this struct was generated from the following file:

- liblava/resource/texture.hpp

4.67 lava::log_config Struct Reference

Log configuration.

```
#include <log.hpp>
```

Public Attributes

- name **logger** = _lava_

  Logger name.
- name **file** = "lava.log"

  Log file.
- i32 **level** = -1

  Log level.
- bool **debug** = false

  Log to console, else file.

4.67.1 Detailed Description

Log configuration.

The documentation for this struct was generated from the following file:

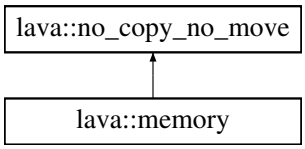
- liblava/util/log.hpp

4.68 lava::memory Struct Reference

Vulkan memory.

```
#include <memory.hpp>
```

Inheritance diagram for lava::memory:



Public Member Functions

- **memory** ()

  Construct a new memory.
- VkAllocationCallbacks * alloc ()

  Get allocation callback.
- void set_callbacks (VkAllocationCallbacks const &callbacks)

  Set the callbacks object.
- void set_use_custom_cpu_callbacks (bool value)

  Set use custom cpu callbacks.

Static Public Member Functions

- static memory & instance ()

  Get memory instance.

4.68.1 Detailed Description

Vulkan memory.

4.68.2 Member Function Documentation

4.68.2.1 alloc()

```
VkAllocationCallbacks * lava::memory::alloc ( ) [inline]
```

Get allocation callback.

Returns

VkAllocationCallbacks* Allocation callbacks

4.68.2.2 instance()

```
static memory & lava::memory::instance ( ) [inline], [static]
```

Get memory instance.

Returns

memory& Memory

4.68.2.3 set_callbacks()

```
void lava::memory::set_callbacks (
            VkAllocationCallbacks const & callbacks ) [inline]
```

Set the callbacks object.

Parameters

callbacks	Allocation Callbacks

4.68.2.4 set_use_custom_cpu_callbacks()

```
void lava::memory::set_use_custom_cpu_callbacks (
            bool value )  [inline]
```

Set use custom cpu callbacks.

Parameters

value	Value state

The documentation for this struct was generated from the following file:

- liblava/base/memory.hpp

4.69 lava::mesh_meta Struct Reference

Mesh meta.

```
#include <mesh.hpp>
```

Public Attributes

- string **filename**

  Name of file (empty: see type)
- mesh_type **type** = mesh_type::none

  Mesh type.

4.69.1 Detailed Description

Mesh meta.

The documentation for this struct was generated from the following file:

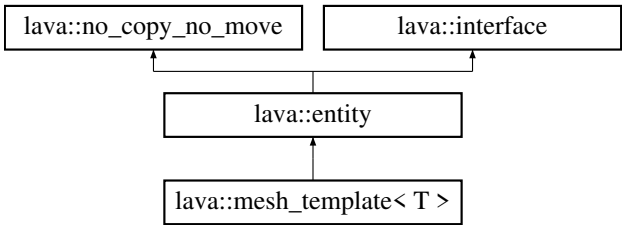
- liblava/resource/mesh.hpp

4.70 lava::mesh_template< T > Struct Template Reference

Temporary templated mesh.

```
#include <mesh.hpp>
```

Inheritance diagram for lava::mesh_template< T >:



Public Types

- using **ptr** = std::shared_ptr< mesh_template< T > >

  Shared pointer to mesh.
- using **map** = std::map< id, ptr >

  Map of meshes.
- using **list** = std::vector< ptr >

  List of meshes.
- using **vertex_list** = std::vector< T >

  List of vertices.

Public Member Functions

- ~**mesh_template** ()

  Destroy the mesh.
- bool create (device_p device, bool mapped=false, VmaMemoryUsage memory_usage=VMA_MEMORY_↩USAGE_CPU_TO_GPU)

  Create a new mesh.
- void **destroy** ()

  Destroy the mesh.
- void bind (VkCommandBuffer cmd_buf) const

  Bind the mesh.
- void draw (VkCommandBuffer cmd_buf) const

  Draw the mesh.
- void bind_draw (VkCommandBuffer cmd_buf) const

  Bind and draw the mesh.
- bool empty () const

  Check if mesh is empty.
- void set_data (mesh_template_data< T > const &value)

  Set the mesh data.
- mesh_template_data< T > & get_data ()

  Get the mesh data.
- void add_data (mesh_template_data< T > const &value)

  Add mesh data to existing data.

- vertex_list & get_vertices ()

  Get the vertices of the mesh.
- vertex_list const & get_vertices () const

  Get the const vertices of the mesh.
- ui32 get_vertices_count () const

  Get the vertices count of the mesh.
- index_list & get_indices ()

  Get the indices of the mesh.
- index_list const & get_indices () const

  Get the const indices of the mesh.
- ui32 get_indices_count () const

  Get the indices count of the mesh.
- bool reload ()

  Reload the mesh data.
- buffer::ptr get_vertex_buffer ()

  Get the vertex buffer of the mesh.
- buffer::ptr get_index_buffer ()

  Get the index buffer of the mesh.

Static Public Member Functions

- static ptr make ()

  Make a new mesh.

4.70.1 Detailed Description

template<typename T = vertex>
struct lava::mesh_template< T >

Temporary templated mesh.

Template Parameters

T	Vertex struct typename

4.70.2 Member Function Documentation

4.70.2.1 add_data()

```
template<typename T = vertex>
void lava::mesh_template< T >::add_data (
            mesh_template_data< T > const & value )  [inline]
```

Add mesh data to existing data.

Parameters

value	Mesh data to add

4.70.2.2 bind()

```
template<typename T >
void lava::mesh_template< T >::bind (
            VkCommandBuffer cmd_buf ) const
```

Bind the mesh.

Parameters

cmd_buf	Command buffer

4.70.2.3 bind_draw()

```
template<typename T = vertex>
void lava::mesh_template< T >::bind_draw (
            VkCommandBuffer cmd_buf ) const  [inline]
```

Bind and draw the mesh.

Parameters

cmd_buf	Command buffer

4.70.2.4 create()

```
template<typename T >
bool lava::mesh_template< T >::create (
            device_p device,
            bool mapped = false,
            VmaMemoryUsage memory_usage = VMA_MEMORY_USAGE_CPU_TO_GPU )
```

Create a new mesh.

Parameters

device	Vulkan device
mapped	Map mesh data
memory_usage	Memory usage

Returns

Create was successful or failed

4.70.2.5 draw()

```
template<typename T >
void lava::mesh_template< T >::draw (
            VkCommandBuffer cmd_buf ) const
```

Draw the mesh.

Parameters

cmd_buf	Command buffer

4.70.2.6 empty()

```
template<typename T = vertex>
bool lava::mesh_template< T >::empty ( ) const  [inline]
```

Check if mesh is empty.

Returns

Mesh is empty or not

4.70.2.7 get_data()

```
template<typename T = vertex>
mesh_template_data< T > & lava::mesh_template< T >::get_data ( )  [inline]
```

Get the mesh data.

Returns

mesh_data& Mesh data

4.70.2.8 get_index_buffer()

```
template<typename T = vertex>
buffer::ptr lava::mesh_template< T >::get_index_buffer ( ) [inline]
```

Get the index buffer of the mesh.

Returns

buffer::ptr Shared pointer to buffer

4.70.2.9 get_indices() [1/2]

```
template<typename T = vertex>
index_list & lava::mesh_template< T >::get_indices ( ) [inline]
```

Get the indices of the mesh.

Returns

index_list& List of indices

4.70.2.10 get_indices() [2/2]

```
template<typename T = vertex>
index_list const & lava::mesh_template< T >::get_indices ( ) const [inline]
```

Get the const indices of the mesh.

Returns

index_list const& List of indices

4.70.2.11 get_indices_count()

```
template<typename T = vertex>
ui32 lava::mesh_template< T >::get_indices_count ( ) const [inline]
```

Get the indices count of the mesh.

Returns

ui32 Number of indices

4.70.2.12 get_vertex_buffer()

```
template<typename T = vertex>
buffer::ptr lava::mesh_template< T >::get_vertex_buffer ( ) [inline]
```

Get the vertex buffer of the mesh.

Returns

buffer::ptr Shared pointer to buffer

4.70.2.13 get_vertices() [1/2]

```
template<typename T = vertex>
vertex_list & lava::mesh_template< T >::get_vertices ( ) [inline]
```

Get the vertices of the mesh.

Returns

vertex::list& List of vertices

4.70.2.14 get_vertices() [2/2]

```
template<typename T = vertex>
vertex_list const & lava::mesh_template< T >::get_vertices ( ) const [inline]
```

Get the const vertices of the mesh.

Returns

vertex::list const& List of vertices

4.70.2.15 get_vertices_count()

```
template<typename T = vertex>
ui32 lava::mesh_template< T >::get_vertices_count ( ) const [inline]
```

Get the vertices count of the mesh.

Returns

ui32 Number of vertices

4.70.2.16 make()

```
template<typename T = vertex>
static ptr lava::mesh_template< T >::make ( ) [inline], [static]
```

Make a new mesh.

Returns

ptr Shared pointer to mesh

4.70.2.17 reload()

```
template<typename T >
bool lava::mesh_template< T >::reload
```

Reload the mesh data.

Returns

Reload was successful or failed

4.70.2.18 set_data()

```
template<typename T = vertex>
void lava::mesh_template< T >::set_data (
            mesh_template_data< T > const & value ) [inline]
```

Set the mesh data.

Parameters

value	Mesh data

The documentation for this struct was generated from the following file:

- liblava/resource/mesh.hpp

4.71 lava::mesh_template_data< T > Struct Template Reference

Templated mesh data.

```
#include <mesh.hpp>
```

Public Member Functions

- template<typename PosType = float>
  void move (std::array< PosType, 3 > offset)

  Move mesh data by offset.
- void scale (auto factor)

  Scale mesh data by factor.
- template<typename PosType = float>
  void scale_vector (std::array< PosType, 3 > factors)

  Scale mesh data by vector.

Public Attributes

- std::vector< T > **vertices**

  List of vertices.
- index_list **indices**

  List of indices.

4.71.1 Detailed Description

template<typename T = vertex>
struct lava::mesh_template_data< T >

Templated mesh data.

Template Parameters

T	Input vertex struct

4.71.2 Member Function Documentation

4.71.2.1 move()

```
template<typename T = vertex>
template<typename PosType = float>
void lava::mesh_template_data< T >::move (
            std::array< PosType, 3 > offset )  [inline]
```

Move mesh data by offset.

Template Parameters

PosType	Coordinate element typename

Parameters

offset	Position offset

4.71.2.2 scale()

```
template<typename T = vertex>
void lava::mesh_template_data< T >::scale (
            auto factor )  [inline]
```

Scale mesh data by factor.

Parameters

factor	Position scaling factor

4.71.2.3 scale_vector()

```
template<typename T = vertex>
template<typename PosType = float>
void lava::mesh_template_data< T >::scale_vector (
            std::array< PosType, 3 > factors )  [inline]
```

Scale mesh data by vector.

Template Parameters

PosType	Coordinate element typename

Parameters

factors	Array of position scaling factors

The documentation for this struct was generated from the following file:

- liblava/resource/mesh.hpp

4.72 lava::texture::mip_level Struct Reference

Texture mip level.

```
#include <texture.hpp>
```

Public Types

- using **list** = std::vector< mip_level >

  List of mip levels.

Public Attributes

- uv2 **extent** {}

  Mip level extent.
- ui32 **size** = 0

  Mip level size.

4.72.1 Detailed Description

Texture mip level.

The documentation for this struct was generated from the following file:

- liblava/resource/texture.hpp

4.73 lava::mouse_active_event Struct Reference

Mouse active event.

```
#include <input.hpp>
```

Public Types

- using **ref** = mouse_active_event const &

  Reference to mouse active event.
- using **func** = std::function< bool(ref)>

  Mouse active event function.
- using **listeners** = std::map< id, func >

  List of mouse active event listeners.
- using **list** = std::vector< mouse_active_event >

  List of mouse active events.

Public Attributes

- id **sender**

  Sender id.
- bool **active** = false

  Active state.

4.73.1 Detailed Description

Mouse active event.

The documentation for this struct was generated from the following file:

- liblava/frame/input.hpp

4.74 lava::mouse_button_event Struct Reference

Mouse button event.

```
#include <input.hpp>
```

Public Types

- using **ref** = mouse_button_event const &

  Reference to mouse button event.
- using **func** = std::function< bool(ref)>

  Mouse button event function.
- using **listeners** = std::map< id, func >

  List of mouse button event listeners.
- using **list** = std::vector< mouse_button_event >

  List of mouse button events.

Public Member Functions

- bool pressed (mouse_button_ref b) const

  Check if mouse button is pressed.
- bool released (mouse_button_ref b) const

  Check if mouse button is released.

Public Attributes

- id **sender**

  Sender id.
- mouse_button **button**

  Input mouse button.
- action_t **action**

  Input action.
- mod_t **mod**

  Input mod.

4.74.1 Detailed Description

Mouse button event.

4.74.2 Member Function Documentation

4.74.2.1 pressed()

```
bool lava::mouse_button_event::pressed (
            mouse_button_ref b ) const  [inline]
```

Check if mouse button is pressed.

Parameters

b	Mouse button to check

Returns

Mouse button is pressed or not

4.74.2.2 released()

```
bool lava::mouse_button_event::released (
            mouse_button_ref b ) const  [inline]
```

Check if mouse button is released.

Parameters

b	Mouse button to check

Returns

Mouse button is released or not

The documentation for this struct was generated from the following file:

- liblava/frame/input.hpp

4.75 lava::mouse_move_event Struct Reference

Mouse move event.

```
#include <input.hpp>
```

Public Types

- using **ref** = mouse_move_event const &

  Reference to mouse move event.
- using **func** = std::function< bool(ref)>

  Mouse move event function.
- using **listeners** = std::map< id, func >

  List of mouse move event listeners.
- using **list** = std::vector< mouse_move_event >

  List of mouse move events.

Public Attributes

- id **sender**

  Sender id.
- mouse_position **position**

  Input mouse position.

4.75.1 Detailed Description

Mouse move event.

The documentation for this struct was generated from the following file:

- liblava/frame/input.hpp

4.76 lava::mouse_position Struct Reference

Input mouse position.

```
#include <input.hpp>
```

Public Attributes

- r64 **x** = 0.0

  X position.
- r64 **y** = 0.0

  Y position.

4.76.1 Detailed Description

Input mouse position.

The documentation for this struct was generated from the following file:

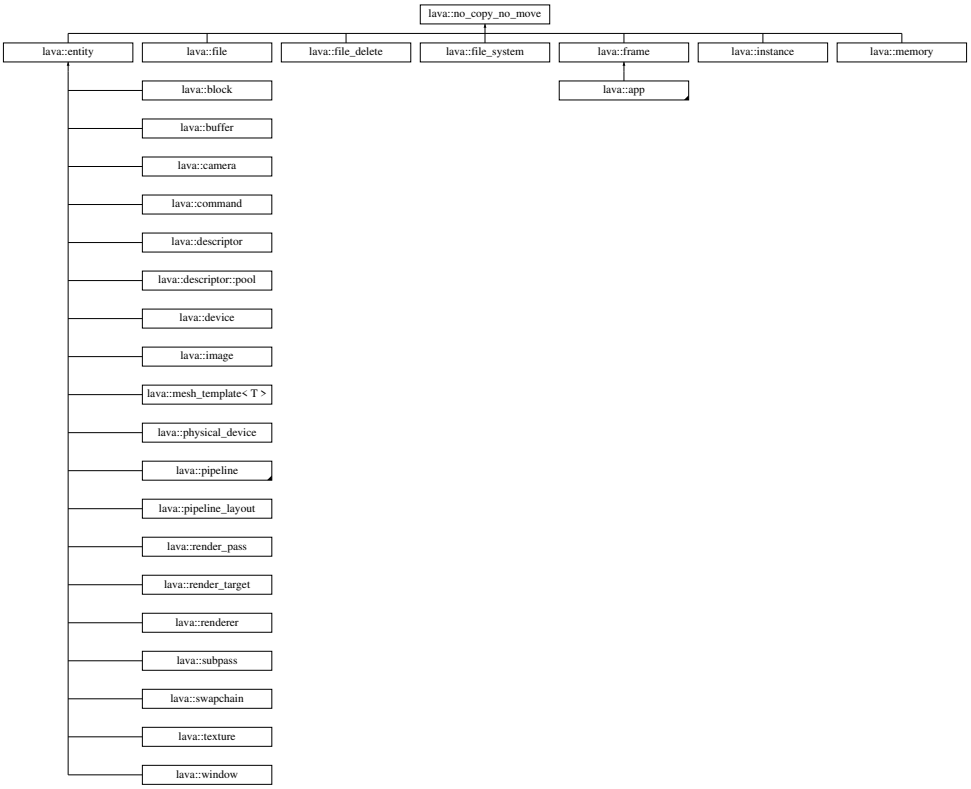
- liblava/frame/input.hpp

4.77 lava::no_copy_no_move Struct Reference

No copy and no move object.

```
#include <types.hpp>
```

Inheritance diagram for lava::no_copy_no_move:



Public Member Functions

- **no_copy_no_move** ()=default

  Construct a new object.
- **no_copy_no_move** (no_copy_no_move const &)=delete

  No copy.
- void **operator=** (no_copy_no_move const &)=delete

  No move.

4.77.1 Detailed Description

No copy and no move object.

The documentation for this struct was generated from the following file:

- liblava/core/types.hpp

4.78 lava::pair_hash Struct Reference

Pair hash.

```
#include <types.hpp>
```

Public Member Functions

- template<class T1 , class T2 >
  size_t operator() (std::pair< T1, T2 > const &p) const

  Hash operator.

4.78.1 Detailed Description

Pair hash.

4.78.2 Member Function Documentation

4.78.2.1 operator()()

```
template<class T1 , class T2 >
size_t lava::pair_hash::operator() (
            std::pair< T1, T2 > const & p ) const  [inline]
```

Hash operator.

Template Parameters

T1	Type of first
T2	Type of second

Parameters

p	Hash pair

Returns

size_t Hash value

The documentation for this struct was generated from the following file:

- liblava/core/types.hpp

4.79 lava::path_drop_event Struct Reference

Path drop event.

```
#include <input.hpp>
```

Public Types

- using **ref** = path_drop_event const &

  Reference to path drop event.
- using **func** = std::function< bool(ref)>

  Path drop event function.
- using **listeners** = std::map< id, func >

  List of path drop event listeners.
- using **list** = std::vector< path_drop_event >

  List of path drop events.

Public Attributes

- id **sender**

  Sender id.
- string_list **files**

  List of files.

4.79.1 Detailed Description

Path drop event.

The documentation for this struct was generated from the following file:

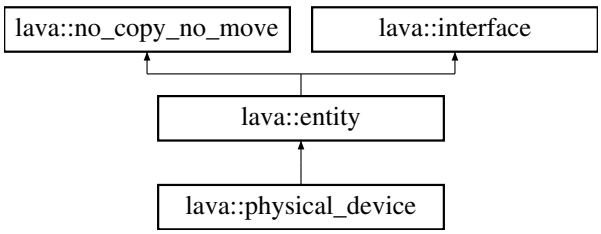
- liblava/frame/input.hpp

4.80 lava::physical_device Struct Reference

Vulkan physical device.

```
#include <physical_device.hpp>
```

Inheritance diagram for lava::physical_device:



Public Types

- using **ptr** = std::shared_ptr< physical_device >

  Shared pointer to physical device.
- using **list** = std::vector< ptr >

  List of physical devices.
- using **ref** = physical_device const &

  Reference to physical device.

Public Member Functions

- **physical_device** ()=default

  Construct a new physical device.
- physical_device (VkPhysicalDevice vk_physical_device)

  Construct and initialize a new physical device.
- void initialize (VkPhysicalDevice vk_physical_device)

  Initialize the physical device.
- bool supported (string_ref extension) const

  Check if extension is supported.
- bool get_queue_family (index &index, VkQueueFlags flags) const

  Get the queue family.
- device::create_param create_default_device_param () const

  Create default device parameters.
- VkPhysicalDeviceProperties const & get_properties () const

  Get the properties.
- VkPhysicalDeviceFeatures const & get_features () const

  Get the features.
- VkPhysicalDeviceMemoryProperties const & get_memory_properties () const

  Get the memory properties.
- VkQueueFamilyPropertiesList const & get_queue_family_properties () const

  Get the queue family properties.
- VkExtensionPropertiesList const & get_extension_properties () const

  Get the extension properties.
- VkPhysicalDevice get () const

  Get the Vulkan physical device.
- name get_device_name () const

  Get the device name.
- string get_device_type_string () const

  Get the device type as string.
- int_version get_driver_version () const

  Get the driver version.
- bool swapchain_supported () const

  Check if swapchain is supported.
- bool surface_supported (index queue_family, VkSurfaceKHR surface) const

  Check if surface is supported.

Static Public Member Functions

- static ptr make (VkPhysicalDevice vk_physical_device)

  Make a new physical device.

4.80.1 Detailed Description

Vulkan physical device.

4.80.2 Constructor & Destructor Documentation

4.80.2.1 physical_device()

```
lava::physical_device::physical_device (
            VkPhysicalDevice vk_physical_device )
```

Construct and initialize a new physical device.

Parameters

vk_physical_device	Vulkan physical device

4.80.3 Member Function Documentation

4.80.3.1 create_default_device_param()

```
device::create_param lava::physical_device::create_default_device_param ( ) const
```

Create default device parameters.

Returns

device::create_param Device create parameters

4.80.3.2 get()

```
VkPhysicalDevice lava::physical_device::get ( ) const  [inline]
```

Get the Vulkan physical device.

Returns

VkPhysicalDevice Vulkan physical device

4.80.3.3 get_device_name()

```
name lava::physical_device::get_device_name ( ) const
```

Get the device name.

Returns

name Name of device

4.80.3.4 get_device_type_string()

```
string lava::physical_device::get_device_type_string ( ) const
```

Get the device type as string.

Returns

string String representation of device type

4.80.3.5 get_driver_version()

```
int_version lava::physical_device::get_driver_version ( ) const
```

Get the driver version.

Returns

int_version Driver version

4.80.3.6 get_extension_properties()

```
VkExtensionPropertiesList const & lava::physical_device::get_extension_properties ( ) const
[inline]
```

Get the extension properties.

Returns

VkExtensionPropertiesList const& List of extension properties

4.80.3.7 get_features()

```
VkPhysicalDeviceFeatures const & lava::physical_device::get_features ( ) const  [inline]
```

Get the features.

Returns

VkPhysicalDeviceFeatures const& Physical device features

4.80.3.8 get_memory_properties()

```
VkPhysicalDeviceMemoryProperties const & lava::physical_device::get_memory_properties ( )
const  [inline]
```

Get the memory properties.

Returns

VkPhysicalDeviceMemoryProperties const& Physical device memory properties

4.80.3.9 get_properties()

```
VkPhysicalDeviceProperties const & lava::physical_device::get_properties ( ) const  [inline]
```

Get the properties.

Returns

VkPhysicalDeviceProperties const& Physical device properties

4.80.3.10 get_queue_family()

```
bool lava::physical_device::get_queue_family (
            index & index,
            VkQueueFlags flags ) const
```

Get the queue family.

Parameters

index	Returned index of queue family
flags	Queue flags that must be set

Returns

Found a queue family or not

4.80.3.11 get_queue_family_properties()

```
VkQueueFamilyPropertiesList const & lava::physical_device::get_queue_family_properties ( )
const [inline]
```

Get the queue family properties.

Returns

VkQueueFamilyPropertiesList const& List of queue family properties

4.80.3.12 initialize()

```
void lava::physical_device::initialize (
            VkPhysicalDevice vk_physical_device )
```

Initialize the physical device.

Parameters

vk_physical_device	Vulkan physical device

4.80.3.13 make()

```
static ptr lava::physical_device::make (
            VkPhysicalDevice vk_physical_device ) [inline], [static]
```

Make a new physical device.

Parameters

vk_physical_device	Vulkan physical device

Returns

ptr Shared pointer to physical device

4.80.3.14 supported()

```
bool lava::physical_device::supported (
            string_ref extension ) const
```

Check if extension is supported.

Parameters

extension	Extension to check

Returns

Extension is supported or not

4.80.3.15 surface_supported()

```
bool lava::physical_device::surface_supported (
            index queue_family,
            VkSurfaceKHR surface ) const
```

Check if surface is supported.

Parameters

queue_family	Index of queue family
surface	Vulkan surface

Returns

Surface is supported or not

4.80.3.16 swapchain_supported()

```
bool lava::physical_device::swapchain_supported ( ) const
```

Check if swapchain is supported.

Returns

Swapchain is supported or not

The documentation for this struct was generated from the following file:

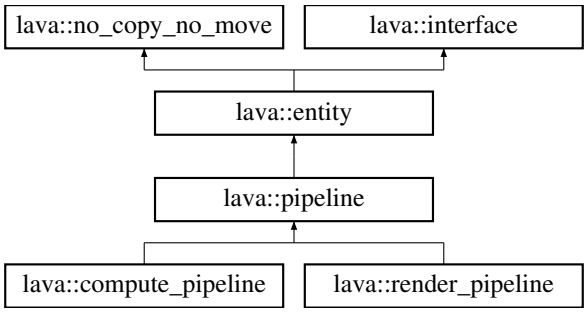
- liblava/base/physical_device.hpp

4.81 lava::pipeline Struct Reference

Pipeline.

```
#include <pipeline.hpp>
```

Inheritance diagram for lava::pipeline:



Classes

- struct shader_stage

  Shader stage.

Public Types

- using **ptr** = std::shared_ptr< pipeline >

  Shared pointer to pipeline.
- using **list** = std::vector< ptr >

  List of pipelines.
- using **process_func** = std::function< void(VkCommandBuffer)>

  Pipeline process function.

Public Member Functions

- pipeline (device_p device, VkPipelineCache pipeline_cache=0)

  Construct a new pipeline.
- ~**pipeline** () override

  Destroy the pipeline.
- bool create ()

  Create a new pipeline.
- void **destroy** ()

  Destroy the pipeline.
- virtual void bind (VkCommandBuffer cmd_buf)=0

  Bind the pipeline.
- void set_active (bool value=true)

  Set pipeline active.
- bool activated () const

  Check if pipeline is active.
- void **toggle** ()

Toggle activation.

- void set_auto_bind (bool value=true)

  Set auto bind.
- bool auto_bind () const

  Check if auto bind is enabled.
- bool ready () const

  Check if pipeline is ready.
- VkPipeline get () const

  Get the pipeline.
- device_p get_device ()

  Get the device.
- pipeline_layout::ptr get_layout () const

  Get the layout.
- void set_layout (pipeline_layout::ptr const &value)

  Set the layout.

Public Attributes

- process_func **on_process**

  Called on pipeline process.

Protected Member Functions

- virtual bool setup ()=0

  Set up the pipeline.
- virtual void **teardown** ()=0

  Tear down the pipeline.

Protected Attributes

- device_p **device** = nullptr

  Vulkan device.
- VkPipeline **vk_pipeline** = VK_NULL_HANDLE

  Vulkan pipeline.
- VkPipelineCache **pipeline_cache** = VK_NULL_HANDLE

  Vulkan pipeline cache.
- pipeline_layout::ptr **layout**

  Pipeline layout.

4.81.1 Detailed Description

Pipeline.

4.81.2 Constructor & Destructor Documentation

4.81.2.1 pipeline()

```
lava::pipeline::pipeline (
            device_p device,
            VkPipelineCache pipeline_cache = 0 )  [explicit]
```

Construct a new pipeline.

Parameters

device	Vulkan device
pipeline_cache	Pipeline cache

4.81.3 Member Function Documentation

4.81.3.1 activated()

```
bool lava::pipeline::activated ( ) const  [inline]
```

Check if pipeline is active.

Returns

Pipeline is active or note

4.81.3.2 auto_bind()

```
bool lava::pipeline::auto_bind ( ) const  [inline]
```

Check if auto bind is enabled.

Returns

Auto bind is enabled or not

4.81.3.3 bind()

```
virtual void lava::pipeline::bind (
            VkCommandBuffer cmd_buf )  [pure virtual]
```

Bind the pipeline.

Parameters

cmd_buf	Command buffer

Implemented in lava::render_pipeline, and lava::compute_pipeline.

4.81.3.4 create()

```
bool lava::pipeline::create ( )
```

Create a new pipeline.

Returns

Create was successful or failed

4.81.3.5 get()

```
VkPipeline lava::pipeline::get ( ) const  [inline]
```

Get the pipeline.

Returns

VkPipeline Vulkan pipeline

4.81.3.6 get_device()

```
device_p lava::pipeline::get_device ( )  [inline]
```

Get the device.

Returns

device_p Vulkan device

4.81.3.7 get_layout()

```
pipeline_layout::ptr lava::pipeline::get_layout ( ) const  [inline]
```

Get the layout.

Returns

pipeline_layout::ptr Pipeline layout

4.81.3.8 ready()

```
bool lava::pipeline::ready ( ) const  [inline]
```

Check if pipeline is ready.

Returns

Pipeline is ready or not

4.81.3.9 set_active()

```
void lava::pipeline::set_active (
            bool value = true )  [inline]
```

Set pipeline active.

Parameters

value	Active state

4.81.3.10 set_auto_bind()

```
void lava::pipeline::set_auto_bind (
            bool value = true )  [inline]
```

Set auto bind.

Parameters

value	Enable state

4.81.3.11 set_layout()

```
void lava::pipeline::set_layout (
            pipeline_layout::ptr const & value )  [inline]
```

Set the layout.

Parameters

value	Pipeline layout

4.81.3.12 setup()

```
virtual bool lava::pipeline::setup ( )  [protected], [pure virtual]
```

Set up the pipeline.

Returns

Setup was successful or failed

The documentation for this struct was generated from the following file:

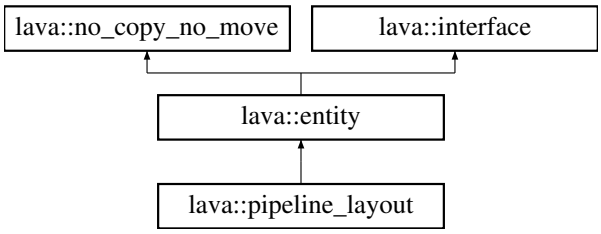
- liblava/block/pipeline.hpp

4.82 lava::pipeline_layout Struct Reference

Pipeline layout.

`#include <pipeline_layout.hpp>`

Inheritance diagram for lava::pipeline_layout:



Public Types

- using **ptr** = std::shared_ptr< pipeline_layout >

  Shared pointer to pipeline layout.
- using **list** = std::vector< ptr >

  List of pipeline layouts.
- using **offset_list** = std::vector< index >

  List of offsets.

Public Member Functions

- void add (descriptor::ptr const &descriptor)
- void add (VkPushConstantRange const &range)
- void add_descriptor (descriptor::ptr const &descriptor)

  Add descriptor.
- void add_push_constant_range (VkPushConstantRange const &range)

  Add push contant range.
- void **clear_descriptors** ()

  Clear descriptors.
- void **clear_ranges** ()

  Clear push constant ranges.
- void **clear** ()

  Clear descriptors and push constant ranges.
- bool create (device_p device)

  Create a new pipeline layout.
- void **destroy** ()

  Destroy the pipeline layout.
- VkPipelineLayout get () const

  Get the Vulkan pipeline layout.
- device_p get_device ()

  Get the device.
- descriptor::list const & get_descriptors () const

  Get the descriptors.
- VkPushConstantRanges const & get_push_constant_ranges () const

  Get the push constant ranges.
- void bind_descriptor_set (VkCommandBuffer cmd_buf, VkDescriptorSet descriptor_set, index first_set=0, offset_list offsets={}, VkPipelineBindPoint bind_point=VK_PIPELINE_BIND_POINT_GRAPHICS)

  Bind descriptor set.
- void bind (VkCommandBuffer cmd_buf, VkDescriptorSet descriptor_set, index first_set=0, offset_list offsets={}, VkPipelineBindPoint bind_point=VK_PIPELINE_BIND_POINT_GRAPHICS)

Static Public Member Functions

- static ptr make ()

  Make a new pipeline layout.

4.82.1 Detailed Description

Pipeline layout.

4.82.2 Member Function Documentation

4.82.2.1 add() [1/2]

```
void lava::pipeline_layout::add (
            descriptor::ptr const & descriptor )  [inline]
```

See also

add_descriptor

4.82.2.2 add() [2/2]

```
void lava::pipeline_layout::add (
            VkPushConstantRange const & range )  [inline]
```

See also

add_push_constant_range

4.82.2.3 add_descriptor()

```
void lava::pipeline_layout::add_descriptor (
            descriptor::ptr const & descriptor )  [inline]
```

Add descriptor.

Parameters

descriptor	Descriptor

4.82.2.4 add_push_constant_range()

```
void lava::pipeline_layout::add_push_constant_range (
            VkPushConstantRange const & range )  [inline]
```

Add push contant range.

Parameters

range	Push contant range

4.82.2.5 bind()

```
void lava::pipeline_layout::bind (
            VkCommandBuffer cmd_buf,
            VkDescriptorSet descriptor_set,
            index first_set = 0,
            offset_list offsets = {},
            VkPipelineBindPoint bind_point = VK_PIPELINE_BIND_POINT_GRAPHICS )  [inline]
```

See also

bind_descriptor_set

4.82.2.6 bind_descriptor_set()

```
void lava::pipeline_layout::bind_descriptor_set (
            VkCommandBuffer cmd_buf,
            VkDescriptorSet descriptor_set,
            index first_set = 0,
            offset_list offsets = {},
            VkPipelineBindPoint bind_point = VK_PIPELINE_BIND_POINT_GRAPHICS )
```

Bind descriptor set.

Parameters

cmd_buf	Command buffer
descriptor_set	Descriptor set
first_set	Index to first descriptor set
offsets	List of offsets
bind_point	Pipeline bind point

4.82.2.7 create()

```
bool lava::pipeline_layout::create (
            device_p device )
```

Create a new pipeline layout.

Parameters

device	Vulkan device

Returns

Create was successful or failed

4.82.2.8 get()

```
VkPipelineLayout lava::pipeline_layout::get ( ) const  [inline]
```

Get the Vulkan pipeline layout.

Returns

VkPipelineLayout Pipeline layout

4.82.2.9 get_descriptors()

```
descriptor::list const & lava::pipeline_layout::get_descriptors ( ) const  [inline]
```

Get the descriptors.

Returns

descriptor::list const& List of descriptors

4.82.2.10 get_device()

```
device_p lava::pipeline_layout::get_device ( ) [inline]
```

Get the device.

Returns

device_p Vulkan device

4.82.2.11 get_push_constant_ranges()

```
VkPushConstantRanges const & lava::pipeline_layout::get_push_constant_ranges ( ) const [inline]
```

Get the push constant ranges.

Returns

VkPushConstantRanges const& List of push constant ranges

4.82.2.12 make()

```
static ptr lava::pipeline_layout::make ( ) [inline], [static]
```

Make a new pipeline layout.

Returns

ptr Shared pointer to pipeline layout

The documentation for this struct was generated from the following file:

- liblava/block/pipeline_layout.hpp

4.83 lava::platform Struct Reference

Stage platform.

```
#include <platform.hpp>
```

Public Types

- using **create_param_func** = std::function< void(device::create_param &)>

  Create parameter function.

Public Member Functions

- device::ptr create (index physical_device=0)

  Create a managed device from a physical device.
- device::ptr create (device::create_param::ref param)

  Create a managed device with create parameters.
- device_p create_device (index physical_device=0)

  Create a managed device.
- device::list const & get_devices () const

  Get all devices.
- void **wait_idle** ()

  Wait for idle on all managed devices.
- bool remove (id::ref device_id)

  Remove device from platform.
- void **clear** ()

  Clear all managed devices.

Public Attributes

- create_param_func **on_create_param**

  Called on create to adjust the create parameters.

4.83.1 Detailed Description

Stage platform.

4.83.2 Member Function Documentation

4.83.2.1 create() [1/2]

```
device::ptr lava::platform::create (
            device::create_param::ref param )
```

Create a managed device with create parameters.

Parameters

param	Create parameters

Returns

device::ptr Vulkan device

4.83.2.2 create() [2/2]

```
device::ptr lava::platform::create (
            index physical_device = 0 )
```

Create a managed device from a physical device.

Parameters

physical_device	Index of physical device

Returns

device::ptr Vulkan device

4.83.2.3 create_device()

```
device_p lava::platform::create_device (
            index physical_device = 0 )
```

Create a managed device.

Parameters

physical_device	Physical device

Returns

device_p Pointer to device

4.83.2.4 get_devices()

```
device::list const & lava::platform::get_devices ( ) const  [inline]
```

Get all devices.

Returns

device::list const& List of devices

4.83.2.5 remove()

```
bool lava::platform::remove (
            id::ref device_id )
```

Remove device from platform.

Parameters

device↩ _id	Id of device

Returns

Remove was successful or failed

The documentation for this struct was generated from the following file:

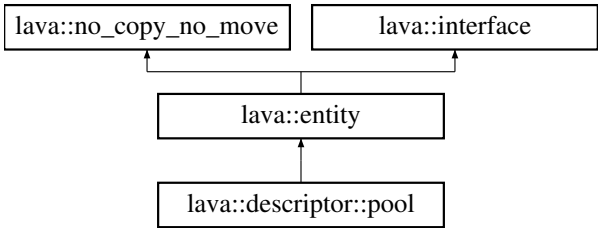
- liblava/base/platform.hpp

4.84 lava::descriptor::pool Struct Reference

Descriptor pool.

```
#include <descriptor.hpp>
```

Inheritance diagram for lava::descriptor::pool:



Public Types

- using **ptr** = std::shared_ptr< pool >

  Shared pointer to pool.
- using **list** = std::vector< ptr >

  List of pools.

Public Member Functions

- bool create (device_p device, VkDescriptorPoolSizesRef sizes, ui32 max=1, VkDescriptorPoolCreateFlags flags=VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_SET_BIT)

  Create a new pool.
- void **destroy** ()

  Destroy the pool.
- VkDescriptorPool get () const

  Get the descriptor pool.
- device_p get_device ()

  Get the device.
- VkDescriptorPoolSizes const & get_sizes () const

  Get the sizes.
- ui32 get_max () const

  Get the max.

Static Public Member Functions

- static ptr make ()

  Make a new descriptor pool.

4.84.1 Detailed Description

Descriptor pool.

4.84.2 Member Function Documentation

4.84.2.1 create()

```
bool lava::descriptor::pool::create (
            device_p device,
            VkDescriptorPoolSizesRef sizes,
            ui32 max = 1,
            VkDescriptorPoolCreateFlags flags = VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_↩
SET_BIT )
```

Create a new pool.

Parameters

device	Vulkan device
sizes	Descriptor pool sizes
max	Number of pools
flags	Create flags

Returns

Create was successful or failed

4.84.2.2 get()

```
VkDescriptorPool lava::descriptor::pool::get ( ) const  [inline]
```

Get the descriptor pool.

Returns

VkDescriptorPool Vulkan descriptor pool

4.84.2.3 get_device()

```
device_p lava::descriptor::pool::get_device ( ) [inline]
```

Get the device.

Returns

device_p Vulkan device

4.84.2.4 get_max()

```
ui32 lava::descriptor::pool::get_max ( ) const [inline]
```

Get the max.

Returns

ui32 Number of pools

4.84.2.5 get_sizes()

```
VkDescriptorPoolSizes const & lava::descriptor::pool::get_sizes ( ) const [inline]
```

Get the sizes.

Returns

VkDescriptorPoolSizes const& Descriptor pool sizes

4.84.2.6 make()

```
static ptr lava::descriptor::pool::make ( ) [inline], [static]
```

Make a new descriptor pool.

Returns

ptr Shared pointer to descriptor pool

The documentation for this struct was generated from the following file:

- liblava/block/descriptor.hpp

4.85 lava::producer Struct Reference

Producer.

```
#include <producer.hpp>
```

Public Types

- enum shader_optimization : index { **none** = 0 , **size** , **performance** }

  Shader optimization level.
- enum shader_language : index { **glsl** = 0 , **hlsl** }

  Shader source language.

Public Member Functions

- mesh::ptr create_mesh (mesh_type mesh_type)

  Create a mesh product.
- mesh::ptr get_mesh (string_ref name)

  Get mesh by prop name.
- bool add_mesh (mesh::ptr mesh)

  Add mesh to products.
- texture::ptr create_texture (uv2 size)

  Create a texture product.
- texture::ptr get_texture (string_ref name)

  Get texture by prop name.
- bool add_texture (texture::ptr product)

  Add texture to products.
- cdata get_shader (string_ref name, bool reload=false)

  Generate shader by prop name.
- cdata reload_shader (string_ref name)

  Regenerate shader by prop name.
- data compile_shader (cdata product, string_ref name, string_ref filename) const

  Compile shader.
- void **destroy** ()

  Destroy all products.
- void **clear** ()

  Clear all products.

Public Attributes

- engine * **context** = nullptr

  Engine.
- id_registry< mesh, string > **meshes**

  Mesh products.
- id_registry< texture, string > **textures**

  Texture products.
- shader_optimization **shader_opt** = shader_optimization::performance

  Shader optimization level.
- shader_language **shader_lang** = shader_language::glsl

  Shader source language.
- bool **shader_debug** = false

  Shader debug information.

4.85.1 Detailed Description

Producer.

4.85.2 Member Function Documentation

4.85.2.1 add_mesh()

```
bool lava::producer::add_mesh (
            mesh::ptr mesh )
```

Add mesh to products.

Parameters

mesh	Mesh

Returns

Added to products or already exists

4.85.2.2 add_texture()

```
bool lava::producer::add_texture (
            texture::ptr product )
```

Add texture to products.

Parameters

product	Texture

Returns

Added to products or already exists

4.85.2.3 compile_shader()

```
data lava::producer::compile_shader (
            cdata product,
```

```
            string_ref name,
            string_ref filename ) const
```

Compile shader.

Parameters

product	Shader data
name	Shader name
filename	Shader filename

Returns

data Compiled shader data

4.85.2.4 create_mesh()

```
mesh::ptr lava::producer::create_mesh (
            mesh_type mesh_type )
```

Create a mesh product.

Parameters

mesh_type	Type of mesh

Returns

mesh::ptr Mesh

4.85.2.5 create_texture()

```
texture::ptr lava::producer::create_texture (
            uv2 size )
```

Create a texture product.

Parameters

size	Size of texture

Returns

texture::ptr Default texture

4.85.2.6 get_mesh()

```
mesh::ptr lava::producer::get_mesh (
            string_ref name )
```

Get mesh by prop name.

Parameters

name	Name of prop

Returns

mesh::ptr Mesh

4.85.2.7 get_shader()

```
cdata lava::producer::get_shader (
            string_ref name,
            bool reload = false )
```

Generate shader by prop name.

Parameters

name	Name of shader
reload	Reload shader

Returns

cdata Shader data

4.85.2.8 get_texture()

```
texture::ptr lava::producer::get_texture (
            string_ref name )
```

Get texture by prop name.

Parameters

name	Name of prop

Returns

texture::ptr Texture

4.85.2.9 reload_shader()

```
cdata lava::producer::reload_shader (
            string_ref name ) [inline]
```

Regenerate shader by prop name.

Parameters

name	Name of shader

Returns

cdata Shader data

The documentation for this struct was generated from the following file:

- liblava/engine/producer.hpp

4.86 lava::property::prop Struct Reference

Prop.

```
#include <property.hpp>
```

Public Types

- using **map** = std::map< string, prop >

  Map of props.

Public Member Functions

- prop (string_ref filename)

  Construct a new prop.

Public Attributes

- string **filename**

  File name of prop.
- file_data **data**

  File data of prop.

4.86.1 Detailed Description

Prop.

4.86.2 Constructor & Destructor Documentation

4.86.2.1 prop()

```
lava::property::prop::prop (
            string_ref filename )  [inline]
```

Construct a new prop.

Parameters

filename	File name of prop

The documentation for this struct was generated from the following file:

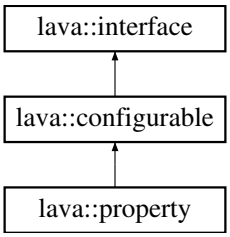
- liblava/engine/property.hpp

4.87 lava::property Struct Reference

Props master.

```
#include <property.hpp>
```

Inheritance diagram for lava::property:



Classes

- struct prop

  Prop.

Public Member Functions

- void add (string_ref name, string_ref filename)

  Add a prop.
- cdata operator() (string_ref name)

  Get prop data.
- string_ref get_filename (string_ref name) const

  Get file name of prop.
- void set_filename (string_ref name, string_ref filename)

  Set filename of prop.
- bool exists (string_ref name) const

  Check if prop exists.
- bool empty (string_ref name) const

  Check if prop data is empty.
- bool load (string_ref name)

  Load prop data (reload if loaded)
- void unload (string_ref name)

  Unload prop data.
- bool load_all ()

  Load all prop data (reload if loaded)
- void **unload_all** ()

  Unload all prop data.
- bool check ()

  Check whether all props are available.
- void parse (cmd_line cmd_line)

  Parse prop overloads.
- void **clear** ()

  Clear all props.
- prop::map const & get_all () const

  Get all props.
- void set_config (json_ref j) override

  Set config.
- json get_config () const override

  Get config.

Public Attributes

- engine * **context** = nullptr

  Engine.

4.87.1 Detailed Description

Props master.

4.87.2 Member Function Documentation

4.87.2.1 add()

```
void lava::property::add (
            string_ref name,
            string_ref filename )
```

Add a prop.

Parameters

name	Name of prop
filename	File name of prop

4.87.2.2 check()

```
bool lava::property::check ( )
```

Check whether all props are available.

Returns

All props are there or are missing (see log)

4.87.2.3 empty()

```
bool lava::property::empty (
            string_ref name ) const  [inline]
```

Check if prop data is empty.

Parameters

name	Name of prop

Returns

Prop data is empty or not

4.87.2.4 exists()

```
bool lava::property::exists (
            string_ref name ) const  [inline]
```

Check if prop exists.

Parameters

name	Name of prop to check

Returns

Prop exists or not

4.87.2.5 get_all()

```
prop::map const & lava::property::get_all ( ) const  [inline]
```

Get all props.

Returns

prop::map const& Map of props

4.87.2.6 get_config()

```
json lava::property::get_config ( ) const  [override], [virtual]
```

Get config.

Returns

json Json file

Implements lava::configurable.

4.87.2.7 get_filename()

```
string_ref lava::property::get_filename (
            string_ref name ) const  [inline]
```

Get file name of prop.

Parameters

name	Name of prop

Returns

string_ref File name

4.87.2.8 load()

```
bool lava::property::load (
            string_ref name )
```

Load prop data (reload if loaded)

Parameters

name	Name of prop

Returns

Load was successful or failed

4.87.2.9 load_all()

```
bool lava::property::load_all ( )
```

Load all prop data (reload if loaded)

Returns

Load was successful or failed

4.87.2.10 operator()()

```
cdata lava::property::operator() (
            string_ref name )
```

Get prop data.

Parameters

name	Name of prop

Returns

cdata Prop const data

4.87.2.11 parse()

```
void lava::property::parse (
            cmd_line cmd_line )
```

Parse prop overloads.

Parameters

cmd_line	Command line arguments

4.87.2.12 set_config()

```
void lava::property::set_config (
            json_ref j )  [override], [virtual]
```

Set config.

Parameters

j	Json file

Implements lava::configurable.

4.87.2.13 set_filename()

```
void lava::property::set_filename (
            string_ref name,
            string_ref filename )  [inline]
```

Set filename of prop.

Parameters

name	Name of prop
filename	File name

4.87.2.14 unload()

```
void lava::property::unload (
            string_ref name )  [inline]
```

Unload prop data.

Parameters

name	Name of prop

The documentation for this struct was generated from the following file:

- liblava/engine/property.hpp

4.88 lava::pseudo_random_generator Struct Reference

Pseudo random generator.

```
#include <random.hpp>
```

Public Member Functions

- pseudo_random_generator (ui32 seed)

  Construct a new pseudo random generator.
- void set_seed (ui32 value)

  Set the seed.
- ui32 get ()

  Get next pseudo random number.

4.88.1 Detailed Description

Pseudo random generator.

4.88.2 Constructor & Destructor Documentation

4.88.2.1 pseudo_random_generator()

```
lava::pseudo_random_generator::pseudo_random_generator (
            ui32 seed )  [inline], [explicit]
```

Construct a new pseudo random generator.

Parameters

seed	Seed for generator

4.88.3 Member Function Documentation

4.88.3.1 get()

```
ui32 lava::pseudo_random_generator::get ( ) [inline]
```

Get next pseudo random number.

Returns

ui32 Random number

4.88.3.2 set_seed()

```
void lava::pseudo_random_generator::set_seed (
            ui32 value ) [inline]
```

Set the seed.

Parameters

value	Generator seed

The documentation for this struct was generated from the following file:

- liblava/util/random.hpp

4.89 lava::queue Struct Reference

Device queue.

```
#include <queue.hpp>
```

Public Types

- using **list** = std::deque< queue >

  List of queues.
- using **ref** = queue const &

  Reference to queue.

Public Member Functions

- bool valid () const

  Check if queue is valid.
- bool operator< (queue const &other) const

  Queue priority compare operator.

Public Attributes

- VkQueue **vk_queue** = nullptr

  Vulkan queue.
- VkQueueFlags **flags** = 0

  Queue flags.
- index **family** = 0

  Queue family index.
- r32 **priority** = 1.f

  Queue priority.

4.89.1 Detailed Description

Device queue.

4.89.2 Member Function Documentation

4.89.2.1 operator<()

```
bool lava::queue::operator< (
            queue const & other ) const  [inline]
```

Queue priority compare operator.

Parameters

other	Queue to compare

Returns

Priority of queue is higher or lower and equal

4.89.2.2 valid()

```
bool lava::queue::valid ( ) const  [inline]
```

Check if queue is valid.

Returns

Queue is valid or not

The documentation for this struct was generated from the following file:

- liblava/base/queue.hpp

4.90 lava::queue_family_info Struct Reference

Queue family information.

```
#include <queue.hpp>
```

Public Types

- using **list** = std::deque< queue_family_info >

  List of queue family informations.

Public Member Functions

- void add (VkQueueFlags flags, ui32 count=1, r32 priority=1.f)

  Add a queue family information.
- ui32 count () const

  Get the count of queues.
- void **clear** ()

  Clear the queue information.

Public Attributes

- index **family_index** = 0

  Queue family index.
- queue_info::list **queues**

  List of queue informations.

4.90.1 Detailed Description

Queue family information.

4.90.2 Member Function Documentation

4.90.2.1 add()

```
void lava::queue_family_info::add (
            VkQueueFlags flags,
            ui32 count = 1,
            r32 priority = 1.f )  [inline]
```

Add a queue family information.

Parameters

flags	Queue flags
count	Number of queues
priority	Queue priority

4.90.2.2 count()

```
ui32 lava::queue_family_info::count ( ) const [inline]
```

Get the count of queues.

Returns

ui32 Count of queues

The documentation for this struct was generated from the following file:

- liblava/base/queue.hpp

4.91 lava::queue_info Struct Reference

Queue information.

```
#include <queue.hpp>
```

Public Types

- using **list** = std::deque< queue_info >

  List of queue informations.

Public Attributes

- VkQueueFlags **flags** = default_queue_flags

  Queue flags.
- r32 **priority** = 1.f

  Queue priority.

4.91.1 Detailed Description

Queue information.

The documentation for this struct was generated from the following file:

- liblava/base/queue.hpp

4.92 lava::random_generator Struct Reference

Random generator.

```
#include <random.hpp>
```

Public Member Functions

- **random_generator** ()

  Construct a new random generator.
- i32 get (i32 low, i32 high)

  Get next random number.
- template<typename T = real>
  T get (T low, T high)

  Get next real random number.

4.92.1 Detailed Description

Random generator.

4.92.2 Member Function Documentation

4.92.2.1 get() [1/2]

```
i32 lava::random_generator::get (
            i32 low,
            i32 high )  [inline]
```

Get next random number.

Parameters

low	Lowest number
high	Highest number

Returns

i32 Random number

4.92.2.2 get() [2/2]

```
template<typename T = real>
T lava::random_generator::get (
```

```
T low,
T high )  [inline]
```

Get next real random number.

Template Parameters

T	Type of number

Parameters

low	Lowest number
high	Highest number

Returns

T Random number

The documentation for this struct was generated from the following file:

- liblava/util/random.hpp

4.93 lava::rect Struct Reference

Rectangle.

`#include <math.hpp>`

Public Types

- using **ref** = rect const &

  Reference to rect.

Public Member Functions

- **rect** ()=default

  Construct a new rectangle.
- rect (i32 left, i32 top, ui32 width, ui32 height)

  Construct a new rectangle.
- rect (iv2 const &left_top, ui32 width, ui32 height)

  Construct a new rectangle.
- rect (iv2 const &left_top, uv2 const &size)

  Construct a new rectangle.
- iv2 const & get_origin () const

  Get the origin.
- iv2 const & get_end_point () const

  Get the end point.
- uv2 get_size () const

  Get the size.
- void set_size (uv2 const &size)

  Set the size.
- void move (iv2 const &offset)

  Move the rectangle.
- bool contains (iv2 point) const

  Check if point is inside the rectangle.

4.93.1 Detailed Description

Rectangle.

4.93.2 Constructor & Destructor Documentation

4.93.2.1 rect() [1/3]

```
lava::rect::rect (
            i32 left,
            i32 top,
            ui32 width,
            ui32 height )  [inline]
```

Construct a new rectangle.

Parameters

left	Left position
top	Top position
width	Rectangle width
height	Rectangle height

4.93.2.2 rect() [2/3]

```
lava::rect::rect (
            iv2 const & left_top,
            ui32 width,
            ui32 height )  [inline]
```

Construct a new rectangle.

Parameters

left_top	Left top position
width	Rectangle width
height	Rectangle height

4.93.2.3 rect() [3/3]

```
lava::rect::rect (
```

```
            iv2 const & left_top,
            uv2 const & size )  [inline]
```

Construct a new rectangle.

Parameters

left_top	Left top position
size	Size of rectangle

4.93.3 Member Function Documentation

4.93.3.1 contains()

```
bool lava::rect::contains (
            iv2 point ) const  [inline]
```

Check if point is inside the rectangle.

Parameters

point	Point to check

Returns

Point is inside or out

4.93.3.2 get_end_point()

```
iv2 const & lava::rect::get_end_point ( ) const  [inline]
```

Get the end point.

Returns

iv2 const& Right bottom position

4.93.3.3 get_origin()

```
iv2 const & lava::rect::get_origin ( ) const  [inline]
```

Get the origin.

Returns

iv2 const& Left top position

4.93.3.4 get_size()

```
uv2 lava::rect::get_size ( ) const  [inline]
```

Get the size.

Returns

uv2 Width and height

4.93.3.5 move()

```
void lava::rect::move (
            iv2 const & offset )  [inline]
```

Move the rectangle.

Parameters

offset	Offset to move

4.93.3.6 set_size()

```
void lava::rect::set_size (
            uv2 const & size )  [inline]
```

Set the size.

Parameters

size	Width and height

The documentation for this struct was generated from the following file:

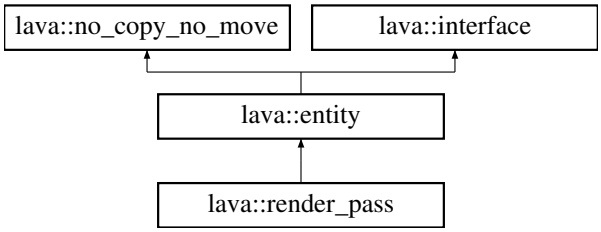
- liblava/util/math.hpp

4.94 lava::render_pass Struct Reference

Render pass.

```
#include <render_pass.hpp>
```

Inheritance diagram for lava::render_pass:



Public Types

- using **ptr** = std::shared_ptr< render_pass >

  Shared pointer to render pass.
- using **list** = std::vector< ptr >

  List of render passes.

Public Member Functions

- render_pass (device_p device)

  Construct a new render pass.
- bool create (VkAttachmentsRef target_attachments, rect::ref area)

  Create a new render pass.
- void **destroy** ()

  Destroy the render pass.
- void process (VkCommandBuffer cmd_buf, index frame)

  Process the render pass.
- device_p get_device ()

  Get the device.
- VkRenderPass get () const

  Get the render pass.
- ui32 get_subpass_count () const

  Get the subpass count.
- bool exists_subpass (index index=0) const

  Check if subpass exists.
- subpass * get_subpass (index index=0)

  Get the subpass.
- subpass::list const & get_subpasses () const

Get the subpasses.

- void add (attachment::ptr const &attachment)

  Add an attachment.
- void add (subpass_dependency::ptr const &dependency)

  Add a subpass dependency.
- void add (subpass::ptr const &subpass)

  Add a subpass.
- void set_clear_values (VkClearValues const &values)

  Set the clear values.
- VkClearValues const & get_clear_values () const

  Get the clear values.
- void set_clear_color (v3 value={})

  Set the clear color.
- v3 get_clear_color () const

  Get the clear color.
- void add (render_pipeline::ptr pipeline, index subpass=0)

  Add a render pipeline to the back of subpass.
- void add_front (render_pipeline::ptr pipeline, index subpass=0)

  Add a render pipeline to the front of subpass.
- void remove (render_pipeline::ptr pipeline, index subpass=0)

  Remove a render pipeline from the subpass.
- target_callback const & get_target_callback () const

  Get the target callback.

Static Public Member Functions

- static ptr make (device_p device)

  Make a new render pass.

4.94.1 Detailed Description

Render pass.

4.94.2 Constructor & Destructor Documentation

4.94.2.1 render_pass()

```
lava::render_pass::render_pass (
            device_p device ) [explicit]
```

Construct a new render pass.

Parameters

device	Vulkan device

4.94.3 Member Function Documentation

4.94.3.1 add() [1/4]

```
void lava::render_pass::add (
            attachment::ptr const & attachment )  [inline]
```

Add an attachment.

Parameters

attachment	Attachment

4.94.3.2 add() [2/4]

```
void lava::render_pass::add (
            render_pipeline::ptr pipeline,
            index subpass = 0 )  [inline]
```

Add a render pipeline to the back of subpass.

Parameters

pipeline	Render pipeline
subpass	Subpass

4.94.3.3 add() [3/4]

```
void lava::render_pass::add (
            subpass::ptr const & subpass )  [inline]
```

Add a subpass.

Parameters

subpass	Subpass

4.94.3.4 add() [4/4]

```
void lava::render_pass::add (
```

```
            subpass_dependency::ptr const & dependency )  [inline]
```

Add a subpass dependency.

Parameters

dependency	Subpass dependency

4.94.3.5 add_front()

```
void lava::render_pass::add_front (
            render_pipeline::ptr pipeline,
            index subpass = 0 )  [inline]
```

Add a render pipeline to the front of subpass.

Parameters

pipeline	Render pipeline
subpass	Subpass

4.94.3.6 create()

```
bool lava::render_pass::create (
            VkAttachmentsRef target_attachments,
            rect::ref area )
```

Create a new render pass.

Parameters

target_attachments	List of target attachments
area	Rectangle area

Returns

Create was successful or failed

4.94.3.7 exists_subpass()

```
bool lava::render_pass::exists_subpass (
            index index = 0 ) const  [inline]
```

Check if subpass exists.

Parameters

index	Index to check

Returns

Subpass exists or not

4.94.3.8 get()

```
VkRenderPass lava::render_pass::get ( ) const  [inline]
```

Get the render pass.

Returns

VkRenderPass Vulkan render pass

4.94.3.9 get_clear_color()

```
v3 lava::render_pass::get_clear_color ( ) const
```

Get the clear color.

Returns

v3 Clear color

4.94.3.10 get_clear_values()

```
VkClearValues const & lava::render_pass::get_clear_values ( ) const  [inline]
```

Get the clear values.

Returns

VkClearValues const& List of clear values

4.94.3.11 get_device()

```
device_p lava::render_pass::get_device ( )  [inline]
```

Get the device.

Returns

device_p Vulkan device

4.94.3.12 get_subpass()

```
subpass * lava::render_pass::get_subpass (
            index index = 0 )  [inline]
```

Get the subpass.

Parameters

index	Index of subpass

Returns

subpass* Subpass

4.94.3.13 get_subpass_count()

```
ui32 lava::render_pass::get_subpass_count ( ) const [inline]
```

Get the subpass count.

Returns

ui32 Number of subpasses

4.94.3.14 get_subpasses()

```
subpass::list const & lava::render_pass::get_subpasses ( ) const [inline]
```

Get the subpasses.

Returns

subpass::list const& List of subpasses

4.94.3.15 get_target_callback()

```
target_callback const & lava::render_pass::get_target_callback ( ) const [inline]
```

Get the target callback.

Returns

target_callback const& Target callback

4.94.3.16 make()

```
static ptr lava::render_pass::make (
            device_p device ) [inline], [static]
```

Make a new render pass.

Parameters

device	Vulkan device

Returns

ptr Shared pointer to render pass

4.94.3.17 process()

```
void lava::render_pass::process (
            VkCommandBuffer cmd_buf,
            index frame )
```

Process the render pass.

Parameters

cmd_buf	Command buffer
frame	Frame index

4.94.3.18 remove()

```
void lava::render_pass::remove (
            render_pipeline::ptr pipeline,
            index subpass = 0 )  [inline]
```

Remove a render pipeline from the subpass.

Parameters

pipeline	Render pipeline
subpass	Subpass

4.94.3.19 set_clear_color()

```
void lava::render_pass::set_clear_color (
            v3 value = {} )
```

Set the clear color.

Parameters

value	Clear color

4.94.3.20 set_clear_values()

```
void lava::render_pass::set_clear_values (
            VkClearValues const & values )  [inline]
```

Set the clear values.

Parameters

values	List of clear values

The documentation for this struct was generated from the following file:

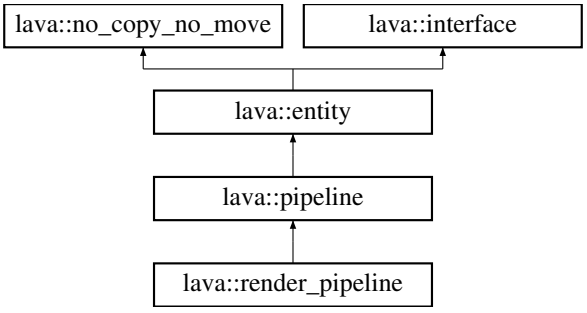
- liblava/block/render_pass.hpp

4.95 lava::render_pipeline Struct Reference

Render pipeline (Graphics)

```
#include <render_pipeline.hpp>
```

Inheritance diagram for lava::render_pipeline:



Classes

- struct create_info

  Render pipeline create information.

Public Types

- enum class sizing_mode : index { **input** = 0 , **absolute** , **relative** }

  Sizing modes.
- using **ptr** = std::shared_ptr< render_pipeline >

  Shared pointer to render pipeline.
- using **map** = std::map< id, ptr >

  Map of render pipelines.
- using **list** = std::vector< ptr >

  List of render pipelines.
- using **create_func** = std::function< bool(create_info &)>

  Create function.

Public Member Functions

- render_pipeline (device_p device, VkPipelineCache pipeline_cache)

  Construct a new render pipeline.
- void bind (VkCommandBuffer cmd_buf) override

  Bind the pipeline.
- void set_viewport_and_scissor (VkCommandBuffer cmd_buf, uv2 size)

  Set the viewport and scissor.
- void set_render_pass (VkRenderPass pass)

  Set the render pass.
- void set (VkRenderPass pass)
- VkRenderPass get_render_pass () const

  Get the render pass.
- index get_subpass () const

  Get the subpass.
- void set_subpass (index value)

  Set the subpass.
- bool create (VkRenderPass pass)

  Create a new render pipeline.
- void set_vertex_input_binding (VkVertexInputBindingDescription const &description)

  Set the vertex input binding.
- void set_vertex_input_bindings (VkVertexInputBindingDescriptions const &descriptions)

  Set the vertex input bindings.
- void set_vertex_input_attribute (VkVertexInputAttributeDescription const &attribute)

  Set the vertex input attribute.
- void set_vertex_input_attributes (VkVertexInputAttributeDescriptions const &attributes)

  Set the vertex input attributes.
- void set_input_topology (VkPrimitiveTopology const &topology)

  Set the input assembler's topology.
- void set_depth_test_and_write (bool test_enable=true, bool write_enable=true)

  Set the depth test and write.
- void set_depth_compare_op (VkCompareOp compare_op)

  Set the depth compare operation.
- void set_rasterization_cull_mode (VkCullModeFlags cull_mode)

  Set the rasterization cull mode.
- void set_rasterization_front_face (VkFrontFace front_face)

  Set the rasterization front face.

- void set_rasterization_polygon_mode (VkPolygonMode polygon_mode)

  Set the rasterization polygon mode.
- void add_color_blend_attachment (VkPipelineColorBlendAttachmentState const &attachment)

  Add color blend attachment.
- void **add_color_blend_attachment** ()

  Add color blend attachment (default)
- void **clear_color_blend_attachment** ()

  Clear color blend attachment.
- void set_dynamic_states (VkDynamicStates const &states)

  Set the dynamic states.
- void add_dynamic_state (VkDynamicState state)

  Add a dynamic state.
- void **clear_dynamic_states** ()

  Clear dynamic states.
- bool add_shader_stage (cdata::ref data, VkShaderStageFlagBits stage)

  Add shader stage.
- bool add_shader (cdata::ref data, VkShaderStageFlagBits stage)

  Add shader.
- void add (shader_stage::ptr const &shader_stage)

  Add shader stage.
- shader_stage::list const & get_shader_stages () const

  Get the shader stages.
- void **clear_shader_stages** ()

  Clear the shader stages.
- void **clear** ()

  Clear the render pipeline.
- void set_auto_size (bool value=true)

  Set the auto size.
- bool auto_sizing () const

  Get the auto sizing state.
- VkViewport get_viewport () const

  Get the viewport.
- void set_viewport (VkViewport value)

  Set the viewport.
- VkRect2D get_scissor () const

  Get the scissor.
- void set_scissor (VkRect2D value)

  Set the scissor.
- sizing_mode get_sizing () const

  Get the sizing.
- void set_sizing (sizing_mode value)

  Set the sizing.
- void copy_to (render_pipeline *target) const

  Copy pipeline configuration to target.
- void copy_from (ptr const &source)

  Copy pipeline configuration from source.
- void set_line_width (r32 value)

  Set the line width.
- r32 get_line_width () const

  Get the line width.
- bool auto_line_width () const

Check if auto line width is active.

- void set_auto_line_width (bool value=true)

  Set the auto line width.

- void set_line_width (VkCommandBuffer cmd_buf)

  Set the line width.

Static Public Member Functions

- static ptr make (device_p device, VkPipelineCache pipeline_cache=0)

  Make a new render pipeline.

Public Attributes

- create_func **on_create**

  Called on render pipeline create.

Additional Inherited Members

4.95.1 Detailed Description

Render pipeline (Graphics)

4.95.2 Constructor & Destructor Documentation

4.95.2.1 render_pipeline()

```
lava::render_pipeline::render_pipeline (
            device_p device,
            VkPipelineCache pipeline_cache ) [explicit]
```

Construct a new render pipeline.

Parameters

device	Vulkan device
pipeline_cache	Pipeline cache

4.95.3 Member Function Documentation

4.95.3.1 add()

```
void lava::render_pipeline::add (
            shader_stage::ptr const & shader_stage )  [inline]
```

Add shader stage.

Parameters

shader_stage	Shader stage

4.95.3.2 add_color_blend_attachment()

```
void lava::render_pipeline::add_color_blend_attachment (
            VkPipelineColorBlendAttachmentState const & attachment )
```

Add color blend attachment.

Parameters

attachment	Pipeline color blend attachment state

4.95.3.3 add_dynamic_state()

```
void lava::render_pipeline::add_dynamic_state (
            VkDynamicState state )
```

Add a dynamic state.

Parameters

state	Dynamic state

4.95.3.4 add_shader()

```
bool lava::render_pipeline::add_shader (
            cdata::ref data,
            VkShaderStageFlagBits stage )  [inline]
```

Add shader.

Parameters

data	Shader data
stage	Shader stage flag bits

Returns

Add was successful or failed

4.95.3.5 add_shader_stage()

```
bool lava::render_pipeline::add_shader_stage (
            cdata::ref data,
            VkShaderStageFlagBits stage )
```

Add shader stage.

Parameters

data	Shader data
stage	Shader stage flag bits

Returns

Add was successful or failed

4.95.3.6 auto_line_width()

```
bool lava::render_pipeline::auto_line_width ( ) const  [inline]
```

Check if auto line width is active.

Returns

Auto line width is enabled or not

4.95.3.7 auto_sizing()

```
bool lava::render_pipeline::auto_sizing ( ) const  [inline]
```

Get the auto sizing state.

Returns

Auto sizing is enabled or not

4.95.3.8 bind()

```
void lava::render_pipeline::bind (
            VkCommandBuffer cmd_buf )  [override], [virtual]
```

Bind the pipeline.

Parameters

cmd_buf	Command buffer

Implements lava::pipeline.

4.95.3.9 copy_from()

```
void lava::render_pipeline::copy_from (
            ptr const & source )  [inline]
```

Copy pipeline configuration from source.

Parameters

source	Render pipeline

4.95.3.10 copy_to()

```
void lava::render_pipeline::copy_to (
            render_pipeline * target ) const
```

Copy pipeline configuration to target.

Parameters

target	Render pipeline

4.95.3.11 create()

```
bool lava::render_pipeline::create (
            VkRenderPass pass )  [inline]
```

Create a new render pipeline.

Parameters

pass	Vulkan render pass

Returns

Create was successful or failed

4.95.3.12 get_line_width()

```
r32 lava::render_pipeline::get_line_width ( ) const  [inline]
```

Get the line width.

Returns

r32 Line width

4.95.3.13 get_render_pass()

```
VkRenderPass lava::render_pipeline::get_render_pass ( ) const  [inline]
```

Get the render pass.

Returns

VkRenderPass Render pass

4.95.3.14 get_scissor()

```
VkRect2D lava::render_pipeline::get_scissor ( ) const  [inline]
```

Get the scissor.

Returns

VkRect2D Scissor rectangle

4.95.3.15 get_shader_stages()

```
shader_stage::list const & lava::render_pipeline::get_shader_stages ( ) const  [inline]
```

Get the shader stages.

Returns

shader_stage::list const& List of shader stages

4.95.3.16 get_sizing()

```
sizing_mode lava::render_pipeline::get_sizing ( ) const  [inline]
```

Get the sizing.

Returns

sizing_mode Sizing mode

4.95.3.17 get_subpass()

```
index lava::render_pipeline::get_subpass ( ) const  [inline]
```

Get the subpass.

Returns

index Index of subpass

4.95.3.18 get_viewport()

```
VkViewport lava::render_pipeline::get_viewport ( ) const  [inline]
```

Get the viewport.

Returns

VkViewport Vulkan viewport

4.95.3.19 make()

```
static ptr lava::render_pipeline::make (
            device_p device,
            VkPipelineCache pipeline_cache = 0 )  [inline], [static]
```

Make a new render pipeline.

Parameters

device	Vulkan device
pipeline_cache	Pipeline cache

Returns

ptr Shared pointer to render pipeline

4.95.3.20 set()

```
void lava::render_pipeline::set (
            VkRenderPass pass )  [inline]
```

See also

set_render_pass

4.95.3.21 set_auto_line_width()

```
void lava::render_pipeline::set_auto_line_width (
            bool value = true )  [inline]
```

Set the auto line width.

Parameters

value	Enable state

4.95.3.22 set_auto_size()

```
void lava::render_pipeline::set_auto_size (
            bool value = true )  [inline]
```

Set the auto size.

Parameters

value	Enable state

4.95.3.23 set_depth_compare_op()

```
void lava::render_pipeline::set_depth_compare_op (
            VkCompareOp compare_op )
```

Set the depth compare operation.

Parameters

compare_op	Depth compare operation

4.95.3.24 set_depth_test_and_write()

```
void lava::render_pipeline::set_depth_test_and_write (
            bool test_enable = true,
            bool write_enable = true )
```

Set the depth test and write.

Parameters

test_enable	Enable depth test
write_enable	Enable depth write

4.95.3.25 set_dynamic_states()

```
void lava::render_pipeline::set_dynamic_states (
            VkDynamicStates const & states )
```

Set the dynamic states.

Parameters

states	List of dynamic states

4.95.3.26 set_input_topology()

```
void lava::render_pipeline::set_input_topology (
            VkPrimitiveTopology const & topology )
```

Set the input assembler's topology.

Parameters

topology	Enum describing polygon primitives

4.95.3.27 set_line_width() [1/2]

```
void lava::render_pipeline::set_line_width (
            r32 value )  [inline]
```

Set the line width.

Parameters

value	Line width

4.95.3.28 set_line_width() [2/2]

```
void lava::render_pipeline::set_line_width (
            VkCommandBuffer cmd_buf )  [inline]
```

Set the line width.

Parameters

cmd_buf	Command buffer

4.95.3.29 set_rasterization_cull_mode()

```
void lava::render_pipeline::set_rasterization_cull_mode (
            VkCullModeFlags cull_mode )
```

Set the rasterization cull mode.

Parameters

cull_mode	Cull mode flags

4.95.3.30 set_rasterization_front_face()

```
void lava::render_pipeline::set_rasterization_front_face (
            VkFrontFace front_face )
```

Set the rasterization front face.

Parameters

front_face	Front face

4.95.3.31 set_rasterization_polygon_mode()

```
void lava::render_pipeline::set_rasterization_polygon_mode (
            VkPolygonMode polygon_mode )
```

Set the rasterization polygon mode.

Parameters

polygon_mode	Polygon mode

4.95.3.32 set_render_pass()

```
void lava::render_pipeline::set_render_pass (
            VkRenderPass pass )  [inline]
```

Set the render pass.

Parameters

pass	Render pass

4.95.3.33 set_scissor()

```
void lava::render_pipeline::set_scissor (
            VkRect2D value )  [inline]
```

Set the scissor.

Parameters

value	Scissor rectangle

4.95.3.34 set_sizing()

```
void lava::render_pipeline::set_sizing (
            sizing_mode value ) [inline]
```

Set the sizing.

Parameters

value	Sizing mode

4.95.3.35 set_subpass()

```
void lava::render_pipeline::set_subpass (
            index value ) [inline]
```

Set the subpass.

Parameters

value	Index of subpass

4.95.3.36 set_vertex_input_attribute()

```
void lava::render_pipeline::set_vertex_input_attribute (
            VkVertexInputAttributeDescription const & attribute )
```

Set the vertex input attribute.

Parameters

attribute	Vertex input attribute description

4.95.3.37 set_vertex_input_attributes()

```
void lava::render_pipeline::set_vertex_input_attributes (
            VkVertexInputAttributeDescriptions const & attributes )
```

Set the vertex input attributes.

Parameters

attributes	List of vertex input attributes descriptions

4.95.3.38 set_vertex_input_binding()

```
void lava::render_pipeline::set_vertex_input_binding (
            VkVertexInputBindingDescription const & description )
```

Set the vertex input binding.

Parameters

description	Vertex input binding description

4.95.3.39 set_vertex_input_bindings()

```
void lava::render_pipeline::set_vertex_input_bindings (
            VkVertexInputBindingDescriptions const & descriptions )
```

Set the vertex input bindings.

Parameters

descriptions	List of vertex input binding descriptions

4.95.3.40 set_viewport()

```
void lava::render_pipeline::set_viewport (
            VkViewport value )  [inline]
```

Set the viewport.

Parameters

value	Vulkan viewport

4.95.3.41 set_viewport_and_scissor()

```
void lava::render_pipeline::set_viewport_and_scissor (
            VkCommandBuffer cmd_buf,
            uv2 size )
```

Set the viewport and scissor.

Parameters

cmd_buf	Command buffer
size	Viewport and scissor size

The documentation for this struct was generated from the following file:

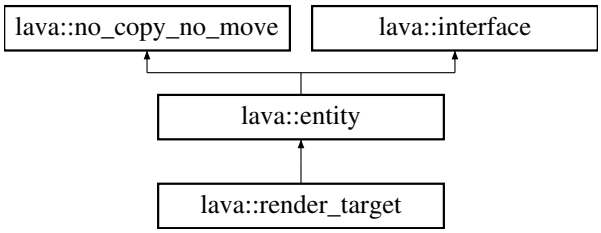
- liblava/block/render_pipeline.hpp

4.96 lava::render_target Struct Reference

Render target.

`#include <render_target.hpp>`

Inheritance diagram for lava::render_target:



Public Types

- using **ptr** = std::shared_ptr< render_target >

  Shared pointer to render target.
- using **swapchain_start_func** = std::function< bool()>

  Swapchain start function.
- using **swapchain_stop_func** = std::function< void()>

  Swapchain stop function.
- using **create_attachments_func** = std::function< VkAttachments()>

  Create attachments function.
- using **destroy_attachments_func** = std::function< void()>

  Destroy attachments function.

Public Member Functions

- bool create (device_p device, VkSurfaceKHR surface, VkSurfaceFormatKHR format, uv2 size, bool v_↩ sync=false)

  Create a new render target.
- void **destroy** ()

  Destroy the render target.
- uv2 get_size () const

  Get the size of the render target.
- bool resize (uv2 new_size)

  Resize the render target.
- ui32 get_frame_count () const

  Get the frame count.
- bool reload_request () const

  Check if render target requests a reload.
- void **reload** ()

  Reload the render target.
- device_p get_device ()

  Get the device.
- swapchain ∗ get_swapchain ()

  Get the swapchain.
- VkFormat get_format () const

  Get the format.
- image::list const & get_backbuffers () const

  Get the backbuffers.
- image::ptr get_backbuffer (index index)

  Get the backbuffer by frame index.
- VkImage get_backbuffer_image (index index)

  Get the backbuffer image by index.
- VkImage get_image (index index)
- void add_callback (target_callback::cptr callback)

  Add callback.
- void remove_callback (target_callback::cptr callback)

  Remove callback.

Static Public Member Functions

- static ptr make ()

  Make a new render target.

Public Attributes

- swapchain_start_func **on_swapchain_start**

  Called on swapchain start.
- swapchain_stop_func **on_swapchain_stop**

  Called on swapchain stop.
- create_attachments_func **on_create_attachments**

  Called on create attachments.
- destroy_attachments_func **on_destroy_attachments**

  Called on destroy attachments.

4.96.1 Detailed Description

Render target.

4.96.2 Member Function Documentation

4.96.2.1 add_callback()

```
void lava::render_target::add_callback (
            target_callback::cptr callback )  [inline]
```

Add callback.

Parameters

callback	Target callback

4.96.2.2 create()

```
bool lava::render_target::create (
            device_p device,
            VkSurfaceKHR surface,
            VkSurfaceFormatKHR format,
            uv2 size,
            bool v_sync = false )
```

Create a new render target.

Parameters

device	Vulkan device
surface	Vulkan surface
format	Surface format
size	Size of target
v_sync	V-Sync enabled

Returns

Create was successful or failed

4.96.2.3 get_backbuffer()

```
image::ptr lava::render_target::get_backbuffer (
            index index )  [inline]
```

Get the backbuffer by frame index.

Parameters

index	Frame index

Returns

image::ptr Backbuffer image

4.96.2.4 get_backbuffer_image()

```
VkImage lava::render_target::get_backbuffer_image (
            index index )  [inline]
```

Get the backbuffer image by index.

Parameters

index	Frame index

Returns

VkImage Vulkan image

4.96.2.5 get_backbuffers()

```
image::list const & lava::render_target::get_backbuffers ( ) const  [inline]
```

Get the backbuffers.

Returns

image::list const& List of backbuffer images

4.96.2.6 get_device()

```
device_p lava::render_target::get_device ( ) [inline]
```

Get the device.

Returns

device_p Vulkan device

4.96.2.7 get_format()

```
VkFormat lava::render_target::get_format ( ) const [inline]
```

Get the format.

Returns

VkFormat Target format

4.96.2.8 get_frame_count()

```
ui32 lava::render_target::get_frame_count ( ) const [inline]
```

Get the frame count.

Returns

ui32 Number of frames

4.96.2.9 get_image()

```
VkImage lava::render_target::get_image (
            index index ) [inline]
```

See also

get_backbuffer_image

4.96.2.10 get_size()

```
uv2 lava::render_target::get_size ( ) const  [inline]
```

Get the size of the render target.

Returns

uv2 Size of render target

4.96.2.11 get_swapchain()

```
swapchain * lava::render_target::get_swapchain ( )  [inline]
```

Get the swapchain.

Returns

swapchain* Target swapchain

4.96.2.12 make()

```
static ptr lava::render_target::make ( )  [inline], [static]
```

Make a new render target.

Returns

ptr Shared pointer to render target

4.96.2.13 reload_request()

```
bool lava::render_target::reload_request ( ) const  [inline]
```

Check if render target requests a reload.

Returns

Request reload or not

4.96.2.14 remove_callback()

```
void lava::render_target::remove_callback (
            target_callback::cptr callback )  [inline]
```

Remove callback.

Parameters

callback	Target callback

4.96.2.15 resize()

```
bool lava::render_target::resize (
            uv2 new_size ) [inline]
```

Resize the render target.

Parameters

new_size	New render target size

Returns

Resize was successful or failed

The documentation for this struct was generated from the following file:

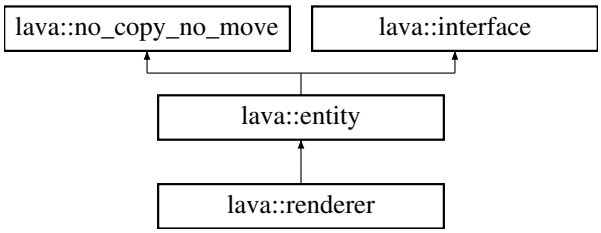
- liblava/frame/render_target.hpp

4.97 lava::renderer Struct Reference

Plain renderer.

```
#include <renderer.hpp>
```

Inheritance diagram for lava::renderer:



Public Types

- using **destroy_func** = std::function< void()>

  Destroy function.

Public Member Functions

- bool create (swapchain *target)

  Create a new renderer.
- void **destroy** ()

  Destroy the renderer.
- optional_index begin_frame ()

  Begin to render a frame.
- bool end_frame (VkCommandBuffers const &cmd_buffers)

  End of frame rendering.
- bool frame (VkCommandBuffers const &cmd_buffers)

  Render a frame.
- index get_frame () const

  Get the current frame index.
- device_p get_device ()

  Get the device.

Public Attributes

- destroy_func **on_destroy**

  Called on renderer destroy.
- bool **active** = true

  Active state.

4.97.1 Detailed Description

Plain renderer.

4.97.2 Member Function Documentation

4.97.2.1 begin_frame()

```
optional_index lava::renderer::begin_frame ( )
```

Begin to render a frame.

Returns

optional_index Frame index

4.97.2.2 create()

```
bool lava::renderer::create (
            swapchain * target )
```

Create a new renderer.

Parameters

target	Swapchain target

Returns

Create was successful or failed

4.97.2.3 end_frame()

```
bool lava::renderer::end_frame (
            VkCommandBuffers const & cmd_buffers )
```

End of frame rendering.

Parameters

cmd_buffers	List of command buffers

Returns

End was successful or failed

4.97.2.4 frame()

```
bool lava::renderer::frame (
            VkCommandBuffers const & cmd_buffers )  [inline]
```

Render a frame.

Parameters

cmd_buffers	List of command buffers

Returns

Render was successful or failed

4.97.2.5 get_device()

```
device_p lava::renderer::get_device ( )  [inline]
```

Get the device.

Returns

device_p Vulkan device

4.97.2.6 get_frame()

```
index lava::renderer::get_frame ( ) const [inline]
```

Get the current frame index.

Returns

index Frame index

The documentation for this struct was generated from the following file:

- liblava/frame/renderer.hpp

4.98 lava::driver::result Struct Reference

Driver result.

```
#include <driver.hpp>
```

Public Attributes

- i32 **driver** = 0

  Run result.
- i32 **selected** = 0

  Selected stage.

4.98.1 Detailed Description

Driver result.

The documentation for this struct was generated from the following file:

- liblava/frame/driver.hpp

4.99 lava::reversion_wrapper< T > Struct Template Reference

Reversion Wrapper.

```
#include <misc.hpp>
```

Public Attributes

- T & **iterable**

  Iterable to wrap.

4.99.1 Detailed Description

template<typename T>
struct lava::reversion_wrapper< T >

Reversion Wrapper.

Template Parameters

T	Type to iterate

The documentation for this struct was generated from the following file:

- liblava/core/misc.hpp

4.100 lava::run_time Struct Reference

Run time.

```
#include <time.hpp>
```

Public Attributes

- ms **current** { 0 }

  Current milliseconds.
- ms **clock** { 16 }

  Clock milliseconds.
- ms **system** { 0 }

  System milliseconds.
- ms **delta** { 0 }

  Delta milliseconds.
- ms **fix_delta** { 0 }

  Fix delta milliseconds (0 = deactivated)
- r32 **speed** = 1.f

  Speed factor.
- bool **paused** = false

  Paused run time.

4.100.1 Detailed Description

Run time.

The documentation for this struct was generated from the following file:

- liblava/core/time.hpp

4.101 lava::scoped_label< T > Struct Template Reference

Scoped debug util label.

```
#include <debug_utils.hpp>
```

Public Member Functions

- scoped_label (T scope, name label, v4 color=v4(0.f))

  Construct a new scoped label.
- ~**scoped_label** ()

  Destroy the scoped label.

4.101.1 Detailed Description

template<typename T>
struct lava::scoped_label< T >

Scoped debug util label.

Template Parameters

T	VkCommandBuffer or VkQueue

4.101.2 Constructor & Destructor Documentation

4.101.2.1 scoped_label()

```
template<typename T >
lava::scoped_label< T >::scoped_label (
            T scope,
            name label,
            v4 color = v4(0.f) )  [inline]
```

Construct a new scoped label.

Parameters

scope	Scoped label
label	Name of label
color	Color of label

The documentation for this struct was generated from the following file:

- liblava/base/debug_utils.hpp

4.102 lava::scroll_event Struct Reference

Scroll event.

```
#include <input.hpp>
```

Public Types

- using **ref** = scroll_event const &

  Reference to scroll event.
- using **func** = std::function< bool(ref)>

  Scroll event function.
- using **listeners** = std::map< id, func >

  List of scroll event listeners.
- using **list** = std::vector< scroll_event >

  List of scroll events.

Public Attributes

- id **sender**

  Sender id.
- scroll_offset **offset**

  Input scroll offset.

4.102.1 Detailed Description

Scroll event.

The documentation for this struct was generated from the following file:

- liblava/frame/input.hpp

4.103 lava::scroll_offset Struct Reference

Input scroll offset.

```
#include <input.hpp>
```

Public Attributes

- r64 **x** = 0.0

  X offset.
- r64 **y** = 0.0

  Y offset.

4.103.1 Detailed Description

Input scroll offset.

The documentation for this struct was generated from the following file:

- liblava/frame/input.hpp

4.104 lava::pipeline::shader_stage Struct Reference

Shader stage.

```
#include <pipeline.hpp>
```

Public Types

- using **ptr** = std::shared_ptr< shader_stage >

  Shared pointer to shader stage.
- using **list** = std::vector< ptr >

  List of shader stages.

Public Member Functions

- **shader_stage** ()

  Construct a new shader stage.
- ~**shader_stage** ()

  Destroy the shader stage.
- void set_stage (VkShaderStageFlagBits stage)

  Set the stage.
- void add_specialization_entry (VkSpecializationMapEntry const &specialization)

  Add specialization entry.
- bool create (device_p device, cdata::ref shader_data, cdata::ref specialization_data=data())

  Create a new shader stage.
- void **destroy** ()

  Destroy the shader stage.
- VkPipelineShaderStageCreateInfo const & get_create_info () const

  Get the create info.

Static Public Member Functions

- static ptr make (VkShaderStageFlagBits stage)

  Make a new pipline shader stage.

4.104.1 Detailed Description

Shader stage.

4.104.2 Member Function Documentation

4.104.2.1 add_specialization_entry()

```
void lava::pipeline::shader_stage::add_specialization_entry (
            VkSpecializationMapEntry const & specialization )
```

Add specialization entry.

Parameters

specialization	Specialization map entry

4.104.2.2 create()

```
bool lava::pipeline::shader_stage::create (
            device_p device,
            cdata::ref shader_data,
            cdata::ref specialization_data = data() )
```

Create a new shader stage.

Parameters

device	Vulkan device
shader_data	Shader data
specialization_data	Specialization data

Returns

Create was successful or failed

4.104.2.3 get_create_info()

```
VkPipelineShaderStageCreateInfo const & lava::pipeline::shader_stage::get_create_info ( )
const [inline]
```

Get the create info.

Returns

VkPipelineShaderStageCreateInfo const& Pipeline shader stage create information

4.104.2.4 make()

```
static ptr lava::pipeline::shader_stage::make (
            VkShaderStageFlagBits stage ) [inline], [static]
```

Make a new pipline shader stage.

Parameters

stage	Shader stage flag bits

Returns

ptr Shared pointer to shader stage

4.104.2.5 set_stage()

```
void lava::pipeline::shader_stage::set_stage (
            VkShaderStageFlagBits stage )  [inline]
```

Set the stage.

Parameters

stage	Shader stage flag bits

The documentation for this struct was generated from the following file:

- liblava/block/pipeline.hpp

4.105 lava::stage Struct Reference

Stage.

```
#include <driver.hpp>
```

Public Types

- using **map** = std::map< index, stage * >

  Map of stages.
- using **func** = std::function< i32(argh::parser)>

  Stage function.

Public Member Functions

- stage (ui32 id, string_ref name, func func)

  Construct a new stage.

Public Attributes

- index **id** = 0

  Stage id.
- string **name**

  Stage name.
- func **on_func**

  Called on stage run.

4.105.1 Detailed Description

Stage.

4.105.2 Constructor & Destructor Documentation

4.105.2.1 stage()

```
lava::stage::stage (
            ui32 id,
            string_ref name,
            func func )  [explicit]
```

Construct a new stage.

Parameters

id	Stage id
name	Stage name
func	Stage function

The documentation for this struct was generated from the following file:

- liblava/frame/driver.hpp

4.106 lava::staging Struct Reference

Texture staging.

```
#include <texture.hpp>
```

Public Member Functions

- void add (texture::ptr texture)

  Add texture for staging.
- bool stage (VkCommandBuffer cmd_buf, index frame)

  Stage textures.
- void **clear** ()

  Clear staging.
- bool busy () const

  Check if staging is busy.

4.106.1 Detailed Description

Texture staging.

4.106.2 Member Function Documentation

4.106.2.1 add()

```
void lava::staging::add (
            texture::ptr texture )  [inline]
```

Add texture for staging.

Parameters

texture	Texture to stage

4.106.2.2 busy()

```
bool lava::staging::busy ( ) const  [inline]
```

Check if staging is busy.

Returns

Staging is busy or not

4.106.2.3 stage()

```
bool lava::staging::stage (
            VkCommandBuffer cmd_buf,
            index frame )
```

Stage textures.

Parameters

cmd_buf	Command buffer
frame	Frame index

Returns

Stage was successful or failed

The documentation for this struct was generated from the following file:

- liblava/resource/texture.hpp

4.107 lava::window::state Struct Reference

Window state.

`#include <window.hpp>`

Public Types

- using **ref** = state const &

  Reference to window state.
- using **optional** = std::optional< window::state >

  Optional window state.

Public Member Functions

- **state** ()

  Construct a new state.

Public Attributes

- i32 **x** = 0

  Window X position.
- i32 **y** = 0

  Window Y position.
- ui32 **width** = 0

  Window width.
- ui32 **height** = 0

  Window height.
- bool **fullscreen** = false

  Fullscreen window.
- bool **floating** = false

  Floating window.
- bool **resizable** = true

  Resizable window.
- bool **decorated** = true

  Decorated window.
- bool **maximized** = false

  Maximized window.
- index **monitor** = 0

  Monitor of window.

4.107.1 Detailed Description

Window state.

The documentation for this struct was generated from the following file:

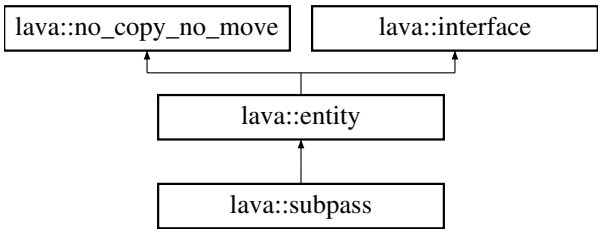
- liblava/frame/window.hpp

4.108 lava::subpass Struct Reference

Subpass.

```
#include <subpass.hpp>
```

Inheritance diagram for lava::subpass:



Public Types

- using **ptr** = std::shared_ptr< subpass >

  Shared pointer to subpass.
- using **list** = std::vector< ptr >

  List of subpasses.

Public Member Functions

- **subpass** ()

  Construct a new subpass.
- void **destroy** ()

  Destroy the subpass.
- void add (render_pipeline::ptr const &pipeline)

  Add a render pipeline to the back of the subpass.
- void add_front (render_pipeline::ptr const &pipeline)

  Add a render pipeline to the fronst of the subpass.
- void remove (render_pipeline::ptr pipeline)

  Remove the render pipeline.
- void **clear_pipelines** ()

  Clear all pipelines.
- void process (VkCommandBuffer cmd_buf, uv2 size)

  Process the subpass.
- VkSubpassDescription const & get_description () const

Get the description.

- void set (VkPipelineBindPoint pipeline_bind_point)

  Set pipeline bind point.
- void set_color_attachment (index attachment, VkImageLayout layout)

  Set the color attachment.
- void set_color_attachment (VkAttachmentReference attachment)

  Set the color attachment.
- void set_color_attachments (VkAttachmentReferences const &attachments)

  Set the color attachments.
- void set_depth_stencil_attachment (index attachment, VkImageLayout layout)

  Set the depth stencil attachment.
- void set_depth_stencil_attachment (VkAttachmentReference attachment)

  Set the depth stencil attachment.
- void set_input_attachment (index attachment, VkImageLayout layout)

  Set the input attachment.
- void set_input_attachment (VkAttachmentReference attachment)

  Set the input attachment.
- void set_input_attachments (VkAttachmentReferences const &attachments)

  Set the input attachments.
- void set_resolve_attachment (index attachment, VkImageLayout layout)

  Set the resolve attachment.
- void set_resolve_attachment (VkAttachmentReference attachment)

  Set the resolve attachment.
- void set_resolve_attachments (VkAttachmentReferences const &attachments)

  Set the resolve attachments.
- void add_preserve_attachment (index attachment)

  Add preserve attachment.
- void set_preserve_attachments (index_list const &attachments)

  Set the preserve attachments.
- void set_active (bool value=true)

  Activate or deactivate the subpass.
- bool activated () const

  Check if subpass is active.

Static Public Member Functions

- static ptr make (VkPipelineBindPoint pipeline_bind_point=VK_PIPELINE_BIND_POINT_GRAPHICS)

  Make a new subpass.

4.108.1 Detailed Description

Subpass.

4.108.2 Member Function Documentation

4.108.2.1 activated()

```
bool lava::subpass::activated ( ) const  [inline]
```

Check if subpass is active.

Returns

Subpass is active or not

4.108.2.2 add()

```
void lava::subpass::add (
            render_pipeline::ptr const & pipeline )  [inline]
```

Add a render pipeline to the back of the subpass.

Parameters

pipeline	Render pipeline

4.108.2.3 add_front()

```
void lava::subpass::add_front (
            render_pipeline::ptr const & pipeline )  [inline]
```

Add a render pipeline to the fronst of the subpass.

Parameters

pipeline	Render pipeline

4.108.2.4 add_preserve_attachment()

```
void lava::subpass::add_preserve_attachment (
            index attachment )
```

Add preserve attachment.

Parameters

attachment	Index of attachment

4.108.2.5 get_description()

```
VkSubpassDescription const & lava::subpass::get_description ( ) const  [inline]
```

Get the description.

Returns

VkSubpassDescription const& Subpass description

4.108.2.6 make()

```
static ptr lava::subpass::make (
            VkPipelineBindPoint pipeline_bind_point = VK_PIPELINE_BIND_POINT_GRAPHICS )  [inline],
[static]
```

Make a new subpass.

Parameters

pipeline_bind_point	Pipeline bind point

Returns

ptr Shared pointer to subpass

4.108.2.7 process()

```
void lava::subpass::process (
            VkCommandBuffer cmd_buf,
            uv2 size )
```

Process the subpass.

Parameters

cmd_buf	Command buffer
size	Size of render pass

4.108.2.8 remove()

```
void lava::subpass::remove (
            render_pipeline::ptr pipeline )
```

Remove the render pipeline.

Parameters

pipeline	Render pipeline

4.108.2.9 set()

```
void lava::subpass::set (
            VkPipelineBindPoint pipeline_bind_point )  [inline]
```

Set pipeline bind point.

Parameters

pipeline_bind_point	Pipeline bind point

4.108.2.10 set_active()

```
void lava::subpass::set_active (
            bool value = true )  [inline]
```

Activate or deactivate the subpass.

Parameters

value	Enable state

4.108.2.11 set_color_attachment() [1/2]

```
void lava::subpass::set_color_attachment (
            index attachment,
            VkImageLayout layout )
```

Set the color attachment.

Parameters

attachment	Index of attachment
layout	Image layout

4.108.2.12 set_color_attachment() [2/2]

```
void lava::subpass::set_color_attachment (
            VkAttachmentReference attachment )
```

Set the color attachment.

Parameters

attachment	Attachment reference

4.108.2.13 set_color_attachments()

```
void lava::subpass::set_color_attachments (
            VkAttachmentReferences const & attachments )
```

Set the color attachments.

Parameters

attachments	List of attachment references

4.108.2.14 set_depth_stencil_attachment() [1/2]

```
void lava::subpass::set_depth_stencil_attachment (
            index attachment,
            VkImageLayout layout )
```

Set the depth stencil attachment.

Parameters

attachment	Index of attachment
layout	Image layout

4.108.2.15 set_depth_stencil_attachment() [2/2]

```
void lava::subpass::set_depth_stencil_attachment (
            VkAttachmentReference attachment )
```

Set the depth stencil attachment.

Parameters

attachment	Attachment reference

4.108.2.16 set_input_attachment() [1/2]

```
void lava::subpass::set_input_attachment (
            index attachment,
            VkImageLayout layout )
```

Set the input attachment.

Parameters

attachment	Index of attachment
layout	Image layout

4.108.2.17 set_input_attachment() [2/2]

```
void lava::subpass::set_input_attachment (
            VkAttachmentReference attachment )
```

Set the input attachment.

Parameters

attachment	Attachment reference

4.108.2.18 set_input_attachments()

```
void lava::subpass::set_input_attachments (
            VkAttachmentReferences const & attachments )
```

Set the input attachments.

Parameters

attachments	List of attachment references

4.108.2.19 set_preserve_attachments()

```
void lava::subpass::set_preserve_attachments (
            index_list const & attachments )
```

Set the preserve attachments.

Parameters

attachments	List of indices

4.108.2.20 set_resolve_attachment() [1/2]

```
void lava::subpass::set_resolve_attachment (
            index attachment,
            VkImageLayout layout )
```

Set the resolve attachment.

Parameters

attachment	Index of attachment
layout	Image layout

4.108.2.21 set_resolve_attachment() [2/2]

```
void lava::subpass::set_resolve_attachment (
            VkAttachmentReference attachment )
```

Set the resolve attachment.

Parameters

attachment	Attachment reference

4.108.2.22 set_resolve_attachments()

```
void lava::subpass::set_resolve_attachments (
            VkAttachmentReferences const & attachments )
```

Set the resolve attachments.

Parameters

attachments	List of attachment references

The documentation for this struct was generated from the following file:

- liblava/block/subpass.hpp

4.109 lava::subpass_dependency Struct Reference

Subpass dependency.

```
#include <subpass.hpp>
```

Public Types

- using **ptr** = std::shared_ptr< subpass_dependency >

  Shared pointer to subpass dependency.
- using **list** = std::vector< ptr >

  List of subpass dependencies.

Public Member Functions

- **subpass_dependency** ()

  Construct a new subpass dependency.
- VkSubpassDependency const & get_dependency () const

  Get the dependency.
- void set_subpass (ui32 src, ui32 dst)

  Set the subpass.
- void set_src_subpass (ui32 src)

  Set the source subpass.
- void set_dst_subpass (ui32 dst)

  Set the dst subpass.
- void set_stage_mask (VkPipelineStageFlags src, VkPipelineStageFlags dst)

  Set the stage mask.
- void set_src_stage_mask (VkPipelineStageFlags mask)

  Set the source stage mask.
- void set_dst_stage_mask (VkPipelineStageFlags mask)

  Set the destination stage mask.
- void set_access_mask (VkAccessFlags src, VkAccessFlags dst)

  Set the access mask.
- void set_src_access_mask (VkAccessFlags mask)

  Set the src access mask.
- void set_dst_access_mask (VkAccessFlags mask)

  Set the dst access mask.
- void set_dependency_flags (VkDependencyFlags flags)

  Set the dependency flags.

Static Public Member Functions

- static ptr make (ui32 src_subpass, ui32 dst_subpass, VkDependencyFlags dependency_flags=VK_↩ DEPENDENCY_BY_REGION_BIT)

  Make a new subpass dependency.

4.109.1 Detailed Description

Subpass dependency.

4.109.2 Member Function Documentation

4.109.2.1 get_dependency()

```
VkSubpassDependency const & lava::subpass_dependency::get_dependency ( ) const  [inline]
```

Get the dependency.

Returns

VkSubpassDependency const& Vulkan subpass dependency

4.109.2.2 make()

```
static ptr lava::subpass_dependency::make (
            ui32 src_subpass,
            ui32 dst_subpass,
            VkDependencyFlags dependency_flags = VK_DEPENDENCY_BY_REGION_BIT )  [inline],
[static]
```

Make a new subpass dependency.

Parameters

src_subpass	Source subpass
dst_subpass	Destination subpass
dependency_flags	Dependency flags

Returns

ptr Shared pointer to subpass dependency

4.109.2.3 set_access_mask()

```
void lava::subpass_dependency::set_access_mask (
            VkAccessFlags src,
            VkAccessFlags dst )  [inline]
```

Set the access mask.

Parameters

src	Source access flags
dst	Destination access flags

4.109.2.4 set_dependency_flags()

```
void lava::subpass_dependency::set_dependency_flags (
            VkDependencyFlags flags )  [inline]
```

Set the dependency flags.

Parameters

flags	Dependency flags

4.109.2.5 set_dst_access_mask()

```
void lava::subpass_dependency::set_dst_access_mask (
            VkAccessFlags mask )  [inline]
```

Set the dst access mask.

Parameters

mask	Access flags

4.109.2.6 set_dst_stage_mask()

```
void lava::subpass_dependency::set_dst_stage_mask (
            VkPipelineStageFlags mask )  [inline]
```

Set the destination stage mask.

Parameters

mask	Pipeline stage flags

4.109.2.7 set_dst_subpass()

```
void lava::subpass_dependency::set_dst_subpass (
            ui32 dst )  [inline]
```

Set the dst subpass.

Parameters

dst	Destination subpass

4.109.2.8 set_src_access_mask()

```
void lava::subpass_dependency::set_src_access_mask (
            VkAccessFlags mask )  [inline]
```

Set the src access mask.

Parameters

mask	Access flags

4.109.2.9 set_src_stage_mask()

```
void lava::subpass_dependency::set_src_stage_mask (
            VkPipelineStageFlags mask )  [inline]
```

Set the source stage mask.

Parameters

mask	Pipeline stage flags

4.109.2.10 set_src_subpass()

```
void lava::subpass_dependency::set_src_subpass (
            ui32 src )  [inline]
```

Set the source subpass.

Parameters

src	Source Subpass

4.109.2.11 set_stage_mask()

```
void lava::subpass_dependency::set_stage_mask (
            VkPipelineStageFlags src,
            VkPipelineStageFlags dst )  [inline]
```

Set the stage mask.

Parameters

src	Source pipeline stage flags
dst	Destination pipeline stage flags

4.109.2.12 set_subpass()

```
void lava::subpass_dependency::set_subpass (
            ui32 src,
            ui32 dst )  [inline]
```

Set the subpass.

Parameters

src	Source Subpass
dst	Destination Subpass

The documentation for this struct was generated from the following file:

- liblava/block/subpass.hpp

4.110 lava::surface_format_request Struct Reference

Surface format request.

```
#include <format.hpp>
```

Public Attributes

- VkFormats formats

  List of formats in request order.
- VkColorSpaceKHR **color_space** = VK_COLOR_SPACE_SRGB_NONLINEAR_KHR

  Color space to request.

4.110.1 Detailed Description

Surface format request.

4.110.2 Member Data Documentation

4.110.2.1 formats

```
VkFormats lava::surface_format_request::formats
```

Initial value:

```
= {
        VK_FORMAT_B8G8R8A8_UNORM,
        VK_FORMAT_R8G8B8A8_UNORM,
        VK_FORMAT_B8G8R8_UNORM,
        VK_FORMAT_R8G8B8_UNORM,
        VK_FORMAT_B8G8R8A8_SRGB,
        VK_FORMAT_R8G8B8A8_SRGB,
        VK_FORMAT_B8G8R8_SRGB,
        VK_FORMAT_R8G8B8_SRGB,
    }
```

List of formats in request order.

The documentation for this struct was generated from the following file:

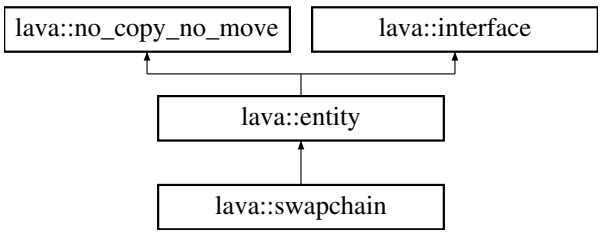
- liblava/resource/format.hpp

4.111 lava::swapchain Struct Reference

Swaphchain.

```
#include <swapchain.hpp>
```

Inheritance diagram for lava::swapchain:



Classes

- struct callback

  Swapchain callback.

Public Member Functions

- bool create (device_p device, VkSurfaceKHR surface, VkSurfaceFormatKHR format, uv2 size, bool v_sync=false)

  Create a new swapchain.
- void **destroy** ()

  Destroy the swapchain.
- bool resize (uv2 new_size)

  Resize the swapchain.
- void **request_reload** ()

  Request a reload of the swapchain.
- bool reload_request () const

  Check if reload of the swapchain is requested.
- device_p get_device ()

  Get the device.
- uv2 get_size () const

  Get the size of the swapchain.
- VkFormat get_format () const

  Get the format of the swapchain.
- VkColorSpaceKHR get_color_space () const

  Get the color space of the swapchain.
- VkSwapchainKHR get () const

  Get the swapchain.
- ui32 get_backbuffer_count () const

  Get the backbuffer count.
- image::list const & get_backbuffers () const

  Get the backbuffers.
- void add_callback (callback *cb)

  Add callback to swapchain.
- void remove_callback (callback *cb)

  Remove callback from swapchain.
- bool v_sync () const

  Check if V-Sync is enabled.
- bool surface_supported (index queue_family) const

  Check if surface is supported by queue family index.

4.111.1 Detailed Description

Swaphchain.

4.111.2 Member Function Documentation

4.111.2.1 add_callback()

```
void lava::swapchain::add_callback (
            callback * cb )
```

Add callback to swapchain.

Parameters

cb	Callback to add

4.111.2.2 create()

```
bool lava::swapchain::create (
            device_p device,
            VkSurfaceKHR surface,
            VkSurfaceFormatKHR format,
            uv2 size,
            bool v_sync = false )
```

Create a new swapchain.

Parameters

device	Vulkan device
surface	Vulkan surface
format	Surface format
size	Size of swapchain
v_sync	V-Sync enabled

Returns

Create was successful or failed

4.111.2.3 get()

```
VkSwapchainKHR lava::swapchain::get ( ) const  [inline]
```

Get the swapchain.

Returns

VkSwapchainKHR Vulkan swapchain

4.111.2.4 get_backbuffer_count()

```
ui32 lava::swapchain::get_backbuffer_count ( ) const  [inline]
```

Get the backbuffer count.

Returns

ui32 Number of backbuffers

4.111.2.5 get_backbuffers()

```
image::list const & lava::swapchain::get_backbuffers ( ) const  [inline]
```

Get the backbuffers.

Returns

image::list const& List of backbuffer images

4.111.2.6 get_color_space()

```
VkColorSpaceKHR lava::swapchain::get_color_space ( ) const  [inline]
```

Get the color space of the swapchain.

Returns

VkColorSpaceKHR Swapchain color space

4.111.2.7 get_device()

```
device_p lava::swapchain::get_device ( )  [inline]
```

Get the device.

Returns

device_p Vulkan device

4.111.2.8 get_format()

```
VkFormat lava::swapchain::get_format ( ) const  [inline]
```

Get the format of the swapchain.

Returns

VkFormat Swapchain format

4.111.2.9 get_size()

```
uv2 lava::swapchain::get_size ( ) const  [inline]
```

Get the size of the swapchain.

Returns

uv2 Swapchain size

4.111.2.10 reload_request()

```
bool lava::swapchain::reload_request ( ) const  [inline]
```

Check if reload of the swapchain is requested.

Returns

Reload is requested or not

4.111.2.11 remove_callback()

```
void lava::swapchain::remove_callback (
            callback * cb )
```

Remove callback from swapchain.

Parameters

cb	Callback to remove

4.111.2.12 resize()

```
bool lava::swapchain::resize (
            uv2 new_size )
```

Resize the swapchain.

Parameters

new_size	New size of swapchain

Returns

Resize was successful or failed

4.111.2.13 surface_supported()

```
bool lava::swapchain::surface_supported (
            index queue_family ) const
```

Check if surface is supported by queue family index.

Parameters

queue_family	Queue family index

Returns

Surface is supported by queue family or not

4.111.2.14 v_sync()

```
bool lava::swapchain::v_sync ( ) const  [inline]
```

Check if V-Sync is enabled.

Returns

V-Sync is active or not

The documentation for this struct was generated from the following file:

- liblava/frame/swapchain.hpp

4.112 lava::target_callback Struct Reference

Target callback.

```
#include <base.hpp>
```

Public Types

- using **cptr** = target_callback const ∗

  Const pointer to target callback.
- using **list** = std::vector< target_callback ∗ >

  List of target callbacks.
- using **clist** = std::vector< cptr >

  Const list of target callbacks.
- using **created_func** = std::function< bool(VkAttachmentsRef, rect::ref)>

  Created function.
- using **destroyed_func** = std::function< void()>

  Destroy function.

Public Attributes

- created_func **on_created**

  Called on target created.
- destroyed_func **on_destroyed**

  Called on target destroyed.

4.112.1 Detailed Description

Target callback.

The documentation for this struct was generated from the following file:

- liblava/base/base.hpp

4.113 lava::telegram Struct Reference

Telegram.

```
#include <telegram.hpp>
```

Public Types

- using **ref** = telegram const &

  Reference to telegram.

Public Member Functions

- telegram (id::ref sender, id::ref receiver, id::ref msg, ms dispatch_time={}, any info={})

  Construct a new telegram.
- bool operator== (ref rhs) const

  Equal operator.
- bool operator< (ref rhs) const

  Time order operator.

Public Attributes

- id **sender**

  Sender id.
- id **receiver**

  Receiver id.
- id **msg**

  Telegram message.
- ms **dispatch_time**

  Dispatch time.
- any **info**

  Telegram information.

4.113.1 Detailed Description

Telegram.

4.113.2 Constructor & Destructor Documentation

4.113.2.1 telegram()

```
lava::telegram::telegram (
            id::ref sender,
            id::ref receiver,
            id::ref msg,
            ms dispatch_time = {},
            any info = {} )  [inline], [explicit]
```

Construct a new telegram.

Parameters

sender	Sender id
receiver	Receiver id
msg	Telegram message
dispatch_time	Dispatch time
info	Telegram information

4.113.3 Member Function Documentation

4.113.3.1 operator<()

```
bool lava::telegram::operator< (
            ref rhs ) const  [inline]
```

Time order operator.

Parameters

rhs	Another telegram

Returns

Telegram is earlier or later

4.113.3.2 operator==()

```
bool lava::telegram::operator== (
            ref rhs ) const  [inline]
```

Equal operator.

Parameters

rhs	Another telegram

Returns

Telegram is equal or not

The documentation for this struct was generated from the following file:

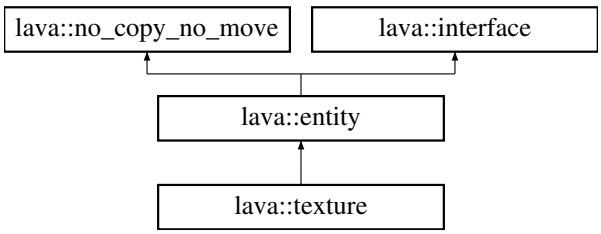
- liblava/util/telegram.hpp

4.114 lava::texture Struct Reference

Texture.

```
#include <texture.hpp>
```

Inheritance diagram for lava::texture:



Classes

- struct layer

  Texture layer.
- struct mip_level

  Texture mip level.

Public Types

- using **ptr** = std::shared_ptr< texture >

  Shared pointer to texture.
- using **map** = std::map< id, ptr >

  Map of textures.
- using **list** = std::vector< ptr >

  List of textures.

Public Member Functions

- ~**texture** ()

  Destroy the texture.
- bool create (device_p device, uv2 size, VkFormat format, layer::list const &layers={}, texture_type type=texture_type::tex_2d)

  Create a new texture.
- void **destroy** ()

  Destroy the texture.
- bool upload (void const *data, size_t data_size)

  Upload data to texture.
- bool stage (VkCommandBuffer cmd_buffer)

  Stage the texture.
- void **destroy_upload_buffer** ()

  Destroy the upload buffer.
- VkDescriptorImageInfo const * get_descriptor_info () const

  Get the descriptor information.
- image::ptr get_image ()

  Get the image of the texture.
- uv2 get_size () const

  Get the size of the texture.
- texture_type get_type () const

  Get the type of the texture.
- VkFormat get_format () const

  Get the format of the texture.

Static Public Member Functions

- static ptr make ()

  Make a new texture.

4.114.1 Detailed Description

Texture.

4.114.2 Member Function Documentation

4.114.2.1 create()

```
bool lava::texture::create (
            device_p device,
            uv2 size,
            VkFormat format,
            layer::list const & layers = {},
            texture_type type = texture_type::tex_2d )
```

Create a new texture.

Parameters

device	Vulkan device
size	Texture size
format	Texture format
layers	List of layers
type	Texture type

Returns

Create was successful or failed

4.114.2.2 get_descriptor_info()

```
VkDescriptorImageInfo const * lava::texture::get_descriptor_info ( ) const  [inline]
```

Get the descriptor information.

Returns

VkDescriptorImageInfo const∗ Descriptor image information

4.114.2.3 get_format()

```
VkFormat lava::texture::get_format ( ) const  [inline]
```

Get the format of the texture.

Returns

VkFormat Texture format

4.114.2.4 get_image()

```
image::ptr lava::texture::get_image ( )  [inline]
```

Get the image of the texture.

Returns

image::ptr Shared pointer to image

4.114.2.5 get_size()

```
uv2 lava::texture::get_size ( ) const  [inline]
```

Get the size of the texture.

Returns

uv2 Texture size

4.114.2.6 get_type()

```
texture_type lava::texture::get_type ( ) const  [inline]
```

Get the type of the texture.

Returns

texture_type Texture type

4.114.2.7 make()

```
static ptr lava::texture::make ( )  [inline], [static]
```

Make a new texture.

Returns

ptr Shared pointer to texture

4.114.2.8 stage()

```
bool lava::texture::stage (
            VkCommandBuffer cmd_buffer )
```

Stage the texture.

Parameters

cmd_buffer	Command buffer

Returns

Stage was successful or failed

4.114.2.9 upload()

```
bool lava::texture::upload (
            void const * data,
            size_t data_size )
```

Upload data to texture.

Parameters

data	Data to upload
data_size	Size of data

Returns

Upload was successful or failed

The documentation for this struct was generated from the following file:

- liblava/resource/texture.hpp

4.115 lava::thread_pool Struct Reference

Thread pool.

```
#include <thread.hpp>
```

Public Types

- using **task** = std::function< void(id::ref)>

  Task function (with thread id)

Public Member Functions

- void setup (ui32 count=2)

  Set up the thread pool.
- void **teardown** ()

  Tear down the thread pool.
- void enqueue (auto f)

  Enqueue a task.

4.115.1 Detailed Description

Thread pool.

4.115.2 Member Function Documentation

4.115.2.1 enqueue()

```
void lava::thread_pool::enqueue (
            auto f )  [inline]
```

Enqueue a task.

Parameters

f	Task function

4.115.2.2 setup()

```
void lava::thread_pool::setup (
            ui32 count = 2 )  [inline]
```

Set up the thread pool.

Parameters

count	Number of threads

The documentation for this struct was generated from the following file:

- liblava/util/thread.hpp

4.116 lava::timer Struct Reference

Timer.

```
#include <time.hpp>
```

Public Member Functions

- **timer** ()

  Construct a new timer.
- void **reset** ()

  Reset the timer.
- ms elapsed () const

  Get the elapsed time.

4.116.1 Detailed Description

Timer.

4.116.2 Member Function Documentation

4.116.2.1 elapsed()

```
ms lava::timer::elapsed ( ) const  [inline]
```

Get the elapsed time.

Returns

ms Elapsed milliseconds

The documentation for this struct was generated from the following file:

- liblava/core/time.hpp

4.117 lava::tooltip Struct Reference

Tooltip.

```
#include <input.hpp>
```

Public Types

- using **list** = std::vector< tooltip >

  List of tooltips.

Public Member Functions

- tooltip (string_ref name, key_t key, mod_t mod)

  Construct a new tooltip.

Public Attributes

- string **name**

  Name of tooltip.
- key_t **key**

  Input key.
- mod_t **mod**

  Input mod.

4.117.1 Detailed Description

Tooltip.

4.117.2 Constructor & Destructor Documentation

4.117.2.1 tooltip()

```
lava::tooltip::tooltip (
            string_ref name,
            key_t key,
            mod_t mod )  [inline]
```

Construct a new tooltip.

Parameters

name	Name of tooltip
key	Input key
mod	Input mod

The documentation for this struct was generated from the following file:

- liblava/frame/input.hpp

4.118 lava::tooltip_list Struct Reference

Tooltip list.

```
#include <input.hpp>
```

Public Member Functions

- void add (string_ref name, key_t key, mod_t mod=mod::none)

  Add a tooltip.
- void **clear** ()

  Clear tooltips.
- tooltip::list const & get_list () const

  Get tooltips.
- void set (tooltip::list const &list)

  Set a new tooltip list.
- string format_string () const

  Convert tooltips to string.

4.118.1 Detailed Description

Tooltip list.

4.118.2 Member Function Documentation

4.118.2.1 add()

```
void lava::tooltip_list::add (
            string_ref name,
            key_t key,
            mod_t mod = mod::none )  [inline]
```

Add a tooltip.

Parameters

name	Name of tooltip
key	Input key
mod	Input mod (default: none)

4.118.2.2 format_string()

```
string lava::tooltip_list::format_string ( ) const
```

Convert tooltips to string.

Returns

string String representation

4.118.2.3 get_list()

```
tooltip::list const & lava::tooltip_list::get_list ( ) const  [inline]
```

Get tooltips.

Returns

tooltip::list List of tooltips

4.118.2.4 set()

```
void lava::tooltip_list::set (
            tooltip::list const & list )  [inline]
```

Set a new tooltip list.

Parameters

list	List of tooltips

The documentation for this struct was generated from the following file:

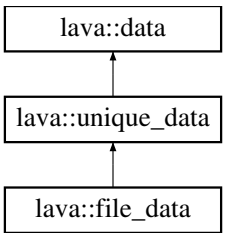
- liblava/frame/input.hpp

4.119 lava::unique_data Struct Reference

Unique data wrapper.

```
#include <data.hpp>
```

Inheritance diagram for lava::unique_data:



Public Types

- using **ref** = unique_data const &

  Reference to unique data wrapper.

Public Member Functions

- unique_data (size_t length=0, data_mode mode=data_mode::alloc)

  Construct a new unique data.
- unique_data (data::ref data)

  Construct a new unique data from another data.
- ~**unique_data** ()

  Destroy the unique data.

Additional Inherited Members

4.119.1 Detailed Description

Unique data wrapper.

4.119.2 Constructor & Destructor Documentation

4.119.2.1 unique_data() [1/2]

```
lava::unique_data::unique_data (
            size_t length = 0,
            data_mode mode = data_mode::alloc )  [inline]
```

Construct a new unique data.

Parameters

length	Length of data
mode	Data mode

4.119.2.2 unique_data() [2/2]

```
lava::unique_data::unique_data (
            data::ref data )  [inline], [explicit]
```

Construct a new unique data from another data.

Parameters

data	Source data

The documentation for this struct was generated from the following file:

- liblava/core/data.hpp

4.120 lava::version Struct Reference

Version.

```
#include <version.hpp>
```

Public Attributes

- i32 **year** = 2022

  Version year.
- i32 **release** = 0

  Version release.
- version_stage **stage** = version_stage::release

  Version stage.
- i32 **rev** = 0

  Version revision.

4.120.1 Detailed Description

Version.

The documentation for this struct was generated from the following file:

- liblava/core/version.hpp

4.121 lava::vertex Struct Reference

Vertex.

```
#include <primitive.hpp>
```

Public Types

- using **list** = std::vector< vertex >

  List of vertices.

Public Member Functions

- bool operator== (vertex const &other) const

  Equal compare operator.

Public Attributes

- v3 **position**

  Vertex position.
- v4 **color**

  Vertex color.
- v2 **uv**

  Vertex uv.
- v3 **normal**

  Vertex normal.

4.121.1 Detailed Description

Vertex.

4.121.2 Member Function Documentation

4.121.2.1 operator==()

```
bool lava::vertex::operator== (
            vertex const & other ) const  [inline]
```

Equal compare operator.

Parameters

other	Another vertex

Returns

Another vertex is equal or not

The documentation for this struct was generated from the following file:

- liblava/resource/primitive.hpp

4.122 lava::vk_result Struct Reference

Vulkan result.

```
#include <base.hpp>
```

Public Member Functions

- operator bool ()

  Check result.

Public Attributes

- bool **state** = false

  State of result.
- VkResult **value** = VK_NOT_READY

  Value of result.

4.122.1 Detailed Description

Vulkan result.

4.122.2 Member Function Documentation

4.122.2.1 operator bool()

```
lava::vk_result::operator bool ( )  [inline]
```

Check result.

Returns

Okay or error

The documentation for this struct was generated from the following file:

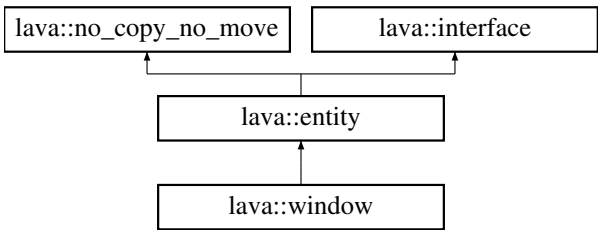
- liblava/base/base.hpp

4.123 lava::window Struct Reference

Window.

```
#include <window.hpp>
```

Inheritance diagram for lava::window:



Classes

- struct state

  Window state.

Public Types

- using **ptr** = std::shared_ptr< window >

  Shared pointer to window.
- using **event** = std::function< void(ptr)>

  Window event function.
- using **map** = std::map< id, ptr >

  Map of windows.
- using **ref** = window const &

  Reference to window.
- using **resize_func** = std::function< bool(ui32, ui32)>

  Resize window function.

Public Member Functions

- **window** ()=default

  Construct a new window.
- window (name title)

  Construct a new window.
- bool create (state::optional state={})

  Create a new window with optional state.
- void **destroy** ()

  Destroy the window.
- state get_state () const

  Get the window state.
- void set_title (string_ref text)

  Set the window title.
- string_ref get_title () const

  Get the window title.
- void set_save_name (string_ref save)

  Set the save name.
- string_ref get_save_name () const

  Get the save name.
- void set_position (i32 x, i32 y)

  Set the position of window.
- void get_position (i32 &x, i32 &y) const

  Get the position of window.
- void set_size (ui32 width, ui32 height)

  Set the size of window.
- void get_size (ui32 &width, ui32 &height) const

  Get the size of window.
- void get_framebuffer_size (ui32 &width, ui32 &height) const

  Get the framebuffer size.
- uv2 get_size () const

Get the size.

- uv2 get_framebuffer_size () const

  Get the framebuffer size.
- void set_mouse_position (r64 x, r64 y)

  Set the mouse position.
- void get_mouse_position (r64 &x, r64 &y) const

  Get the mouse position.
- mouse_position get_mouse_position () const

  Get the mouse position in window.
- void **hide_mouse_cursor** ()

  Hide mouse cursor.
- void **show_mouse_cursor** ()

  Show mouse cursor.
- float get_aspect_ratio () const

  Get the aspect ratio of window.
- void **show** ()

  Show the window.
- void **hide** ()

  Hide the window.
- bool visible () const

  Check if window is visible.
- void **iconify** ()

  Iconify the window.
- bool iconified () const

  Check if the window is iconified.
- void **restore** ()

  Restore the window.
- void **maximize** ()

  Maximize the window.
- bool maximized () const

  Check if the window is maximized.
- void **focus** ()

  Focus the window.
- bool focused () const

  Check if the window is focused.
- void set_fullscreen (bool active)

  Set the window to fullscreen.
- bool fullscreen () const

  Check if the window is fullscreen.
- bool hovered () const

  Check if mouse hovered over the window.
- bool resizable () const

  Check if the window is resizable.
- void set_resizable (bool value)

  Set the window resizable.
- bool decorated () const

  Check if the window is decorated.
- void set_decorated (bool value)

  Set the window decorated.
- bool floating () const

  Check if the window is floating.

- void set_floating (bool value)

  Set the window floating.
- bool close_request () const

  Check if the window request to close.
- bool switch_mode_request () const

  Check if the window request to switch mode.
- bool switch_mode (state::optional state={})

  Switch mode of the window.
- GLFWwindow * get () const

  Get GLFW handle.
- bool resize_request () const

  Check if the window request to resize.
- bool handle_resize ()

  Handle window resize.
- void **update_state** ()

  Update window state.
- void assign (input *callback)

  Assign input callback.
- void show_save_title (bool value=true)

  Show the save title in the window.
- bool save_title () const

  Check the show save title state.
- void **update_title** ()

  Update the window title.
- VkSurfaceKHR create_surface ()

  Create a surface.
- void set_icon (data_cptr data, uv2 size)

  Set the window icon.
- index get_monitor () const

  Get the monitor index of the window.
- void **center** ()

  Center the window on the monitor.

Public Attributes

- resize_func **on_resize**

  Called on window resize.

4.123.1 Detailed Description

Window.

4.123.2 Constructor & Destructor Documentation

4.123.2.1 window()

```
lava::window::window (
            name title )  [inline], [explicit]
```

Construct a new window.

Parameters

title	Title of window

4.123.3 Member Function Documentation

4.123.3.1 assign()

```
void lava::window::assign (
            input * callback )  [inline]
```

Assign input callback.

Parameters

callback	Input callbacl

4.123.3.2 close_request()

```
bool lava::window::close_request ( ) const
```

Check if the window request to close.

Returns

Window has close request or not

4.123.3.3 create()

```
bool lava::window::create (
            state::optional state = {} )
```

Create a new window with optional state.

Parameters

state	Window state

Returns

Create was successful or failed

4.123.3.4 create_surface()

```
VkSurfaceKHR lava::window::create_surface ( )
```

Create a surface.

Returns

VkSurfaceKHR Vulkan surface

4.123.3.5 decorated()

```
bool lava::window::decorated ( ) const
```

Check if the window is decorated.

Returns

Window is decorated or not

4.123.3.6 floating()

```
bool lava::window::floating ( ) const
```

Check if the window is floating.

Returns

Window is floating or not

4.123.3.7 focused()

```
bool lava::window::focused ( ) const
```

Check if the window is focused.

Returns

Window is focused or not

4.123.3.8 fullscreen()

```
bool lava::window::fullscreen ( ) const  [inline]
```

Check if the window is fullscreen.

Returns

Window is fullscreen or not

4.123.3.9 get()

```
GLFWwindow * lava::window::get ( ) const  [inline]
```

Get GLFW handle.

Returns

GLFWwindow* GLFW window handle

4.123.3.10 get_aspect_ratio()

```
float lava::window::get_aspect_ratio ( ) const
```

Get the aspect ratio of window.

Returns

float Aspect ratio

4.123.3.11 get_framebuffer_size() [1/2]

```
uv2 lava::window::get_framebuffer_size ( ) const
```

Get the framebuffer size.

Returns

uv2 Size of framebuffer

4.123.3.12 get_framebuffer_size() [2/2]

```
void lava::window::get_framebuffer_size (
            ui32 & width,
            ui32 & height ) const
```

Get the framebuffer size.

Parameters

width	Framebuffer width
height	Framebuffer height

4.123.3.13 get_monitor()

```
index lava::window::get_monitor ( ) const
```

Get the monitor index of the window.

Returns

index Monitor index

4.123.3.14 get_mouse_position() [1/2]

```
mouse_position lava::window::get_mouse_position ( ) const
```

Get the mouse position in window.

Returns

mouse_position Position of mouse

4.123.3.15 get_mouse_position() [2/2]

```
void lava::window::get_mouse_position (
            r64 & x,
            r64 & y ) const
```

Get the mouse position.

Parameters

x	Mouse X position
y	Mouse Y position

4.123.3.16 get_position()

```
void lava::window::get_position (
            i32 & x,
            i32 & y ) const
```

Get the position of window.

Parameters

x	X position
y	Y position

4.123.3.17 get_save_name()

```
string_ref lava::window::get_save_name ( ) const  [inline]
```

Get the save name.

Returns

name Save name of window

4.123.3.18 get_size() [1/2]

```
uv2 lava::window::get_size ( ) const
```

Get the size.

Returns

uv2 Size of window

4.123.3.19 get_size() [2/2]

```
void lava::window::get_size (
            ui32 & width,
            ui32 & height ) const
```

Get the size of window.

Parameters

width	Window width
height	Window height

4.123.3.20 get_state()

```
state lava::window::get_state ( ) const
```

Get the window state.

Returns

state Window state

4.123.3.21 get_title()

```
string_ref lava::window::get_title ( ) const [inline]
```

Get the window title.

Returns

name Title of window

4.123.3.22 handle_resize()

```
bool lava::window::handle_resize ( ) [inline]
```

Handle window resize.

Returns

Resize was successful or failed

4.123.3.23 hovered()

```
bool lava::window::hovered ( ) const
```

Check if mouse hovered over the window.

Returns

Mouse hovered or not

4.123.3.24 iconified()

```
bool lava::window::iconified ( ) const
```

Check if the window is iconified.

Returns

Window is iconified or not

4.123.3.25 maximized()

```
bool lava::window::maximized ( ) const
```

Check if the window is maximized.

Returns

Window is maximized or not

4.123.3.26 resizable()

```
bool lava::window::resizable ( ) const
```

Check if the window is resizable.

Returns

Window is resizable or not

4.123.3.27 resize_request()

```
bool lava::window::resize_request ( ) const  [inline]
```

Check if the window request to resize.

Returns

Window has resize request or not

4.123.3.28 save_title()

```
bool lava::window::save_title ( ) const  [inline]
```

Check the show save title state.

Returns

Save title is active or not

4.123.3.29 set_decorated()

```
void lava::window::set_decorated (
            bool value )
```

Set the window decorated.

Parameters

value	Decorated state

4.123.3.30 set_floating()

```
void lava::window::set_floating (
            bool value )
```

Set the window floating.

Parameters

value	Floating state

4.123.3.31 set_fullscreen()

```
void lava::window::set_fullscreen (
            bool active )  [inline]
```

Set the window to fullscreen.

Parameters

active	Fullscreen or windowed mode

4.123.3.32 set_icon()

```
void lava::window::set_icon (
            data_cptr data,
            uv2 size )
```

Set the window icon.

Parameters

data	Image data
size	Image size

4.123.3.33 set_mouse_position()

```
void lava::window::set_mouse_position (
            r64 x,
            r64 y )
```

Set the mouse position.

Parameters

x	Mouse X position
y	Mouse Y position

4.123.3.34 set_position()

```
void lava::window::set_position (
```

```
            i32 x,
            i32 y )
```

Set the position of window.

Parameters

x	X positoin
y	Y position

4.123.3.35 set_resizable()

```
void lava::window::set_resizable (
            bool value )
```

Set the window resizable.

Parameters

value	Resizable state

4.123.3.36 set_save_name()

```
void lava::window::set_save_name (
            string_ref save )  [inline]
```

Set the save name.

Parameters

save	Save name of window

4.123.3.37 set_size()

```
void lava::window::set_size (
            ui32 width,
            ui32 height )
```

Set the size of window.

Parameters

width	Window width
height	Window height

4.123.3.38 set_title()

```
void lava::window::set_title (
            string_ref text )
```

Set the window title.

Parameters

text	Title of window

4.123.3.39 show_save_title()

```
void lava::window::show_save_title (
            bool value = true )  [inline]
```

Show the save title in the window.

Parameters

value	Save title state

4.123.3.40 switch_mode()

```
bool lava::window::switch_mode (
            state::optional state = {} )
```

Switch mode of the window.

Parameters

state	Target window state

Returns

Switch was successful or failed

4.123.3.41 switch_mode_request()

```
bool lava::window::switch_mode_request ( ) const  [inline]
```

Check if the window request to switch mode.

Returns

Window has switch mode request or not

4.123.3.42 visible()

```
bool lava::window::visible ( ) const
```

Check if window is visible.

Returns

Window is visible or not

The documentation for this struct was generated from the following file:

- liblava/frame/window.hpp

Chapter 5

File Documentation

5.1 liblava/app.hpp File Reference

App module.

```
#include <liblava/app/app.hpp>
#include <liblava/app/benchmark.hpp>
#include <liblava/app/camera.hpp>
#include <liblava/app/config.hpp>
#include <liblava/app/def.hpp>
#include <liblava/app/forward_shading.hpp>
#include <liblava/app/imgui.hpp>
```

5.1.1 Detailed Description

App module.

Author

Lava Block OÜ and contributors

Copyright

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5.2 app.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/app/app.hpp>
#include <liblava/app/benchmark.hpp>
#include <liblava/app/camera.hpp>
#include <liblava/app/config.hpp>
#include <liblava/app/def.hpp>
#include <liblava/app/forward_shading.hpp>
#include <liblava/app/imgui.hpp>
```

5.3 liblava/app/app.hpp File Reference

Application with basic functionality.

```
#include <liblava/app/benchmark.hpp>
#include <liblava/app/camera.hpp>
#include <liblava/app/config.hpp>
#include <liblava/app/forward_shading.hpp>
#include <liblava/block.hpp>
#include <liblava/frame.hpp>
```

Classes

- struct lava::app

  Application with basic functionality.

Variables

- constexpr bool const **lava::draw_separator** = true

  Draw with separator.
- constexpr bool const **lava::draw_no_separator** = false

  Draw without separator.
- constexpr bool const **lava::draw_fps** = true

  Draw with fps.
- constexpr bool const **lava::draw_no_fps** = false

  Draw without fps.
- constexpr bool const **lava::draw_spacing** = true

  Draw with spacing.
- constexpr bool const **lava::draw_no_spacing** = false

  Draw without spacing.

5.3.1 Detailed Description

Application with basic functionality.

Authors

Lava Block OÜ and contributors

Copyright

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5.4 app.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/app/benchmark.hpp>
#include <liblava/app/camera.hpp>
#include <liblava/app/config.hpp>
#include <liblava/app/forward_shading.hpp>
#include <liblava/block.hpp>
#include <liblava/frame.hpp>

namespace lava {

constexpr bool const draw_separator = true;

constexpr bool const draw_no_separator = false;

constexpr bool const draw_fps = true;

constexpr bool const draw_no_fps = false;

constexpr bool const draw_spacing = true;

constexpr bool const draw_no_spacing = false;

struct app : frame {
    explicit app(frame_env::ref env);

    explicit app(name name, argh::parser cmd_line = {});

    virtual bool setup();

    window_t window;

    input_t input;

    imgui_t imgui;

    imgui::config imgui_config;

    tooltip_list tooltips;

    device_p device = nullptr;

    camera_t camera;

    gamepad pad;

    staging_t staging;

    block_t block;

    renderer_t renderer;

    forward_shading shading;

    render_target::ptr target;

    run_time_t run_time;

    file_system fs;

    VkPipelineCache pipeline_cache = nullptr;

    using update_func = std::function<bool(delta)>;

    update_func on_update;

    using create_func = std::function<bool()>;

    create_func on_create;

    using destroy_func = std::function<void()>;

    destroy_func on_destroy;

    bool v_sync() const {
        return config.v_sync;
    }

    ui32 get_frame_counter() const {
        return frame_counter;
    }
```

```
    void draw_about(bool separator = draw_separator,
                    bool fps = draw_fps,
                    bool spacing = draw_spacing) const;

    app_config config;

    json_file config_file;

    using process_func = std::function<void(VkCommandBuffer, index)>;

    process_func on_process;

    id::ref block_cmd() const {
        return block_command;
    }

    using setup_func = std::function<bool()>;

    setup_func on_setup;

    string screenshot();

    void parse_config(argh::parser cmd_line);

private:
    void mount_resource();

    bool setup_file_system();

    bool setup_window();

    bool setup_device();

    bool setup_render();

    void setup_run();

    void handle_config();

    void handle_input();

    void handle_keys();

    void handle_window();

    void update();

    void render();

    bool create_imgui();

    void destroy_imgui();

    bool create_target();

    void destroy_target();

    bool create_block();

    bool create_pipeline_cache();

    void destroy_pipeline_cache();

    texture::ptr imgui_fonts;

    bool toggle_v_sync = false;

    ui32 frame_counter = 0;

    json_file::callback config_callback;

    id block_command;

    benchmark_data frames;
};

} // namespace lava
```

5.5 liblava/app/benchmark.hpp File Reference

Benchmark.

```
#include <liblava/app/def.hpp>
#include <liblava/frame/frame.hpp>
```

Classes

- struct lava::benchmark_data

  Benchmark data.

Functions

- bool lava::parse (cmd_line cmd_line, benchmark_data &data)

  Parse command line arguments and set benchmark data.
- void lava::benchmark (frame &app, benchmark_data &data)

  Start a benchmark run.
- bool lava::write_frames_json (benchmark_data &data)

  Write frames to json file.

5.5.1 Detailed Description

Benchmark.

Authors

Lava Block OÜ and contributors

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5.5.2 Function Documentation

5.5.2.1 benchmark()

```
void lava::benchmark (
            frame & app,
            benchmark_data & data )
```

Start a benchmark run.

Parameters

app	App to benchmark
data	Benchmark data setting

5.5.2.2 parse()

```
bool lava::parse (
            cmd_line cmd_line,
            benchmark_data & data )
```

Parse command line arguments and set benchmark data.

Parameters

cmd_line	Command line arguments
data	Benchmark data

Returns

Benchmark data is parsed or not ready

5.5.2.3 write_frames_json()

```
bool lava::write_frames_json (
            benchmark_data & data )
```

Write frames to json file.

Parameters

data	Benchmark data setting

Returns

Write was successful or failed

5.6 benchmark.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/app/def.hpp>
11 #include <liblava/frame/frame.hpp>
12
13 namespace lava {
14
18 struct benchmark_data {
20     ms time = ms{ 10000 };
21
23     ms offset = ms{ 5000 };
24
```

```
    string file = _benchmark_json_;

    string path;

    bool exit = true;

    ui32 buffer_size = 100000;

    using list = std::vector<ui32>;

    list values;

    index current = 0;

    ms start_timestamp = ms{ 0 };
};

bool parse(cmd_line cmd_line, benchmark_data& data);

void benchmark(frame& app, benchmark_data& data);

bool write_frames_json(benchmark_data& data);

} // namespace lava
```

5.7 liblava/app/camera.hpp File Reference

First Person / Look At camera.

```
#include <liblava/frame/gamepad.hpp>
#include <liblava/frame/input.hpp>
#include <liblava/resource/buffer.hpp>
```

Classes

- struct lava::camera

  First Person / Look At camera.

Typedefs

- using **lava::camera_t** = camera

  Camera type.

Enumerations

- enum class lava::camera_mode : index { **first_person** = 0 , **look_at** }

  Camera modes.

5.7.1 Detailed Description

First Person / Look At camera.

Authors

Lava Block OÜ and contributors

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5.8 camera.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/frame/gamepad.hpp>
#include <liblava/frame/input.hpp>
#include <liblava/resource/buffer.hpp>

namespace lava {

enum class camera_mode : index {
    first_person = 0,
    look_at,
};

struct camera : entity {
    using ptr = std::shared_ptr<camera>;

    using map = std::map<id, ptr>;

    using list = std::vector<ptr>;

    bool create(device_p device);

    void destroy();

    void update_projection();

    void update_view(delta dt, mouse_position mouse_pos);

    void update_view(delta dt, gamepad::ref pad);

    mat4 get_view() const;

    mat4 get_projection() const;

    mat4 get_view_projection() const;

    bool handle(key_event::ref event);

    bool handle(mouse_button_event::ref event,
                mouse_position mouse_pos);

    bool handle(scroll_event::ref event);

    bool valid() const {
        return data ? data->valid() : false;
    }

    VkDescriptorBufferInfo const* get_descriptor_info() const {
        return data ? data->get_descriptor_info() : nullptr;
    }

    void upload();

    void stop();

    void reset();

    void set_active(bool value = true) {
        active = value;
    }

    bool activated() const {
        return active;
    }

    bool moving() const {
        return move_up || move_down || move_left || move_right;
    }

    void set_movement_keys(keys_ref up, keys_ref down,
                           keys_ref left, keys_ref right) {
        up_keys = up;
        down_keys = down;
        left_keys = left;
        right_keys = right;
    }

    v3 position = v3(0.f);

    v3 rotation = v3(0.f);

```

```
    r32 rotation_speed = 20.f;

    r32 movement_speed = 1.f;

    r32 zoom_speed = 20.f;

    r32 fov = 60.f;

    r32 z_near = 0.1f;

    r32 z_far = 256.f;

    r32 aspect_ratio = 1.77f;

    camera_mode mode = camera_mode::first_person;

    bool lock_z = false;

    bool lock_rotation = false;

private:
    void move_first_person(delta dt);

    bool active = true;

    bool move_up = false;

    bool move_down = false;

    bool move_left = false;

    bool move_right = false;

    bool rotate = false;

    bool translate = false;

    r64 mouse_pos_x = 0.0;

    r64 mouse_pos_y = 0.0;

    r64 scroll_pos = 0.0;

    keys up_keys{ key::w };

    keys down_keys{ key::s };

    keys left_keys{ key::a };

    keys right_keys{ key::d };

    buffer::ptr data;

    size_t size = sizeof(mat4) * 2;

    mat4 projection = mat4(0.f);

    mat4 view = mat4(0.f);
};

using camera_t = camera;

} // namespace lava
```

5.9 liblava/app/config.hpp File Reference

Application configuration.

```
#include <liblava/app/imgui.hpp>
#include <liblava/frame/window.hpp>
#include <liblava/fwd.hpp>
#include <liblava/resource/format.hpp>
```

Classes

- struct lava::app_config

  Application configuration.

Functions

- void lava::set_window_icon (window &window, string_ref icon_file="icon.png")

  Set the window icon.

5.9.1 Detailed Description

Application configuration.

Authors

Lava Block OÜ and contributors

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5.9.2 Function Documentation

5.9.2.1 set_window_icon()

```
void lava::set_window_icon (
            window & window,
            string_ref icon_file = "icon.png" )
```

Set the window icon.

Parameters

window	Target window
icon_file	Icon file

5.10 config.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/app/imgui.hpp>
11 #include <liblava/frame/window.hpp>
12 #include <liblava/fwd.hpp>
13 #include <liblava/resource/format.hpp>
14
15 namespace lava {
16
20 struct app_config : configurable {
22     app* context = nullptr;
23
25     name org = _liblava_;
```

```

    name ext = "zip";

    bool save_window = true;

    bool handle_key_events = true;

    bool v_sync = false;

    surface_format_request surface;

    index physical_device = 0;

    imgui::font imgui_font;

    string id = _default_;

    void set_config(json_ref j) override;

    json get_config() const override;

    void update_window_state();

    window::state::optional window_state;
};

void set_window_icon(window& window, string_ref icon_file = "icon.png");

} // namespace lava
```

5.11 liblava/app/forward_shading.hpp File Reference

Forward shading.

```
#include <liblava/block/render_pass.hpp>
#include <liblava/frame/render_target.hpp>
```

Classes

- struct lava::forward_shading

  Forward shading.

5.11.1 Detailed Description

Forward shading.

Authors

Lava Block OÜ and contributors

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5.12 forward_shading.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/block/render_pass.hpp>
#include <liblava/frame/render_target.hpp>

namespace lava {

struct forward_shading {
    explicit forward_shading() = default;

    ~forward_shading() {
        destroy();
    }

    bool create(render_target::ptr target);

    void destroy();

    render_pass::ptr get_pass() const {
        return pass;
    }

    VkRenderPass get_vk_pass() const {
        return pass->get();
    }

    image::ptr get_depth_stencil() const {
        return depth_stencil;
    }

private:
    render_target::ptr target;

    render_pass::ptr pass;

    image::ptr depth_stencil;
};

} // namespace lava
```

5.13 liblava/app/icon.hpp File Reference

App default icon data.

`#include <liblava/core/types.hpp>`

Variables

- constexpr uchar **lava::icon_png** [ ]

  App default icon data Generate: xxd -i res/icon.png.
- constexpr ui32 **lava::icon_png_len** = 13773

  App default icon data length.

5.13.1 Detailed Description

App default icon data.

Authors

Lava Block OÜ and contributors

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5.14 icon.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/core/types.hpp>

namespace lava {

constexpr uchar icon_png[] = {
    0x89, 0x50, 0x4e, 0x47, 0x0d, 0x0a, 0x1a, 0x0a, 0x00, 0x00, 0x00, 0x0d,
    0x49, 0x48, 0x44, 0x52, 0x00, 0x00, 0x01, 0x00, 0x00, 0x00, 0x01, 0x00,
    0x08, 0x06, 0x00, 0x00, 0x00, 0x5c, 0x72, 0xa8, 0x66, 0x00, 0x00, 0x00,
    0x06, 0x62, 0x4b, 0x47, 0x44, 0x00, 0xff, 0x00, 0xff, 0x00, 0xff, 0xa0,
    0xbd, 0xa7, 0x93, 0x00, 0x00, 0x00, 0x09, 0x70, 0x48, 0x59, 0x73, 0x00,
    0x00, 0x00, 0x01, 0x00, 0x00, 0x00, 0x01, 0x01, 0x38, 0x22, 0xf4, 0x40,
    0x00, 0x00, 0x00, 0x07, 0x74, 0x49, 0x4d, 0x45, 0x07, 0xe6, 0x07, 0x17,
    0x0a, 0x01, 0x20, 0x5c, 0xc5, 0x75, 0x56, 0x00, 0x00, 0x20, 0x00, 0x49,
    0x44, 0x41, 0x54, 0x78, 0xda, 0xed, 0x9d, 0x79, 0x7c, 0xcc, 0xd7, 0xf7,
    0xff, 0x9f, 0xd9, 0xf7, 0x5d, 0x42, 0x82, 0x88, 0x9d, 0x24, 0x76, 0xaa,
    0x14, 0x45, 0x29, 0x2d, 0x45, 0xab, 0x25, 0x76, 0xad, 0xda, 0x5b, 0x54,
    0xed, 0xb5, 0xd6, 0x56, 0x94, 0x0a, 0x4a, 0xed, 0x6a, 0x2b, 0xa9, 0xa5,
    0xd6, 0xaa, 0xda, 0x77, 0x8a, 0x24, 0x62, 0xc9, 0x82, 0x48, 0x44, 0x16,
    0xd9, 0xd7, 0xc9, 0x7e, 0x7f, 0x7f, 0xa4, 0xbe, 0xbf, 0xcf, 0xe7, 0xd3,
    0xcc, 0x7b, 0x46, 0x11, 0x93, 0xcc, 0x7d, 0x3e, 0x1e, 0xf3, 0x0f, 0x67,
    0x26, 0x33, 0xf7, 0x9e, 0xf3, 0x7a, 0xdf, 0xe5, 0x9c, 0x7b, 0x41, 0x22,
    0x91, 0x48, 0x24, 0x12, 0x89, 0x44, 0x22, 0x91, 0x48, 0x24, 0x12, 0x89,
    0x44, 0x22, 0x91, 0x48, 0x24, 0x12, 0x89, 0x44, 0x22, 0x91, 0x48, 0x24,
    0x12, 0x89, 0x44, 0x22, 0x91, 0x48, 0x24, 0x12, 0x89, 0x44, 0x22, 0x91,
    0x48, 0x24, 0x12, 0x89, 0x44, 0x22, 0x91, 0x48, 0x24, 0x12, 0x89, 0x44,
    0xa2, 0x9b, 0x38, 0x01, 0xab, 0x80, 0x1b, 0x40, 0x07, 0xd9, 0x1c, 0x12,
    0x89, 0x7e, 0x60, 0x0a, 0x8c, 0x05, 0x12, 0x01, 0xf1, 0x1f, 0xaf, 0x7d,
    0x40, 0x0d, 0xd9, 0x3c, 0x12, 0x49, 0xd9, 0xe5, 0x5d, 0xe0, 0xce, 0xff,
    0x04, 0xfe, 0x7f, 0xbe, 0x54, 0xc0, 0x02, 0xc0, 0x56, 0x36, 0x95, 0x44,
    0x52, 0x76, 0xa8, 0x09, 0x1c, 0x50, 0x08, 0xfc, 0xff, 0x7d, 0xc5, 0x01,
    0x83, 0x01, 0x43, 0xd9, 0x74, 0x12, 0x49, 0xe9, 0xc5, 0x0e, 0x58, 0x02,
    0xe4, 0x3c, 0x47, 0xf0, 0xff, 0xe7, 0xeb, 0x2a, 0xd0, 0x52, 0x36, 0xa3,
    0x44, 0x52, 0xba, 0x30, 0x02, 0x86, 0x00, 0x31, 0xff, 0x32, 0xf0, 0xff,
    0xf3, 0x55, 0x08, 0x6c, 0x05, 0x2a, 0xca, 0x66, 0x95, 0x48, 0x74, 0x9f,
    0xd6, 0xc0, 0x75, 0x4d, 0x81, 0xed, 0xea, 0xea, 0x2a, 0x76, 0xee, 0xdc,
    0x29, 0xae, 0x5d, 0xbb, 0x26, 0x7a, 0xf9, 0xf4, 0xd2, 0x46, 0x08, 0x32,
    0x81, 0x19, 0x80, 0xa5, 0x6c, 0x62, 0x89, 0x44, 0xf7, 0x70, 0x07, 0x76,
    0xfc, 0xfd, 0xc4, 0x56, 0x0c, 0xe6, 0x39, 0x73, 0xe6, 0x88, 0x27, 0x4f,
    0x9e, 0x88, 0x67, 0xe4, 0xe4, 0xe4, 0x88, 0x63, 0xc7, 0x8e, 0x89, 0xa6,
    0xcd, 0x9a, 0x6a, 0x23, 0x04, 0xe1, 0x40, 0x2f, 0xd9, 0xdc, 0x12, 0x89,
    0x6e, 0x60, 0x09, 0xcc, 0x06, 0xb2, 0x34, 0x05, 0xef, 0xc0, 0x41, 0x03,
    0x45, 0x40, 0x40, 0x80, 0x28, 0x2c, 0x2c, 0x14, 0xc5, 0x91, 0x92, 0x92,
    0x22, 0xd6, 0xad, 0x5b, 0x27, 0x8c, 0x8c, 0x8c, 0xb4, 0x11, 0x82, 0xd3,
    0x40, 0x03, 0xd9, 0xfc, 0x12, 0xc9, 0xeb, 0xc1, 0x00, 0xf0, 0x01, 0x22,
    0x35, 0x05, 0xeb, 0x1b, 0x6f, 0x36, 0x17, 0xc7, 0x8f, 0x1f, 0x17, 0x39,
    0x39, 0x39, 0x42, 0x1b, 0x1e, 0x3e, 0x7c, 0x28, 0xc6, 0x7f, 0x3d, 0x5e,
    0x1b, 0x11, 0x28, 0x00, 0xd6, 0x03, 0x2e, 0xb2, 0x3b, 0x24, 0x92, 0x92,
    0xa3, 0x31, 0x70, 0x4e, 0x53, 0x80, 0x5a, 0xda, 0x5b, 0x8a, 0xf5, 0xeb,
    0xd7, 0x8b, 0xd4, 0xd4, 0x54, 0xf1, 0xbc, 0x14, 0x14, 0x14, 0x88, 0xab,
    0x57, 0xaf, 0x8a, 0xee, 0x1f, 0x76, 0xd7, 0x46, 0x08, 0x52, 0x80, 0xaf,
    0x29, 0x4a, 0x32, 0x92, 0x48, 0x24, 0xaf, 0x88, 0x0a, 0xc0, 0x06, 0x6d,
    0xe6, 0xf9, 0x93, 0x26, 0x4d, 0x12, 0xe1, 0xe1, 0xe1, 0xe2, 0x45, 0x51,
    0xa9, 0x54, 0xe2, 0xc0, 0x81, 0x03, 0xa2, 0x6e, 0xdd, 0xba, 0xda, 0x08,
    0x41, 0x08, 0xd0, 0x59, 0x76, 0x93, 0x44, 0xf2, 0x72, 0x31, 0x03, 0x26,
    0x01, 0x69, 0x9a, 0x82, 0xf0, 0x93, 0x5e, 0x9f, 0x88, 0x6b, 0xd7, 0xae,
    0x89, 0x82, 0xc2, 0x02, 0xf1, 0x32, 0x89, 0x8f, 0x8f, 0x17, 0xcb, 0x96,
    0x2d, 0x13, 0x06, 0x06, 0x06, 0xda, 0x08, 0xc1, 0x51, 0xc0, 0x53, 0x76,
    0x9b, 0x44, 0xf2, 0xe2, 0x74, 0x01, 0x42, 0x35, 0x05, 0x9d, 0x57, 0x3d,
    0x2f, 0x71, 0xf0, 0xe0, 0x41, 0x91, 0x9d, 0x9d, 0x2d, 0x5e, 0x25, 0xc1,
    0xc1, 0xc1, 0x62, 0xcc, 0xb8, 0x31, 0xda, 0x88, 0x40, 0x0e, 0xe0, 0x0b,
    0x38, 0xc8, 0x2e, 0x94, 0x48, 0x9e, 0x1f, 0x2f, 0xe0, 0x4f, 0x4d, 0x81,
    0x66, 0x61, 0x61, 0x21, 0x96, 0x2f, 0x5f, 0x2e, 0x12, 0x12, 0x12, 0x44,
    0x49, 0x91, 0x9f, 0x9f, 0x2f, 0x2e, 0x5c, 0xb8, 0x20, 0x3a, 0x75, 0xea,
    0xa4, 0x8d, 0x10, 0xc4, 0x03, 0x43, 0x01, 0x63, 0xd9, 0xa5, 0x12, 0x89,
    0x66, 0x1c, 0x81, 0x1f, 0x81, 0x3c, 0x4d, 0xc1, 0x35, 0x6e, 0xdc, 0x38,
    0x11, 0x1a, 0x1a, 0x2a, 0x5e, 0x17, 0x99, 0x99, 0x99, 0x62, 0xe7, 0xce,
    0x9d, 0xa2, 0x72, 0xe5, 0xca, 0xda, 0x08, 0x81, 0x3f, 0xd0, 0x4e, 0x76,
    0xaf, 0x44, 0x52, 0x3c, 0x46, 0xc0, 0x48, 0x20, 0x41, 0x53, 0x30, 0x75,
    0xee, 0xdc, 0x59, 0x5c, 0xba, 0x74, 0x49, 0x14, 0x14, 0x14, 0x08, 0x5d,
    0x20, 0x3a, 0x3a, 0x5a, 0x2c, 0x58, 0xb0, 0x40, 0xdb, 0xd4, 0xe2, 0xbd,
    0x40, 0x35, 0xd9, 0xdd, 0x12, 0xc9, 0xff, 0xe7, 0x1d, 0xe0, 0xb6, 0xa6,
    0xe0, 0x71, 0x72, 0x72, 0x12, 0xbb, 0x76, 0xed, 0x12, 0x99, 0x99, 0x99,
    0x42, 0x17, 0x09, 0x0a, 0x0a, 0x12, 0x43, 0x86, 0x0c, 0xd1, 0x46, 0x04,
    0x54, 0xc0, 0x3c, 0xc0, 0x5a, 0x76, 0xbd, 0x44, 0x9f, 0xa9, 0x0e, 0xfc,
    0xa6, 0xcd, 0x93, 0x73, 0xf1, 0xe2, 0xc5, 0x22, 0x26, 0x36, 0x46, 0xe8,
    0x3a, 0x79, 0x79, 0x79, 0xe2, 0xc4, 0x89, 0x13, 0xa2, 0x45, 0x8b, 0x16,
    0xda, 0x08, 0xc1, 0x63, 0xa0, 0x1f, 0xb2, 0xec, 0x58, 0xa2, 0x67, 0x58,
    0x53, 0x74, 0xf0, 0x46, 0xb6, 0xa6, 0x20, 0x19, 0x3e, 0x7c, 0xb8, 0xb8,
    0x7b, 0xef, 0xae, 0xda, 0xf4, 0x5d, 0x5d, 0x25, 0x25, 0x35, 0x45, 0x6c,
```

```
    0xdc, 0xb8, 0x51, 0x54, 0x70, 0xad, 0xa0, 0x6d, 0xd9, 0xf1, 0x9b, 0xd2,
    0x2d, 0x24, 0xfa, 0x30, 0xcf, 0x1f, 0x0c, 0x44, 0x6b, 0x0a, 0x8a, 0x37,
    0xdf, 0x7c, 0x53, 0x9c, 0x3c, 0x79, 0x52, 0xe4, 0xe5, 0xe5, 0x89, 0xd2,
    0x4c, 0x64, 0x64, 0xa4, 0x98, 0x31, 0x63, 0xc6, 0xf3, 0x94, 0x1d, 0xbb,
    0x4a, 0x37, 0x91, 0x94, 0x45, 0x5a, 0x01, 0xd7, 0x34, 0x05, 0x42, 0x95,
    0x2a, 0x55, 0xc4, 0x96, 0x2d, 0x5b, 0x44, 0x5a, 0x5a, 0x9a, 0x28, 0x2b,
    0x14, 0x16, 0x16, 0x0a, 0x7f, 0x7f, 0x7f, 0xf1, 0xc9, 0x27, 0x9f, 0x68,
     0x23, 0x04, 0x69, 0xc0, 0x34, 0x8a, 0x92, 0x9f, 0x24, 0x92, 0x52, 0x8f,
     0x1b, 0xb0, 0x5d, 0x9b, 0x79, 0xfe, 0xec, 0xd9, 0xb3, 0x45, 0x54, 0x54,
     0x94, 0x28, 0xab, 0x64, 0x67, 0x67, 0x8b, 0x43, 0x87, 0x0e, 0x09, 0x2f,
     0x2f, 0x2f, 0x6d, 0x84, 0xe0, 0x11, 0xf0, 0xa1, 0x74, 0x1f, 0x49, 0x69,
     0xc5, 0x12, 0x98, 0x0e, 0x64, 0x68, 0x72, 0xf6, 0xde, 0xbd, 0x7b, 0x8b,
     0x80, 0x80, 0x00, 0xa1, 0x2f, 0x24, 0xa5, 0x24, 0x89, 0x1f, 0x7f, 0xfc,
     0x51, 0x98, 0x9a, 0x9a, 0x6a, 0x5b, 0x76, 0x5c, 0x5f, 0xba, 0x93, 0xa4,
     0x34, 0xe1, 0x43, 0xd1, 0xc1, 0x19, 0x8a, 0xce, 0xdd, 0xbc, 0x79, 0x73,
     0x71, 0xf8, 0xf0, 0x61, 0xad, 0xcb, 0x74, 0xcb, 0x1a, 0xe1, 0xe1, 0xe1,
     0xe2, 0xeb, 0xaf, 0xbf, 0xd6, 0x46, 0x04, 0xf2, 0x80, 0x35, 0x14, 0xdd,
     0x65, 0x20, 0x91, 0xe8, 0x2c, 0x8d, 0x80, 0xb3, 0x9a, 0x1c, 0xda, 0xd6,
     0xd6, 0x56, 0xac, 0x5d, 0xbb, 0x56, 0x24, 0x27, 0x27, 0x0b, 0x7d, 0xa7,
     0xa0, 0xa0, 0x40, 0x5c, 0xba, 0x74, 0x49, 0x74, 0xee, 0xdc, 0x59, 0x1b,
     0x21, 0x48, 0x06, 0xc6, 0x03, 0x26, 0xd2, 0xd5, 0x24, 0xba, 0x84, 0x0b,
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    0x52, 0xb6, 0x79, 0x83, 0xa2, 0x33, 0xf6, 0x9f, 0x05, 0x7e, 0x2a, 0x30,
    0xe1, 0xef, 0x91, 0x82, 0x44, 0x22, 0xd1, 0x13, 0xfa, 0x02, 0x2b, 0x81,
    0x0a, 0xb2, 0x29, 0x24, 0x12, 0x89, 0x44, 0x22, 0x91, 0x48, 0x24, 0x12,
    0x89, 0x44, 0x22, 0x91, 0x48, 0x24, 0x12, 0x89, 0x44, 0x22, 0x91, 0x48,
    0x24, 0x12, 0x89, 0x44, 0x22, 0x91, 0x48, 0x24, 0x12, 0x89, 0x44, 0x22,
    0x91, 0x48, 0x24, 0x12, 0x89, 0x44, 0x22, 0x91, 0x48, 0x74, 0x9f, 0xff,
    0x07, 0x99, 0x3f, 0x80, 0xa7, 0x39, 0xed, 0x2c, 0x27, 0x00, 0x00, 0x00,
    0x00, 0x49, 0x45, 0x4e, 0x44, 0xae, 0x42, 0x60, 0x82
};

constexpr ui32 icon_png_len = 13773;

} // namespace lava
```

5.15 liblava/app/imgui.hpp File Reference

ImGui integration.

```
#include <liblava/block/render_pipeline.hpp>
#include <liblava/file.hpp>
#include <liblava/frame/input.hpp>
#include <liblava/resource/texture.hpp>
```

Classes

- struct lava::imgui

  ImGui integration.
- struct lava::imgui::icon_font

  ImGui icon font settings.
- struct lava::imgui::font

  ImGui font settings.
- struct lava::imgui::config

  ImGui configuration.

Typedefs

- using **lava::imgui_t** = imgui

  Imgui type.

Functions

- void lava::setup_imgui_font (imgui::config &config, imgui::font::ref font)

  Set up ImGui font.
- void lava::setup_imgui_font_icons (imgui::font &font, string filename, ui16 min, ui16 max)

  Set up imgui font icons.
- void lava::imgui_left_spacing (ui32 top=1)

  ImGui left spacing with top offset.

Variables

- constexpr const r32 **lava::default_imgui_font_size** = 18.f

  Default ImGui font size.

5.15.1 Detailed Description

ImGui integration.

Authors

Lava Block OÜ and contributors

Copyright

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5.15.2 Function Documentation

5.15.2.1 imgui_left_spacing()

```
void lava::imgui_left_spacing (
            ui32 top = 1 )
```

ImGui left spacing with top offset.

Parameters

top	Top offset

5.15.2.2 setup_imgui_font()

```
void lava::setup_imgui_font (
```

```
            imgui::config & config,
            imgui::font::ref font )
```

Set up ImGui font.

Parameters

config	ImGui configuration
font	ImGui font

5.15.2.3 setup_imgui_font_icons()

```
void lava::setup_imgui_font_icons (
            imgui::font & font,
            string filename,
            ui16 min,
            ui16 max )
```

Set up imgui font icons.

Parameters

font	Imgui font
filename	Font icon file name
min	Min range
max	Max range

5.16 imgui.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/block/render_pipeline.hpp>
11 #include <liblava/file.hpp>
12 #include <liblava/frame/input.hpp>
13 #include <liblava/resource/texture.hpp>
14
15 // fwd
16 struct GLFWwindow;
17 struct GLFWcursor;
18 struct ImDrawData;
19 struct ImGuiStyle;
20
21 namespace lava {
22
24 constexpr const r32 default_imgui_font_size = 18.f;
25
29 struct imgui {
33     explicit imgui() = default;
34
39     explicit imgui(GLFWwindow* window) {
40         setup(window);
41     }
42
46     ~imgui() {
47         destroy();
```

```
48      }
49
53      struct icon_font {
55          data font_data;
56
58          ui16 range_begin = 0;
59
61          ui16 range_end = 0;
62
64          r32 size = default_imgui_font_size;
65      };
66
70      struct font {
72          using ref = font const&;
73
75          string file;
76
78          r32 size = 21.f;
79
81          string icon_file;
82
84          r32 icon_size = 21.f;
85
87          ui16 icon_range_begin = 0;
88
90          ui16 icon_range_end = 0;
91      };
92
96      struct config {
98          data font_data;
99
101         r32 font_size = default_imgui_font_size;
102
104         std::shared_ptr<ImGuiStyle> style;
105
107         icon_font icon;
108
110         std::filesystem::path ini_file_dir;
111     };
112
118     void setup(GLFWwindow* window, config config);
119
124     void setup(GLFWwindow* window) {
125         setup(window, config());
126     }
127
134     bool create(render_pipeline::ptr pipeline, index max_frames);
135
143     bool create(device_p device,
144                 index max_frames,
145                 VkPipelineCache pipeline_cache) {
146         return create(render_pipeline::make(device, pipeline_cache),
147                       max_frames);
148     }
149
158     bool create(device_p device,
159                 index max_frames,
160                 VkRenderPass pass,
161                 VkPipelineCache pipeline_cache = 0) {
162         if (!create(device, max_frames, pipeline_cache))
163             return false;
164
165         return pipeline->create(pass);
166     }
167
173     bool upload_fonts(texture::ptr texture);
174
178     void destroy();
179
184     bool ready() const {
185         return initialized;
186     }
187
192     render_pipeline::ptr get_pipeline() {
193         return pipeline;
194     }
195
197     using draw_func = std::function<void()>;
198
200     draw_func on_draw;
201
206     bool capture_mouse() const;
207
212     bool capture_keyboard() const;
213
218     void set_active(bool value = true) {
219         active = value;
```

```
    }

    bool activated() const {
        return active;
    }

    void toggle() {
        active = !active;
    }

    void set_ini_file(std::filesystem::path dir);

    std::filesystem::path get_ini_file() const {
        return std::filesystem::path(ini_file);
    }

    void convert_style_to_srgb();

    input_callback const& get_input_callback() const {
        return callback;
    }

private:
    void handle_key_event(i32 key, i32 scancode, i32 action, i32 mods);

    void handle_mouse_button_event(i32 button, i32 action, i32 mods);

    void handle_scroll_event(r64 x_offset, r64 y_offset);

    void prepare_draw_lists(ImDrawData* draw_data);

    void render_draw_lists(VkCommandBuffer cmd_buf);

    void invalidate_device_objects();

    void update_mouse_pos_and_buttons();

    void update_mouse_cursor();

    void new_frame();

    void render(VkCommandBuffer cmd_buf);

    device_p device = nullptr;

    // Initialized state
    bool initialized = false;

    render_pipeline::ptr pipeline;

    pipeline_layout::ptr layout;

    size_t buffer_memory_alignment = 256;

    index frame = 0;

    index max_frames = 4;

    buffer::list vertex_buffers;

    buffer::list index_buffers;

    descriptor_ptr descriptor;

    descriptor::pool::ptr descriptor_pool;

    VkDescriptorSet descriptor_set = VK_NULL_HANDLE;

    GLFWwindow* window = nullptr;

    bool mouse_just_pressed[5] = { false, false, false, false, false };

    r64 current_time = 0.0;

    std::vector<GLFWcursor*> mouse_cursors;

    string ini_file;

    bool active = true;

    input_callback callback;

    std::array<ui16, 3> icons_range;
};

using imgui_t = imgui;

```

```
void setup_imgui_font(imgui::config& config,
                      imgui::font::ref font);

void setup_imgui_font_icons(imgui::font& font,
                            string filename,
                            ui16 min, ui16 max);

void imgui_left_spacing(ui32 top = 1);

} // namespace lava
```

5.17 liblava/asset.hpp File Reference

Asset module.

```
#include <liblava/asset/image_loader.hpp>
#include <liblava/asset/load_mesh.hpp>
#include <liblava/asset/load_texture.hpp>
#include <liblava/asset/write_image.hpp>
```

5.17.1 Detailed Description

Asset module.

Author

Lava Block OÜ and contributors

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5.18 asset.hpp

Go to the documentation of this file.

```

#pragma once

#include <liblava/asset/image_loader.hpp>
#include <liblava/asset/load_mesh.hpp>
#include <liblava/asset/load_texture.hpp>
#include <liblava/asset/write_image.hpp>
```

5.19 liblava/asset/image_loader.hpp File Reference

Load image data from file and memory.

```
#include <liblava/file/file.hpp>
#include <liblava/util/math.hpp>
```

Classes

- struct lava::image_loader

  Load image data from file and memory.

5.19.1 Detailed Description

Load image data from file and memory.

Authors

Lava Block OÜ and contributors

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5.20 image_loader.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/file/file.hpp>
11 #include <liblava/util/math.hpp>
12
13 namespace lava {
14
18 struct image_loader {
23     explicit image_loader(string_ref filename);
24
29     explicit image_loader(cdata::ref image);
30
34     ~image_loader();
35
40     bool ready() const {
41         return data != nullptr;
42     }
43
48     data_cptr get() const {
49         return data;
50     }
51
56     size_t size() const {
57         return channels * dimensions.x * dimensions.y;
58     }
59
64     uv2 get_dimensions() const {
65         return dimensions;
66     }
67
72     ui32 get_channels() const {
73         return channels;
74     }
75
76 private:
78     data_ptr data = nullptr;
79
81     uv2 dimensions = uv2(0, 0);
82
84     ui32 channels = 0;
85 };
86
87 } // namespace lava
```

5.21 liblava/asset/load_mesh.hpp File Reference

Load mesh from file.

```
#include <liblava/resource/mesh.hpp>
```

Functions

- mesh::ptr lava::load_mesh (device_p device, string_ref filename, string_ref temp_dir)

  Load mesh from file.

5.21.1 Detailed Description

Load mesh from file.

Authors

Lava Block OÜ and contributors

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5.21.2 Function Documentation

5.21.2.1 load_mesh()

```
mesh::ptr lava::load_mesh (
            device_p device,
            string_ref filename,
            string_ref temp_dir )
```

Load mesh from file.

Parameters

device	Vulkan device
filename	File to load
temp_dir	Temporary directory

Returns

mesh::ptr Loaded mesh

5.22 load_mesh.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/resource/mesh.hpp>

namespace lava {

mesh::ptr load_mesh(device_p device,
                    string_ref filename,
                    string_ref temp_dir);

} // namespace lava
```

5.23 liblava/asset/load_texture.hpp File Reference

Load texture from file.

`#include <liblava/resource/texture.hpp>`

Functions

- texture::ptr lava::load_texture (device_p device, file_format file_format, texture_type type=texture_type::tex↩_2d)

  Load texture from file.
- texture::ptr lava::load_texture (device_p device, string_ref filename, VkFormat format=VK_FORMAT_R8G8↩B8A8_SRGB, texture_type type=texture_type::tex_2d)

  Load texture from file with default format (sRGB)
- texture::ptr lava::create_default_texture (device_p device, uv2 size={ 512, 512 }, v3 color=v3(1.f), r32 alpha=0.7529f)

  Create a default texture with checkerboard pattern.

5.23.1 Detailed Description

Load texture from file.

Authors

Lava Block OÜ and contributors

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5.23.2 Function Documentation

5.23.2.1 create_default_texture()

```
texture::ptr lava::create_default_texture (
            device_p device,
            uv2 size = { 512, 512 },
            v3 color = v3(1.f),
            r32 alpha = 0.7529f )
```

Create a default texture with checkerboard pattern.

Parameters

device	Vulkan device
size	Size of texture
color	Color of texture
alpha	Alpha value of texture

Returns

texture::ptr Loaded texture

5.23.2.2 load_texture() [1/2]

```
texture::ptr lava::load_texture (
            device_p device,
            file_format file_format,
            texture_type type = texture_type::tex_2d )
```

Load texture from file.

Parameters

device	Vulkan device
file_format	File and format
type	Type of texture

Returns

texture::ptr Loaded texture

5.23.2.3 load_texture() [2/2]

```
texture::ptr lava::load_texture (
            device_p device,
            string_ref filename,
            VkFormat format = VK_FORMAT_R8G8B8A8_SRGB,
            texture_type type = texture_type::tex_2d )  [inline]
```

Load texture from file with default format (sRGB)

Parameters

device	Vulkan device
filename	File to load
format	Format of texture
type	Type of texture

Returns

texture::ptr Loaded texture

5.24 load_texture.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/resource/texture.hpp>

namespace lava {

texture::ptr load_texture(device_p device,
                          file_format file_format,
                          texture_type type = texture_type::tex_2d);

inline texture::ptr load_texture(device_p device,
                                 string_ref filename,
                                 VkFormat format = VK_FORMAT_R8G8B8A8_SRGB,
                                 texture_type type = texture_type::tex_2d) {
    return load_texture(device, { filename, format }, type);
}

texture::ptr create_default_texture(device_p device,
                                    uv2 size = { 512, 512 },
                                    v3 color = v3(1.f),
                                    r32 alpha = 0.7529f);

} // namespace lava
```

5.25 liblava/asset/write_image.hpp File Reference

Write image data to file.

`#include <liblava/resource/image.hpp>`

Functions

- bool lava::write_image_png (device_p device, image::ptr image, string_ref filename, bool swizzle)

  Write image data to png file.

5.25.1 Detailed Description

Write image data to file.

Authors

Lava Block OÜ and contributors

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5.25.2 Function Documentation

5.25.2.1 write_image_png()

```
bool lava::write_image_png (
            device_p device,
            image::ptr image,
            string_ref filename,
            bool swizzle )
```

Write image data to png file.

Parameters

device	Vulkan device
image	Image to write
filename	File to write
swizzle	Swizzle data

Returns

Write was successful or failed

5.26 write_image.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/resource/image.hpp>

namespace lava {

bool write_image_png(device_p device,
                     image::ptr image,
                     string_ref filename,
                     bool swizzle);

} // namespace lava
```

5.27 liblava/base.hpp File Reference

Base module.

```
#include <liblava/base/base.hpp>
#include <liblava/base/debug_utils.hpp>
#include <liblava/base/device.hpp>
#include <liblava/base/device_table.hpp>
#include <liblava/base/instance.hpp>
#include <liblava/base/memory.hpp>
#include <liblava/base/physical_device.hpp>
#include <liblava/base/platform.hpp>
#include <liblava/base/queue.hpp>
```

5.27.1 Detailed Description

Base module.

Author

Lava Block OÜ and contributors

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5.28 base.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/base/base.hpp>
11 #include <liblava/base/debug_utils.hpp>
12 #include <liblava/base/device.hpp>
13 #include <liblava/base/device_table.hpp>
14 #include <liblava/base/instance.hpp>
15 #include <liblava/base/memory.hpp>
16 #include <liblava/base/physical_device.hpp>
17 #include <liblava/base/platform.hpp>
18 #include <liblava/base/queue.hpp>
```

5.29 liblava/base/base.hpp File Reference

Vulkan base types.

```
#include <liblava/core/version.hpp>
#include <liblava/util/math.hpp>
#include <vulkan/vulkan.h>
#include <volk.h>
```

Classes

- struct lava::vk_result

  Vulkan result.
- struct lava::target_callback

  Target callback.

Typedefs

- using **lava::VkObjectHandle** = ui64

  Vulkan object handle.
- using **lava::VkFormats** = std::vector< VkFormat >

  List of Vulkan formats.
- using **lava::VkImages** = std::vector< VkImage >

  List of Vulkan images.
- using **lava::VkImagesRef** = VkImages const &

  Reference to list of Vulkan images.
- using **lava::VkImageViews** = std::vector< VkImageView >

  List of Vulkan images views.
- using **lava::VkFramebuffers** = std::vector< VkFramebuffer >

  List of Vulkan frame buffers.
- using **lava::VkCommandPools** = std::vector< VkCommandPool >

  List of Vulkan command pools.
- using **lava::VkCommandBuffers** = std::vector< VkCommandBuffer >

  List of Vulkan command buffers.
- using **lava::VkFences** = std::vector< VkFence >

  List of Vulkan fences.
- using **lava::VkSemaphores** = std::vector< VkSemaphore >

  List of Vulkan semaphores.
- using **lava::VkPresentModeKHRs** = std::vector< VkPresentModeKHR >

  List of Vulkan present modes.
- using **lava::VkDescriptorSets** = std::vector< VkDescriptorSet >

  List of Vulkan descriptor sets.
- using **lava::VkDescriptorSetLayouts** = std::vector< VkDescriptorSetLayout >

  List of Vulkan descriptor set layouts.
- using **lava::VkDescriptorSetLayoutBindings** = std::vector< VkDescriptorSetLayoutBinding >

  List of Vulkan descriptor set layout bindings.
- using **lava::VkDescriptorPoolSizes** = std::vector< VkDescriptorPoolSize >

  List of Vulkan descriptor pool sizes.
- using **lava::VkDescriptorPoolSizesRef** = VkDescriptorPoolSizes const &

  Reference to a list of Vulkan descritpr pool sizes.
- using **lava::VkPushConstantRanges** = std::vector< VkPushConstantRange >

  List of Vulkan push constant ranges.
- using **lava::VkAttachmentReferences** = std::vector< VkAttachmentReference >

  List of Vulkan attachment references.
- using **lava::VkClearValues** = std::vector< VkClearValue >

  List of Vulkan clear values.
- using **lava::VkPipelineShaderStageCreateInfos** = std::vector< VkPipelineShaderStageCreateInfo >

  List of Vulkan pipeline shader stage create infos.
- using **lava::VkSpecializationMapEntries** = std::vector< VkSpecializationMapEntry >

  List of Vulkan specialization map entries.
- using **lava::VkVertexInputBindingDescriptions** = std::vector< VkVertexInputBindingDescription >

  List of Vulkan vertex input binding descriptions.
- using **lava::VkVertexInputAttributeDescriptions** = std::vector< VkVertexInputAttributeDescription >

  List of Vulkan vertex input attribute descriptions.
- using **lava::VkPipelineColorBlendAttachmentStates** = std::vector< VkPipelineColorBlendAttachment↩State >

  List of Vulkan pipeline color blend attachment states.

- using **lava::VkDynamicStates** = std::vector< VkDynamicState >

  List of Vulkan dynamic states.
- using **lava::VkQueueFamilyPropertiesList** = std::vector< VkQueueFamilyProperties >

  List of Vulkan queue family properties.
- using **lava::VkExtensionPropertiesList** = std::vector< VkExtensionProperties >

  List of Vulkan extension properties.
- using **lava::VkLayerPropertiesList** = std::vector< VkLayerProperties >

  List of Vulkan layer properties.
- using **lava::VkPhysicalDevices** = std::vector< VkPhysicalDevice >

  List of Vulkan physical devices.
- using **lava::VkAttachments** = std::vector< VkImageViews >

  List of Vulkan attachments (image views)
- using **lava::VkAttachmentsRef** = VkAttachments const &

  Reference of Vulkan attachments (image views)

Enumerations

- enum class lava::api_version : index { **v1_0** = 0 , **v1_1** , **v1_2** , **v1_3** }

  Vulkan api versions.

Functions

- bool lava::check (VkResult result)

  Check a Vulkan result.
- bool lava::failed (VkResult result)

  Check if a Vulkan result failed.
- string lava::to_string (VkResult result)

  Convert a Vulkan result to string.
- string lava::version_to_string (ui32 version)

  Convert a version to string.
- int_version lava::to_version (ui32 version)

  Convert a version to internal version.
- ui32 lava::to_version (int_version version)

  Convert a version to Vulkan version.
- api_version lava::to_api_version (ui32 version)

  Convert a version to Vulkan API version.

Variables

- constexpr bool const **lava::build_failed** = false

  Build failed.
- constexpr bool const **lava::build_done** = true

  Build done.
- constexpr ui32 const **lava::Vk_Limit_DescriptorSets** = 4

  Limit of Vulkan description sets.
- constexpr ui32 const **lava::Vk_Limit_Bindings** = 16

  Limit of Vulkan bindings.
- constexpr ui32 const **lava::Vk_Limit_Attachments** = 8

Limit of Vulkan attachments.

- constexpr ui32 const **lava::Vk_Limit_VertexAttribs** = 16

  Limit of Vulkan vertex attributes.
- constexpr ui32 const **lava::Vk_Limit_VertexBuffers** = 4

  Limit of Vulkan vertex buffers.
- constexpr ui32 const **lava::Vk_Limit_PushConstant_Size** = 128

  Limit of Vulkan push constant size.
- constexpr ui32 const **lava::Vk_Limit_UBO_Size** = 16 ∗ 1024

  Limit of Vulkan UBO size.

5.29.1 Detailed Description

Vulkan base types.

Authors

Lava Block OÜ and contributors

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5.29.2 Function Documentation

5.29.2.1 check()

```
bool lava::check (
            VkResult result )
```

Check a Vulkan result.

Parameters

result	Result to check

Returns

Okay or error

5.29.2.2 failed()

```
bool lava::failed (
            VkResult result )  [inline]
```

Check if a Vulkan result failed.

Parameters

result	Result to check

Returns

Error or okay

5.29.2.3 to_api_version()

```
api_version lava::to_api_version (
            ui32 version )
```

Convert a version to Vulkan API version.

Parameters

version	Vulkan version to convert

Returns

api_version Converted API version

5.29.2.4 to_string()

```
string lava::to_string (
            VkResult result )
```

Convert a Vulkan result to string.

Parameters

result	Result to convert

Returns

string String of result

5.29.2.5 to_version() [1/2]

```
ui32 lava::to_version (
            int_version version )
```

Convert a version to Vulkan version.

Parameters

version	Internal version to convert

Returns

ui32 Converted Vulkan version

5.29.2.6 to_version() [2/2]

```
int_version lava::to_version (
            ui32 version )
```

Convert a version to internal version.

Parameters

version	Version to convert

Returns

int_version Converted version

5.29.2.7 version_to_string()

```
string lava::version_to_string (
            ui32 version )
```

Convert a version to string.

Parameters

version	Version to convert

Returns

string String of version

5.30 base.hpp

Go to the documentation of this file.

```
1
8 #pragma once
```

```

#include <liblava/core/version.hpp>
#include <liblava/util/math.hpp>

// clang-format off

#define VK_NO_PROTOTYPES
#include <vulkan/vulkan.h>
#include <volk.h>

// clang-format on

namespace lava {

using VkObjectHandle = ui64;

using VkFormats = std::vector<VkFormat>;

using VkImages = std::vector<VkImage>;

using VkImagesRef = VkImages const&;

using VkImageViews = std::vector<VkImageView>;

using VkFramebuffers = std::vector<VkFramebuffer>;

using VkCommandPools = std::vector<VkCommandPool>;

using VkCommandBuffers = std::vector<VkCommandBuffer>;

using VkFences = std::vector<VkFence>;

using VkSemaphores = std::vector<VkSemaphore>;

using VkPresentModeKHRs = std::vector<VkPresentModeKHR>;

using VkDescriptorSets = std::vector<VkDescriptorSet>;

using VkDescriptorSetLayouts = std::vector<VkDescriptorSetLayout>;

using VkDescriptorSetLayoutBindings = std::vector<VkDescriptorSetLayoutBinding>;

using VkDescriptorPoolSizes = std::vector<VkDescriptorPoolSize>;

using VkDescriptorPoolSizesRef = VkDescriptorPoolSizes const&;

using VkPushConstantRanges = std::vector<VkPushConstantRange>;

using VkAttachmentReferences = std::vector<VkAttachmentReference>;

using VkClearValues = std::vector<VkClearValue>;

using VkPipelineShaderStageCreateInfos = std::vector<VkPipelineShaderStageCreateInfo>;

using VkSpecializationMapEntries = std::vector<VkSpecializationMapEntry>;

using VkVertexInputBindingDescriptions = std::vector<VkVertexInputBindingDescription>;

using VkVertexInputAttributeDescriptions = std::vector<VkVertexInputAttributeDescription>;

using VkPipelineColorBlendAttachmentStates = std::vector<VkPipelineColorBlendAttachmentState>;

using VkDynamicStates = std::vector<VkDynamicState>;

using VkQueueFamilyPropertiesList = std::vector<VkQueueFamilyProperties>;

using VkExtensionPropertiesList = std::vector<VkExtensionProperties>;

using VkLayerPropertiesList = std::vector<VkLayerProperties>;

using VkExtensionPropertiesList = std::vector<VkExtensionProperties>;

using VkPhysicalDevices = std::vector<VkPhysicalDevice>;

bool check(VkResult result);

inline bool failed(VkResult result) {
    return !check(result);
}

string to_string(VkResult result);

string version_to_string(ui32 version);

int_version to_version(ui32 version);

ui32 to_version(int_version version);
```

```

enum class api_version : index {
    v1_0 = 0,
    v1_1,
    v1_2,
    v1_3
};

api_version to_api_version(ui32 version);

struct vk_result {
    bool state = false;

    VkResult value = VK_NOT_READY;

    operator bool() {
        return state;
    }
};

constexpr bool const build_failed = false;

constexpr bool const build_done = true;

using VkAttachments = std::vector<VkImageViews>;

using VkAttachmentsRef = VkAttachments const&;

struct target_callback {
    using cptr = target_callback const*;

    using list = std::vector<target_callback*>;

    using clist = std::vector<cptr>;

    using created_func = std::function<bool(VkAttachmentsRef, rect::ref)>;

    created_func on_created;

    using destroyed_func = std::function<void()>;

    destroyed_func on_destroyed;
};

constexpr ui32 const Vk_Limit_DescriptorSets = 4;

constexpr ui32 const Vk_Limit_Bindings = 16;

constexpr ui32 const Vk_Limit_Attachments = 8;

constexpr ui32 const Vk_Limit_VertexAttribs = 16;

constexpr ui32 const Vk_Limit_VertexBuffers = 4;

constexpr ui32 const Vk_Limit_PushConstant_Size = 128;

constexpr ui32 const Vk_Limit_UBO_Size = 16 * 1024;

} // namespace lava
```

5.31 liblava/base/debug_utils.hpp File Reference

Debug utilities.

```
#include <liblava/base/base.hpp>
#include <liblava/core/def.hpp>
```

Classes

- struct lava::scoped_label< T >

  Scoped debug util label.

Macros

- #define **LAVA_DEBUG_UTILS** LAVA_DEBUG

  Only active in debug - enable for release profiling.

Functions

- void lava::begin_label (VkCommandBuffer cmd_buf, name label, v4 color)
- void lava::end_label (VkCommandBuffer cmd_buf)
- void lava::insert_label (VkCommandBuffer cmd_buf, name label, v4 color)
- void lava::begin_label (VkQueue queue, name label, v4 color)
- void lava::end_label (VkQueue queue)
- void lava::insert_label (VkQueue queue, name label, v4 color)
- void lava::set_object_name (VkDevice device, VkObjectType type, VkObjectHandle handle, name object)
- void lava::set_object_tag (VkDevice device, VkObjectType type, VkObjectHandle handle, ui64 name, void_cp tag, size_t size)
- void lava::set_name (VkDevice device, VkObjectHandle handle, name object)

  Set the object name.
- void lava::set_tag (VkDevice device, VkObjectHandle handle, ui64 name, void_cp tag, size_t size)

  Set the object tag.
- void lava::set_instance_name (VkDevice device, VkInstance handle, name object)
- void lava::set_instance_tag (VkDevice device, VkInstance handle, ui64 name, void_cp tag, size_t size)
- void lava::set_physical_device_name (VkDevice device, VkPhysicalDevice handle, name object)
- void lava::set_physical_device_tag (VkDevice device, VkPhysicalDevice handle, ui64 name, void_cp tag, size_t size)
- void lava::set_device_name (VkDevice device, name object)
- void lava::set_device_tag (VkDevice device, ui64 name, void_cp tag, size_t size)
- void lava::set_queue_name (VkDevice device, VkQueue handle, name object)
- void lava::set_queue_tag (VkDevice device, VkQueue handle, ui64 name, void_cp tag, size_t size)
- void lava::set_semaphore_name (VkDevice device, VkSemaphore handle, name object)
- void lava::set_semaphore_tag (VkDevice device, VkSemaphore handle, ui64 name, void_cp tag, size_t size)
- void lava::set_command_buffer_name (VkDevice device, VkCommandBuffer handle, name object)
- void lava::set_command_buffer_tag (VkDevice device, VkCommandBuffer handle, ui64 name, void_cp tag, size_t size)
- void lava::set_fence_name (VkDevice device, VkFence handle, name object)
- void lava::set_fence_tag (VkDevice device, VkFence handle, ui64 name, void_cp tag, size_t size)
- void lava::set_device_memory_name (VkDevice device, VkDeviceMemory handle, name object)
- void lava::set_device_memory_tag (VkDevice device, VkDeviceMemory handle, ui64 name, void_cp tag, size_t size)
- void lava::set_buffer_name (VkDevice device, VkBuffer handle, name object)
- void lava::set_buffer_tag (VkDevice device, VkBuffer handle, ui64 name, void_cp tag, size_t size)
- void lava::set_image_name (VkDevice device, VkImage handle, name object)
- void lava::set_image_tag (VkDevice device, VkImage handle, ui64 name, void_cp tag, size_t size)
- void lava::set_event_name (VkDevice device, VkEvent handle, name object)
- void lava::set_event_tag (VkDevice device, VkEvent handle, ui64 name, void_cp tag, size_t size)
- void lava::set_query_pool_name (VkDevice device, VkQueryPool handle, name object)
- void lava::set_query_pool_tag (VkDevice device, VkQueryPool handle, ui64 name, void_cp tag, size_t size)
- void lava::set_buffer_view_name (VkDevice device, VkBufferView handle, name object)
- void lava::set_buffer_view_tag (VkDevice device, VkBufferView handle, ui64 name, void_cp tag, size_t size)
- void lava::set_image_view_name (VkDevice device, VkImageView handle, name object)
- void lava::set_image_view_tag (VkDevice device, VkImageView handle, ui64 name, void_cp tag, size_t size)
- void lava::set_shader_module_name (VkDevice device, VkShaderModule handle, name object)
- void lava::set_shader_module_tag (VkDevice device, VkShaderModule handle, ui64 name, void_cp tag, size_t size)

- void lava::set_pipeline_cache_name (VkDevice device, VkPipelineCache handle, name object)
- void lava::set_pipeline_cache_tag (VkDevice device, VkPipelineCache handle, ui64 name, void_cp tag, size_t size)
- void lava::set_pipeline_layout_name (VkDevice device, VkPipelineLayout handle, name object)
- void lava::set_pipeline_layout_tag (VkDevice device, VkPipelineLayout handle, ui64 name, void_cp tag, size_t size)
- void lava::set_render_pass_name (VkDevice device, VkRenderPass handle, name object)
- void lava::set_render_pass_tag (VkDevice device, VkRenderPass handle, ui64 name, void_cp tag, size_t size)
- void lava::set_pipeline_name (VkDevice device, VkPipeline handle, name object)
- void lava::set_pipeline_tag (VkDevice device, VkPipeline handle, ui64 name, void_cp tag, size_t size)
- void lava::set_descriptor_set_layout_name (VkDevice device, VkDescriptorSetLayout handle, name object)
- void lava::set_descriptor_set_layout_tag (VkDevice device, VkDescriptorSetLayout handle, ui64 name, void↩_cp tag, size_t size)
- void lava::set_sampler_name (VkDevice device, VkSampler handle, name object)
- void lava::set_sampler_tag (VkDevice device, VkSampler handle, ui64 name, void_cp tag, size_t size)
- void lava::set_descriptor_pool_name (VkDevice device, VkDescriptorPool handle, name object)
- void lava::set_descriptor_pool_tag (VkDevice device, VkDescriptorPool handle, ui64 name, void_cp tag, size_t size)
- void lava::set_descriptor_set_name (VkDevice device, VkDescriptorSet handle, name object)
- void lava::set_descriptor_set_tag (VkDevice device, VkDescriptorSet handle, ui64 name, void_cp tag, size_t size)
- void lava::set_framebuffer_name (VkDevice device, VkFramebuffer handle, name object)
- void lava::set_framebuffer_tag (VkDevice device, VkFramebuffer handle, ui64 name, void_cp tag, size_t size)
- void lava::set_command_pool_name (VkDevice device, VkCommandPool handle, name object)
- void lava::set_command_pool_tag (VkDevice device, VkCommandPool handle, ui64 name, void_cp tag, size_t size)
- void lava::set_sampler_ycbcr_conversion_name (VkDevice device, VkSamplerYcbcrConversion handle, name object)
- void lava::set_sampler_ycbcr_conversion_tag (VkDevice device, VkSamplerYcbcrConversion handle, ui64 name, void_cp tag, size_t size)
- void lava::set_descriptor_update_template_name (VkDevice device, VkDescriptorUpdateTemplate handle, name object)
- void lava::set_descriptor_update_template_tag (VkDevice device, VkDescriptorUpdateTemplate handle, ui64 name, void_cp tag, size_t size)
- void lava::set_surface_name (VkDevice device, VkSurfaceKHR handle, name object)
- void lava::set_surface_tag (VkDevice device, VkSurfaceKHR handle, ui64 name, void_cp tag, size_t size)
- void lava::set_swapchain_name (VkDevice device, VkSwapchainKHR handle, name object)
- void lava::set_swapchain_tag (VkDevice device, VkSwapchainKHR handle, ui64 name, void_cp tag, size_t size)
- void lava::set_display_name (VkDevice device, VkDisplayKHR handle, name object)
- void lava::set_display_tag (VkDevice device, VkDisplayKHR handle, ui64 name, void_cp tag, size_t size)
- void lava::set_display_mode_name (VkDevice device, VkDisplayModeKHR handle, name object)
- void lava::set_display_mode_tag (VkDevice device, VkDisplayModeKHR handle, ui64 name, void_cp tag, size_t size)
- void lava::set_debug_report_callback_name (VkDevice device, VkDebugReportCallbackEXT handle, name object)
- void lava::set_debug_report_callback_tag (VkDevice device, VkDebugReportCallbackEXT handle, ui64 name, void_cp tag, size_t size)
- void lava::set_indirect_commands_layout_name (VkDevice device, VkIndirectCommandsLayoutNV handle, name object)
- void lava::set_indirect_commands_layout_tag (VkDevice device, VkIndirectCommandsLayoutNV handle, ui64 name, void_cp tag, size_t size)
- void lava::set_debug_utils_messenger_name (VkDevice device, VkDebugUtilsMessengerEXT handle, name object)

- void lava::set_debug_utils_messenger_tag (VkDevice device, VkDebugUtilsMessengerEXT handle, ui64 name, void_cp tag, size_t size)
- void lava::set_validation_cache_name (VkDevice device, VkValidationCacheEXT handle, name object)
- void lava::set_validation_cache_tag (VkDevice device, VkValidationCacheEXT handle, ui64 name, void_cp tag, size_t size)
- void lava::set_acceleration_structure_nv_name (VkDevice device, VkAccelerationStructureNV handle, name object)
- void lava::set_acceleration_structure_nv_tag (VkDevice device, VkAccelerationStructureNV handle, ui64 name, void_cp tag, size_t size)
- void lava::set_acceleration_structure_name (VkDevice device, VkAccelerationStructureKHR handle, name object)
- void lava::set_acceleration_structure_tag (VkDevice device, VkAccelerationStructureKHR handle, ui64 name, void_cp tag, size_t size)
- void lava::set_performance_configuration_name (VkDevice device, VkPerformanceConfigurationINTEL handle, name object)
- void lava::set_performance_configuration_tag (VkDevice device, VkPerformanceConfigurationINTEL handle, ui64 name, void_cp tag, size_t size)
- void lava::set_deferred_operation_name (VkDevice device, VkDeferredOperationKHR handle, name object)
- void lava::set_deferred_operation_tag (VkDevice device, VkDeferredOperationKHR handle, ui64 name, void_cp tag, size_t size)
- void lava::set_private_data_slot_name (VkDevice device, VkPrivateDataSlotEXT handle, name object)
- void lava::set_private_data_slot_tag (VkDevice device, VkPrivateDataSlotEXT handle, ui64 name, void_cp tag, size_t size)

5.31.1 Detailed Description

Debug utilities.

Authors

Lava Block OÜ and contributors

Copyright

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5.31.2 Function Documentation

5.31.2.1 begin_label() [1/2]

```
void lava::begin_label (
            VkCommandBuffer cmd_buf,
            name label,
            v4 color )  [inline]
```

See also

begin_label

5.31.2.2 begin_label() [2/2]

```
void lava::begin_label (
            VkQueue queue,
            name label,
            v4 color )  [inline]
```

See also

begin_label

5.31.2.3 end_label() [1/2]

```
void lava::end_label (
            VkCommandBuffer cmd_buf )  [inline]
```

See also

end_label

5.31.2.4 end_label() [2/2]

```
void lava::end_label (
            VkQueue queue )  [inline]
```

See also

end_label

5.31.2.5 insert_label() [1/2]

```
void lava::insert_label (
            VkCommandBuffer cmd_buf,
            name label,
            v4 color )  [inline]
```

See also

insert_label

5.31.2.6 insert_label() [2/2]

```
void lava::insert_label (
            VkQueue queue,
            name label,
            v4 color )  [inline]
```

See also

insert_label

5.31.2.7 set_acceleration_structure_name()

```
void lava::set_acceleration_structure_name (
            VkDevice device,
            VkAccelerationStructureKHR handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.8 set_acceleration_structure_nv_name()

```
void lava::set_acceleration_structure_nv_name (
            VkDevice device,
            VkAccelerationStructureNV handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.9 set_acceleration_structure_nv_tag()

```
void lava::set_acceleration_structure_nv_tag (
            VkDevice device,
            VkAccelerationStructureNV handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.10 set_acceleration_structure_tag()

```
void lava::set_acceleration_structure_tag (
            VkDevice device,
            VkAccelerationStructureKHR handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.11 set_buffer_name()

```
void lava::set_buffer_name (
            VkDevice device,
            VkBuffer handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.12 set_buffer_tag()

```
void lava::set_buffer_tag (
            VkDevice device,
            VkBuffer handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.13 set_buffer_view_name()

```
void lava::set_buffer_view_name (
            VkDevice device,
            VkBufferView handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.14 set_buffer_view_tag()

```
void lava::set_buffer_view_tag (
            VkDevice device,
            VkBufferView handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.15 set_command_buffer_name()

```
void lava::set_command_buffer_name (
            VkDevice device,
            VkCommandBuffer handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.16 set_command_buffer_tag()

```
void lava::set_command_buffer_tag (
            VkDevice device,
            VkCommandBuffer handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.17 set_command_pool_name()

```
void lava::set_command_pool_name (
            VkDevice device,
            VkCommandPool handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.18 set_command_pool_tag()

```
void lava::set_command_pool_tag (
            VkDevice device,
            VkCommandPool handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.19 set_debug_report_callback_name()

```
void lava::set_debug_report_callback_name (
            VkDevice device,
            VkDebugReportCallbackEXT handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.20 set_debug_report_callback_tag()

```
void lava::set_debug_report_callback_tag (
            VkDevice device,
            VkDebugReportCallbackEXT handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.21 set_debug_utils_messenger_name()

```
void lava::set_debug_utils_messenger_name (
            VkDevice device,
            VkDebugUtilsMessengerEXT handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.22 set_debug_utils_messenger_tag()

```
void lava::set_debug_utils_messenger_tag (
            VkDevice device,
            VkDebugUtilsMessengerEXT handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.23 set_deferred_operation_name()

```
void lava::set_deferred_operation_name (
            VkDevice device,
            VkDeferredOperationKHR handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.24 set_deferred_operation_tag()

```
void lava::set_deferred_operation_tag (
            VkDevice device,
            VkDeferredOperationKHR handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.25 set_descriptor_pool_name()

```
void lava::set_descriptor_pool_name (
            VkDevice device,
            VkDescriptorPool handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.26 set_descriptor_pool_tag()

```
void lava::set_descriptor_pool_tag (
            VkDevice device,
            VkDescriptorPool handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.27 set_descriptor_set_layout_name()

```
void lava::set_descriptor_set_layout_name (
            VkDevice device,
            VkDescriptorSetLayout handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.28 set_descriptor_set_layout_tag()

```
void lava::set_descriptor_set_layout_tag (
            VkDevice device,
            VkDescriptorSetLayout handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.29 set_descriptor_set_name()

```
void lava::set_descriptor_set_name (
            VkDevice device,
            VkDescriptorSet handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.30 set_descriptor_set_tag()

```
void lava::set_descriptor_set_tag (
            VkDevice device,
            VkDescriptorSet handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.31 set_descriptor_update_template_name()

```
void lava::set_descriptor_update_template_name (
            VkDevice device,
            VkDescriptorUpdateTemplate handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.32 set_descriptor_update_template_tag()

```
void lava::set_descriptor_update_template_tag (
            VkDevice device,
            VkDescriptorUpdateTemplate handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.33 set_device_memory_name()

```
void lava::set_device_memory_name (
            VkDevice device,
            VkDeviceMemory handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.34 set_device_memory_tag()

```
void lava::set_device_memory_tag (
            VkDevice device,
            VkDeviceMemory handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.35 set_device_name()

```
void lava::set_device_name (
            VkDevice device,
            name object )  [inline]
```

See also

set_name()

5.31.2.36 set_device_tag()

```
void lava::set_device_tag (
            VkDevice device,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.37 set_display_mode_name()

```
void lava::set_display_mode_name (
            VkDevice device,
            VkDisplayModeKHR handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.38 set_display_mode_tag()

```
void lava::set_display_mode_tag (
            VkDevice device,
            VkDisplayModeKHR handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.39 set_display_name()

```
void lava::set_display_name (
            VkDevice device,
            VkDisplayKHR handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.40 set_display_tag()

```
void lava::set_display_tag (
            VkDevice device,
            VkDisplayKHR handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.41 set_event_name()

```
void lava::set_event_name (
            VkDevice device,
            VkEvent handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.42 set_event_tag()

```
void lava::set_event_tag (
            VkDevice device,
            VkEvent handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.43 set_fence_name()

```
void lava::set_fence_name (
            VkDevice device,
            VkFence handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.44 set_fence_tag()

```
void lava::set_fence_tag (
            VkDevice device,
            VkFence handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.45 set_framebuffer_name()

```
void lava::set_framebuffer_name (
            VkDevice device,
            VkFramebuffer handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.46 set_framebuffer_tag()

```
void lava::set_framebuffer_tag (
            VkDevice device,
            VkFramebuffer handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.47 set_image_name()

```
void lava::set_image_name (
            VkDevice device,
            VkImage handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.48 set_image_tag()

```
void lava::set_image_tag (
            VkDevice device,
            VkImage handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.49 set_image_view_name()

```
void lava::set_image_view_name (
            VkDevice device,
            VkImageView handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.50 set_image_view_tag()

```
void lava::set_image_view_tag (
            VkDevice device,
            VkImageView handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.51 set_indirect_commands_layout_name()

```
void lava::set_indirect_commands_layout_name (
            VkDevice device,
            VkIndirectCommandsLayoutNV handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.52 set_indirect_commands_layout_tag()

```
void lava::set_indirect_commands_layout_tag (
            VkDevice device,
            VkIndirectCommandsLayoutNV handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.53 set_instance_name()

```
void lava::set_instance_name (
            VkDevice device,
            VkInstance handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.54 set_instance_tag()

```
void lava::set_instance_tag (
            VkDevice device,
            VkInstance handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.55 set_name()

```
void lava::set_name (
            VkDevice device,
            VkObjectHandle handle,
            name object )  [inline]
```

Set the object name.

See also

https://www.khronos.org/registry/vulkan/specs/1.3-extensions/man/html/↩ VkObjectType.html

Parameters

device	Vulkan device
handle	Object handle
object	Name of object

5.31.2.56 set_object_name()

```
void lava::set_object_name (
            VkDevice device,
            VkObjectType type,
            VkObjectHandle handle,
            name object )  [inline]
```

See also

set_object_name

5.31.2.57 set_object_tag()

```
void lava::set_object_tag (
            VkDevice device,
            VkObjectType type,
            VkObjectHandle handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_object_tag

5.31.2.58 set_performance_configuration_name()

```
void lava::set_performance_configuration_name (
            VkDevice device,
            VkPerformanceConfigurationINTEL handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.59 set_performance_configuration_tag()

```
void lava::set_performance_configuration_tag (
            VkDevice device,
            VkPerformanceConfigurationINTEL handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.60 set_physical_device_name()

```
void lava::set_physical_device_name (
            VkDevice device,
            VkPhysicalDevice handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.61 set_physical_device_tag()

```
void lava::set_physical_device_tag (
            VkDevice device,
            VkPhysicalDevice handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.62 set_pipeline_cache_name()

```
void lava::set_pipeline_cache_name (
            VkDevice device,
            VkPipelineCache handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.63 set_pipeline_cache_tag()

```
void lava::set_pipeline_cache_tag (
            VkDevice device,
            VkPipelineCache handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.64 set_pipeline_layout_name()

```
void lava::set_pipeline_layout_name (
            VkDevice device,
            VkPipelineLayout handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.65 set_pipeline_layout_tag()

```
void lava::set_pipeline_layout_tag (
            VkDevice device,
            VkPipelineLayout handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.66 set_pipeline_name()

```
void lava::set_pipeline_name (
            VkDevice device,
            VkPipeline handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.67 set_pipeline_tag()

```
void lava::set_pipeline_tag (
            VkDevice device,
            VkPipeline handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.68 set_private_data_slot_name()

```
void lava::set_private_data_slot_name (
            VkDevice device,
            VkPrivateDataSlotEXT handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.69 set_private_data_slot_tag()

```
void lava::set_private_data_slot_tag (
            VkDevice device,
            VkPrivateDataSlotEXT handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.70 set_query_pool_name()

```
void lava::set_query_pool_name (
            VkDevice device,
            VkQueryPool handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.71 set_query_pool_tag()

```
void lava::set_query_pool_tag (
            VkDevice device,
            VkQueryPool handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.72 set_queue_name()

```
void lava::set_queue_name (
            VkDevice device,
            VkQueue handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.73 set_queue_tag()

```
void lava::set_queue_tag (
            VkDevice device,
            VkQueue handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.74 set_render_pass_name()

```
void lava::set_render_pass_name (
            VkDevice device,
            VkRenderPass handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.75 set_render_pass_tag()

```
void lava::set_render_pass_tag (
            VkDevice device,
            VkRenderPass handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.76 set_sampler_name()

```
void lava::set_sampler_name (
            VkDevice device,
            VkSampler handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.77 set_sampler_tag()

```
void lava::set_sampler_tag (
            VkDevice device,
            VkSampler handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.78 set_sampler_ycbcr_conversion_name()

```
void lava::set_sampler_ycbcr_conversion_name (
            VkDevice device,
            VkSamplerYcbcrConversion handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.79 set_sampler_ycbcr_conversion_tag()

```
void lava::set_sampler_ycbcr_conversion_tag (
            VkDevice device,
            VkSamplerYcbcrConversion handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.80 set_semaphore_name()

```
void lava::set_semaphore_name (
            VkDevice device,
            VkSemaphore handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.81 set_semaphore_tag()

```
void lava::set_semaphore_tag (
            VkDevice device,
            VkSemaphore handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.82 set_shader_module_name()

```
void lava::set_shader_module_name (
            VkDevice device,
            VkShaderModule handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.83 set_shader_module_tag()

```
void lava::set_shader_module_tag (
            VkDevice device,
            VkShaderModule handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.84 set_surface_name()

```
void lava::set_surface_name (
            VkDevice device,
            VkSurfaceKHR handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.85 set_surface_tag()

```
void lava::set_surface_tag (
            VkDevice device,
            VkSurfaceKHR handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.86 set_swapchain_name()

```
void lava::set_swapchain_name (
            VkDevice device,
            VkSwapchainKHR handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.87 set_swapchain_tag()

```
void lava::set_swapchain_tag (
            VkDevice device,
            VkSwapchainKHR handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.31.2.88 set_tag()

```
void lava::set_tag (
            VkDevice device,
            VkObjectHandle handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

Set the object tag.

See also

https://www.khronos.org/registry/vulkan/specs/1.3-extensions/man/html/↩ VkObjectType.html

Parameters

device	Vulkan device
handle	Object handle
name	Name of tag
tag	Tag data
size	Size of tag data

5.31.2.89 set_validation_cache_name()

```
void lava::set_validation_cache_name (
            VkDevice device,
            VkValidationCacheEXT handle,
            name object )  [inline]
```

See also

set_name()

5.31.2.90 set_validation_cache_tag()

```
void lava::set_validation_cache_tag (
            VkDevice device,
            VkValidationCacheEXT handle,
            ui64 name,
            void_cp tag,
            size_t size )  [inline]
```

See also

set_tag()

5.32 debug_utils.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/base/base.hpp>
11 #include <liblava/core/def.hpp>
12
14 #ifndef LAVA_DEBUG_UTILS
15     #define LAVA_DEBUG_UTILS LAVA_DEBUG
16 #endif
17
18 namespace lava {
19
20 #if LAVA_DEBUG_UTILS
21
28 void begin_label(VkCommandBuffer cmd_buf,
29                  name label,
```

```
                 v4 color);

void end_label(VkCommandBuffer cmd_buf);

void insert_label(VkCommandBuffer cmd_buf,
                  name label,
                  v4 color);

void begin_label(VkQueue queue,
                 name label,
                 v4 color);

void end_label(VkQueue queue);

void insert_label(VkQueue queue,
                  name label,
                  v4 color);

void set_object_name(VkDevice device,
                     VkObjectType type,
                     VkObjectHandle handle,
                     name object);

void set_object_tag(VkDevice device,
                    VkObjectType type,
                    VkObjectHandle handle,
                    ui64 name,
                    void_cp tag,
                     size_t size);

#else

inline void begin_label(VkCommandBuffer cmd_buf,
                        name label,
                        v4 color) {}

inline void end_label(VkCommandBuffer cmd_buf) {}

inline void insert_label(VkCommandBuffer cmd_buf,
                         name label,
                         v4 color) {}

inline void begin_label(VkQueue queue,
                        name label,
                        v4 color) {}

inline void end_label(VkQueue queue) {}

inline void insert_label(VkQueue queue,
                         name label,
                         v4 color) {}

inline void set_object_name(VkDevice device,
                            VkObjectType type,
                            VkObjectHandle handle,
                            name object) {}

inline void set_object_tag(VkDevice device,
                           VkObjectType type,
                           VkObjectHandle handle,
                           ui64 name,
                           void_cp tag,
                           size_t size) {}

#endif

template<typename T>
struct scoped_label {
    scoped_label(T scope, name label, v4 color = v4(0.f))
    : scope(scope) {
        begin_label(scope, label, color);
    }

    ~scoped_label() {
        end_label(scope);
    }

private:
    T scope;
};

inline void set_name(VkDevice device,
                     VkObjectHandle handle,
                     name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_UNKNOWN,
                    handle,
```

```
                    object);
}

inline void set_tag(VkDevice device,
                    VkObjectHandle handle,
                    ui64 name,
                    void_cp tag,
                    size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_UNKNOWN,
                   handle,
                   name,
                   tag,
                   size);
}

// ---

inline void set_instance_name(VkDevice device,
                              VkInstance handle,
                              name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_INSTANCE,
                    VkObjectHandle(handle),
                    object);
}

inline void set_instance_tag(VkDevice device,
                             VkInstance handle,
                             ui64 name,
                             void_cp tag,
                             size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_INSTANCE,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_physical_device_name(VkDevice device,
                                     VkPhysicalDevice handle,
                                     name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_PHYSICAL_DEVICE,
                    VkObjectHandle(handle),
                    object);
}

inline void set_physical_device_tag(VkDevice device,
                                    VkPhysicalDevice handle,
                                    ui64 name,
                                    void_cp tag,
                                    size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_PHYSICAL_DEVICE,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_device_name(VkDevice device,
                            name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_DEVICE,
                    VkObjectHandle(device),
                    object);
}

inline void set_device_tag(VkDevice device,
                           ui64 name,
                           void_cp tag,
                           size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_DEVICE,
                   VkObjectHandle(device),
                   name,
                   tag,
                   size);
}

// ---
```

```
inline void set_queue_name(VkDevice device,
                           VkQueue handle,
                           name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_QUEUE,
                    VkObjectHandle(handle),
                    object);
}

inline void set_queue_tag(VkDevice device,
                          VkQueue handle,
                          ui64 name,
                          void_cp tag,
                          size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_QUEUE,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_semaphore_name(VkDevice device,
                               VkSemaphore handle,
                               name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_SEMAPHORE,
                    VkObjectHandle(handle),
                    object);
}

inline void set_semaphore_tag(VkDevice device,
                              VkSemaphore handle,
                              ui64 name,
                              void_cp tag,
                              size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_SEMAPHORE,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_command_buffer_name(VkDevice device,
                                    VkCommandBuffer handle,
                                    name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_COMMAND_BUFFER,
                    VkObjectHandle(handle),
                    object);
}

inline void set_command_buffer_tag(VkDevice device,
                                   VkCommandBuffer handle,
                                   ui64 name,
                                   void_cp tag,
                                   size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_COMMAND_BUFFER,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_fence_name(VkDevice device,
                           VkFence handle,
                           name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_FENCE,
                    VkObjectHandle(handle),
                    object);
}

inline void set_fence_tag(VkDevice device,
                          VkFence handle,
                          ui64 name,
                          void_cp tag,
                          size_t size) {
```

```
    set_object_tag(device,
                   VK_OBJECT_TYPE_FENCE,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_device_memory_name(VkDevice device,
                                   VkDeviceMemory handle,
                                   name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_DEVICE_MEMORY,
                    VkObjectHandle(handle),
                    object);
}

inline void set_device_memory_tag(VkDevice device,
                                  VkDeviceMemory handle,
                                  ui64 name,
                                  void_cp tag,
                                  size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_DEVICE_MEMORY,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_buffer_name(VkDevice device,
                            VkBuffer handle,
                            name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_BUFFER,
                    VkObjectHandle(handle),
                    object);
}

inline void set_buffer_tag(VkDevice device,
                           VkBuffer handle,
                           ui64 name,
                           void_cp tag,
                           size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_BUFFER,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_image_name(VkDevice device,
                           VkImage handle,
                           name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_IMAGE,
                    VkObjectHandle(handle),
                    object);
}

inline void set_image_tag(VkDevice device,
                          VkImage handle,
                          ui64 name,
                          void_cp tag,
                          size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_IMAGE,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_event_name(VkDevice device,
                           VkEvent handle,
                           name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_EVENT,
```

```
                    VkObjectHandle(handle),
                    object);
}

inline void set_event_tag(VkDevice device,
                          VkEvent handle,
                          ui64 name,
                          void_cp tag,
                          size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_EVENT,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_query_pool_name(VkDevice device,
                                VkQueryPool handle,
                                name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_QUERY_POOL,
                    VkObjectHandle(handle),
                    object);
}

inline void set_query_pool_tag(VkDevice device,
                               VkQueryPool handle,
                               ui64 name,
                               void_cp tag,
                               size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_QUERY_POOL,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_buffer_view_name(VkDevice device,
                                 VkBufferView handle,
                                 name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_BUFFER_VIEW,
                    VkObjectHandle(handle),
                    object);
}

inline void set_buffer_view_tag(VkDevice device,
                                VkBufferView handle,
                                ui64 name,
                                void_cp tag,
                                size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_BUFFER_VIEW,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_image_view_name(VkDevice device,
                                VkImageView handle,
                                name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_IMAGE_VIEW,
                    VkObjectHandle(handle),
                    object);
}

inline void set_image_view_tag(VkDevice device,
                               VkImageView handle,
                               ui64 name,
                               void_cp tag,
                               size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_IMAGE_VIEW,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
```

```
}

// ---

inline void set_shader_module_name(VkDevice device,
                                   VkShaderModule handle,
                                   name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_SHADER_MODULE,
                    VkObjectHandle(handle),
                    object);
}

inline void set_shader_module_tag(VkDevice device,
                                  VkShaderModule handle,
                                  ui64 name,
                                  void_cp tag,
                                  size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_SHADER_MODULE,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_pipeline_cache_name(VkDevice device,
                                    VkPipelineCache handle,
                                    name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_PIPELINE_CACHE,
                    VkObjectHandle(handle),
                    object);
}

inline void set_pipeline_cache_tag(VkDevice device,
                                   VkPipelineCache handle,
                                   ui64 name,
                                   void_cp tag,
                                   size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_PIPELINE_CACHE,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_pipeline_layout_name(VkDevice device,
                                     VkPipelineLayout handle,
                                     name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_PIPELINE_LAYOUT,
                    VkObjectHandle(handle),
                    object);
}

inline void set_pipeline_layout_tag(VkDevice device,
                                    VkPipelineLayout handle,
                                    ui64 name,
                                    void_cp tag,
                                    size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_PIPELINE_LAYOUT,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_render_pass_name(VkDevice device,
                                 VkRenderPass handle,
                                 name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_RENDER_PASS,
                    VkObjectHandle(handle),
                    object);
}

inline void set_render_pass_tag(VkDevice device,
                                VkRenderPass handle,
```

```
                                ui64 name,
                                void_cp tag,
                                size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_RENDER_PASS,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_pipeline_name(VkDevice device,
                              VkPipeline handle,
                              name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_PIPELINE,
                    VkObjectHandle(handle),
                    object);
}

inline void set_pipeline_tag(VkDevice device,
                             VkPipeline handle,
                             ui64 name,
                             void_cp tag,
                             size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_PIPELINE,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_descriptor_set_layout_name(VkDevice device,
                                           VkDescriptorSetLayout handle,
                                           name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_DESCRIPTOR_SET_LAYOUT,
                    VkObjectHandle(handle),
                    object);
}

inline void set_descriptor_set_layout_tag(VkDevice device,
                                          VkDescriptorSetLayout handle,
                                          ui64 name,
                                          void_cp tag,
                                          size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_DESCRIPTOR_SET_LAYOUT,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_sampler_name(VkDevice device,
                             VkSampler handle,
                             name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_SAMPLER,
                    VkObjectHandle(handle),
                    object);
}

inline void set_sampler_tag(VkDevice device,
                            VkSampler handle,
                            ui64 name,
                            void_cp tag,
                            size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_SAMPLER,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_descriptor_pool_name(VkDevice device,
                                     VkDescriptorPool handle,
```

```
                                    name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_DESCRIPTOR_POOL,
                    VkObjectHandle(handle),
                    object);
}

inline void set_descriptor_pool_tag(VkDevice device,
                                    VkDescriptorPool handle,
                                    ui64 name,
                                    void_cp tag,
                                    size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_DESCRIPTOR_POOL,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_descriptor_set_name(VkDevice device,
                                    VkDescriptorSet handle,
                                    name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_DESCRIPTOR_SET,
                    VkObjectHandle(handle),
                    object);
}

inline void set_descriptor_set_tag(VkDevice device,
                                   VkDescriptorSet handle,
                                   ui64 name,
                                   void_cp tag,
                                   size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_DESCRIPTOR_SET,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_framebuffer_name(VkDevice device,
                                 VkFramebuffer handle,
                                 name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_FRAMEBUFFER,
                    VkObjectHandle(handle),
                    object);
}

inline void set_framebuffer_tag(VkDevice device,
                                VkFramebuffer handle,
                                ui64 name,
                                void_cp tag,
                                size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_FRAMEBUFFER,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_command_pool_name(VkDevice device,
                                  VkCommandPool handle,
                                  name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_COMMAND_POOL,
                    VkObjectHandle(handle),
                    object);
}

inline void set_command_pool_tag(VkDevice device,
                                 VkCommandPool handle,
                                 ui64 name,
                                 void_cp tag,
                                 size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_COMMAND_POOL,
                   VkObjectHandle(handle),
```

```
                    name,
                    tag,
                    size);
}

// ---

inline void set_sampler_ycbcr_conversion_name(VkDevice device,
                                              VkSamplerYcbcrConversion handle,
                                              name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION,
                    VkObjectHandle(handle),
                    object);
}

inline void set_sampler_ycbcr_conversion_tag(VkDevice device,
                                             VkSamplerYcbcrConversion handle,
                                             ui64 name,
                                             void_cp tag,
                                             size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_descriptor_update_template_name(VkDevice device,
                                                VkDescriptorUpdateTemplate handle,
                                                name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_DESCRIPTOR_UPDATE_TEMPLATE,
                    VkObjectHandle(handle),
                    object);
}

inline void set_descriptor_update_template_tag(VkDevice device,
                                               VkDescriptorUpdateTemplate handle,
                                               ui64 name,
                                               void_cp tag,
                                               size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_DESCRIPTOR_UPDATE_TEMPLATE,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_surface_name(VkDevice device,
                             VkSurfaceKHR handle,
                             name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_SURFACE_KHR,
                    VkObjectHandle(handle),
                    object);
}

inline void set_surface_tag(VkDevice device,
                            VkSurfaceKHR handle,
                            ui64 name,
                            void_cp tag,
                            size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_SURFACE_KHR,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_swapchain_name(VkDevice device,
                               VkSwapchainKHR handle,
                               name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_SWAPCHAIN_KHR,
                    VkObjectHandle(handle),
                    object);
}
```

```
inline void set_swapchain_tag(VkDevice device,
                              VkSwapchainKHR handle,
                              ui64 name,
                              void_cp tag,
                              size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_SWAPCHAIN_KHR,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_display_name(VkDevice device,
                             VkDisplayKHR handle,
                             name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_DISPLAY_KHR,
                    VkObjectHandle(handle),
                    object);
}

inline void set_display_tag(VkDevice device,
                            VkDisplayKHR handle,
                            ui64 name,
                            void_cp tag,
                            size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_DISPLAY_KHR,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_display_mode_name(VkDevice device,
                                  VkDisplayModeKHR handle,
                                  name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_DISPLAY_MODE_KHR,
                    VkObjectHandle(handle),
                    object);
}

inline void set_display_mode_tag(VkDevice device,
                                 VkDisplayModeKHR handle,
                                 ui64 name,
                                 void_cp tag,
                                 size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_DISPLAY_MODE_KHR,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_debug_report_callback_name(VkDevice device,
                                           VkDebugReportCallbackEXT handle,
                                           name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_DEBUG_REPORT_CALLBACK_EXT,
                    VkObjectHandle(handle),
                    object);
}

inline void set_debug_report_callback_tag(VkDevice device,
                                          VkDebugReportCallbackEXT handle,
                                          ui64 name,
                                          void_cp tag,
                                          size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_DEBUG_REPORT_CALLBACK_EXT,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---
```

```
inline void set_indirect_commands_layout_name(VkDevice device,
                                              VkIndirectCommandsLayoutNV handle,
                                              name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_INDIRECT_COMMANDS_LAYOUT_NV,
                    VkObjectHandle(handle),
                    object);
}

inline void set_indirect_commands_layout_tag(VkDevice device,
                                             VkIndirectCommandsLayoutNV handle,
                                             ui64 name,
                                             void_cp tag,
                                             size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_INDIRECT_COMMANDS_LAYOUT_NV,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_debug_utils_messenger_name(VkDevice device,
                                           VkDebugUtilsMessengerEXT handle,
                                           name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_DEBUG_UTILS_MESSENGER_EXT,
                    VkObjectHandle(handle),
                    object);
}

inline void set_debug_utils_messenger_tag(VkDevice device,
                                          VkDebugUtilsMessengerEXT handle,
                                          ui64 name,
                                          void_cp tag,
                                          size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_DEBUG_UTILS_MESSENGER_EXT,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_validation_cache_name(VkDevice device,
                                      VkValidationCacheEXT handle,
                                      name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_VALIDATION_CACHE_EXT,
                    VkObjectHandle(handle),
                    object);
}

inline void set_validation_cache_tag(VkDevice device,
                                     VkValidationCacheEXT handle,
                                     ui64 name,
                                     void_cp tag,
                                     size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_VALIDATION_CACHE_EXT,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_acceleration_structure_nv_name(VkDevice device,
                                               VkAccelerationStructureNV handle,
                                               name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_ACCELERATION_STRUCTURE_NV,
                    VkObjectHandle(handle),
                    object);
}

inline void set_acceleration_structure_nv_tag(VkDevice device,
                                              VkAccelerationStructureNV handle,
                                              ui64 name,
                                              void_cp tag,
                                              size_t size) {
```

```
    set_object_tag(device,
                   VK_OBJECT_TYPE_ACCELERATION_STRUCTURE_NV,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_acceleration_structure_name(VkDevice device,
                                            VkAccelerationStructureKHR handle,
                                            name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_ACCELERATION_STRUCTURE_KHR,
                    VkObjectHandle(handle),
                    object);
}

inline void set_acceleration_structure_tag(VkDevice device,
                                           VkAccelerationStructureKHR handle,
                                           ui64 name,
                                           void_cp tag,
                                           size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_ACCELERATION_STRUCTURE_KHR,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_performance_configuration_name(VkDevice device,
                                               VkPerformanceConfigurationINTEL handle,
                                               name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_PERFORMANCE_CONFIGURATION_INTEL,
                    VkObjectHandle(handle),
                    object);
}

inline void set_performance_configuration_tag(VkDevice device,
                                              VkPerformanceConfigurationINTEL handle,
                                              ui64 name,
                                              void_cp tag,
                                              size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_PERFORMANCE_CONFIGURATION_INTEL,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_deferred_operation_name(VkDevice device,
                                        VkDeferredOperationKHR handle,
                                        name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_DEFERRED_OPERATION_KHR,
                    VkObjectHandle(handle),
                    object);
}

inline void set_deferred_operation_tag(VkDevice device,
                                       VkDeferredOperationKHR handle,
                                       ui64 name,
                                       void_cp tag,
                                       size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_DEFERRED_OPERATION_KHR,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

// ---

inline void set_private_data_slot_name(VkDevice device,
                                       VkPrivateDataSlotEXT handle,
                                       name object) {
    set_object_name(device,
                    VK_OBJECT_TYPE_PRIVATE_DATA_SLOT_EXT,
```

```
                    VkObjectHandle(handle),
                    object);
}

inline void set_private_data_slot_tag(VkDevice device,
                                      VkPrivateDataSlotEXT handle,
                                      ui64 name,
                                      void_cp tag,
                                      size_t size) {
    set_object_tag(device,
                   VK_OBJECT_TYPE_PRIVATE_DATA_SLOT_EXT,
                   VkObjectHandle(handle),
                   name,
                   tag,
                   size);
}

} // namespace lava
```

5.33 liblava/base/device.hpp File Reference

Vulkan device.

```
#include <liblava/base/device_table.hpp>
#include <liblava/base/queue.hpp>
#include <liblava/core/data.hpp>
#include <liblava/core/id.hpp>
#include <liblava/fwd.hpp>
```

Classes

- struct lava::device

  Vulkan device.
- struct lava::device::create_param

  Device create parameters.

Typedefs

- using **lava::device_p** = device ∗

  Pointer to device.
- using **lava::device_cp** = device const ∗

  Const pointer to device.
- using **lava::physical_device_cp** = physical_device const ∗

  Const pointer to a physical device.
- using **lava::one_time_command_func** = std::function< void(VkCommandBuffer)>

  One time command function.

Functions

- VkShaderModule lava::create_shader_module (device_p device, cdata::ref data)

  Create a shader module.
- bool lava::one_time_submit_pool (device_p device, VkCommandPool pool, queue::ref queue, one_time_command_func callback)

  Submit one time command function with pool.
- bool lava::one_time_submit (device_p device, queue::ref queue, one_time_command_func callback)

  Submit one time command function.

5.33.1 Detailed Description

Vulkan device.

Authors

Lava Block OÜ and contributors

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5.33.2 Function Documentation

5.33.2.1 create_shader_module()

```
VkShaderModule lava::create_shader_module (
            device_p device,
            cdata::ref data )
```

Create a shader module.

Parameters

device	Vulkan device
data	Shader data

Returns

VkShaderModule Shader module

5.33.2.2 one_time_submit()

```
bool lava::one_time_submit (
            device_p device,
            queue::ref queue,
            one_time_command_func callback )  [inline]
```

Submit one time command function.

Parameters

device	Vulkan device
queue	Target queue
callback	Function to be called

Returns

Submit was successful or failed

5.33.2.3 one_time_submit_pool()

```
bool lava::one_time_submit_pool (
            device_p device,
            VkCommandPool pool,
            queue::ref queue,
            one_time_command_func callback )
```

Submit one time command function with pool.

Parameters

device	Vulkan device
pool	Command pool (VK_NULL_HANDLE: managed)
queue	Target queue
callback	Function to be called

Returns

Submit was successful or failed

5.34 device.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/base/device_table.hpp>
11 #include <liblava/base/queue.hpp>
12 #include <liblava/core/data.hpp>
13 #include <liblava/core/id.hpp>
14 #include <liblava/fwd.hpp>
15
16 namespace lava {
17
19 using device_p = device*;
20
22 using device_cp = device const*;
23
25 using physical_device_cp = physical_device const*;
26
30 struct device : device_table, entity {
32     using ptr = std::shared_ptr<device>;
33
35     using list = std::vector<ptr>;
36
40     struct create_param {
42         using ref = create_param const&;
43
45         physical_device_cp physical_device = nullptr;
46
48         VmaAllocatorCreateFlags vma_flags = 0;
49
51         names extensions;
52
54         VkPhysicalDeviceFeatures features{};
55
```

```
        bool has_features_2 = false;

        void const* next = nullptr;

        queue_family_info::list queue_family_infos;

        void add_swapchain_extension() {
            extensions.push_back("VK_KHR_swapchain");
        }

        void add_portability_subset_extension() {
            extensions.push_back("VK_KHR_portability_subset");
        }

        void set_default_queues() {
            lava::set_default_queues(queue_family_infos);
        }

        void set_all_queues();

        bool add_queue(VkQueueFlags flags,
                       r32 priority = 1.f) {
            return add_queues(flags, 1, priority);
        }

        bool add_queues(VkQueueFlags flags,
                        ui32 count,
                        r32 priority = 1.f);

        bool add_dedicated_queues(r32 priority = 1.f);

        verify_queues_result verify_queues() const;
    };

    static ptr make() {
        return std::make_shared<device>();
    }

    ~device() {
        destroy();
    }

    bool create(create_param::ref param);

    void destroy();

    queue::ref get_graphics_queue(index index = 0) const {
        return get_graphics_queues().at(index);
    }

    queue::ref graphics_queue(index index = 0) const {
        return get_graphics_queue(index);
    }

    queue::ref get_compute_queue(index index = 0) const {
        return get_compute_queues().at(index);
    }

    queue::ref compute_queue(index index = 0) const {
        return get_compute_queue(index);
    }

    queue::ref get_transfer_queue(index index = 0) const {
        return get_transfer_queues().at(index);
    }

    queue::ref transfer_queue(index index = 0) const {
        return get_transfer_queue(index);
    }

    queue::list const& get_graphics_queues() const {
        return graphics_queue_list;
    }

    queue::list const& graphics_queues() const {
        return get_graphics_queues();
    }

    queue::list const& get_compute_queues() const {
        return compute_queue_list;
    }

    queue::list const& compute_queues() const {
        return get_compute_queues();
    }

    queue::list const& get_transfer_queues() const {
```

```
        return transfer_queue_list;
    }

    queue::list const& transfer_queues() const {
        return get_transfer_queues();
    }

    queue::list const& get_queues() const {
        return queue_list;
    }

    queue::list const& queues() const {
        return get_queues();
    }

    VkDevice get() const {
        return vk_device;
    }

    VolkDeviceTable const& call() const {
        return table;
    }

    bool wait_for_idle() const {
        return check(call().vkDeviceWaitIdle(vk_device));
    }

    physical_device_cp get_physical_device() const {
        return physical_device;
    }

    VkPhysicalDevice get_vk_physical_device() const;

    VkPhysicalDeviceFeatures const& get_features() const;

    VkPhysicalDeviceProperties const& get_properties() const;

    bool surface_supported(VkSurfaceKHR surface) const;

    void set_allocator(allocator::ptr value) {
        mem_allocator = value;
    }

    allocator::ptr get_allocator() {
        return mem_allocator;
    }

    VmaAllocator alloc() const {
        return mem_allocator != nullptr
                   ? mem_allocator->get()
                   : nullptr;
    }

private:
    physical_device_cp physical_device = nullptr;

    queue::list graphics_queue_list;

    queue::list compute_queue_list;

    queue::list transfer_queue_list;

    queue::list queue_list;

    VkPhysicalDeviceFeatures features{};

    allocator::ptr mem_allocator;
};

VkShaderModule create_shader_module(device_p device,
                                    cdata::ref data);

using one_time_command_func = std::function<void(VkCommandBuffer)>;

bool one_time_submit_pool(device_p device,
                          VkCommandPool pool,
                          queue::ref queue,
                          one_time_command_func callback);

inline bool one_time_submit(device_p device,
                            queue::ref queue,
                            one_time_command_func callback) {
    return one_time_submit_pool(device,
                                VK_NULL_HANDLE,
                                queue,
                                callback);
}
```

```
408
409 } // namespace lava
```

5.35 liblava/base/device_table.hpp File Reference

Device functions.

```
#include <liblava/base/memory.hpp>
```

Classes

- struct lava::device_table

  Device functions.

5.35.1 Detailed Description

Device functions.

Authors

Lava Block OÜ and contributors

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5.36 device_table.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/base/memory.hpp>
11
12 namespace lava {
13
17 struct device_table {
21     void load_table() {
22         volkLoadDeviceTable(&table, vk_device);
23     }
24
28     vk_result vkCreateImageView(const VkImageViewCreateInfo* pCreateInfo,
29                                 const VkAllocationCallbacks* pAllocator,
30                                 VkImageView* pView) {
31         auto result = table.vkCreateImageView(vk_device,
32                                               pCreateInfo,
33                                               pAllocator,
34                                               pView);
35         return { check(result), result };
36     }
37
41     vk_result vkCreateImageView(const VkImageViewCreateInfo* pCreateInfo,
42                                 VkImageView* pView) {
43         return vkCreateImageView(pCreateInfo,
44                                  memory::instance().alloc(),
45                                  pView);
46     }
47
```

```
vk_result vkCreateSampler(const VkSamplerCreateInfo* pCreateInfo,
                          const VkAllocationCallbacks* pAllocator,
                          VkSampler* pSampler) {
    auto result = table.vkCreateSampler(vk_device,
                                        pCreateInfo,
                                        pAllocator,
                                        pSampler);
    return { check(result), result };
}

vk_result vkCreateSampler(const VkSamplerCreateInfo* pCreateInfo,
                          VkSampler* pSampler) {
    return vkCreateSampler(pCreateInfo,
                           memory::instance().alloc(),
                           pSampler);
}

vk_result vkCreateShaderModule(const VkShaderModuleCreateInfo* pCreateInfo,
                               const VkAllocationCallbacks* pAllocator,
                               VkShaderModule* pShaderModule) {
    auto result = table.vkCreateShaderModule(vk_device,
                                             pCreateInfo,
                                             pAllocator,
                                             pShaderModule);
    return { check(result), result };
}

vk_result vkCreateShaderModule(const VkShaderModuleCreateInfo* pCreateInfo,
                               VkShaderModule* pShaderModule) {
    return vkCreateShaderModule(pCreateInfo,
                                memory::instance().alloc(),
                                pShaderModule);
}

vk_result vkCreateFence(const VkFenceCreateInfo* pCreateInfo,
                        const VkAllocationCallbacks* pAllocator,
                        VkFence* pFence) {
    auto result = table.vkCreateFence(vk_device,
                                      pCreateInfo,
                                      pAllocator,
                                      pFence);
    return { check(result), result };
}

vk_result vkCreateFence(const VkFenceCreateInfo* pCreateInfo,
                        VkFence* pFence) {
    return vkCreateFence(pCreateInfo,
                         memory::instance().alloc(),
                         pFence);
}

vk_result vkCreateSemaphore(const VkSemaphoreCreateInfo* pCreateInfo,
                            const VkAllocationCallbacks* pAllocator,
                            VkSemaphore* pSemaphore) {
    auto result = table.vkCreateSemaphore(vk_device,
                                          pCreateInfo,
                                          pAllocator,
                                          pSemaphore);
    return { check(result), result };
}

vk_result vkCreateSemaphore(const VkSemaphoreCreateInfo* pCreateInfo,
                            VkSemaphore* pSemaphore) {
    return vkCreateSemaphore(pCreateInfo,
                             memory::instance().alloc(),
                             pSemaphore);
}

vk_result vkWaitForFences(uint32_t fenceCount,
                          const VkFence* pFences,
                          VkBool32 waitAll,
                          uint64_t timeout) {
    auto result = table.vkWaitForFences(vk_device,
                                        fenceCount,
                                        pFences,
                                        waitAll,
                                        timeout);
    if ((result == VK_TIMEOUT) && (timeout != UINT64_MAX))
        return { false, result };

    return { check(result), result };
}

vk_result vkResetFences(uint32_t fenceCount,
                        const VkFence* pFences) {
    auto result = table.vkResetFences(vk_device,
                                      fenceCount,
```

```
                                        pFences);
        return { check(result), result };
    }

    vk_result vkQueueSubmit(VkQueue queue,
                            uint32_t submitCount,
                            const VkSubmitInfo* pSubmits,
                            VkFence fence) {
        auto result = table.vkQueueSubmit(queue,
                                          submitCount,
                                          pSubmits,
                                          fence);
        return { check(result), result };
    }

    vk_result vkAcquireNextImageKHR(VkSwapchainKHR swapchain,
                                    uint64_t timeout,
                                    VkSemaphore semaphore,
                                    VkFence fence,
                                    uint32_t* pImageIndex) {
        auto result = table.vkAcquireNextImageKHR(vk_device,
                                                  swapchain,
                                                  timeout,
                                                  semaphore,
                                                  fence,
                                                  pImageIndex);
        return { check(result), result };
    }

    vk_result vkQueuePresentKHR(VkQueue queue,
                                const VkPresentInfoKHR* pPresentInfo) {
        auto result = table.vkQueuePresentKHR(queue,
                                              pPresentInfo);
        return { check(result), result };
    }

    vk_result vkCreateSwapchainKHR(const VkSwapchainCreateInfoKHR* pCreateInfo,
                                   const VkAllocationCallbacks* pAllocator,
                                   VkSwapchainKHR* pSwapchain) {
        auto result = table.vkCreateSwapchainKHR(vk_device,
                                                 pCreateInfo,
                                                 pAllocator,
                                                 pSwapchain);
        return { check(result), result };
    }

    vk_result vkCreateSwapchainKHR(const VkSwapchainCreateInfoKHR* pCreateInfo,
                                   VkSwapchainKHR* pSwapchain) {
        return vkCreateSwapchainKHR(pCreateInfo,
                                    memory::instance().alloc(),
                                    pSwapchain);
    }

    void vkDestroySwapchainKHR(VkSwapchainKHR swapchain,
                               const VkAllocationCallbacks* pAllocator = memory::instance().alloc()) {
        table.vkDestroySwapchainKHR(vk_device,
                                    swapchain,
                                    pAllocator);
    }

    vk_result vkGetSwapchainImagesKHR(VkSwapchainKHR swapchain,
                                      uint32_t* pSwapchainImageCount, VkImage* pSwapchainImages) {
        auto result = table.vkGetSwapchainImagesKHR(vk_device,
                                                     swapchain,
                                                     pSwapchainImageCount,
                                                     pSwapchainImages);
        return { check(result), result };
    }

    vk_result vkCreateCommandPool(const VkCommandPoolCreateInfo* pCreateInfo,
                                  const VkAllocationCallbacks* pAllocator,
                                  VkCommandPool* pCommandPool) {
        auto result = table.vkCreateCommandPool(vk_device,
                                                pCreateInfo,
                                                pAllocator,
                                                pCommandPool);
        return { check(result), result };
    }

    vk_result vkCreateCommandPool(const VkCommandPoolCreateInfo* pCreateInfo,
                                  VkCommandPool* pCommandPool) {
        return vkCreateCommandPool(pCreateInfo,
                                   memory::instance().alloc(),
                                   pCommandPool);
    }

    vk_result vkCreateCommandPool(index queue_family,
```

```
                                VkCommandPool* pCommandPool) {
    VkCommandPoolCreateInfo create_info{
        .sType = VK_STRUCTURE_TYPE_COMMAND_POOL_CREATE_INFO,
        .queueFamilyIndex = queue_family,
    };
    return vkCreateCommandPool(&create_info,
                               pCommandPool);
}

vk_result vkAllocateCommandBuffers(const VkCommandBufferAllocateInfo* pAllocateInfo,
                                   VkCommandBuffer* pCommandBuffers) {
    auto result = table.vkAllocateCommandBuffers(vk_device,
                                                 pAllocateInfo,
                                                 pCommandBuffers);
    return { check(result), result };
}

vk_result vkAllocateCommandBuffers(VkCommandPool commandPool,
                                   uint32_t commandBufferCount,
                                   VkCommandBuffer* pCommandBuffers,
                                   VkCommandBufferLevel level = VK_COMMAND_BUFFER_LEVEL_PRIMARY) {
    VkCommandBufferAllocateInfo alloc_info{
        .sType = VK_STRUCTURE_TYPE_COMMAND_BUFFER_ALLOCATE_INFO,
        .commandPool = commandPool,
        .level = level,
        .commandBufferCount = commandBufferCount,
    };
    return vkAllocateCommandBuffers(&alloc_info,
                                    pCommandBuffers);
}

void vkDestroyImageView(VkImageView imageView,
                        const VkAllocationCallbacks* pAllocator = memory::instance().alloc()) {
    table.vkDestroyImageView(vk_device,
                             imageView,
                             pAllocator);
}

void vkDestroyFence(VkFence fence,
                    const VkAllocationCallbacks* pAllocator = memory::instance().alloc()) {
    table.vkDestroyFence(vk_device,
                         fence,
                         pAllocator);
}

void vkDestroySemaphore(VkSemaphore semaphore,
                        const VkAllocationCallbacks* pAllocator = memory::instance().alloc()) {
    table.vkDestroySemaphore(vk_device,
                             semaphore,
                             pAllocator);
}

void vkFreeCommandBuffers(VkCommandPool commandPool,
                          uint32_t commandBufferCount,
                          const VkCommandBuffer* pCommandBuffers) {
    table.vkFreeCommandBuffers(vk_device,
                               commandPool,
                               commandBufferCount,
                               pCommandBuffers);
}

void vkDestroyCommandPool(VkCommandPool commandPool,
                          const VkAllocationCallbacks* pAllocator = memory::instance().alloc()) {
    table.vkDestroyCommandPool(vk_device,
                               commandPool,
                               pAllocator);
}

void vkDestroySampler(VkSampler sampler,
                      const VkAllocationCallbacks* pAllocator = memory::instance().alloc()) {
    table.vkDestroySampler(vk_device,
                           sampler,
                           pAllocator);
}

void vkUpdateDescriptorSets(uint32_t descriptorWriteCount,
                            const VkWriteDescriptorSet* pDescriptorWrites,
                            uint32_t descriptorCopyCount = 0,
                            const VkCopyDescriptorSet* pDescriptorCopies = nullptr) {
    table.vkUpdateDescriptorSets(vk_device,
                                 descriptorWriteCount,
                                 pDescriptorWrites,
                                 descriptorCopyCount,
                                 pDescriptorCopies);
}

template<std::size_t SIZE>
```

```
    void vkUpdateDescriptorSets(std::array<VkWriteDescriptorSet, SIZE> const& descriptor_writes) {
        vkUpdateDescriptorSets(to_i32(descriptor_writes.size()),
                               descriptor_writes.data());
    }

    template<std::size_t SIZE>
    void vkUpdateDescriptorSets(std::array<VkCopyDescriptorSet, SIZE> const& descriptor_copies) {
        vkUpdateDescriptorSets(0,
                               nullptr,
                               to_i32(descriptor_copies.size()),
                               descriptor_copies.data());
    }

    template<std::size_t WRITE_SIZE, std::size_t COPY_SIZE>
    void vkUpdateDescriptorSets(std::array<VkWriteDescriptorSet, WRITE_SIZE> const& descriptor_writes,
                                std::array<VkCopyDescriptorSet, COPY_SIZE> const& descriptor_copies) {
        vkUpdateDescriptorSets(to_i32(descriptor_writes.size()),
                               descriptor_writes.data(),
                               to_i32(descriptor_copies.size()),
                               descriptor_copies.data());
    }

    void vkUpdateDescriptorSets(std::initializer_list<VkWriteDescriptorSet> descriptor_writes) {
        vkUpdateDescriptorSets(to_i32(descriptor_writes.size()),
                               descriptor_writes.begin());
    }

    void vkUpdateDescriptorSets(std::initializer_list<VkCopyDescriptorSet> descriptor_copies) {
        vkUpdateDescriptorSets(0,
                               nullptr,
                               to_i32(descriptor_copies.size()),
                               descriptor_copies.begin());
    }

    void vkUpdateDescriptorSets(std::initializer_list<VkWriteDescriptorSet> descriptor_writes,
                                std::initializer_list<VkCopyDescriptorSet> descriptor_copies) {
        vkUpdateDescriptorSets(to_i32(descriptor_writes.size()),
                               descriptor_writes.begin(),
                               to_i32(descriptor_copies.size()),
                               descriptor_copies.begin());
    }

    VkDevice vk_device = nullptr;

    VolkDeviceTable table = {};
};

} // namespace lava
```

5.37 liblava/base/instance.hpp File Reference

Vulkan instance.

```
#include <liblava/base/physical_device.hpp>
```

Classes

- struct lava::instance_info

  Vulkan instance information.
- struct lava::instance

  Vulkan instance.
- struct lava::instance::create_param

  Instance create parameters.
- struct lava::instance::debug_config

  Debug configuration.

Functions

- bool lava::check (instance::create_param::ref param)

  Check instance create parameters.
- int_version lava::get_instance_version ()

  Get the instance version.
- VkLayerPropertiesList lava::enumerate_layer_properties ()

  Enumerate enabled layer properties.
- VkExtensionPropertiesList lava::enumerate_extension_properties (name layer_name=nullptr)

  Enumerate enabled extension properties.

5.37.1 Detailed Description

Vulkan instance.

Authors

Lava Block OÜ and contributors

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5.37.2 Function Documentation

5.37.2.1 check()

```
bool lava::check (
            instance::create_param::ref param )
```

Check instance create parameters.

Parameters

param	Create parameters

Returns

Check was successful or failed

5.37.2.2 enumerate_extension_properties()

```
VkExtensionPropertiesList lava::enumerate_extension_properties (
            name layer_name = nullptr )
```

Enumerate enabled extension properties.

Parameters

layer_name	Name of layer

Returns

VkExtensionPropertiesList List of extension properties

5.37.2.3 enumerate_layer_properties()

```
VkLayerPropertiesList lava::enumerate_layer_properties ( )
```

Enumerate enabled layer properties.

Returns

VkLayerPropertiesList List of layer properties

5.37.2.4 get_instance_version()

```
int_version lava::get_instance_version ( )
```

Get the instance version.

Returns

int_version Version

5.38 instance.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/base/physical_device.hpp>
11
12 namespace lava {
13
17 struct instance_info {
19     using ref = instance_info const&;
20
22     name app_name = _lava_;
23
25     name engine_name = _liblava_;
26
28     int_version app_version;
29
31     int_version engine_version;
32
34     api_version req_api_version = api_version::v1_0;
35 };
36
40 struct instance : no_copy_no_move {
```

```
    struct create_param {
        using ref = create_param const&;

        names layers{};

        names extensions{};
    };

    struct debug_config {
        using ref = debug_config const&;

        bool validation = false;

        bool render_doc = false;

        bool verbose = false;

        bool utils = false;
    };

    static instance& singleton() {
        static instance instance;
        return instance;
    }

    bool create(create_param& param,
                debug_config::ref debug,
                instance_info::ref info);

    void destroy();

    physical_device::list const& get_physical_devices() const {
        return physical_devices;
    }

    physical_device::ref get_first_physical_device() const {
        return *physical_devices.front().get();
    }

    VkInstance get() const {
        return vk_instance;
    }

    debug_config::ref get_debug_config() const {
        return debug;
    }

    instance_info::ref get_info() const {
        return info;
    }

private:
    explicit instance() = default;

    ~instance();

    bool check_debug(create_param& param) const;

    bool enumerate_physical_devices();

    bool create_validation_report();

    void destroy_validation_report();

    VkInstance vk_instance = nullptr;

    physical_device::list physical_devices;

    debug_config debug;

    instance_info info;

    VkDebugUtilsMessengerEXT debug_messenger = VK_NULL_HANDLE;
};

bool check(instance::create_param::ref param);

int_version get_instance_version();

VkLayerPropertiesList enumerate_layer_properties();

VkExtensionPropertiesList enumerate_extension_properties(name layer_name = nullptr);

} // namespace lava
```

5.39 liblava/base/memory.hpp File Reference

Vulkan allocator.

```
#include <liblava/base/base.hpp>
#include <liblava/fwd.hpp>
#include <vk_mem_alloc.h>
#include <memory>
```

Classes

- struct lava::allocator

  Vulkan allocator.
- struct lava::memory

  Vulkan memory.

Typedefs

- using **lava::device_cptr** = device const ∗

  Const pointer to device.

Functions

- allocator::ptr lava::create_allocator (device_cptr device, VmaAllocatorCreateFlags flags=0)

  Create a allocator.
- index lava::find_memory_type_with_properties (VkPhysicalDeviceMemoryProperties properties, ui32 type↩_bits, VkMemoryPropertyFlags required_properties)

  Find the memory type with properties.
- index lava::find_memory_type (VkPhysicalDevice gpu, VkMemoryPropertyFlags properties, ui32 type_bits)

  Find the memory type.

5.39.1 Detailed Description

Vulkan allocator.

Authors

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5.39.2 Function Documentation

5.39.2.1 create_allocator()

```
allocator::ptr lava::create_allocator (
            device_cptr device,
            VmaAllocatorCreateFlags flags = 0 )  [inline]
```

Create a allocator.

Parameters

device	Vulkan device
flags	VMA allocator create flags

Returns

allocator::ptr Allocator

5.39.2.2 find_memory_type()

```
index lava::find_memory_type (
            VkPhysicalDevice gpu,
            VkMemoryPropertyFlags properties,
            ui32 type_bits )
```

Find the memory type.

Parameters

gpu	Physical device
properties	Memory properties flags
type_bits	Type bits

Returns

type Result type

5.39.2.3 find_memory_type_with_properties()

```
index lava::find_memory_type_with_properties (
            VkPhysicalDeviceMemoryProperties properties,
            ui32 type_bits,
            VkMemoryPropertyFlags required_properties )
```

Find the memory type with properties.

Parameters

properties	Physical device memory properties
type_bits	Type bits
required_properties	Memory property flags

Returns

type Result type

5.40 memory.hpp

Go to the documentation of this file.

```
#pragma once

// clang-format off

#include <liblava/base/base.hpp>
#include <liblava/fwd.hpp>
#include <vk_mem_alloc.h>
#include <memory>

// clang-format on

namespace lava {

using device_cptr = device const*;

struct allocator {
    using ptr = std::shared_ptr<allocator>;

    static ptr make() {
        return std::make_shared<allocator>();
    }

    allocator() = default;

    explicit allocator(VmaAllocator allocator)
    : vma_allocator(allocator) {}

    bool create(device_cptr device,
                VmaAllocatorCreateFlags flags = 0);

    void destroy();

    bool valid() const {
        return vma_allocator != nullptr;
    }

    VmaAllocator get() const {
        return vma_allocator;
    }

private:
    VmaAllocator vma_allocator = nullptr;
};

inline allocator::ptr create_allocator(device_cptr device,
                                       VmaAllocatorCreateFlags flags = 0) {
    auto result = allocator::make();
    if (!result->create(device, flags))
        return nullptr;

    return result;
}

struct memory : no_copy_no_move {
    memory();

    static memory& instance() {
        static memory memory;
        return memory;
    }

    VkAllocationCallbacks* alloc() {
        if (use_custom_cpu_callbacks)
            return &vk_callbacks;

        return nullptr;
    }

    void set_callbacks(VkAllocationCallbacks const& callbacks) {
        vk_callbacks = callbacks;
    }

    void set_use_custom_cpu_callbacks(bool value) {
```

```
        use_custom_cpu_callbacks = value;
    }

private:
    bool use_custom_cpu_callbacks = true;

    VkAllocationCallbacks vk_callbacks = {};
};

index find_memory_type_with_properties(VkPhysicalDeviceMemoryProperties properties,
                                       ui32 type_bits,
                                       VkMemoryPropertyFlags required_properties);

index find_memory_type(VkPhysicalDevice gpu,
                       VkMemoryPropertyFlags properties,
                       ui32 type_bits);

} // namespace lava
```

5.41 liblava/base/physical_device.hpp File Reference

Vulkan physical device.

```
#include <liblava/base/device.hpp>
```

Classes

- struct lava::physical_device

  Vulkan physical device.

5.41.1 Detailed Description

Vulkan physical device.

Authors

Lava Block OÜ and contributors

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5.42 physical_device.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/base/device.hpp>
11
12 namespace lava {
13
17 struct physical_device : entity {
19     using ptr = std::shared_ptr<physical_device>;
20
22     using list = std::vector<ptr>;
23
25     using ref = physical_device const&;
```

```
    static ptr make(VkPhysicalDevice vk_physical_device) {
        return std::make_shared<physical_device>(vk_physical_device);
    }

    physical_device() = default;

    physical_device(VkPhysicalDevice vk_physical_device);

    void initialize(VkPhysicalDevice vk_physical_device);

    bool supported(string_ref extension) const;

    bool get_queue_family(index& index, VkQueueFlags flags) const;

    device::create_param create_default_device_param() const;

    VkPhysicalDeviceProperties const& get_properties() const {
        return properties;
    }

    VkPhysicalDeviceFeatures const& get_features() const {
        return features;
    }

    VkPhysicalDeviceMemoryProperties const& get_memory_properties() const {
        return memory_properties;
    }

    VkQueueFamilyPropertiesList const& get_queue_family_properties() const {
        return queue_family_properties;
    }

    VkExtensionPropertiesList const& get_extension_properties() const {
        return extension_properties;
    }

    VkPhysicalDevice get() const {
        return vk_physical_device;
    }

    name get_device_name() const;

    string get_device_type_string() const;

    int_version get_driver_version() const;

    bool swapchain_supported() const;

    bool surface_supported(index queue_family,
                           VkSurfaceKHR surface) const;

private:
    VkPhysicalDevice vk_physical_device = nullptr;

    VkPhysicalDeviceProperties properties = {};

    VkPhysicalDeviceFeatures features = {};

    VkPhysicalDeviceMemoryProperties memory_properties = {};

    VkQueueFamilyPropertiesList queue_family_properties;

    VkExtensionPropertiesList extension_properties;
};

} // namespace lava
```

5.43 liblava/base/platform.hpp File Reference

Stage platform.

```
#include <liblava/base/device.hpp>
```

Classes

- struct lava::platform

  Stage platform.

Typedefs

- using **lava::platform_t** = platform

  Platform type.

5.43.1 Detailed Description

Stage platform.

Authors

Lava Block OÜ and contributors

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5.44 platform.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/base/device.hpp>
11
12 namespace lava {
13
17 struct platform {
23     device::ptr create(index physical_device = 0);
24
30     device::ptr create(device::create_param::ref param);
31
37     device_p create_device(index physical_device = 0);
38
43     device::list const& get_devices() const {
44         return devices;
45     }
46
50     void wait_idle();
51
57     bool remove(id::ref device_id);
58
62     void clear();
63
65     using create_param_func = std::function<void(device::create_param&)>;
66
68     create_param_func on_create_param;
69
70 private:
72     device::list devices;
73 };
74
76 using platform_t = platform;
77
78 } // namespace lava
```

5.45 liblava/base/queue.hpp File Reference

Device queue.

```
#include <deque>
#include <liblava/base/base.hpp>
```

Classes

- struct lava::queue

  Device queue.
- struct lava::queue_info

  Queue information.
- struct lava::queue_family_info

  Queue family information.

Enumerations

- enum class lava::verify_queues_result : index {
  ok = 0 , **empty_list** , **no_properties** , **duplicate_family_index** ,
  no_family_index , **no_queues** , **too_many_queues** , **no_compatible_flags** }

  Result of queue verifications.

Functions

- void lava::set_default_queues (queue_family_info::list &list)

  Set the default queues.
- void lava::set_all_queues (queue_family_info::list &list, VkQueueFamilyPropertiesList const &properties)

  Set all queues.
- bool lava::add_queues (queue_family_info::list &list, VkQueueFamilyPropertiesList const &properties, Vk↩QueueFlags flags, ui32 count, r32 priority=1.f)

  Add queues.
- bool lava::add_dedicated_queues (queue_family_info::list &list, VkQueueFamilyPropertiesList const &properties, r32 priority=1.f)

  Add dedicated queues.
- verify_queues_result lava::verify_queues (queue_family_info::list const &list, VkQueueFamilyPropertiesList const &properties)

  Verify queues.

Variables

- constexpr VkQueueFlags const lava::default_queue_flags

  Default queue flags.

5.45.1 Detailed Description

Device queue.

Authors

Lava Block OÜ and contributors

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5.45.2 Function Documentation

5.45.2.1 add_dedicated_queues()

```
bool lava::add_dedicated_queues (
            queue_family_info::list & list,
            VkQueueFamilyPropertiesList const & properties,
            r32 priority = 1.f )
```

Add dedicated queues.

Parameters

list	List of queue family informations
properties	List of queue family properties
priority	Queue priority

Returns

Add was successful or failed

5.45.2.2 add_queues()

```
bool lava::add_queues (
            queue_family_info::list & list,
            VkQueueFamilyPropertiesList const & properties,
            VkQueueFlags flags,
            ui32 count,
            r32 priority = 1.f )
```

Add queues.

Parameters

list	List of queue family informations
properties	List of queue family properties
flags	Queue flags
count	Number of queues
priority	Queue priority

Returns

Add was successful or failed

5.45.2.3 set_all_queues()

```
void lava::set_all_queues (
            queue_family_info::list & list,
            VkQueueFamilyPropertiesList const & properties )
```

Set all queues.

Parameters

list	List of queue family informations
properties	List of queue family properties

5.45.2.4 set_default_queues()

```
void lava::set_default_queues (
            queue_family_info::list & list )
```

Set the default queues.

Parameters

list	List of queue family informations

5.45.2.5 verify_queues()

```
verify_queues_result lava::verify_queues (
            queue_family_info::list const & list,
            VkQueueFamilyPropertiesList const & properties )
```

Verify queues.

Parameters

list	List of queue family informations
properties	List of queue family properties

Returns

verify_queues_result Verification result

5.45.3 Variable Documentation

5.45.3.1 default_queue_flags

```
constexpr VkQueueFlags const lava::default_queue_flags  [constexpr]
```

Initial value:

```
= VK_QUEUE_GRAPHICS_BIT
                                                        | VK_QUEUE_COMPUTE_BIT
                                                        | VK_QUEUE_TRANSFER_BIT
```

Default queue flags.

5.46 queue.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <deque>
11 #include <liblava/base/base.hpp>
12
13 namespace lava {
14
18 struct queue {
20     using list = std::deque<queue>;
21
23     using ref = queue const&;
24
26     VkQueue vk_queue = nullptr;
27
29     VkQueueFlags flags = 0;
30
32     index family = 0;
33
35     r32 priority = 1.f;
36
41     bool valid() const {
42         return vk_queue != nullptr;
43     }
44
50     bool operator<(queue const& other) const {
51         return priority < other.priority;
52     }
53 };
54
56 constexpr VkQueueFlags const default_queue_flags = VK_QUEUE_GRAPHICS_BIT
57                                                   | VK_QUEUE_COMPUTE_BIT
58                                                   | VK_QUEUE_TRANSFER_BIT;
59
63 struct queue_info {
65     using list = std::deque<queue_info>;
66
68     VkQueueFlags flags = default_queue_flags;
69
71     r32 priority = 1.f;
72 };
73
77 struct queue_family_info {
79     using list = std::deque<queue_family_info>;
80
82     index family_index = 0;
83
85     queue_info::list queues;
86
93     void add(VkQueueFlags flags,
94              ui32 count = 1,
95              r32 priority = 1.f) {
96         for (auto i = 0u; i < count; ++i) {
97             queue_info info{ flags, priority };
98             queues.push_back(info);
99         }
100     }
101
106     ui32 count() const {
107         return to_ui32(queues.size());
108     }
109
113     void clear() {
```

```
        queues.clear();
    }
};

void set_default_queues(queue_family_info::list& list);

void set_all_queues(queue_family_info::list& list,
                    VkQueueFamilyPropertiesList const& properties);

bool add_queues(queue_family_info::list& list,
                VkQueueFamilyPropertiesList const& properties,
                VkQueueFlags flags,
                ui32 count,
                r32 priority = 1.f);

bool add_dedicated_queues(queue_family_info::list& list,
                          VkQueueFamilyPropertiesList const& properties,
                          r32 priority = 1.f);

enum class verify_queues_result : index {
    ok = 0,
    empty_list,
    no_properties,
    duplicate_family_index,
    no_family_index,
    no_queues,
    too_many_queues,
    no_compatible_flags
};

verify_queues_result verify_queues(queue_family_info::list const& list,
                                   VkQueueFamilyPropertiesList const& properties);

} // namespace lava
```

5.47 liblava/block/attachment.hpp File Reference

Attachment description.

```
#include <liblava/base/base.hpp>
#include <memory>
```

Classes

- struct lava::attachment

  Attachment description.

5.47.1 Detailed Description

Attachment description.

Authors

Lava Block OÜ and contributors

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5.48 attachment.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/base/base.hpp>
#include <memory>

namespace lava {

struct attachment {
    using ptr = std::shared_ptr<attachment>;

    using list = std::vector<ptr>;

    static ptr make(VkFormat format = VK_FORMAT_UNDEFINED,
                    VkSampleCountFlagBits samples = VK_SAMPLE_COUNT_1_BIT) {
        return std::make_shared<attachment>(format, samples);
    }

    explicit attachment(VkFormat format = VK_FORMAT_UNDEFINED,
                        VkSampleCountFlagBits samples = VK_SAMPLE_COUNT_1_BIT) {
        description.flags = 0;
        description.format = format;
        description.samples = samples;
        description.loadOp = VK_ATTACHMENT_LOAD_OP_LOAD;
        description.storeOp = VK_ATTACHMENT_STORE_OP_STORE;
        description.stencilLoadOp = VK_ATTACHMENT_LOAD_OP_LOAD;
        description.stencilStoreOp = VK_ATTACHMENT_STORE_OP_STORE;
        description.initialLayout = VK_IMAGE_LAYOUT_UNDEFINED;
        description.finalLayout = VK_IMAGE_LAYOUT_UNDEFINED;
    }

    VkAttachmentDescription const& get_description() const {
        return description;
    }

    void set_format(VkFormat format) {
        description.format = format;
    }

    void set_samples(VkSampleCountFlagBits samples) {
        description.samples = samples;
    }

    void set_op(VkAttachmentLoadOp load_op, VkAttachmentStoreOp store_op) {
        set_load_op(load_op);
        set_store_op(store_op);
    }

    void set_load_op(VkAttachmentLoadOp load_op) {
        description.loadOp = load_op;
    }

    void set_store_op(VkAttachmentStoreOp store_op) {
        description.storeOp = store_op;
    }

    void set_stencil_op(VkAttachmentLoadOp load_op,
                        VkAttachmentStoreOp store_op) {
        set_stencil_load_op(load_op);
        set_stencil_store_op(store_op);
    }

    void set_stencil_load_op(VkAttachmentLoadOp load_op) {
        description.stencilLoadOp = load_op;
    }

    void set_stencil_store_op(VkAttachmentStoreOp store_op) {
        description.stencilStoreOp = store_op;
    }

    void set_layouts(VkImageLayout initial,
                     VkImageLayout final) {
        set_initial_layout(initial);
        set_final_layout(final);
    }

    void set_initial_layout(VkImageLayout layout) {
        description.initialLayout = layout;
    }

    void set_final_layout(VkImageLayout layout) {
        description.finalLayout = layout;
```

```
    }

private:
    VkAttachmentDescription description;
};

} // namespace lava
```

5.49 liblava/block.hpp File Reference

Block module.

```
#include <liblava/block/attachment.hpp>
#include <liblava/block/block.hpp>
#include <liblava/block/compute_pipeline.hpp>
#include <liblava/block/def.hpp>
#include <liblava/block/descriptor.hpp>
#include <liblava/block/pipeline.hpp>
#include <liblava/block/pipeline_layout.hpp>
#include <liblava/block/render_pass.hpp>
#include <liblava/block/render_pipeline.hpp>
#include <liblava/block/subpass.hpp>
```

5.49.1 Detailed Description

Block module.

Author

Lava Block OÜ and contributors

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5.50 block.hpp

Go to the documentation of this file.

```

#pragma once

#include <liblava/block/attachment.hpp>
#include <liblava/block/block.hpp>
#include <liblava/block/compute_pipeline.hpp>
#include <liblava/block/def.hpp>
#include <liblava/block/descriptor.hpp>
#include <liblava/block/pipeline.hpp>
#include <liblava/block/pipeline_layout.hpp>
#include <liblava/block/render_pass.hpp>
#include <liblava/block/render_pipeline.hpp>
#include <liblava/block/subpass.hpp>
```

5.51 liblava/block/block.hpp File Reference

Command buffer model.

```
#include <liblava/base/device.hpp>
```

Classes

- struct lava::command

  Block command.
- struct lava::block

  Block of commands.

Typedefs

- using **lava::block_t** = block

  Block type.

5.51.1 Detailed Description

Command buffer model.

Authors

Lava Block OÜ and contributors

Copyright

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5.52 block.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/base/device.hpp>
11
12 namespace lava {
13
17 struct command : entity {
19     using ptr = std::shared_ptr<command>;
20
22     using cptr = command const*;
23
25     using map = std::map<id, ptr>;
26
28     using clist = std::vector<cptr>;
29
31     VkCommandBuffers buffers = {};
32
34     using process_func = std::function<void(VkCommandBuffer)>;
35
37     process_func on_process;
38
```

```
    bool active = true;

    static ptr make() {
        return std::make_shared<command>();
    }

    bool create(device_p device,
                index frame_count,
                VkCommandPools command_pools);

    void destroy(device_p device,
                 VkCommandPools command_pools);
};

struct block : entity {
    using ptr = std::shared_ptr<block>;

    using cptr = command const*;

    using map = std::map<id, ptr>;

    using clist = std::vector<cptr>;

    static ptr make() {
        return std::make_shared<block>();
    }

    ~block() {
        destroy();
    }

    bool create(device_p device,
                index frame_count,
                index queue_family);

    void destroy();

    index get_frame_count() const {
        return to_index(cmd_pools.size());
    }

    id add_cmd(command::process_func func, bool active = true);

    id add_command(command::process_func func, bool active = true) {
        return add_cmd(func, active);
    }

    void remove_cmd(id::ref cmd_id);

    void remove_command(id::ref cmd_id) {
        remove_cmd(cmd_id);
    }

    bool process(index frame);

    index get_current_frame() const {
        return current_frame;
    }

    VkCommandBuffer get_command_buffer(id::ref cmd_id) const {
        return commands.at(cmd_id)->buffers.at(current_frame);
    }

    VkCommandBuffer get_command_buffer(id::ref cmd_id, index frame) const {
        return commands.at(cmd_id)->buffers.at(frame);
    }

    VkCommandBuffers get_buffers() {
        VkCommandBuffers result;

        for (auto& cmd : cmd_order)
            if (cmd->active)
                result.push_back(cmd->buffers.at(current_frame));

        return result;
    }

    command::map const& get_commands() const {
        return commands;
    }

    command::clist const& get_cmd_order() const {
        return cmd_order;
    }

    bool activated(id::ref cmd_id);

```

```
    bool set_active(id::ref cmd_id, bool active = true);

    device_p get_device() {
        return device;
    }

private:
    device_p device = nullptr;

    index current_frame = 0;

    VkCommandPools cmd_pools = {};

    command::map commands;

    command::clist cmd_order;
};

using block_t = block;

} // namespace lava
```

5.53 liblava/block/compute_pipeline.hpp File Reference

Compute pipeline.

```
#include <liblava/block/pipeline.hpp>
```

Classes

- struct lava::compute_pipeline

  Compute pipeline.

5.53.1 Detailed Description

Compute pipeline.

Authors

Lava Block OÜ and contributors

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5.54 compute_pipeline.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/block/pipeline.hpp>

namespace lava {

struct compute_pipeline : pipeline {
    using ptr = std::shared_ptr<compute_pipeline>;

    using map = std::map<id, ptr>;

    using list = std::vector<ptr>;

    static ptr make(device_p device,
                    VkPipelineCache pipeline_cache = 0) {
        return std::make_shared<compute_pipeline>(device, pipeline_cache);
    }

    using pipeline::pipeline;

    void bind(VkCommandBuffer cmdBuffer) override;

    bool set_shader_stage(cdata::ref data,
                          VkShaderStageFlagBits stage);

    void set(shader_stage::ptr const& stage) {
        shader_stage = stage;
    }

    shader_stage::ptr const& get_shader_stage() const {
        return shader_stage;
    }

    void copy_to(compute_pipeline* target) const;

    void copy_from(ptr const& source) {
        source->copy_to(this);
    }

private:
    bool setup() override;

    void teardown() override;

    shader_stage::ptr shader_stage;
};

} // namespace lava
```

5.55 liblava/block/descriptor.hpp File Reference

Descriptor definition.

```
#include <liblava/base/device.hpp>
```

Classes

- struct lava::descriptor

  Descriptor.
- struct lava::descriptor::binding

  Descriptor binding.
- struct lava::descriptor::pool

  Descriptor pool.

Typedefs

- using **lava::descriptor_ptr** = descriptor::ptr

  Shared pointer to descriptor.

5.55.1 Detailed Description

Descriptor definition.

Authors

Lava Block OÜ and contributors

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5.56 descriptor.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/base/device.hpp>
11
12 namespace lava {
13
17 struct descriptor : entity {
21     struct binding {
23         using ptr = std::shared_ptr<binding>;
24
26         using list = std::vector<ptr>;
27
33         static ptr make(index index) {
34             auto result = std::make_shared<binding>();
35             result->set(index);
36             result->set_count(1);
37             return result;
38         }
39
43         explicit binding();
44
49         VkDescriptorSetLayoutBinding const& get() const {
50             return vk_binding;
51         }
52
57         void set(index index) {
58             vk_binding.binding = index;
59         }
60
65         void set_type(VkDescriptorType descriptor_type) {
66             vk_binding.descriptorType = descriptor_type;
67         }
68
73         void set_count(ui32 descriptor_count) {
74             vk_binding.descriptorCount = descriptor_count;
75         }
76
81         void set_stage_flags(VkShaderStageFlags stage_flags) {
82             vk_binding.stageFlags = stage_flags;
83         }
84
89         void set_samplers(VkSampler const* immutable_samplers) {
90             vk_binding.pImmutableSamplers = immutable_samplers;
91         }
92
93     private:
95         VkDescriptorSetLayoutBinding vk_binding;
96     };
```

```
97
101     struct pool : entity {
103         using ptr = std::shared_ptr<pool>;
104
106         using list = std::vector<ptr>;
107
112         static ptr make() {
113             return std::make_shared<descriptor::pool>();
114         }
115
124         bool create(device_p device,
125                     VkDescriptorPoolSizesRef sizes,
126                     ui32 max = 1,
127                     VkDescriptorPoolCreateFlags flags =
128                         VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_SET_BIT);
129
133         void destroy();
134
139         VkDescriptorPool get() const {
140             return vk_pool;
141         }
142
147         device_p get_device() {
148             return device;
149         }
150
155         VkDescriptorPoolSizes const& get_sizes() const {
156             return sizes;
157         }
158
163         ui32 get_max() const {
164             return max;
165         }
166
167     private:
169         device_p device = nullptr;
170
172         VkDescriptorPool vk_pool = VK_NULL_HANDLE;
173
175         VkDescriptorPoolSizes sizes;
176
178         ui32 max = 0;
179     };
180
182     using ptr = std::shared_ptr<descriptor>;
183
185     using list = std::vector<ptr>;
186
191     static ptr make() {
192         return std::make_shared<descriptor>();
193     }
194
201     void add_binding(index binding,
202                      VkDescriptorType descriptor_type,
203                      VkShaderStageFlags stage_flags);
204
208     void clear_bindings() {
209         bindings.clear();
210     }
211
216     void add(binding::ptr const& binding) {
217         bindings.push_back(binding);
218     }
219
225     bool create(device_p device);
226
230     void destroy();
231
236     ui32 get_binding_count() const {
237         return to_ui32(bindings.size());
238     }
239
244     binding::list const& get_bindings() {
245         return bindings;
246     }
247
252     VkDescriptorSetLayout get() const {
253         return layout;
254     }
255
260     device_p get_device() {
261         return device;
262     }
263
269     VkDescriptorSet allocate_set(VkDescriptorPool pool);
270
274     VkDescriptorSet allocate(VkDescriptorPool pool) {
```

```
        return allocate_set(pool);
    }

    bool free_set(VkDescriptorSet& descriptor_set,
                  VkDescriptorPool pool);

    bool free(VkDescriptorSet& descriptor_set,
              VkDescriptorPool pool) {
        return free_set(descriptor_set, pool);
    }

    VkDescriptorSets allocate_sets(ui32 size,
                                   VkDescriptorPool pool);

    VkDescriptorSets allocate(ui32 size,
                              VkDescriptorPool pool) {
        return allocate_sets(size, pool);
    }

    bool free_sets(VkDescriptorSets& descriptor_sets,
                   VkDescriptorPool pool);

    bool free(VkDescriptorSets& descriptor_sets,
              VkDescriptorPool pool) {
        return free_sets(descriptor_sets, pool);
    }

private:
    device_p device = nullptr;

    VkDescriptorSetLayout layout = VK_NULL_HANDLE;

    binding::list bindings;
};

using descriptor_ptr = descriptor::ptr;

} // namespace lava
```

5.57 liblava/block/pipeline.hpp File Reference

Pipeline.

```
#include <liblava/block/pipeline_layout.hpp>
```

Classes

- struct lava::pipeline

  Pipeline.
- struct lava::pipeline::shader_stage

  Shader stage.

Functions

- pipeline::shader_stage::ptr lava::create_pipeline_shader_stage (device_p device, cdata::ref data, Vk↩ ShaderStageFlagBits stage)

  Create a new pipeline shader stage.

5.57.1 Detailed Description

Pipeline.

Authors

Lava Block OÜ and contributors

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5.57.2 Function Documentation

5.57.2.1 create_pipeline_shader_stage()

```
pipeline::shader_stage::ptr lava::create_pipeline_shader_stage (
            device_p device,
            cdata::ref data,
            VkShaderStageFlagBits stage )
```

Create a new pipeline shader stage.

Parameters

device	Vulkan device
data	Shader data
stage	Shader stage flag bits

Returns

pipeline::shader_stage::ptr Shared pointer to pipeline shader stage

5.58 pipeline.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/block/pipeline_layout.hpp>
11
12 namespace lava {
13
17 struct pipeline : entity {
19     using ptr = std::shared_ptr<pipeline>;
20
22     using list = std::vector<ptr>;
23
25     using process_func = std::function<void(VkCommandBuffer)>;
26
28     process_func on_process;
```

```
29
35      explicit pipeline(device_p device,
36                        VkPipelineCache pipeline_cache = 0);
37
41      ~pipeline() override;
42
47      bool create();
48
52      void destroy();
53
58      virtual void bind(VkCommandBuffer cmd_buf) = 0;
59
64      void set_active(bool value = true) {
65          active = value;
66      }
67
72      bool activated() const {
73          return active;
74      }
75
79      void toggle() {
80          active = !active;
81      }
82
87      void set_auto_bind(bool value = true) {
88          auto_bind_active = value;
89      }
90
95      bool auto_bind() const {
96          return auto_bind_active;
97      }
98
103      bool ready() const {
104          return vk_pipeline != VK_NULL_HANDLE;
105      }
106
111      VkPipeline get() const {
112          return vk_pipeline;
113      }
114
119      device_p get_device() {
120          return device;
121      }
122
127      pipeline_layout::ptr get_layout() const {
128          return layout;
129      }
130
135      void set_layout(pipeline_layout::ptr const& value) {
136          layout = value;
137      }
138
142      struct shader_stage {
144          using ptr = std::shared_ptr<shader_stage>;
145
147          using list = std::vector<ptr>;
148
154          static ptr make(VkShaderStageFlagBits stage) {
155              auto result = std::make_shared<shader_stage>();
156              result->set_stage(stage);
157              return result;
158          }
159
163          explicit shader_stage();
164
168          ~shader_stage();
169
174          void set_stage(VkShaderStageFlagBits stage) {
175              create_info.stage = stage;
176          }
177
182          void add_specialization_entry(VkSpecializationMapEntry const& specialization);
183
191          bool create(device_p device,
192                      cdata::ref shader_data,
193                      cdata::ref specialization_data = data());
194
198          void destroy();
199
204          VkPipelineShaderStageCreateInfo const& get_create_info() const {
205              return create_info;
206          }
207
208      private:
210          device_p device = nullptr;
211
213          VkPipelineShaderStageCreateInfo create_info;
```

```
        VkSpecializationInfo specialization_info;

        VkSpecializationMapEntries specialization_entries;

        data specialization_data_copy;
    };

protected:
    virtual bool setup() = 0;

    virtual void teardown() = 0;

    device_p device = nullptr;

    VkPipeline vk_pipeline = VK_NULL_HANDLE;

    VkPipelineCache pipeline_cache = VK_NULL_HANDLE;

    pipeline_layout::ptr layout;

private:
    bool active = true;

    bool auto_bind_active = true;
};

pipeline::shader_stage::ptr create_pipeline_shader_stage(device_p device,
                                                         cdata::ref data,
                                                         VkShaderStageFlagBits stage);

} // namespace lava
```

5.59 liblava/block/pipeline_layout.hpp File Reference

Pipeline layout.

```
#include <liblava/block/descriptor.hpp>
```

Classes

- struct lava::pipeline_layout

  Pipeline layout.

5.59.1 Detailed Description

Pipeline layout.

Authors

Lava Block OÜ and contributors

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5.60 pipeline_layout.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/block/descriptor.hpp>

namespace lava {

struct pipeline_layout : entity {
    using ptr = std::shared_ptr<pipeline_layout>;

    using list = std::vector<ptr>;

    static ptr make() {
        return std::make_shared<pipeline_layout>();
    }

    void add(descriptor::ptr const& descriptor) {
        descriptors.push_back(descriptor);
    }

    void add(VkPushConstantRange const& range) {
        push_constant_ranges.push_back(range);
    }

    void add_descriptor(descriptor::ptr const& descriptor) {
        add(descriptor);
    }

    void add_push_constant_range(VkPushConstantRange const& range) {
        add(range);
    }

    void clear_descriptors() {
        descriptors.clear();
    }

    void clear_ranges() {
        push_constant_ranges.clear();
    }

    void clear() {
        clear_descriptors();
        clear_ranges();
    }

    bool create(device_p device);

    void destroy();

    VkPipelineLayout get() const {
        return layout;
    }

    device_p get_device() {
        return device;
    }

    descriptor::list const& get_descriptors() const {
        return descriptors;
    }

    VkPushConstantRanges const& get_push_constant_ranges() const {
        return push_constant_ranges;
    }

    using offset_list = std::vector<index>;

    void bind_descriptor_set(VkCommandBuffer cmd_buf,
                             VkDescriptorSet descriptor_set,
                             index first_set = 0,
                             offset_list offsets = {},
                             VkPipelineBindPoint bind_point = VK_PIPELINE_BIND_POINT_GRAPHICS);

    void bind(VkCommandBuffer cmd_buf,
              VkDescriptorSet descriptor_set,
              index first_set = 0,
              offset_list offsets = {},
              VkPipelineBindPoint bind_point = VK_PIPELINE_BIND_POINT_GRAPHICS) {
        bind_descriptor_set(cmd_buf,
                            descriptor_set,
                            first_set,
                            offsets,
```

```
                                bind_point);
    }

private:
    device_p device = nullptr;

    VkPipelineLayout layout = VK_NULL_HANDLE;

    descriptor::list descriptors;

    VkPushConstantRanges push_constant_ranges;
};

} // namespace lava
```

5.61 liblava/block/render_pass.hpp File Reference

Render pass.

```
#include <liblava/base/device.hpp>
#include <liblava/block/attachment.hpp>
#include <liblava/block/subpass.hpp>
```

Classes

- struct lava::render_pass

  Render pass.

5.61.1 Detailed Description

Render pass.

Authors

Lava Block OÜ and contributors

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5.62 render_pass.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/base/device.hpp>
#include <liblava/block/attachment.hpp>
#include <liblava/block/subpass.hpp>

namespace lava {

struct render_pass : entity {
    using ptr = std::shared_ptr<render_pass>;

    using list = std::vector<ptr>;

```

```
    static ptr make(device_p device) {
        return std::make_shared<render_pass>(device);
    }

    explicit render_pass(device_p device);

    bool create(VkAttachmentsRef target_attachments,
                rect::ref area);

    void destroy();

    void process(VkCommandBuffer cmd_buf,
                 index frame);

    device_p get_device() {
        return device;
    }

    VkRenderPass get() const {
        return vk_render_pass;
    }

    ui32 get_subpass_count() const {
        return to_ui32(subpasses.size());
    }

    bool exists_subpass(index index = 0) const {
        return index < subpasses.size();
    }

    subpass* get_subpass(index index = 0) {
        return subpasses.at(index).get();
    }

    subpass::list const& get_subpasses() const {
        return subpasses;
    }

    void add(attachment::ptr const& attachment) {
        attachments.push_back(attachment);
    }

    void add(subpass_dependency::ptr const& dependency) {
        dependencies.push_back(dependency);
    }

    void add(subpass::ptr const& subpass) {
        subpasses.push_back(subpass);
    }

    void set_clear_values(VkClearValues const& values) {
        clear_values = values;
    }

    VkClearValues const& get_clear_values() const {
        return clear_values;
    }

    void set_clear_color(v3 value = {});

    v3 get_clear_color() const;

    void add(render_pipeline::ptr pipeline,
             index subpass = 0) {
        subpasses.at(subpass)->add(pipeline);
    }

    void add_front(render_pipeline::ptr pipeline,
                   index subpass = 0) {
        subpasses.at(subpass)->add_front(pipeline);
    }

    void remove(render_pipeline::ptr pipeline,
                index subpass = 0) {
        subpasses.at(subpass)->remove(pipeline);
    }

    target_callback const& get_target_callback() const {
        return callback;
    }

private:
    device_p device = nullptr;

    VkRenderPass vk_render_pass = VK_NULL_HANDLE;

    VkFramebuffers framebuffers = {};
```

```
    attachment::list attachments;

    subpass_dependency::list dependencies;

    subpass::list subpasses;

    VkClearValues clear_values = {};

    rect area;

    target_callback callback;

    void begin(VkCommandBuffer cmd_buf,
               index frame);

    void end(VkCommandBuffer cmd_buf);

    bool on_target_created(VkAttachmentsRef target_attachments,
                           rect::ref area);

    void on_target_destroyed();
};

} // namespace lava
```

5.63 liblava/block/render_pipeline.hpp File Reference

Render pipeline (Graphics)

```
#include <liblava/block/pipeline.hpp>
```

Classes

- struct lava::render_pipeline

  Render pipeline (Graphics)
- struct lava::render_pipeline::create_info

  Render pipeline create information.

Functions

- VkPipelineColorBlendAttachmentState lava::create_pipeline_color_blend_attachment ()

  Create a color blend attachment.

5.63.1 Detailed Description

Render pipeline (Graphics)

Authors

Lava Block OÜ and contributors

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5.63.2 Function Documentation

5.63.2.1 create_pipeline_color_blend_attachment()

```
VkPipelineColorBlendAttachmentState lava::create_pipeline_color_blend_attachment ( )
```

Create a color blend attachment.

Returns

VkPipelineColorBlendAttachmentState Pipeline color blend attachment state

5.64 render_pipeline.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/block/pipeline.hpp>
11
12 namespace lava {
13
17 struct render_pipeline : pipeline {
19     using ptr = std::shared_ptr<render_pipeline>;
20
22     using map = std::map<id, ptr>;
23
25     using list = std::vector<ptr>;
26
30     enum class sizing_mode : index {
31         input = 0,
32         absolute,
33         relative
34     };
35
39     struct create_info {
41         VkPipelineVertexInputStateCreateInfo vertex_input_state;
42
44         VkPipelineInputAssemblyStateCreateInfo input_assembly_state;
45
47         VkPipelineViewportStateCreateInfo viewport_state;
48
50         VkPipelineMultisampleStateCreateInfo multisample_state;
51
53         VkPipelineDepthStencilStateCreateInfo depth_stencil_state;
54
56         VkPipelineRasterizationStateCreateInfo rasterization_state;
57     };
58
65     static ptr make(device_p device,
66                     VkPipelineCache pipeline_cache = 0) {
67         return std::make_shared<render_pipeline>(device, pipeline_cache);
68     }
69
75     explicit render_pipeline(device_p device,
76                              VkPipelineCache pipeline_cache);
77
82     void bind(VkCommandBuffer cmd_buf) override;
83
89     void set_viewport_and_scissor(VkCommandBuffer cmd_buf,
90                                   uv2 size);
91
96     void set_render_pass(VkRenderPass pass) {
97         render_pass = pass;
98     }
99
103     void set(VkRenderPass pass) {
104         set_render_pass(pass);
105     }
106
```

```
    VkRenderPass get_render_pass() const {
        return render_pass;
    }

    index get_subpass() const {
        return subpass;
    }

    void set_subpass(index value) {
        subpass = value;
    }

    bool create(VkRenderPass pass) {
        set(pass);

        return pipeline::create();
    }

    void set_vertex_input_binding(VkVertexInputBindingDescription const& description);

    void set_vertex_input_bindings(VkVertexInputBindingDescriptions const& descriptions);

    void set_vertex_input_attribute(VkVertexInputAttributeDescription const& attribute);

    void set_vertex_input_attributes(VkVertexInputAttributeDescriptions const& attributes);

    void set_input_topology(VkPrimitiveTopology const& topology);

    void set_depth_test_and_write(bool test_enable = true,
                                  bool write_enable = true);

    void set_depth_compare_op(VkCompareOp compare_op);

    void set_rasterization_cull_mode(VkCullModeFlags cull_mode);

    void set_rasterization_front_face(VkFrontFace front_face);

    void set_rasterization_polygon_mode(VkPolygonMode polygon_mode);

    void add_color_blend_attachment(VkPipelineColorBlendAttachmentState const& attachment);

    void add_color_blend_attachment();

    void clear_color_blend_attachment();

    void set_dynamic_states(VkDynamicStates const& states);

    void add_dynamic_state(VkDynamicState state);

    void clear_dynamic_states();

    bool add_shader_stage(cdata::ref data,
                          VkShaderStageFlagBits stage);

    bool add_shader(cdata::ref data,
                    VkShaderStageFlagBits stage) {
        return add_shader_stage(data, stage);
    }

    void add(shader_stage::ptr const& shader_stage) {
        shader_stages.push_back(shader_stage);
    }

    shader_stage::list const& get_shader_stages() const {
        return shader_stages;
    }

    void clear_shader_stages() {
        shader_stages.clear();
    }

    void clear() {
        clear_color_blend_attachment();
        clear_shader_stages();
    }

    void set_auto_size(bool value = true) {
        auto_size = value;
    }

    bool auto_sizing() const {
        return auto_size;
    }

    VkViewport get_viewport() const {
        return viewport;
    }
```

```
311
316     void set_viewport(VkViewport value) {
317         viewport = value;
318     }
319
324     VkRect2D get_scissor() const {
325         return scissor;
326     }
327
332     void set_scissor(VkRect2D value) {
333         scissor = value;
334     }
335
340     sizing_mode get_sizing() const {
341         return sizing;
342     }
343
348     void set_sizing(sizing_mode value) {
349         sizing = value;
350     }
351
356     void copy_to(render_pipeline* target) const;
357
362     void copy_from(ptr const& source) {
363         source->copy_to(this);
364     }
365
370     void set_line_width(r32 value) {
371         line_width = value;
372     }
373
378     r32 get_line_width() const {
379         return line_width;
380     }
381
386     bool auto_line_width() const {
387         return auto_line_width_state;
388     }
389
394     void set_auto_line_width(bool value = true) {
395         auto_line_width_state = value;
396     }
397
402     void set_line_width(VkCommandBuffer cmd_buf) {
403         vkCmdSetLineWidth(cmd_buf, line_width);
404     }
405
407     using create_func = std::function<bool(create_info&)>;
408
410     create_func on_create;
411
412 private:
417     bool setup() override;
418
422     void teardown() override;
423
425     VkRenderPass render_pass = VK_NULL_HANDLE;
426
428     index subpass = 0;
429
431     create_info info;
432
434     VkVertexInputBindingDescriptions vertex_input_bindings;
435
437     VkVertexInputAttributeDescriptions vertex_input_attributes;
438
440     VkPipelineColorBlendAttachmentStates color_blend_attachment_states;
441
443     VkPipelineColorBlendStateCreateInfo color_blend_state;
444
446     VkPipelineDynamicStateCreateInfo dynamic_state;
447
449     VkDynamicStates dynamic_states;
450
452     shader_stage::list shader_stages;
453
455     sizing_mode sizing = sizing_mode::input;
456
458     VkViewport viewport;
459
461     VkRect2D scissor;
462
464     bool auto_size = true;
465
467     bool auto_line_width_state = false;
468
470     r32 line_width = 1.f;
```

```
};

VkPipelineColorBlendAttachmentState create_pipeline_color_blend_attachment();

} // namespace lava
```

5.65 liblava/block/subpass.hpp File Reference

Subpass.

`#include <liblava/block/render_pipeline.hpp>`

Classes

- struct lava::subpass

  Subpass.
- struct lava::subpass_dependency

  Subpass dependency.

5.65.1 Detailed Description

Subpass.

Authors

Lava Block OÜ and contributors

Copyright

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5.66 subpass.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/block/render_pipeline.hpp>

namespace lava {

struct subpass : entity {
    using ptr = std::shared_ptr<subpass>;

    using list = std::vector<ptr>;

    static ptr make(VkPipelineBindPoint pipeline_bind_point =
                        VK_PIPELINE_BIND_POINT_GRAPHICS) {
        auto result = std::make_shared<subpass>();
        result->set(pipeline_bind_point);
        return result;
    }

    explicit subpass();

    void destroy();
```

```

    void add(render_pipeline::ptr const& pipeline) {
        pipelines.push_back(pipeline);
    }

    void add_front(render_pipeline::ptr const& pipeline) {
        pipelines.insert(pipelines.begin(), pipeline);
    }

    void remove(render_pipeline::ptr pipeline);

    void clear_pipelines();

    void process(VkCommandBuffer cmd_buf,
                 uv2 size);

    VkSubpassDescription const& get_description() const {
        return description;
    }

    void set(VkPipelineBindPoint pipeline_bind_point) {
        description.pipelineBindPoint = pipeline_bind_point;
    }

    void set_color_attachment(index attachment,
                              VkImageLayout layout);

    void set_color_attachment(VkAttachmentReference attachment);

    void set_color_attachments(VkAttachmentReferences const& attachments);

    void set_depth_stencil_attachment(index attachment,
                                      VkImageLayout layout);

    void set_depth_stencil_attachment(VkAttachmentReference attachment);

    void set_input_attachment(index attachment,
                              VkImageLayout layout);

    void set_input_attachment(VkAttachmentReference attachment);

    void set_input_attachments(VkAttachmentReferences const& attachments);

    void set_resolve_attachment(index attachment, VkImageLayout layout);

    void set_resolve_attachment(VkAttachmentReference attachment);

    void set_resolve_attachments(VkAttachmentReferences const& attachments);

    void add_preserve_attachment(index attachment);

    void set_preserve_attachments(index_list const& attachments);

    void set_active(bool value = true) {
        active = value;
    }

    bool activated() const {
        return active;
    }

private:
    VkSubpassDescription description;

    bool active = true;

    VkAttachmentReferences color_attachments;

    VkAttachmentReference depth_stencil_attachment{};

    VkAttachmentReferences input_attachments;

    VkAttachmentReferences resolve_attachments;

    index_list preserve_attachments;

    render_pipeline::list pipelines;
};

struct subpass_dependency {
    using ptr = std::shared_ptr<subpass_dependency>;

    using list = std::vector<ptr>;

    static ptr make(ui32 src_subpass,
                    ui32 dst_subpass,
                    VkDependencyFlags dependency_flags =
```

```
                        VK_DEPENDENCY_BY_REGION_BIT) {
        auto result = std::make_shared<subpass_dependency>();
        result->set_subpass(src_subpass, dst_subpass);
        result->set_dependency_flags(dependency_flags);
        return result;
    }

    explicit subpass_dependency();

    VkSubpassDependency const& get_dependency() const {
        return dependency;
    }

    void set_subpass(ui32 src, ui32 dst) {
        set_src_subpass(src);
        set_dst_subpass(dst);
    }

    void set_src_subpass(ui32 src) {
        dependency.srcSubpass = src;
    }

    void set_dst_subpass(ui32 dst) {
        dependency.dstSubpass = dst;
    }

    void set_stage_mask(VkPipelineStageFlags src,
                        VkPipelineStageFlags dst) {
        set_src_stage_mask(src);
        set_dst_stage_mask(dst);
    }

    void set_src_stage_mask(VkPipelineStageFlags mask) {
        dependency.srcStageMask = mask;
    }

    void set_dst_stage_mask(VkPipelineStageFlags mask) {
        dependency.dstStageMask = mask;
    }

    void set_access_mask(VkAccessFlags src,
                         VkAccessFlags dst) {
        set_src_access_mask(src);
        set_dst_access_mask(dst);
    }

    void set_src_access_mask(VkAccessFlags mask) {
        dependency.srcAccessMask = mask;
    }

    void set_dst_access_mask(VkAccessFlags mask) {
        dependency.dstAccessMask = mask;
    }

    void set_dependency_flags(VkDependencyFlags flags) {
        dependency.dependencyFlags = flags;
    }

private:
    VkSubpassDependency dependency;
};

} // namespace lava
```

5.67 liblava/core.hpp File Reference

Core module.

```
#include <liblava/core/data.hpp>
#include <liblava/core/def.hpp>
#include <liblava/core/id.hpp>
#include <liblava/core/misc.hpp>
#include <liblava/core/time.hpp>
#include <liblava/core/types.hpp>
#include <liblava/core/version.hpp>
```

5.67.1 Detailed Description

Core module.

Author

Lava Block OÜ and contributors

Copyright

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5.68 core.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/core/data.hpp>
11 #include <liblava/core/def.hpp>
12 #include <liblava/core/id.hpp>
13 #include <liblava/core/misc.hpp>
14 #include <liblava/core/time.hpp>
15 #include <liblava/core/types.hpp>
16 #include <liblava/core/version.hpp>
```

5.69 liblava/core/data.hpp File Reference

Data wrapper.

```
#include <string.h>
#include <liblava/core/types.hpp>
```

Classes

- struct lava::data_provider

  Data provider.
- struct lava::data

  Data wrapper.
- struct lava::cdata

  Const data wrapper.
- struct lava::unique_data

  Unique data wrapper.

Macros

- #define **strndup**(p, n) _strdup(p)

  Duplicate string.

Typedefs

- using **lava::data_ptr** = char ∗

  Data pointer.
- using **lava::data_cptr** = char const ∗

  Const data pointer.

Enumerations

- enum class lava::data_mode : index { **alloc** = 0 , **no_alloc** }

  Data modes.

Functions

- data_ptr lava::as_ptr (auto ∗value)

  Cast to data pointer.
- auto lava::align_up (auto value, auto align)

  Align value up.
- size_t lava::align (size_t size, size_t min=0)

  Align a size.
- template<typename T >
  size_t lava::align (size_t min=0)

  Get alignment of type.
- void ∗ lava::alloc_data (size_t size, size_t alignment=sizeof(c8))

  Allocate data.
- void lava::free_data (void ∗data)

  Free data.
- void ∗ lava::realloc_data (void ∗data, size_t size, size_t alignment)

  Reallocate data.
- size_t lava::next_pow_2 (size_t x)

  Get next power of two.
- char ∗ lava::human_readable (size_t const sz)

  Convert data size to human readable string.

5.69.1 Detailed Description

Data wrapper.

Authors

Lava Block OÜ and contributors

Copyright

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5.69.2 Function Documentation

5.69.2.1 align() [1/2]

```
template<typename T >
size_t lava::align (
            size_t min = 0 )  [inline]
```

Get alignment of type.

Template Parameters

T	Type to align

Parameters

min	Minimal alignment

Returns

size_t Aligned size

5.69.2.2 align() [2/2]

```
size_t lava::align (
            size_t size,
            size_t min = 0 )  [inline]
```

Align a size.

Parameters

size	Site to align
min	Minimal alignment

Returns

size_t Aligned size

5.69.2.3 align_up()

```
auto lava::align_up (
            auto value,
            auto align )  [inline]
```

Align value up.

Parameters

value	Value to align
align	Target alignment

Returns

auto Aligned value

5.69.2.4 alloc_data()

```
void * lava::alloc_data (
            size_t size,
            size_t alignment = sizeof(c8) )  [inline]
```

Allocate data.

Parameters

size	Size of data
alignment	Target alignment

Returns

void∗ Allocated data

5.69.2.5 as_ptr()

```
data_ptr lava::as_ptr (
            auto * value )  [inline]
```

Cast to data pointer.

Parameters

value	Value to cast

Returns

data_ptr Data pointer

5.69.2.6 free_data()

```
void lava::free_data (
            void * data )  [inline]
```

Free data.

Parameters

data	Data to free

5.69.2.7 human_readable()

```
char * lava::human_readable (
            size_t const sz )  [inline]
```

Convert data size to human readable string.

Parameters

sz	Size to convert

Returns

char∗ Human readable size string

5.69.2.8 next_pow_2()

```
size_t lava::next_pow_2 (
            size_t x )  [inline]
```

Get next power of two.

Parameters

x	Source value

Returns

size_t Next power of two

5.69.2.9 realloc_data()

```
void * lava::realloc_data (
            void * data,
            size_t size,
            size_t alignment )  [inline]
```

Reallocate data.

Parameters

data	Data to reallocate
size	Size of data
alignment	Target alignment

Returns

void∗ Reallocated data

5.70 data.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <string.h>
11 #include <liblava/core/types.hpp>
12
13 namespace lava {
14
16 using data_ptr = char*;
17
19 using data_cptr = char const*;
20
24 struct data_provider {
28     using alloc_func = std::function<data_ptr(size_t, size_t)>;
29
31     alloc_func on_alloc;
32
36     using free_func = std::function<void()>;
37
39     free_func on_free;
40
44     using realloc_func = std::function<data_ptr(data_ptr, size_t, size_t)>;
45
47     realloc_func on_realloc;
48 };
49
55 inline data_ptr as_ptr(auto* value) {
56     return (data_ptr) value;
57 }
58
65 inline auto align_up(auto value,
66                       auto align) {
67     return (value + align - 1) / align * align;
68 }
69
76 inline size_t align(size_t size,
77                     size_t min = 0) {
78     if (min == 0)
79         return align_up(size, sizeof(void*));
80
81     return align_up((size + min - 1) & ~(min - 1), sizeof(void*));
82 }
83
90 template<typename T>
91 inline size_t align(size_t min = 0) {
92     return align(sizeof(T), min);
93 }
94
101 inline void* alloc_data(size_t size,
102                         size_t alignment = sizeof(c8)) {
103 #if _WIN32
104     return _aligned_malloc(size, alignment);
105 #else
106     if (size % alignment == 0) {
107         return aligned_alloc(alignment, size);
108     } else {
109         return aligned_alloc(alignment, ((size / alignment) + 1) * alignment);
110     }
111 #endif
112 }
113
118 inline void free_data(void* data) {
```

```
#if _WIN32
    _aligned_free(data);
#else
    free(data);
#endif
}

inline void* realloc_data(void* data,
                          size_t size,
                          size_t alignment) {
#if _WIN32
    return _aligned_realloc(data, size, alignment);
#else
    return realloc(data, align(size, alignment));
#endif
}

enum class data_mode : index {
    alloc = 0,
    no_alloc
};

struct data {
    using ref = data const&;

    data() = default;

    data(auto* ptr, size_t size)
    : ptr(as_ptr(ptr)), size(size) {}

    void set(size_t length,
             data_mode mode = data_mode::alloc) {
        size = length;
        alignment = align<data_ptr>();

        if (mode == data_mode::alloc)
            allocate();
    }

    bool allocate() {
        ptr = as_ptr(alloc_data(size, alignment));
        return ptr != nullptr;
    }

    void free() {
        if (!ptr)
            return;

        free_data(ptr);
        ptr = nullptr;
    }

    data_ptr end() const {
        return ptr + size;
    }

    data_ptr ptr = nullptr;

    size_t size = 0;

    size_t alignment = 0;
};

struct cdata {
    using ref = cdata const&;

    cdata() = default;

    cdata(void const* ptr,
          size_t length)
    : ptr(as_ptr(ptr)), size(length) {}

    cdata(data::ref data)
    : cdata(data.ptr, data.size) {}

    data_cptr ptr = nullptr;

    size_t size = 0;
};

struct unique_data : data {
    using ref = unique_data const&;

    unique_data(size_t length = 0,
                data_mode mode = data_mode::alloc) {
        if (length)
            set(length, mode);
```

```
    }

    explicit unique_data(data::ref data) {
        ptr = data.ptr;
        size = data.size;
        alignment = data.alignment;
    }

    ~unique_data() {
        free();
    }
};

inline size_t next_pow_2(size_t x) {
    x--;
    x |= x » 1;
    x |= x » 2;
    x |= x » 4;
    x |= x » 8;
    x |= x » 16;
    x++;
    return x;
}

#ifndef __GNUC__
    #define strndup(p, n) _strdup(p)
#endif

inline char* human_readable(size_t const sz) {
    auto const buffer_size = 32;

    char const prefixes[] = "KMGTPEZY";
    char buf[buffer_size];
    auto which = -1;

    auto result = to_r64(sz);
    while (result > 1024 && which < 7) {
        result /= 1024;
        ++which;
    }

    char unit[] = "\0i";
    if (which >= 0)
        unit[0] = prefixes[which];

    snprintf(buf, buffer_size, "%.2f %sB", result, unit);
    return strndup(buf, buffer_size);
}

#ifndef __GNUC__
    #undef strndup
#endif

} // namespace lava
```

5.71 liblava/core/id.hpp File Reference

Object Identification.

```
#include <atomic>
#include <deque>
#include <liblava/core/types.hpp>
#include <memory>
#include <mutex>
#include <set>
```

Classes

- struct lava::id

  Identification.
- struct lava::ids

Id factory.

- struct lava::id_listeners< T >

  Id listeners.
- struct lava::entity

  Entity.
- struct lava::id_registry< T, Meta >

  Id registry.

Typedefs

- using **lava::string_id_map** = std::map< string, id >

  Map of string ids.

Functions

- id lava::to_id (auto value)

  Convert to id.
- template<typename T >
  id lava::add_id_map (T const &object, std::map< id, T > &map)

  Add object to id map.
- template<typename T >
  bool lava::remove_id_map (id::ref object_id, std::map< id, T > &map)

  Remove object from id map.

Variables

- constexpr id const **lava::undef_id** = id()

  Undefined id.

5.71.1 Detailed Description

Object Identification.

Authors

Lava Block OÜ and contributors

Copyright

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5.71.2 Function Documentation

5.71.2.1 add_id_map()

```
template<typename T >
id lava::add_id_map (
            T const & object,
            std::map< id, T > & map )  [inline]
```

Add object to id map.

Template Parameters

T	Type of object

Parameters

object	Object to add
map	Target map

Returns

id Id of object in map

5.71.2.2 remove_id_map()

```
template<typename T >
bool lava::remove_id_map (
            id::ref object_id,
            std::map< id, T > & map )  [inline]
```

Remove object from id map.

Template Parameters

T	Type of object

Parameters

object↩ _id	Object to remove
map	Target map

Returns

Removed object from map or object not found

5.71.2.3 to_id()

```
id lava::to_id (
            auto value )  [inline]
```

Convert to id.

Parameters

value	Source value

Returns

id Converted value

5.72 id.hpp

Go to the documentation of this file.

```
#pragma once

#include <atomic>
#include <deque>
#include <liblava/core/types.hpp>
#include <memory>
#include <mutex>
#include <set>

namespace lava {

struct id {
    using ref = id const&;

    using set = std::set<id>;

    using set_ref = set const&;

    using list = std::vector<id>;

    using map = std::map<id, id>;

    using index_map = std::map<id, index>;

    using string_map = std::map<id, string>;

    index value = no_index;

    bool valid() const {
        return value != no_index;
    }

    string to_string() const {
        return std::to_string(value);
    }

    void invalidate() {
        *this = {};
    }

    auto operator<=>(id const&) const = default;
};

using string_id_map = std::map<string, id>;

constexpr id const undef_id = id();

inline id to_id(auto value) {
    return { static_cast<index>(value) };
}

struct ids {
    static ids& instance() {
        static ids ids;
        return ids;
    }

    id next() {
        return { ++next_id };
    }

private:
    std::atomic<index> next_id = { no_index };
};

```

```
template<typename T>
inline id add_id_map(T const& object,
                     std::map<id, T>& map) {
    auto next = ids::instance().next();
    map.emplace(next, std::move(object));
    return next;
}

template<typename T>
inline bool remove_id_map(id::ref object_id,
                          std::map<id, T>& map) {
    if (!map.count(object_id))
        return false;

    map.erase(object_id);

    return true;
}

template<typename T>
struct id_listeners {
    id add(typename T::func const& listener) {
        return add_id_map(listener, list);
    }

    void remove(id& id) {
        if (remove_id_map(id, list))
            id.invalidate();
    }

    typename T::listeners const& get_list() const {
        return list;
    }

private:
    typename T::listeners list = {};
};

struct entity : no_copy_no_move, interface {
    entity()
    : entity_id(ids::instance().next()) {}

    id::ref get_id() const {
        return entity_id;
    }

private:
    id entity_id;
};

template<typename T, typename Meta>
struct id_registry {
    using ptr = std::shared_ptr<T>;

    using map = std::map<id, ptr>;

    using meta_map = std::map<id, Meta>;

    id create(Meta info = {}) {
        auto object = std::make_shared<T>();
        add(object, info);

        return object->get_id();
    }

    void add(ptr object,
             Meta info = {}) {
        objects.emplace(object->get_id(), object);
        meta.emplace(object->get_id(), info);
    }

    bool exists(id::ref object_id) const {
        return objects.count(object_id);
    }

    ptr get(id::ref object_id) const {
        return objects.at(object_id);
    }

    Meta const& get_meta(id::ref object_id) const {
        return meta.at(object_id);
    }

    map const& get_all() const {
        return objects;
    }

```

```
    meta_map const& get_all_meta() const {
        return meta;
    }

    bool update(id::ref object_id,
                Meta const& meta) {
        if (!exists(object_id))
            return false;

        meta.at(object_id) = meta;
        return true;
    }

    void remove(id::ref object_id) {
        objects.erase(object_id);
        meta.erase(object_id);
    }

    void clear() {
        objects.clear();
        meta.clear();
    }

private:
    map objects;

    meta_map meta;
};

} // namespace lava
```

5.73 liblava/core/misc.hpp File Reference

Miscellaneous helpers.

```
#include <cstring>
#include <liblava/core/types.hpp>
#include <utility>
```

Classes

- struct lava::reversion_wrapper< T >

  Reversion Wrapper.

Functions

- bool lava::exists (names_ref list, name item)

  Check if name exists in name list.
- template<typename T >
  void lava::remove (std::vector< T > &list, T item)

  Remove item from list.
- template<typename T >
  bool lava::contains (std::vector< T > &list, T item)

  Check if item is included in list.
- template<typename T >
  void lava::append (std::vector< T > &list, std::vector< T > &items)

  Append a list of items to another list.
- void lava::ltrim (string &s)

  Trim string only from start (in place)
- void lava::rtrim (string &s)

Trim string only from end (in place)

- void lava::trim (string &s)

  Trim string from both ends (in place)
- string lava::ltrim_copy (string s)

  Trim string only from start (copying)
- string lava::rtrim_copy (string s)

  Trim string only from end (copying)
- string lava::trim_copy (string s)

  Trim string from both ends (copying)
- string & lava::remove_chars (string &s, string_ref chars)

  Remove chars in string.
- string & lava::remove_punctuation_marks (string &s)

  Remove punctuation marks in string.
- string lava::remove_chars_copy (string s, string_ref chars)

  Remove chars in string (copying)
- string & lava::remove_nondigit (string &s)

  Remove all non digit chars in string.
- string lava::remove_nondigit_copy (string s)

  Remove all non digit chars in string (copying)
- string & lava::remove_chars_if_not (string &s, string_ref allowed)

  Remove all chars in string which are not allowed.
- string lava::remove_chars_if_not_copy (string s, string_ref allowed)

  Remove all chars in string which are not allowed (copying)
- template<typename T >
  auto lava::begin (reversion_wrapper< T > w)

  Begin the iterator.
- template<typename T >
  auto lava::end (reversion_wrapper< T > w)

  End the iterator.
- template<typename T >
  reversion_wrapper< T > lava::reverse (T &&iterable)

  Reverse iteration.

Variables

- constexpr name **lava::_punctuation_marks_** = "\"\'"

  Punctuation marks.

5.73.1 Detailed Description

Miscellaneous helpers.

Authors

Lava Block OÜ and contributors

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5.73.2 Function Documentation

5.73.2.1 append()

```
template<typename T >
void lava::append (
            std::vector< T > & list,
            std::vector< T > & items )  [inline]
```

Append a list of items to another list.

Template Parameters

T	Type of list

Parameters

list	List of items
items	Items to append

5.73.2.2 begin()

```
template<typename T >
auto lava::begin (
            reversion_wrapper< T > w )  [inline]
```

Begin the iterator.

Template Parameters

T	Type of iterable

Parameters

w	Reversion wrapper

Returns

auto Iterator

5.73.2.3 contains()

```
template<typename T >
bool lava::contains (
            std::vector< T > & list,
            T item )  [inline]
```

Check if item is included in list.

Template Parameters

T	Type of list

Parameters

list	List of items
item	Item to check

Returns

Item exists or not found

5.73.2.4 end()

```
template<typename T >
auto lava::end (
            reversion_wrapper< T > w )  [inline]
```

End the iterator.

Template Parameters

T	Type of iterable

Parameters

w	Reversion wrapper

Returns

auto Iterator

5.73.2.5 exists()

```
bool lava::exists (
            names_ref list,
            name item )  [inline]
```

Check if name exists in name list.

Parameters

list	List of names
item	Item to check

Returns

Item exists or not found

5.73.2.6 ltrim()

```
void lava::ltrim (
            string & s )  [inline]
```

Trim string only from start (in place)

Parameters

s	String to trim

5.73.2.7 ltrim_copy()

```
string lava::ltrim_copy (
            string s )  [inline]
```

Trim string only from start (copying)

Parameters

s	String to trim

Returns

string Trimmed string

5.73.2.8 remove()

```
template<typename T >
void lava::remove (
```

```
std::vector< T > & list,
T item )  [inline]
```

Remove item from list.

Template Parameters

T	Type of list

Parameters

list	List of items
item	Item to remove

5.73.2.9 remove_chars()

```
string & lava::remove_chars (
            string & s,
            string_ref chars )  [inline]
```

Remove chars in string.

Parameters

s	Target string
chars	Chars to remove

Returns

string& Cleared string

5.73.2.10 remove_chars_copy()

```
string lava::remove_chars_copy (
            string s,
            string_ref chars )  [inline]
```

Remove chars in string (copying)

Parameters

s	Target string
chars	Chars to remove

Returns

string Cleared string

5.73.2.11 remove_chars_if_not()

```
string & lava::remove_chars_if_not (
            string & s,
            string_ref allowed )  [inline]
```

Remove all chars in string which are not allowed.

Parameters

s	Target string
allowed	Allowed chars

Returns

string& Cleared string

5.73.2.12 remove_chars_if_not_copy()

```
string lava::remove_chars_if_not_copy (
            string s,
            string_ref allowed )  [inline]
```

Remove all chars in string which are not allowed (copying)

Parameters

s	Target string
allowed	Allowed chars

Returns

string Cleared string

5.73.2.13 remove_nondigit()

```
string & lava::remove_nondigit (
            string & s )  [inline]
```

Remove all non digit chars in string.

Parameters

s	Target string

Returns

string& Cleared string

5.73.2.14 remove_nondigit_copy()

```
string lava::remove_nondigit_copy (
            string s )  [inline]
```

Remove all non digit chars in string (copying)

Parameters

s	Target string

Returns

string Cleared string

5.73.2.15 remove_punctuation_marks()

```
string & lava::remove_punctuation_marks (
            string & s )  [inline]
```

Remove punctuation marks in string.

Parameters

s	Target string

Returns

string& Cleared string

5.73.2.16 reverse()

```
template<typename T >
reversion_wrapper< T > lava::reverse (
            T && iterable )  [inline]
```

Reverse iteration.

Template Parameters

T	Type of iterable

Parameters

iterable	Iterable

Returns

reversion_wrapper<T> Wrapper

5.73.2.17 rtrim()

```
void lava::rtrim (
            string & s )  [inline]
```

Trim string only from end (in place)

Parameters

s	String to trim

5.73.2.18 rtrim_copy()

```
string lava::rtrim_copy (
            string s )  [inline]
```

Trim string only from end (copying)

Parameters

s	String to trim

Returns

string Trimmed string

5.73.2.19 trim()

```
void lava::trim (
            string & s )  [inline]
```

Trim string from both ends (in place)

Parameters

s	String to trim

5.73.2.20 trim_copy()

```
string lava::trim_copy (
            string s )  [inline]
```

Trim string from both ends (copying)

Parameters

s	String to trim

Returns

string Trimmed string

5.74 misc.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <cstring>
11 #include <liblava/core/types.hpp>
12 #include <utility>
13
14 namespace lava {
15
22 inline bool exists(names_ref list, name item) {
23     auto itr = std::find_if(list.begin(), list.end(),
24                             [&](name entry) { return strcmp(entry, item) == 0; });
25     return itr != list.end();
26 }
27
34 template<typename T>
35 inline void remove(std::vector<T>& list, T item) {
36     list.erase(std::remove(list.begin(), list.end(), item), list.end());
37 }
38
46 template<typename T>
47 inline bool contains(std::vector<T>& list, T item) {
48     return std::find(list.begin(), list.end(), item) != list.end();
49 }
50
57 template<typename T>
58 inline void append(std::vector<T>& list, std::vector<T>& items) {
59     list.insert(list.end(), items.begin(), items.end());
60 }
61
66 inline void ltrim(string& s) {
67     s.erase(s.begin(), std::find_if(s.begin(), s.end(), [](uchar ch) {
68                 return !std::isspace(ch);
69             }));
70 }
71
76 inline void rtrim(string& s) {
77     s.erase(std::find_if(s.rbegin(), s.rend(), [](uchar ch) {
```

```
                return !std::isspace(ch);
            }).base(),
            s.end());
}

inline void trim(string& s) {
    ltrim(s);
    rtrim(s);
}

inline string ltrim_copy(string s) {
    ltrim(s);
    return s;
}

inline string rtrim_copy(string s) {
    rtrim(s);
    return s;
}

inline string trim_copy(string s) {
    trim(s);
    return s;
}

inline string& remove_chars(string& s, string_ref chars) {
    s.erase(remove_if(s.begin(), s.end(), [&chars](name_ref c) {
                return chars.find(c) != string::npos;
            }),
            s.end());
    return s;
}

constexpr name _punctuation_marks_ = "\"\'";

inline string& remove_punctuation_marks(string& s) {
    return remove_chars(s, _punctuation_marks_);
}

inline string remove_chars_copy(string s, string_ref chars) {
    return remove_chars(s, chars);
}

inline string& remove_nondigit(string& s) {
    s.erase(remove_if(s.begin(), s.end(), [](name_ref c) {
                return !isdigit(c);
            }),
            s.end());
    return s;
}

inline string remove_nondigit_copy(string s) {
    return remove_nondigit(s);
}

inline string& remove_chars_if_not(string& s, string_ref allowed) {
    s.erase(remove_if(s.begin(), s.end(), [&allowed](name_ref c) {
                return allowed.find(c) == string::npos;
            }),
            s.end());
    return s;
}

inline string remove_chars_if_not_copy(string s, string_ref allowed) {
    return remove_chars_if_not(s, allowed);
}

template<typename T>
struct reversion_wrapper {
    T& iterable;
};

template<typename T>
inline auto begin(reversion_wrapper<T> w) {
    return std::rbegin(w.iterable);
}

template<typename T>
inline auto end(reversion_wrapper<T> w) {
    return std::rend(w.iterable);
}

template<typename T>
inline reversion_wrapper<T> reverse(T&& iterable) {
    return { iterable };
}

```

```
247 } // namespace lava
```

5.75 liblava/core/time.hpp File Reference

Run time.

```
#include <chrono>
#include <iomanip>
#include <liblava/core/types.hpp>
#include <sstream>
```

Classes

- struct lava::timer

  Timer.
- struct lava::run_time

  Run time.

Typedefs

- using **lava::seconds** = std::chrono::seconds

  Seconds.
- using **lava::milliseconds** = std::chrono::milliseconds

  Milliseconds.
- using **lava::ms** = milliseconds

  Milliseconds.
- using **lava::clock** = std::chrono::high_resolution_clock

  Clock.
- using **lava::time_point** = clock::time_point

  Time point.
- using **lava::duration** = clock::duration

  Duration.
- using **lava::run_time_t** = run_time

  Run time type.

Functions

- delta lava::to_delta (milliseconds ms)

  Convert milliseconds to delta.
- delta lava::to_dt (milliseconds ms)
- real lava::to_sec (milliseconds ms)

  Convert milliseconds to seconds.
- ms lava::to_ms (delta dt)

  Convert delta to milliseconds.
- ms lava::to_ms (real sec)

  Convert seconds to milliseconds.

- template<typename CLOCK_TYPE = std::chrono::system_clock>
  string lava::timestamp (const typename CLOCK_TYPE::time_point &time_point, string_ref format="%Y-%m-%d %H-%M-%S")

  Convert timestamp to string.
- string lava::get_current_time ()

  Get the current time as string.
- ms lava::get_current_timestamp_ms ()

  Get the current timestamp in milliseconds.
- ui64 lava::get_current_timestamp ()

  Get the current timestamp in milliseconds (uint)

Variables

- constexpr seconds const **lava::one_second** = seconds(1)

  One second.
- constexpr ms const **lava::one_ms** = ms(1)

  One millisecond.

5.75.1 Detailed Description

Run time.

Authors

Lava Block OÜ and contributors

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5.75.2 Function Documentation

5.75.2.1 get_current_time()

```
string lava::get_current_time ( ) [inline]
```

Get the current time as string.

Returns

string Time and date representation

5.75.2.2 get_current_timestamp()

```
ui64 lava::get_current_timestamp ( ) [inline]
```

Get the current timestamp in milliseconds (uint)

Returns

ui64 Timestamp in ms (uint)

5.75.2.3 get_current_timestamp_ms()

```
ms lava::get_current_timestamp_ms ( ) [inline]
```

Get the current timestamp in milliseconds.

Returns

ms Timestamp in ms

5.75.2.4 timestamp()

```
template<typename CLOCK_TYPE = std::chrono::system_clock>
string lava::timestamp (
            const typename CLOCK_TYPE::time_point & time_point,
            string_ref format = "%Y-%m-%d %H-%M-%S" ) [inline]
```

Convert timestamp to string.

Template Parameters

CLOCK_TYPE	Clock type

Parameters

time_point	Time point
format	String format

Returns

string Converted string

5.75.2.5 to_delta()

```
delta lava::to_delta (
            milliseconds ms )  [inline]
```

Convert milliseconds to delta.

Parameters

ms	Milliseconds to convert

Returns

delta Converted delta

5.75.2.6 to_dt()

```
delta lava::to_dt (
            milliseconds ms )  [inline]
```

See also

to_delta()

5.75.2.7 to_ms() [1/2]

```
ms lava::to_ms (
            delta dt )  [inline]
```

Convert delta to milliseconds.

Parameters

dt	Delta to convert

Returns

ms Converted milliseconds

5.75.2.8 to_ms() [2/2]

```
ms lava::to_ms (
            real sec )  [inline]
```

Convert seconds to milliseconds.

Parameters

sec	Seconds to convert

Returns

ms Converted milliseconds

5.75.2.9 to_sec()

```
real lava::to_sec (
            milliseconds ms )  [inline]
```

Convert milliseconds to seconds.

Parameters

ms	Milliseconds to convert

Returns

real Converted seconds

5.76 time.hpp

Go to the documentation of this file.

```
#pragma once

#include <chrono>
#include <iomanip>
#include <liblava/core/types.hpp>
#include <sstream>

namespace lava {

using seconds = std::chrono::seconds;

using milliseconds = std::chrono::milliseconds;

using ms = milliseconds;

constexpr seconds const one_second = seconds(1);

constexpr ms const one_ms = ms(1);

using clock = std::chrono::high_resolution_clock;

using time_point = clock::time_point;

using duration = clock::duration;

inline delta to_delta(milliseconds ms) {
    return ms.count() / 1000.f;
}

inline delta to_dt(milliseconds ms) {
    return to_delta(ms);
}

```

```
inline real to_sec(milliseconds ms) {
    return ms.count() / 1000.;
}

inline ms to_ms(delta dt) {
    return ms(to_i32(dt * 1000.f));
}

inline ms to_ms(real sec) {
    return ms(to_i32(sec * 1000.));
}

struct timer {
    timer()
    : start_time(clock::now()) {}

    void reset() {
        start_time = clock::now();
    }

    ms elapsed() const {
        return std::chrono::duration_cast<ms>(clock::now()
                                              - start_time);
    }

private:
    time_point start_time;
};

struct run_time {
    ms current{ 0 };

    ms clock{ 16 };

    ms system{ 0 };

    ms delta{ 0 };

    ms fix_delta{ 0 };

    r32 speed = 1.f;

    bool paused = false;
};

using run_time_t = run_time;

#ifdef _WIN32
    #pragma warning(push)
    #pragma warning(disable : 4996) //_CRT_SECURE_NO_WARNINGS
#endif

template<typename CLOCK_TYPE = std::chrono::system_clock>
inline string timestamp(const typename CLOCK_TYPE::time_point& time_point,
                        string_ref format = "%Y-%m-%d %H-%M-%S") {
    auto ms = std::chrono::duration_cast<milliseconds>(
                  time_point.time_since_epoch())
              % 1000;

    auto const t = CLOCK_TYPE::to_time_t(time_point);
    auto const tm = *std::localtime(std::addressof(t));

    std::ostringstream stm;
    stm « std::put_time(std::addressof(tm), str(format))
        « '.' « std::setfill('0') « std::setw(3) « ms.count();
    return stm.str();
}

inline string get_current_time() {
    auto now = std::chrono::system_clock::now();
    return timestamp(now);
}

#ifdef _WIN32
    #pragma warning(pop)
#endif

inline ms get_current_timestamp_ms() {
    return std::chrono::duration_cast<ms>(
        clock::now().time_since_epoch());
}

inline ui64 get_current_timestamp() {
    return get_current_timestamp_ms().count();
}

} // namespace lava
```

5.77 liblava/core/types.hpp File Reference

Basic types.

```
#include <cstdint>
#include <functional>
#include <liblava/core/def.hpp>
#include <map>
#include <string>
#include <string_view>
#include <vector>
```

Classes

- struct lava::no_copy_no_move

  No copy and no move object.
- struct lava::interface

  Interface.
- struct lava::pair_hash

  Pair hash.

Macros

- #define ENUM_FLAG_OPERATORS(T)

  Enum flag operators.

Typedefs

- using **lava::int8** = std::int8_t

  8 bit integer
- using lava::i8 = int8
- using **lava::uint8** = std::uint8_t

  8 bit unsigned integer
- using lava::ui8 = uint8
- using **lava::int16** = std::int16_t

  16 bit integer
- using lava::i16 = int16
- using **lava::uint16** = std::uint16_t

  16 bit unsigned integer
- using lava::ui16 = uint16
- using **lava::int32** = std::int32_t

  32 bit integer
- using lava::i32 = int32
- using **lava::uint32** = std::uint32_t

  32 bit unsigned integer
- using lava::ui32 = uint32
- using **lava::int64** = std::int64_t

  64 bit integer
- using lava::i64 = int64

- using **lava::uint64** = std::uint64_t

  64 bit unsigned integer
- using lava::ui64 = uint64
- using **lava::char8** = std::int_fast8_t

  8 bit char
- using lava::c8 = char8
- using **lava::uchar8** = std::uint_fast8_t

  8 bit unsigened char
- using lava::uc8 = uchar8
- using **lava::char16** = int16

  16 bit char
- using lava::c16 = char16
- using **lava::uchar16** = uint16

  16 bit unsigned char
- using lava::uc16 = uchar16
- using **lava::char32** = int32

  32 bit char
- using lava::c32 = char32
- using **lava::uchar32** = uint32

  32 bit unsigned char
- using lava::uc32 = uchar32
- using **lava::size_t** = std::size_t

  Size.
- using **lava::uchar** = unsigned char

  Unsigned char.
- using **lava::r32** = float

  Single-precision floating-point.
- using **lava::r64** = double

  Double-precision floating-point.
- using **lava::real** = r64

  Real number.
- using **lava::delta** = r32

  Delta.
- using **lava::void_p** = void ∗

  Void pointer.
- using **lava::void_cp** = void const ∗

  Const void pointer.
- using **lava::flag** = ui32

  Flag.
- using **lava::index** = ui32

  Index.
- using **lava::index_t** = index

  Index type.
- using **lava::index_list** = std::vector< index >

  List of indices.
- using **lava::index_map** = std::map< index, index >

  Map of indices.
- using **lava::string** = std::string

  String.
- using **lava::string_ref** = string const &

  Reference to string.

- using **lava::string_list** = std::vector< string >

  List of strings.
- using **lava::string_list_ref** = string_list const &

  Reference to list of strings.
- using **lava::string_view** = std::string_view

  String view.
- using **lava::string_map** = std::map< string, string >

  Map of strings.
- using **lava::string_map_ref** = string_map const &

  Reference to map of strings.
- using **lava::name** = char const *

  Name.
- using **lava::name_ref** = char const &

  Reference to name.
- using **lava::names** = std::vector< name >

  List of names.
- using **lava::names_ref** = names const &

  Reference to list of names.

Functions

- name lava::str (string_ref value)

  Get c-string representation of string.
- r32 lava::to_r32 (auto value)

  Convert to r32.
- r64 lava::to_r64 (auto value)

  Convert to r64.
- i32 lava::to_i32 (auto value)

  Convert to i32.
- i64 lava::to_i64 (auto value)

  Convert to i64.
- ui32 lava::to_ui32 (auto value)

  Convert to ui32.
- ui64 lava::to_ui64 (auto value)

  Convert to ui64.
- size_t lava::to_size_t (auto value)

  Convert to size_t.
- index lava::to_index (auto value)

  Convert to index.
- char const * lava::to_char (auto value)

  Convert to pointer const char.
- template<typename T >
  void lava::hash_combine (size_t &seed, T const &val)

  Combine hash seed with value - from boost (functional/hash)
- template<typename T >
  void lava::hash_val (size_t &seed, T const &val)

  Auxiliary generic functions to create a hash value using a seed.
- template<typename T , typename... Types>
  void lava::hash_val (size_t &seed, T const &val, Types const &... args)
- template<typename... Types>
  size_t lava::hash_val (Types const &... args)

Variables

- constexpr index const **lava::no_index** = 0xffffffff

  No index.
- constexpr name **lava::_lava_** = "lava"

  lava
- constexpr name **lava::_liblava_** = "liblava"

  liblava
- constexpr name **lava::_default_** = "default"

  default

5.77.1 Detailed Description

Basic types.

Authors

Lava Block OÜ and contributors

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5.77.2 Macro Definition Documentation

5.77.2.1 ENUM_FLAG_OPERATORS

```
#define ENUM_FLAG_OPERATORS(
            T )
```

Value:

```
inline T operator~(T a) { \
    return static_cast<T>(~static_cast<std::underlying_type<T>::type>(a)); \
} \
inline T operator|(T a, T b) { \
    return static_cast<T>(static_cast<std::underlying_type<T>::type>(a) |
   static_cast<std::underlying_type<T>::type>(b)); \
} \
inline T operator&(T a, T b) { \
    return static_cast<T>(static_cast<std::underlying_type<T>::type>(a) &
   static_cast<std::underlying_type<T>::type>(b)); \
} \
inline T operator^(T a, T b) { \
    return static_cast<T>(static_cast<std::underlying_type<T>::type>(a) ^
   static_cast<std::underlying_type<T>::type>(b)); \
} \
inline T& operator|=(T& a, T b) { \
    return reinterpret_cast<T&>(reinterpret_cast<std::underlying_type<T>::type&>(a) |=
   static_cast<std::underlying_type<T>::type>(b)); \
} \
inline T& operator&=(T& a, T b) { \
    return reinterpret_cast<T&>(reinterpret_cast<std::underlying_type<T>::type&>(a) &=
   static_cast<std::underlying_type<T>::type>(b)); \
} \
inline T& operator^=(T& a, T b) { \
    return reinterpret_cast<T&>(reinterpret_cast<std::underlying_type<T>::type&>(a) ^=
   static_cast<std::underlying_type<T>::type>(b)); \
}
```

Enum flag operators.

5.77.3 Typedef Documentation

5.77.3.1 c16

```
using lava::c16 = typedef char16
```

See also

char16

5.77.3.2 c32

```
using lava::c32 = typedef char32
```

See also

char32

5.77.3.3 c8

```
using lava::c8 = typedef char8
```

See also

char8

5.77.3.4 i16

```
using lava::i16 = typedef int16
```

See also

int16

5.77.3.5 i32

```
using lava::i32 = typedef int32
```

See also

int32

5.77.3.6 i64

```
using lava::i64 = typedef int64
```

See also

int64

5.77.3.7 i8

```
using lava::i8 = typedef int8
```

See also

int8

5.77.3.8 uc16

```
using lava::uc16 = typedef uchar16
```

See also

uchar16

5.77.3.9 uc32

```
using lava::uc32 = typedef uchar32
```

See also

uchar32

5.77.3.10 uc8

```
using lava::uc8 = typedef uchar8
```

See also

uchar8

5.77.3.11 ui16

```
using lava::ui16 = typedef uint16
```

See also

uint16

5.77.3.12 ui32

```
using lava::ui32 = typedef uint32
```

See also

uint32

5.77.3.13 ui64

```
using lava::ui64 = typedef uint64
```

See also

uint64

5.77.3.14 ui8

```
using lava::ui8 = typedef uint8
```

See also

uint8

5.77.4 Function Documentation

5.77.4.1 hash_combine()

```
template<typename T >
void lava::hash_combine (
            size_t & seed,
            T const & val )  [inline]
```

Combine hash seed with value - from boost (functional/hash)

See also

http://www.boost.org/doc/libs/1_77_0/doc/html/hash/combine.html

Template Parameters

T	Type of value

Parameters

seed	Seed to combine
val	Value to combine

5.77.4.2 hash_val() [1/3]

```
template<typename T >
void lava::hash_val (
            size_t & seed,
            T const & val )  [inline]
```

Auxiliary generic functions to create a hash value using a seed.

Template Parameters

T	Type of value

Parameters

seed	Seed for hash
val	Hash value

5.77.4.3 hash_val() [2/3]

```
template<typename T , typename... Types>
void lava::hash_val (
            size_t & seed,
            T const & val,
            Types const &... args ) [inline]
```

See also

hash_val<T>()

5.77.4.4 hash_val() [3/3]

```
template<typename... Types>
size_t lava::hash_val (
            Types const &... args ) [inline]
```

See also

hash_val<T>()

5.77.4.5 str()

```
name lava::str (
            string_ref value ) [inline]
```

Get c-string representation of string.

Parameters

value	Source string

Returns

name C-string representation

5.77.4.6 to_char()

```
char const * lava::to_char (
            auto value ) [inline]
```

Convert to pointer const char.

Parameters

value	Source value

Returns

char const∗ Converted value

5.77.4.7 to_i32()

```
i32 lava::to_i32 (
            auto value )  [inline]
```

Convert to i32.

Parameters

value	Source value

Returns

i32 Converted value

5.77.4.8 to_i64()

```
i64 lava::to_i64 (
            auto value )  [inline]
```

Convert to i64.

Parameters

value	Source value

Returns

i64 Converted value

5.77.4.9 to_index()

```
index lava::to_index (
            auto value )  [inline]
```

Convert to index.

Parameters

value	Source value

Returns

index Converted value

5.77.4.10 to_r32()

```
r32 lava::to_r32 (
            auto value )  [inline]
```

Convert to r32.

Parameters

value	Source value

Returns

r32 Converted value

5.77.4.11 to_r64()

```
r64 lava::to_r64 (
            auto value )  [inline]
```

Convert to r64.

Parameters

value	Source value

Returns

r64 Converted value

5.77.4.12 to_size_t()

```
size_t lava::to_size_t (
            auto value )  [inline]
```

Convert to size_t.

Parameters

value	Source value

Returns

size_t Converted value

5.77.4.13 to_ui32()

```
ui32 lava::to_ui32 (
            auto value )  [inline]
```

Convert to ui32.

Parameters

value	Source value

Returns

ui32 Converted value

5.77.4.14 to_ui64()

```
ui64 lava::to_ui64 (
            auto value )  [inline]
```

Convert to ui64.

Parameters

value	Source value

Returns

ui64 Converted value

5.78 types.hpp

Go to the documentation of this file.

```
#pragma once
```

```

#include <cstdint>
#include <functional>
#include <liblava/core/def.hpp>
#include <map>
#include <string>
#include <string_view>
#include <vector>

namespace lava {

using int8 = std::int8_t;

using i8 = int8;

using uint8 = std::uint8_t;

using ui8 = uint8;

using int16 = std::int16_t;

using i16 = int16;

using uint16 = std::uint16_t;

using ui16 = uint16;

using int32 = std::int32_t;

using i32 = int32;

using uint32 = std::uint32_t;

using ui32 = uint32;

using int64 = std::int64_t;

using i64 = int64;

using uint64 = std::uint64_t;

using ui64 = uint64;

using char8 = std::int_fast8_t;

using c8 = char8;

using uchar8 = std::uint_fast8_t;

using uc8 = uchar8;

using char16 = int16;

using c16 = char16;

using uchar16 = uint16;

using uc16 = uchar16;

using char32 = int32;

using c32 = char32;

using uchar32 = uint32;

using uc32 = uchar32;

using size_t = std::size_t;

using uchar = unsigned char;

using r32 = float;

using r64 = double;

using real = r64;

using delta = r32;

using void_p = void*;

using void_cp = void const*;

using flag = ui32;

using index = ui32;

```

```
using index_t = index;

constexpr index const no_index = 0xffffffff;

using index_list = std::vector<index>;

using index_map = std::map<index, index>;

using string = std::string;

using string_ref = string const&;

using string_list = std::vector<string>;

using string_list_ref = string_list const&;

using string_view = std::string_view;

using string_map = std::map<string, string>;

using string_map_ref = string_map const&;

using name = char const*;

using name_ref = char const&;

using names = std::vector<name>;

using names_ref = names const&;

constexpr name _lava_ = "lava";

constexpr name _liblava_ = "liblava";

constexpr name _default_ = "default";

inline name str(string_ref value) {
    return value.c_str();
}

inline r32 to_r32(auto value) {
    return static_cast<r32>(value);
}

inline r64 to_r64(auto value) {
    return static_cast<r64>(value);
}

inline i32 to_i32(auto value) {
    return static_cast<i32>(value);
}

inline i64 to_i64(auto value) {
    return static_cast<i64>(value);
}

inline ui32 to_ui32(auto value) {
    return static_cast<ui32>(value);
}

inline ui64 to_ui64(auto value) {
    return static_cast<ui64>(value);
}

inline size_t to_size_t(auto value) {
    return static_cast<size_t>(value);
}

inline index to_index(auto value) {
    return static_cast<index>(value);
}

inline char const* to_char(auto value) {
    return (char const*) value;
}

struct no_copy_no_move {
    no_copy_no_move() = default;

    no_copy_no_move(no_copy_no_move const&) = delete;

    void operator=(no_copy_no_move const&) = delete;
};

struct interface {
    virtual ~interface() = default;
};
```

```

template<typename T>
inline void hash_combine(size_t& seed,
                         T const& val) {
    seed ^= std::hash<T>()(val) + 0x9e3779b9
            + (seed « 6) + (seed » 2);
}

template<typename T>
inline void hash_val(size_t& seed,
                     T const& val) {
    hash_combine(seed, val);
}

template<typename T, typename... Types>
inline void hash_val(size_t& seed,
                     T const& val,
                     Types const&... args) {
    hash_combine(seed, val);
    hash_val(seed, args...);
}

template<typename... Types>
inline size_t hash_val(Types const&... args) {
    size_t seed = 0;
    hash_val(seed, args...);
    return seed;
}

struct pair_hash {
    template<class T1, class T2>
    size_t operator()(std::pair<T1, T2> const& p) const {
        return hash_val(p.first, p.second);
    }
};

#define ENUM_FLAG_OPERATORS(T) \
    inline T operator~(T a) { \
        return static_cast<T>(~static_cast<std::underlying_type<T>::type>(a)); \
    } \
    inline T operator|(T a, T b) { \
        return static_cast<T>(static_cast<std::underlying_type<T>::type>(a) |
    static_cast<std::underlying_type<T>::type>(b)); \
    } \
    inline T operator&(T a, T b) { \
        return static_cast<T>(static_cast<std::underlying_type<T>::type>(a) &
    static_cast<std::underlying_type<T>::type>(b)); \
    } \
    inline T operator^(T a, T b) { \
        return static_cast<T>(static_cast<std::underlying_type<T>::type>(a) ^
    static_cast<std::underlying_type<T>::type>(b)); \
    } \
    inline T& operator|=(T& a, T b) { \
        return reinterpret_cast<T&>(reinterpret_cast<std::underlying_type<T>::type&>(a) |=
    static_cast<std::underlying_type<T>::type>(b)); \
    } \
    inline T& operator&=(T& a, T b) { \
        return reinterpret_cast<T&>(reinterpret_cast<std::underlying_type<T>::type&>(a) &=
    static_cast<std::underlying_type<T>::type>(b)); \
    } \
    inline T& operator^=(T& a, T b) { \
        return reinterpret_cast<T&>(reinterpret_cast<std::underlying_type<T>::type&>(a) ^=
    static_cast<std::underlying_type<T>::type>(b)); \
    }

} // namespace lava
```

5.79 liblava/core/version.hpp File Reference

Version information.

```
#include <liblava/core/types.hpp>
```

Classes

- struct lava::internal_version

Internal version.

- struct lava::version

  Version.

Typedefs

- using **lava::int_version** = internal_version

  Internal version.

Enumerations

- enum class lava::version_stage {
  preview , **alpha** , **beta** , **rc** ,
  release }

  Version stages.

Functions

- int_version lava::to_version (string_ref str)

  Convert string to internal version.

Variables

- constexpr name **lava::_build_date** = LAVA_BUILD_DATE

  Build date.
- constexpr name **lava::_build_time** = LAVA_BUILD_TIME

  Build time.

5.79.1 Detailed Description

Version information.

Authors

Lava Block OÜ and contributors

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5.79.2 Function Documentation

5.79.2.1 to_version()

```
int_version lava::to_version (
            string_ref str ) [inline]
```

Convert string to internal version.

Parameters

str	String to convert

Returns

int_version Internal version

5.80 version.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/core/types.hpp>

namespace lava {

struct internal_version {
    i32 major = LAVA_VERSION_MAJOR;

    i32 minor = LAVA_VERSION_MINOR;

    i32 patch = LAVA_VERSION_PATCH;

    auto operator<=>(internal_version const&) const = default;
};

using int_version = internal_version;

inline int_version to_version(string_ref str) {
    int_version result{ 0, 0, 0 };
    sscanf(str.c_str(), "%d.%d.%d",
           &result.major, &result.minor, &result.patch);
    return result;
}

enum class version_stage {
    preview,
    alpha,
    beta,
    rc,
    release
};

struct version {
    i32 year = 2022;

    i32 release = 0;

    version_stage stage = version_stage::release;

    i32 rev = 0;
};

constexpr name _build_date = LAVA_BUILD_DATE;

constexpr name _build_time = LAVA_BUILD_TIME;

} // namespace lava
```

5.81 liblava/engine.hpp File Reference

Engine module.

```
#include <liblava/engine/engine.hpp>
#include <liblava/engine/producer.hpp>
#include <liblava/engine/property.hpp>
```

5.81.1 Detailed Description

Engine module.

Author

Lava Block OÜ and contributors

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5.82 engine.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/engine/engine.hpp>
11 #include <liblava/engine/producer.hpp>
12 #include <liblava/engine/property.hpp>
```

5.83 liblava/engine/engine.hpp File Reference

Engine.

```
#include <liblava/app/app.hpp>
#include <liblava/engine/producer.hpp>
#include <liblava/engine/property.hpp>
```

Classes

- struct lava::engine

  Engine.

Variables

- constexpr name **lava::_props_** = "props"

  config

5.83.1 Detailed Description

Engine.

Authors

Lava Block OÜ and contributors

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5.84 engine.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/app/app.hpp>
#include <liblava/engine/producer.hpp>
#include <liblava/engine/property.hpp>

namespace lava {

constexpr name _props_ = "props";

struct engine : app {
    using app::app;

    bool setup() override;

    property props;

    producer_t producer;

private:
    void handle_config();

    json_file::callback config_callback;
};

} // namespace lava
```

5.85 liblava/engine/producer.hpp File Reference

Producer.

```
#include <liblava/fwd.hpp>
#include <liblava/resource.hpp>
```

Classes

- struct lava::producer

  Producer.

Typedefs

- using **lava::producer_t** = producer

  Producer type.

Variables

- constexpr name **lava::_shader_path_** = "shader/"

  shader folder
- constexpr name **lava::_temp_path_** = "temp/"

  temp folder
- constexpr name **lava::_hash_json_** = "hash.json"

  hash file

5.85.1 Detailed Description

Producer.

Authors

Lava Block OÜ and contributors

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5.86 producer.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/fwd.hpp>
11 #include <liblava/resource.hpp>
12
13 namespace lava {
14
16 constexpr name _shader_path_ = "shader/";
17
19 constexpr name _temp_path_ = "temp/";
20
22 constexpr name _hash_json_ = "hash.json";
23
27 struct producer {
29     engine* context = nullptr;
30
36     mesh::ptr create_mesh(mesh_type mesh_type);
37
43     mesh::ptr get_mesh(string_ref name);
44
50     bool add_mesh(mesh::ptr mesh);
51
57     texture::ptr create_texture(uv2 size);
58
64     texture::ptr get_texture(string_ref name);
65
71     bool add_texture(texture::ptr product);
72
79     cdata get_shader(string_ref name,
80                      bool reload = false);
81
87     cdata reload_shader(string_ref name) {
88         return get_shader(name, true);
89     }
90
98     data compile_shader(cdata product,
99                         string_ref name,
100                          string_ref filename) const;
101
105     void destroy();
106
110     void clear();
111
113     id_registry<mesh, string> meshes;
114
116     id_registry<texture, string> textures;
117
121     enum shader_optimization : index {
122         none = 0,
123         size,
124         performance
125     };
126
128     shader_optimization shader_opt = shader_optimization::performance;
129
133     enum shader_language : index {
134         glsl = 0,
135         hlsl
```

```
    };

    shader_language shader_lang = shader_language::glsl;

    bool shader_debug = false;

private:
    void update_hash(string_ref name,
                     string_map_ref file_hash_map) const;

    bool valid_shader(string_ref name) const;

    using shader_map = std::map<string, data>;

    shader_map shaders;
};

using producer_t = producer;

} // namespace lava
```

5.87 liblava/engine/property.hpp File Reference

Props master.

```
#include <liblava/file/file_utils.hpp>
#include <liblava/file/json.hpp>
#include <liblava/frame/argh.hpp>
#include <liblava/fwd.hpp>
```

Classes

- struct lava::property

  Props master.
- struct lava::property::prop

  Prop.

5.87.1 Detailed Description

Props master.

Authors

Lava Block OÜ and contributors

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5.88 property.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/file/file_utils.hpp>
11 #include <liblava/file/json.hpp>
12 #include <liblava/frame/argh.hpp>
13 #include <liblava/fwd.hpp>
14
15 namespace lava {
16
20 struct property : configurable {
22     engine* context = nullptr;
23
27     struct prop {
29         using map = std::map<string, prop>;
30
35         prop(string_ref filename)
36         : filename(filename) {}
37
39         string filename;
40
42         file_data data;
43     };
44
50     void add(string_ref name,
51              string_ref filename);
52
58     cdata operator()(string_ref name);
59
65     string_ref get_filename(string_ref name) const {
66         return map.at(name).filename;
67     }
68
74     void set_filename(string_ref name,
75                       string_ref filename) {
76         map.at(name).filename = filename;
77     }
78
84     bool exists(string_ref name) const {
85         return map.count(name);
86     }
87
93     bool empty(string_ref name) const {
94         return map.at(name).data.ptr == nullptr;
95     }
96
102     bool load(string_ref name);
103
108     void unload(string_ref name) {
109         map.at(name).data = {};
110     }
111
116     bool load_all();
117
121     void unload_all() {
122         for (auto& [name, prop] : map)
123             prop.data = {};
124     }
125
130     bool check();
131
136     void parse(cmd_line cmd_line);
137
141     void clear() {
142         map.clear();
143     }
144
149     prop::map const& get_all() const {
150         return map;
151     }
152
157     void set_config(json_ref j) override;
158
163     json get_config() const override;
164
165 private:
167     prop::map map;
168 };
169
170 } // namespace lava
```

5.89 liblava/file.hpp File Reference

File module.

```
#include <liblava/file/file.hpp>
#include <liblava/file/file_system.hpp>
#include <liblava/file/file_utils.hpp>
#include <liblava/file/json.hpp>
#include <liblava/file/json_file.hpp>
```

5.89.1 Detailed Description

File module.

Author

Lava Block OÜ and contributors

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5.90 file.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/file/file.hpp>
11 #include <liblava/file/file_system.hpp>
12 #include <liblava/file/file_utils.hpp>
13 #include <liblava/file/json.hpp>
14 #include <liblava/file/json_file.hpp>
```

5.91 liblava/file/file.hpp File Reference

File access.

```
#include <fstream>
#include <liblava/core/data.hpp>
```

Classes

- struct lava::file

  File.

Enumerations

- enum class lava::file_type : index { **none** = 0 , **fs** , **f_stream** }

  File types.
- enum class lava::file_mode : index { **read** = 0 , **write** }

  File modes.

Functions

- bool lava::file_error (i64 result)

  Check file error result.

Variables

- constexpr i64 const **lava::file_error_result** = -1

  File error result.

5.91.1 Detailed Description

File access.

Authors

Lava Block OÜ and contributors

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5.91.2 Function Documentation

5.91.2.1 file_error()

```
bool lava::file_error (
            i64 result )  [inline]
```

Check file error result.

Parameters

result	Result code to check

Returns

Error result or okay

5.92 file.hpp

Go to the documentation of this file.

```
#pragma once

#include <fstream>
#include <liblava/core/data.hpp>

// fwd
struct PHYSFS_File;

namespace lava {

enum class file_type : index {
    none = 0,
    fs,
    f_stream
};

constexpr i64 const file_error_result = -1;

inline bool file_error(i64 result) {
    return result == file_error_result;
}

enum class file_mode : index {
    read = 0,
    write
};

struct file : no_copy_no_move {
    using ref = file const&;

    explicit file(string_ref path = "",
                  file_mode mode = file_mode::read);

    ~file();

    bool open(string_ref path,
              file_mode mode = file_mode::read);

    void close();

    bool opened() const;

    i64 get_size() const;

    i64 read(data_ptr data) {
        return read(data,
                    to_ui64(get_size()));
    }

    i64 read(data_ptr data, ui64 size);

    i64 write(data_cptr data, ui64 size);

    i64 seek(ui64 position);

    i64 tell() const;

    bool writable() const {
        return mode == file_mode::write;
    }

    file_type get_type() const {
        return type;
    }

    string_ref get_path() const {
        return path;
    }

private:
    file_type type = file_type::none;

    file_mode mode = file_mode::read;
```

```
    string path;

    PHYSFS_File* fs_file = nullptr;

    mutable std::ifstream i_stream;

    mutable std::ofstream o_stream;
};

} // namespace lava
```

5.93 liblava/file/file_system.hpp File Reference

File system.

```
#include <filesystem>
#include <liblava/core/version.hpp>
```

Classes

- struct lava::file_system

  File system.

5.93.1 Detailed Description

File system.

Authors

Lava Block OÜ and contributors

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5.94 file_system.hpp

Go to the documentation of this file.

```
#pragma once

#include <filesystem>
#include <liblava/core/version.hpp>

namespace lava {

struct file_system : no_copy_no_move {
    int_version get_version();

    string get_base_dir();

    string get_full_base_dir(string_ref path);

    string get_pref_dir();

    string get_res_dir();
```

```

    bool mount(string_ref path);

    bool mount_base(string_ref base_dir_path);

    bool exists(string_ref file);

    string get_real_dir(string_ref file);

    string_list enumerate_files(string_ref path);

    bool initialize(string_ref argv_0,
                    string_ref org,
                    string_ref app,
                    string_ref ext);

    void terminate();

    string_list mount_res();

    bool create_folder(string_ref name = "data");

    void clean_pref_dir();

    string_ref get_org() const {
        return org;
    }

    string_ref get_app() const {
        return app;
    }

    string_ref get_ext() const {
        return ext;
    }

    bool ready() const {
        return initialized;
    }

private:
    bool initialized = false;

    string org;

    string app;

    string ext;

    string res_path;
};

} // namespace lava
```

5.95 liblava/file/file_utils.hpp File Reference

File utilities.

```
#include <liblava/core/data.hpp>
```

Classes

- struct lava::file_data

  File data.
- struct lava::file_delete

  File delete guard.

Functions

- bool lava::read_file (std::vector< char > &out, string_ref filename)

  Read data from file.
- bool lava::write_file (string_ref filename, char const *data, size_t data_size)

  Write data to file.
- bool lava::extension (string_ref filename, string_ref extension)

  Check extension of file.
- bool lava::extension (string_ref filename, string_list_ref extensions)

  Check extensions of file.
- string lava::get_filename_from (string_ref path, bool with_extension=false)

  Get the file name from path.
- bool lava::remove_existing_path (string &target, string_ref path)

  Remove existing path.
- bool lava::load_file_data (string_ref filename, data &target)

  Load file data.

5.95.1 Detailed Description

File utilities.

Authors

Lava Block OÜ and contributors

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5.95.2 Function Documentation

5.95.2.1 extension() [1/2]

```
bool lava::extension (
            string_ref filename,
            string_list_ref extensions )
```

Check extensions of file.

Parameters

filename	Name of file
extensions	List of extensions to check

Returns

Extension found or not

5.95.2.2 extension() [2/2]

```
bool lava::extension (
            string_ref filename,
            string_ref extension )
```

Check extension of file.

Parameters

filename	Name of file
extension	Extension to check

Returns

Extension found or not

5.95.2.3 get_filename_from()

```
string lava::get_filename_from (
            string_ref path,
            bool with_extension = false )
```

Get the file name from path.

Parameters

path	Target path
with_extension	Include extension in file name

Returns

string File name

5.95.2.4 load_file_data()

```
bool lava::load_file_data (
            string_ref filename,
            data & target )
```

Load file data.

Parameters

filename	Name of file
target	Target data

Returns

Load was successful or failed

5.95.2.5 read_file()

```
bool lava::read_file (
            std::vector< char > & out,
            string_ref filename )
```

Read data from file.

Parameters

out	File data
filename	Name of file

Returns

Read was successful or failed

5.95.2.6 remove_existing_path()

```
bool lava::remove_existing_path (
            string & target,
            string_ref path )
```

Remove existing path.

Parameters

target	Target path
path	Path to remove

Returns

Remove was successful or failed

5.95.2.7 write_file()

```
bool lava::write_file (
            string_ref filename,
            char const * data,
            size_t data_size )
```

Write data to file.

Parameters

filename	Name of file
data	Data to write
data_size	Size of data

Returns

Write was successful or failed

5.96 file_utils.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/core/data.hpp>

namespace lava {

bool read_file(std::vector<char>& out, string_ref filename);

bool write_file(string_ref filename,
                char const* data,
                size_t data_size);

bool extension(string_ref filename, string_ref extension);

bool extension(string_ref filename,
               string_list_ref extensions);

string get_filename_from(string_ref path,
                         bool with_extension = false);

bool remove_existing_path(string& target,
                          string_ref path);

bool load_file_data(string_ref filename,
                    data& target);

struct file_data : unique_data {
    using ref = file_data const&;

    using unique_data::unique_data;

    explicit file_data(string_ref filename)
    : filename(filename) {
        load_file_data(filename, *this);
    }

    string filename;
};

struct file_delete : no_copy_no_move {
    explicit file_delete(string filename = "")
    : filename(filename) {}

    ~file_delete();

    string filename;

```

```
    bool active = true;
};

} // namespace lava
```

5.97 json.hpp

```

#pragma once

#include <liblava/core/types.hpp>
#include <nlohmann/json.hpp>

namespace lava {

using json = nlohmann::json;

using json_ref = json const&;

struct configurable : interface {
    virtual void set_config(json_ref j) = 0;

    virtual json get_config() const = 0;
};

} // namespace lava
```

5.98 liblava/file/json_file.hpp File Reference

Json file.

`#include <liblava/file/json.hpp>`

Classes

- struct lava::json_file

  Json file.
- struct lava::json_file::callback

  Json file callback.

5.98.1 Detailed Description

Json file.

Authors

Lava Block OÜ and contributors

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5.99 json_file.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/file/json.hpp>

namespace lava {

struct json_file {
    explicit json_file(string_ref path = "config.json");

    struct callback {
        using list = std::vector<callback*>;

        using load_func = std::function<void(json_ref)>;

        load_func on_load;

        using save_func = std::function<json()>;

        save_func on_save;
    };

    void add(callback* callback);

    void remove(callback* callback);

    void clear() {
        callbacks.clear();
    }

    void set(string_ref value) {
        path = value;
    }

    string_ref get() const {
        return path;
    }

    bool load();

    bool save();

private:
    string path;

    callback::list callbacks;
};

} // namespace lava
```

5.100 liblava/frame.hpp File Reference

Frame module.

```
#include <liblava/frame/argh.hpp>
#include <liblava/frame/driver.hpp>
#include <liblava/frame/frame.hpp>
#include <liblava/frame/gamepad.hpp>
#include <liblava/frame/input.hpp>
#include <liblava/frame/render_target.hpp>
#include <liblava/frame/renderer.hpp>
#include <liblava/frame/swapchain.hpp>
#include <liblava/frame/window.hpp>
```

5.100.1 Detailed Description

Frame module.

Author

Lava Block OÜ and contributors

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5.101 frame.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/frame/argh.hpp>
11 #include <liblava/frame/driver.hpp>
12 #include <liblava/frame/frame.hpp>
13 #include <liblava/frame/gamepad.hpp>
14 #include <liblava/frame/input.hpp>
15 #include <liblava/frame/render_target.hpp>
16 #include <liblava/frame/renderer.hpp>
17 #include <liblava/frame/swapchain.hpp>
18 #include <liblava/frame/window.hpp>
```

5.102 liblava/frame/frame.hpp File Reference

Framework.

```
#include <liblava/base/device.hpp>
#include <liblava/base/instance.hpp>
#include <liblava/base/platform.hpp>
#include <liblava/core/time.hpp>
#include <liblava/frame/argh.hpp>
```

Classes

- struct lava::frame_env

  Framework environment.
- struct lava::frame

  Framework.

Enumerations

- enum lava::error {
  not_ready = -1 , **create_failed** = -2 , **init_failed** = -3 , **load_failed** = -4 ,
  run_aborted = -5 , **still_running** = -6 , **program_failed** = -7 }

  Error codes.

Functions

- ms lava::now ()

  Get the current time.
- void lava::handle_events (bool wait=false)

  Handle events.
- void lava::handle_events (ms timeout)

  Handle events.
- void lava::handle_events (seconds timeout)

  Handle events.
- void **lava::post_empty_event** ()

  Post an empty event.

Variables

- constexpr bool const **lava::run_abort** = false

  Run abort result.
- constexpr bool const **lava::run_continue** = true

  Run continue result.

5.102.1 Detailed Description

Framework.

Authors

Lava Block OÜ and contributors

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5.102.2 Function Documentation

5.102.2.1 handle_events() [1/3]

```
void lava::handle_events (
            bool wait = false )
```

Handle events.

Parameters

wait	Wait for events

5.102.2.2 handle_events() [2/3]

```
void lava::handle_events (
            ms timeout )
```

Handle events.

Parameters

timeout	Wait timeout in milliseconds

5.102.2.3 handle_events() [3/3]

```
void lava::handle_events (
            seconds timeout )
```

Handle events.

Parameters

timeout	Wait timeout in seconds

5.102.2.4 now()

```
ms lava::now ( )
```

Get the current time.

Returns

ms Current milliseconds

5.103 frame.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/base/device.hpp>
#include <liblava/base/instance.hpp>
#include <liblava/base/platform.hpp>
#include <liblava/core/time.hpp>
#include <liblava/frame/argh.hpp>

namespace lava {
```

```
17
21 struct frame_env {
23     using ref = frame_env const&;
24
28     explicit frame_env() {
29         set_default();
30     }
31
37     explicit frame_env(name app_name,
38                        argh::parser cmd_line)
39     : cmd_line(cmd_line) {
40         info.app_name = app_name;
41         set_default();
42     }
43
47     void set_default();
48
50     argh::parser cmd_line;
51
53     log_config log;
54
56     instance_info info;
57
59     instance::create_param param;
60
62     instance::debug_config debug;
63 };
64
68 enum error {
69     not_ready = -1,
70     create_failed = -2,
71     init_failed = -3,
72     load_failed = -4,
73     run_aborted = -5,
74     still_running = -6,
75     program_failed = -7,
76 };
77
82 ms now();
83
85 constexpr bool const run_abort = false;
86
88 constexpr bool const run_continue = true;
89
93 struct frame : interface, no_copy_no_move {
95     using ptr = std::shared_ptr<frame>;
96
101     explicit frame(argh::parser cmd_line);
102
107     explicit frame(frame_env env);
108
112     ~frame() override;
113
118     bool ready() const {
119         return initialized;
120     }
121
123     using result = i32; // error < 0
124
129     result run();
130
135     bool shut_down();
136
138     using run_func = std::function<bool(id::ref)>;
139
141     using run_func_ref = run_func const&;
142
148     id add_run(run_func_ref func);
149
151     using run_end_func = std::function<void()>;
152
154     using run_end_func_ref = run_end_func const&;
155
161     id add_run_end(run_end_func_ref func);
162
164     using run_once_func = std::function<bool()>;
165
167     using run_once_func_ref = run_once_func const&;
168
173     void add_run_once(run_once_func_ref func) {
174         run_once_list.push_back(func);
175     }
176
182     bool remove(id::ref func_id);
183
188     ms get_running_time() const {
189         return now() - start_time;
```

```
    }

    r64 get_running_time_sec() const {
        return to_sec(get_running_time());
    }

    cmd_line get_cmd_line() const {
        return env.cmd_line;
    }

    frame_env::ref get_env() const {
        return env;
    }

    name get_name() const {
        return env.info.app_name;
    }

    bool waiting_for_events() const {
        return wait_for_events;
    }

    void set_wait_for_events(bool value = true) {
        wait_for_events = value;
    }

    platform_t platform;

private:
    bool setup();

    void teardown();

    bool run_step();

    void trigger_run_remove();

    void trigger_run_end();

    bool initialized = false;

    frame_env env;

    bool running = false;

    bool wait_for_events = false;

    ms start_time;

    using run_func_map = std::map<id, run_func>;

    run_func_map run_map;

    using run_end_func_map = std::map<id, run_end_func>;

    run_end_func_map run_end_map;

    using run_once_func_list = std::vector<run_once_func>;

    run_once_func_list run_once_list;

    id::list run_remove_list;
};

void handle_events(bool wait = false);

void handle_events(ms timeout);

void handle_events(seconds timeout);

void post_empty_event();

} // namespace lava
```

5.104 liblava/frame/argh.hpp File Reference

Json.

```
#include <argh.h>
#include <liblava/util/log.hpp>
```

Typedefs

- using **lava::cmd_line** = argh::parser const &

  Command line.

Functions

- void lava::log_command_line (cmd_line cmd_line)

  Log command line.

5.104.1 Detailed Description

Json.

Command line arguments.

Authors

Lava Block OÜ and contributors

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5.104.2 Function Documentation

5.104.2.1 log_command_line()

```
void lava::log_command_line (
            cmd_line cmd_line )  [inline]
```

Log command line.

Parameters

cmd_line	Command line arguments

5.105 argh.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <argh.h>
```

```
#include <liblava/util/log.hpp>

namespace lava {

using cmd_line = argh::parser const&;

inline void log_command_line(cmd_line cmd_line) {
    if (!cmd_line.pos_args().empty()) {
        for (auto const& pos_arg : cmd_line.pos_args())
            log()->info("cmd {}", pos_arg);
    }

    if (!cmd_line.flags().empty()) {
        for (auto const& flag : cmd_line.flags())
            log()->info("cmd flag: {}", flag);
    }

    if (!cmd_line.params().empty()) {
        for (auto const& [key, value] : cmd_line.params())
            log()->info("cmd param: {} = {}", key, value);
    }
}

} // namespace lava
```

5.106 liblava/frame/driver.hpp File Reference

Stage driver.

`#include <liblava/frame/argh.hpp>`

Classes

- struct lava::stage

  Stage.
- struct lava::driver

  Stage driver.
- struct lava::driver::result

  Driver result.

Macros

- #define **STAGE_OBJ** stage_

  Stage object.
- #define **STAGE_FUNC** _stage

  Stage function.
- #define **STR_**(n, m) n##m

  String concatenation.
- #define **STR**(n, m) STR_(n, m)

  String concatenation.
- #define LAVA_STAGE(ID, NAME)

  lava stage macro

5.106.1 Detailed Description

Stage driver.

Authors

Lava Block OÜ and contributors

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5.106.2 Macro Definition Documentation

5.106.2.1 LAVA_STAGE

```
#define LAVA_STAGE(
            ID,
            NAME )
```

Value:

```
lava::i32 STR(STAGE_FUNC, ID)(argh::parser argh); \
lava::stage STR(STAGE_OBJ, ID)(ID, NAME, ::STR(STAGE_FUNC, ID)); \
lava::i32 STR(STAGE_FUNC, ID)(argh::parser argh)
```

lava stage macro

5.107 driver.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/frame/argh.hpp>
11
12 namespace lava {
13
17 struct stage {
19     using map = std::map<index, stage*>;
20
22     using func = std::function<i32(argh::parser)>;
23
30     explicit stage(ui32 id,
31                    string_ref name,
32                    func func);
33
35     index id = 0;
36
38     string name;
39
41     func on_func;
42 };
43
47 struct driver {
51     enum error {
52         stages_empty = -1,
53         stage_not_found = -2,
54         undef_run = -3,
55     };
56
```

```
    struct result {
        i32 driver = 0;

        i32 selected = 0;
    };

    static driver& instance() {
        static driver driver;
        return driver;
    }

    void add_stage(stage* stage) {
        assert(!stages.count(stage->id) && "stage id already defined.");
        stages.emplace(stage->id, stage);
    }

    stage::map const& get_stages() const {
        return stages;
    }

    i32 run(argh::parser cmd_line = {});

     using run_func = std::function<result(argh::parser)>;

     run_func on_run;

private:
     driver() = default;

     stage::map stages;
};

} // namespace lava

#define STAGE_OBJ stage_

#define STAGE_FUNC _stage

#define STR_(n, m) n##m

#define STR(n, m) STR_(n, m)

#define LAVA_STAGE(ID, NAME) \
    lava::i32 STR(STAGE_FUNC, ID)(argh::parser argh); \
    lava::stage STR(STAGE_OBJ, ID)(ID, NAME, ::STR(STAGE_FUNC, ID)); \
    lava::i32 STR(STAGE_FUNC, ID)(argh::parser argh)
```

5.108 liblava/frame/gamepad.hpp File Reference

Gamepad manager.

```
#include <liblava/core/id.hpp>
```

Classes

- struct lava::gamepad

  Gamepad.
- struct lava::gamepad_manager

  Gamepad manager.

Typedefs

- using **lava::gamepad_id_ref** = gamepad_id const &

  Reference to gamepad id.
- using **lava::gamepad_button_ref** = gamepad_button const &

  Reference to gamepad button.
- using **lava::gamepad_axis_ref** = gamepad_axis const &

  Reference to gamepad axis.

Enumerations

- enum class lava::gamepad_id : ui32 {
  _1 = 0 , **_2** , **_3** , **_4** ,
  _5 , **_6** , **_7** , **_8** ,
  _9 , **_10** , **_11** , **_12** ,
  _13 , **_14** , **_15** , **_16** ,
  last = _16 }

  Gamepad ids.
- enum class lava::gamepad_button : index {
  a = 0 , **b** , **x** , **y** ,
  left_bumper , **right_bumper** , **back** , **start** ,
  guide , **left_thumb** , **right_thumb** , **dpad_up** ,
  dpad_right , **dpad_down** , **dpad_left** , **last** = dpad_left ,
  cross = a , **circle** = b , **square** = x , **triangle** = y }

  Gamepad buttons.
- enum class lava::gamepad_axis : index {
  left_x = 0 , **left_y** , **right_x** , **right_y** ,
  left_trigger , **right_trigger** , **last** = right_trigger }

  Gamepad axis.

Functions

- gamepad::list lava::gamepads ()

  Get list of all gamepads.

5.108.1 Detailed Description

Gamepad manager.

Authors

Lava Block OÜ and contributors

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5.108.2 Function Documentation

5.108.2.1 gamepads()

```
gamepad::list lava::gamepads ( )
```

Get list of all gamepads.

Returns

gamepad::list List of gamepads

5.109 gamepad.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/core/id.hpp>

namespace lava {

enum class gamepad_id : ui32 {
    _1 = 0,
    _2,
    _3,
    _4,
    _5,
    _6,
    _7,
    _8,
    _9,
    _10,
    _11,
    _12,
    _13,
    _14,
    _15,
    _16,

    last = _16,
};

using gamepad_id_ref = gamepad_id const&;

enum class gamepad_button : index {
    a = 0,
    b,
    x,
    y,

    left_bumper,
    right_bumper,

    back,
    start,
    guide,

    left_thumb,
    right_thumb,

    dpad_up,
    dpad_right,
    dpad_down,
    dpad_left,

    last = dpad_left,

    cross = a,
    circle = b,
    square = x,
    triangle = y,
};

using gamepad_button_ref = gamepad_button const&;

enum class gamepad_axis : index {
    left_x = 0,
    left_y,

    right_x,
    right_y,

    left_trigger,
    right_trigger,

    last = right_trigger,
};

using gamepad_axis_ref = gamepad_axis const&;

struct gamepad {
    using list = std::vector<gamepad>;

    using ref = gamepad const&;

    explicit gamepad(gamepad_id_ref pad_id = gamepad_id::_1);
```

```
    bool ready() const;

    bool update();

    bool pressed(gamepad_button_ref button) const {
        return state.buttons[to_ui32(button)];
    }

    r32 value(gamepad_axis_ref axis) const {
        return state.axes[to_ui32(axis)];
    }

    gamepad_id_ref get_pad_id() const {
        return pad_id;
    }

    ui32 get_id() const {
        return to_ui32(get_pad_id());
    }

    name get_name() const;

private:
    gamepad_id pad_id;

    struct state {
        uchar buttons[15];

        r32 axes[6];
    };

    gamepad::state state;
};

gamepad::list gamepads();

struct gamepad_manager {
    using listener_func = std::function<bool(gamepad, bool)>;

    static gamepad_manager& singleton() {
        static gamepad_manager manager;
        return manager;
    }

    id add(listener_func listener);

    void remove(id::ref func_id);

private:
    explicit gamepad_manager();

    using listener_map = std::map<id, listener_func>;

    listener_map map;
};

} // namespace lava
```

5.110 liblava/frame/input.hpp File Reference

Input handling.

```
#include <liblava/core/id.hpp>
#include <liblava/core/misc.hpp>
```

Classes

- struct lava::key_event

  Key event.
- struct lava::scroll_offset

  Input scroll offset.

- struct lava::scroll_event

  Scroll event.
- struct lava::mouse_position

  Input mouse position.
- struct lava::mouse_move_event

  Mouse move event.
- struct lava::mouse_button_event

  Mouse button event.
- struct lava::path_drop_event

  Path drop event.
- struct lava::mouse_active_event

  Mouse active event.
- struct lava::input_callback

  Input callback.
- struct lava::input_events< T >

  List of input events.
- struct lava::input

  Input handling.
- struct lava::tooltip

  Tooltip.
- struct lava::tooltip_list

  Tooltip list.

Typedefs

- using **lava::key_t** = key

  Key type.
- using **lava::key_ref** = key const &

  Reference to key.
- using **lava::keys** = std::vector< key >

  List of keys.
- using **lava::keys_ref** = keys const &

  Reference to list of keys.
- using **lava::action_t** = action

  Action type.
- using **lava::action_ref** = action const &

  Refernece to action.
- using **lava::mod_t** = mod

  Mod type.
- using **lava::mod_ref** = mod const &

  Reference to mod.
- using **lava::mouse_position_ref** = mouse_position const &

  Reference to mouse position.
- using **lava::mouse_button_ref** = mouse_button const &

  Reference to mouse button.
- using **lava::input_key_events** = input_events< key_event >

  List of key events.
- using **lava::input_scroll_events** = input_events< scroll_event >

  List of scroll events.
- using **lava::input_mouse_move_events** = input_events< mouse_move_event >

List of mouse move events.

- using **lava::input_mouse_button_events** = input_events< mouse_button_event >

  List of mouse button events.
- using **lava::input_mouse_active_events** = input_events< mouse_active_event >

  List of mouse active events.
- using **lava::input_path_drop_events** = input_events< path_drop_event >

  List of path drop events.
- using **lava::input_t** = input

  Input type.

Enumerations

- enum class lava::key : i32 {
  unknown = -1 , **space** = 32 , **apostrophe** = 29 , **comma** = 44 ,
  minus = 45 , **period** = 46 , **slash** = 47 , **_0** = 48 ,
  _1 , **_2** , **_3** , **_4** ,
  _5 , **_6** , **_7** , **_8** ,
  _9 , **semicolon** = 59 , **equal** = 61 , **a** = 65 ,
  b , **c** , **d** , **e** ,
  f , **g** , **h** , **i** ,
  j , **k** , **l** , **m** ,
  n , **o** , **p** , **q** ,
  r , **s** , **t** , **u** ,
  v , **w** , **x** , **y** ,
  z , **left_bracket** = 91 , **backslash** = 92 , **right_bracket** = 93 ,
  grave_accent = 96 , **world_1** = 161 , **world_2** = 162 , **escape** = 256 ,
  enter , **tab** , **backspace** , **insert** ,
  del , **right** , **left** , **down** ,
  up , **page_up** , **page_down** , **home** ,
  end , **caps_lock** = 280 , **scroll_lock** , **num_lock** ,
  print_screen , **pause** , **f1** = 290 , **f2** ,
  f3 , **f4** , **f5** , **f6** ,
  f7 , **f8** , **f9** , **f10** ,
  f11 , **f12** , **f13** , **f14** ,
  f15 , **f16** , **f17** , **f18** ,
  f19 , **f20** , **f21** , **f22** ,
  f23 , **f24** , **f25** , **kp_0** = 320 ,
  kp_1 , **kp_2** , **kp_3** , **kp_4** ,
  kp_5 , **kp_6** , **kp_7** , **kp_8** ,
  kp_9 , **kp_decimal** , **kp_divide** , **kp_multiply** ,
  kp_subtract , **kp_add** , **kp_enter** , **kp_equal** ,
  left_shift = 340 , **left_control** , **left_alt** , **left_super** ,
  right_shift , **right_control** , **right_alt** , **right_super** ,
  menu , **last** = menu }

  Input keys.
- enum class lava::action : index { **release** = 0 , **press** , **repeat** }

  Input actions.
- enum class lava::mod : flag {
  none = 0 << 0 , **shift** = 1 << 0 , **control** = 1 << 1 , **alt** = 1 << 2 ,
  super = 1 << 3 , **caps_lock** = 1 << 4 , **num_lock** = 1 << 5 }

  Input mods.
- enum class lava::mouse_button : index {
  _1 = 0 , **_2** , **_3** , **_4** ,
  _5 , **_6** , **_7** , **_8** ,
  last = _8 , **left** = _1 , **right** = _2 , **middle** = _3 }

  Input mouse buttons.

Functions

- string lava::to_string (key k)

  Convert input key to string.
- i32 lava::get_scancode (key key)

  Get scancode based on key.
- bool lava::check_mod (mod m, mod c)

  Check if mod is active.
- string lava::to_string (mod m)

  Convert input mod to string.

Variables

- constexpr bool const **lava::input_ignore** = false

  Input ignore result.
- constexpr bool const **lava::input_done** = true

  Input done result.

5.110.1 Detailed Description

Input handling.

Authors

Lava Block OÜ and contributors

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5.110.2 Function Documentation

5.110.2.1 check_mod()

```
bool lava::check_mod (
            mod m,
            mod c )  [inline]
```

Check if mod is active.

Parameters

m	Target mod
c	Mod to check

Returns

Mod is active or not

5.110.2.2 get_scancode()

```
i32 lava::get_scancode (
            key key )
```

Get scancode based on key.

Parameters

key	Input key

Returns

i32 Input scan code

5.110.2.3 to_string() [1/2]

```
string lava::to_string (
            key k )
```

Convert input key to string.

Parameters

k	Input key

Returns

string String representation

5.110.2.4 to_string() [2/2]

```
string lava::to_string (
            mod m )
```

Convert input mod to string.

Parameters

m	Input mod

Returns

string String representation

5.111 input.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/core/id.hpp>
#include <liblava/core/misc.hpp>

namespace lava {

enum class key : i32 {
    unknown = -1,

    /* printable keys */

    space = 32,
    apostrophe = 29, /* ' */
    comma = 44, /* , */
    minus = 45, /* - */
    period = 46, /* . */
    slash = 47, /* / */

    _0 = 48,
    _1,
    _2,
    _3,
    _4,
    _5,
    _6,
    _7,
    _8,
    _9,

    semicolon = 59, /* ; */
    equal = 61, /* = */

    a = 65,
    b,
    c,
    d,
    e,
    f,
    g,
    h,
    i,
    j,
    k,
    l,
    m,
    n,
    o,
    p,
    q,
    r,
    s,
    t,
    u,
    v,
    w,
    x,
    y,
    z,

    left_bracket = 91, /* [ */
    backslash = 92, /* \ */
    right_bracket = 93, /* ] */
    grave_accent = 96, /* ` */
```

```

    world_1 = 161, /* non-US #1 */
    world_2 = 162, /* non-US #2 */

    /* function keys */

    escape = 256,
    enter,
    tab,
    backspace,
    insert,
    del, /* delete */

    right,
    left,
    down,
    up,

    page_up,
    page_down,
    home,
    end,

    caps_lock = 280,
    scroll_lock,
     num_lock,
     print_screen,
     pause,

     f1 = 290,
     f2,
     f3,
     f4,
     f5,
     f6,
     f7,
     f8,
     f9,
     f10,
     f11,
     f12,
     f13,
     f14,
     f15,
     f16,
     f17,
     f18,
     f19,
     f20,
     f21,
     f22,
     f23,
     f24,
     f25,

     kp_0 = 320,
     kp_1,
     kp_2,
     kp_3,
     kp_4,
     kp_5,
     kp_6,
     kp_7,
     kp_8,
     kp_9,

     kp_decimal,
     kp_divide,
     kp_multiply,
     kp_subtract,
     kp_add,
     kp_enter,
     kp_equal,

     left_shift = 340,
     left_control,
     left_alt,
     left_super,

     right_shift,
     right_control,
     right_alt,
     right_super,

     menu,

     last = menu,
```

```
};

using key_t = key;

using key_ref = key const&;

using keys = std::vector<key>;

using keys_ref = keys const&;

string to_string(key k);

i32 get_scancode(key key);

enum class action : index {
    release = 0,
    press,
    repeat
};

using action_t = action;

using action_ref = action const&;

enum class mod : flag {
    none = 0 « 0,
    shift = 1 « 0,
    control = 1 « 1,
    alt = 1 « 2,
    super = 1 « 3,
    caps_lock = 1 « 4,
    num_lock = 1 « 5,
};

ENUM_FLAG_OPERATORS(mod)


using mod_t = mod;

inline bool check_mod(mod m, mod c) {
    return (m & c) != mod::none;
}

using mod_ref = mod const&;

string to_string(mod m);

struct key_event {
    using ref = key_event const&;

    using func = std::function<bool(ref)>;

    using listeners = std::map<id, func>;

    using list = std::vector<key_event>;

    id sender;

    key_t key;

    action_t action;

    mod_t mod;

    i32 scancode = 0;

    bool pressed(key_ref k) const {
        return (action == action::press) && (key == k);
    }

    bool released(key_ref k) const {
        return (action == action::release) && (key == k);
    }

    bool repeated(key_ref k) const {
        return (action == action::repeat) && (key == k);
    }

    bool active() const {
        return (action == action::press) || (action == action::repeat);
    }

    bool pressed(key_ref k, mod_ref m) const {
        return pressed(k) && (mod == m);
    }
};

```

```
struct scroll_offset {
    r64 x = 0.0;

    r64 y = 0.0;
};

struct scroll_event {
    using ref = scroll_event const&;

    using func = std::function<bool(ref)>;

    using listeners = std::map<id, func>;

    using list = std::vector<scroll_event>;

    id sender;

    scroll_offset offset;
};

struct mouse_position {
    r64 x = 0.0;

    r64 y = 0.0;
};

using mouse_position_ref = mouse_position const&;

struct mouse_move_event {
    using ref = mouse_move_event const&;

    using func = std::function<bool(ref)>;

    using listeners = std::map<id, func>;

    using list = std::vector<mouse_move_event>;

    id sender;

    mouse_position position;
};

enum class mouse_button : index {
    _1 = 0,
    _2,
    _3,
    _4,
    _5,
    _6,
    _7,
    _8,

    last = _8,

    left = _1,
    right = _2,
    middle = _3,
};

using mouse_button_ref = mouse_button const&;

struct mouse_button_event {
    using ref = mouse_button_event const&;

    using func = std::function<bool(ref)>;

    using listeners = std::map<id, func>;

    using list = std::vector<mouse_button_event>;

    id sender;

    mouse_button button;

    action_t action;

    mod_t mod;

    bool pressed(mouse_button_ref b) const {
        return action == action::press && button == b;
    }

    bool released(mouse_button_ref b) const {
        return action == action::release && button == b;
    }
};

```

```
struct path_drop_event {
    using ref = path_drop_event const&;

    using func = std::function<bool(ref)>;

    using listeners = std::map<id, func>;

    using list = std::vector<path_drop_event>;

    id sender;

    string_list files;
};

struct mouse_active_event {
    using ref = mouse_active_event const&;

    using func = std::function<bool(ref)>;

    using listeners = std::map<id, func>;

    using list = std::vector<mouse_active_event>;

    id sender;

    bool active = false;
};

struct input_callback {
    using cptr = input_callback const*;

    using list = std::vector<input_callback*>;

    using clist = std::vector<cptr>;

    template<typename T>
    using func = std::function<bool(typename T::ref)>;

    key_event::func on_key_event;

    scroll_event::func on_scroll_event;

    mouse_move_event::func on_mouse_move_event;

    mouse_button_event::func on_mouse_button_event;

    mouse_active_event::func on_mouse_active_event;

    path_drop_event::func on_path_drop_event;
};

template<typename T>
struct input_events : T::list {
    void add(typename T::ref event) {
        this->push_back(event);
    }

    id_listeners<T> listeners;
};

using input_key_events = input_events<key_event>;

using input_scroll_events = input_events<scroll_event>;

using input_mouse_move_events = input_events<mouse_move_event>;

using input_mouse_button_events = input_events<mouse_button_event>;

using input_mouse_active_events = input_events<mouse_active_event>;

using input_path_drop_events = input_events<path_drop_event>;

constexpr bool const input_ignore = false;

constexpr bool const input_done = true;

struct input {
    input_key_events key;

    input_scroll_events scroll;

    input_mouse_move_events mouse_move;

    input_mouse_button_events mouse_button;

    input_mouse_active_events mouse_active;

```

```
    input_path_drop_events path_drop;

    void handle_events();

    void add(input_callback::cptr callback) {
        callbacks.push_back(callback);
    }

    void remove(input_callback::cptr callback) {
        lava::remove(callbacks, callback);
    }

    mouse_position_ref get_mouse_position() const {
        return current_position;
    }

    void set_mouse_position(mouse_position_ref position) {
        current_position = position;
    }

private:
    void handle_mouse_events();

    mouse_position current_position;

    input_callback::clist callbacks;
};

using input_t = input;

struct tooltip {
    tooltip(string_ref name,
            key_t key,
            mod_t mod)
    : name(name), key(key), mod(mod) {
    }

    using list = std::vector<tooltip>;

    string name;

    key_t key;

    mod_t mod;
};

struct tooltip_list {
    void add(string_ref name,
             key_t key,
             mod_t mod = mod::none) {
        tooltips.emplace_back(name, key, mod);
    }

    void clear() {
        tooltips.clear();
    }

    tooltip::list const& get_list() const {
        return tooltips;
    }

    void set(tooltip::list const& list) {
        tooltips = list;
    }

    string format_string() const;

private:
    tooltip::list tooltips;
};

} // namespace lava
```

5.112 liblava/frame/render_target.hpp File Reference

Render target.

```
#include <liblava/core/misc.hpp>
#include <liblava/frame/swapchain.hpp>
```

```
#include <liblava/fwd.hpp>
#include <liblava/resource/format.hpp>
```

Classes

- struct lava::render_target

  Render target.

Functions

- render_target::ptr lava::create_target (window *window, device_p device, bool v_sync=false, surface_format_request request={})

  Create a new render target.
- render_target::ptr lava::create_target_v_sync (window *window, device_p device, surface_format_request request={})

  Create a new render target with V-Sync enabled.

5.112.1 Detailed Description

Render target.

Authors

Lava Block OÜ and contributors

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5.112.2 Function Documentation

5.112.2.1 create_target()

```
render_target::ptr lava::create_target (
            window * window,
            device_p device,
            bool v_sync = false,
            surface_format_request request = {} )
```

Create a new render target.

Parameters

window	Target window
device	Vulkan device
v_sync	V-Sync enabled
request	Surface format request

Returns

render_target::ptr Shared pointer to render target

5.112.2.2 create_target_v_sync()

```
render_target::ptr lava::create_target_v_sync (
            window * window,
            device_p device,
            surface_format_request request = {} )  [inline]
```

Create a new render target with V-Sync enabled.

Parameters

window	Target window
device	Vulkan device
request	Surface format request

Returns

render_target::ptr Shared pointer to render target

5.113 render_target.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/core/misc.hpp>
11 #include <liblava/frame/swapchain.hpp>
12 #include <liblava/fwd.hpp>
13 #include <liblava/resource/format.hpp>
14
15 namespace lava {
16
20 struct render_target : entity {
22     using ptr = std::shared_ptr<render_target>;
23
28     static ptr make() {
29         return std::make_shared<render_target>();
30     }
31
41     bool create(device_p device,
42                 VkSurfaceKHR surface,
43                 VkSurfaceFormatKHR format,
44                 uv2 size,
45                 bool v_sync = false);
46
50     void destroy();
51
56     uv2 get_size() const {
57         return target.get_size();
58     }
59
65     bool resize(uv2 new_size) {
66         return target.resize(new_size);
67     }
68
73     ui32 get_frame_count() const {
74         return target.get_backbuffer_count();
75     }
```

```

    bool reload_request() const {
        return target.reload_request();
    }

    void reload() {
        target.resize(target.get_size());
    }

    device_p get_device() {
        return target.get_device();
    }

    swapchain* get_swapchain() {
        return &target;
    }

    VkFormat get_format() const {
        return target.get_format();
    }

    image::list const& get_backbuffers() const {
        return target.get_backbuffers();
    }

    inline image::ptr get_backbuffer(index index) {
        auto& backbuffers = get_backbuffers();
        if (index >= backbuffers.size())
            return nullptr;

        return backbuffers.at(index);
    }

    inline VkImage get_backbuffer_image(index index) {
        auto result = get_backbuffer(index);
        return result ? result->get() : 0;
    }

    inline VkImage get_image(index index) {
        return get_backbuffer_image(index);
    }

    void add_callback(target_callback::cptr callback) {
        target_callbacks.push_back(callback);
    }

    void remove_callback(target_callback::cptr callback) {
        remove(target_callbacks, callback);
    }

    using swapchain_start_func = std::function<bool()>;

    swapchain_start_func on_swapchain_start;

    using swapchain_stop_func = std::function<void()>;

    swapchain_stop_func on_swapchain_stop;

    using create_attachments_func = std::function<VkAttachments()>;

    create_attachments_func on_create_attachments;

    using destroy_attachments_func = std::function<void()>;

    destroy_attachments_func on_destroy_attachments;

private:
    swapchain target;

    swapchain::callback swapchain_callback;

    target_callback::clist target_callbacks;
};

render_target::ptr create_target(window* window,
                                 device_p device,
                                 bool v_sync = false,
                                 surface_format_request request = {});

inline render_target::ptr create_target_v_sync(window* window,
                                               device_p device,
                                               surface_format_request request = {}) {
    return create_target(window,
                         device,
                         true,
                         request);
}
```

```
233
234 } // namespace lava
```

5.114 liblava/frame/renderer.hpp File Reference

Plain renderer.

```
#include <liblava/frame/swapchain.hpp>
#include <optional>
```

Classes

- struct lava::renderer

  Plain renderer.

Typedefs

- using **lava::optional_index** = std::optional< index >

  Optional frame index.
- using **lava::renderer_t** = renderer

  Renderer type.

5.114.1 Detailed Description

Plain renderer.

Authors

Lava Block OÜ and contributors

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5.115 renderer.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/frame/swapchain.hpp>
#include <optional>

namespace lava {

using optional_index = std::optional<index>;

struct renderer : entity {
    bool create(swapchain* target);

    void destroy();

    optional_index begin_frame();

    bool end_frame(VkCommandBuffers const& cmd_buffers);

    bool frame(VkCommandBuffers const& cmd_buffers) {
        if (!begin_frame())
            return false;

        return end_frame(cmd_buffers);
    }

    index get_frame() const {
        return current_frame;
    }

    device_p get_device() {
        return device;
    }

    using destroy_func = std::function<void()>;

    destroy_func on_destroy;

    bool active = true;

private:
    device_p device = nullptr;

    queue graphics_queue;

    swapchain* target = nullptr;

    index current_frame = 0;

    ui32 queued_frames = 2;

    ui32 current_sync = 0;

    VkFences fences = {};

    VkFences fences_in_use = {};

    VkSemaphores image_acquired_semaphores = {};

    VkSemaphores render_complete_semaphores = {};
};

using renderer_t = renderer;

} // namespace lava
```

5.116 liblava/frame/swapchain.hpp File Reference

Swapchain.

```
#include <liblava/resource/image.hpp>
```

Classes

- struct lava::swapchain

  Swaphchain.
- struct lava::swapchain::callback

  Swapchain callback.

5.116.1 Detailed Description

Swapchain.

Authors

Lava Block OÜ and contributors

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5.117 swapchain.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/resource/image.hpp>
11
12 namespace lava {
13
17 struct swapchain : entity {
27     bool create(device_p device,
28                 VkSurfaceKHR surface,
29                 VkSurfaceFormatKHR format,
30                 uv2 size,
31                 bool v_sync = false);
32
36     void destroy();
37
43     bool resize(uv2 new_size);
44
48     void request_reload() {
49         reload_request_active = true;
50     }
51
56     bool reload_request() const {
57         return reload_request_active;
58     }
59
64     device_p get_device() {
65         return device;
66     }
67
72     uv2 get_size() const {
73         return size;
74     }
75
80     VkFormat get_format() const {
81         return format.format;
82     }
83
88     VkColorSpaceKHR get_color_space() const {
89         return format.colorSpace;
90     }
91
96     VkSwapchainKHR get() const {
97         return vk_swapchain;
98     }
99
```

```
    ui32 get_backbuffer_count() const {
        return to_ui32(backbuffers.size());
    }

    image::list const& get_backbuffers() const {
        return backbuffers;
    }

    struct callback {
        using list = std::vector<callback*>;

        using created_func = std::function<bool()>;

        created_func on_created;

        using destroyed_func = std::function<void()>;

        destroyed_func on_destroyed;
    };

    void add_callback(callback* cb);

    void remove_callback(callback* cb);

    bool v_sync() const {
        return v_sync_active;
    }

    bool surface_supported(index queue_family) const;

private:
    VkPresentModeKHR choose_present_mode(VkPresentModeKHRs const& present_modes) const;

    VkSwapchainCreateInfoKHR create_info(VkPresentModeKHRs present_modes);

    bool setup();

    void teardown();

    void destroy_backbuffer_views();

    device_p device = nullptr;

    VkSurfaceKHR surface = VK_NULL_HANDLE;

    VkSurfaceFormatKHR format = {};

    VkSwapchainKHR vk_swapchain = VK_NULL_HANDLE;

    image::list backbuffers;

    uv2 size;

    bool reload_request_active = false;

    bool v_sync_active = false;

    callback::list callbacks;
};

} // namespace lava
```

5.118 liblava/frame/window.hpp File Reference

Window.

```
#include <liblava/core/data.hpp>
#include <liblava/frame/input.hpp>
#include <liblava/util/math.hpp>
#include <vulkan/vulkan.h>
#include <optional>
```

Classes

- struct lava::window

Window.

- struct lava::window::state

  Window state.

Typedefs

- using **lava::window_t** = window

  Window type.

Functions

- VkSurfaceKHR lava::create_surface (GLFWwindow ∗window)

  Create a new surface.
- window ∗ lava::get_window (GLFWwindow ∗handle)

  Get the window by GLFW handle.

5.118.1 Detailed Description

Window.

Authors

Lava Block OÜ and contributors

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5.118.2 Function Documentation

5.118.2.1 create_surface()

```
VkSurfaceKHR lava::create_surface (
            GLFWwindow * window )
```

Create a new surface.

Parameters

window	GLFW window handle

Returns

VkSurfaceKHR Vulkan surface

5.118.2.2 get_window()

```
window * lava::get_window (
            GLFWwindow * handle )
```

Get the window by GLFW handle.

Parameters

handle	GLFW window handle

Returns

window* Assigned Window

5.119 window.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/core/data.hpp>
11 #include <liblava/frame/input.hpp>
12 #include <liblava/util/math.hpp>
13
14 #define VK_NO_PROTOTYPES
15 #include <vulkan/vulkan.h>
16 #include <optional>
17
19 struct GLFWwindow;
20
21 namespace lava {
22
26 struct window : entity {
30     struct state {
32         using ref = state const&;
33
35         using optional = std::optional<window::state>;
36
40         explicit state() {}
41
43         i32 x = 0;
44
46         i32 y = 0;
47
49         ui32 width = 0;
50
52         ui32 height = 0;
53
55         bool fullscreen = false;
56
58         bool floating = false;
59
61         bool resizable = true;
62
64         bool decorated = true;
65
67         bool maximized = false;
68
70         index monitor = 0;
71     };
72
74     using ptr = std::shared_ptr<window>;
75
77     using event = std::function<void(ptr)>;
78
80     using map = std::map<id, ptr>;
```

```
    using ref = window const&;

    window() = default;

    explicit window(name title)
    : title(title) {}

    bool create(state::optional state = {});

    void destroy();

    state get_state() const;

    void set_title(string_ref text);

    string_ref get_title() const {
        return title;
    }

    void set_save_name(string_ref save) {
        save_name = save;
    }

    string_ref get_save_name() const {
        return save_name;
    }

    void set_position(i32 x, i32 y);

    void get_position(i32& x, i32& y) const;

    void set_size(ui32 width, ui32 height);

    void get_size(ui32& width, ui32& height) const;

    void get_framebuffer_size(ui32& width, ui32& height) const;

    uv2 get_size() const;

    uv2 get_framebuffer_size() const;

    void set_mouse_position(r64 x, r64 y);

    void get_mouse_position(r64& x, r64& y) const;

    mouse_position get_mouse_position() const;

    void hide_mouse_cursor();

    void show_mouse_cursor();

    float get_aspect_ratio() const;

    void show();

    void hide();

    bool visible() const;

    void iconify();

    bool iconified() const;

    void restore();

    void maximize();

    bool maximized() const;

    void focus();

    bool focused() const;

    void set_fullscreen(bool active) {
        if (fullscreen_active != active)
            switch_mode_request_active = true;
    }

    bool fullscreen() const {
        return fullscreen_active;
    }

    bool hovered() const;

    bool resizable() const;

```

```
    void set_resizable(bool value);

    bool decorated() const;

    void set_decorated(bool value);

    bool floating() const;

    void set_floating(bool value);

    bool close_request() const;

    bool switch_mode_request() const {
        return switch_mode_request_active;
    }

    bool switch_mode(state::optional state = {});

    GLFWwindow* get() const {
        return handle;
    }

    bool resize_request() const {
        return resize_request_active;
    }

    bool handle_resize() {
        if (on_resize)
            if (!on_resize(framebuffer_width,
                           framebuffer_height))
                return false;

        resize_request_active = false;
        return true;
    }

    void update_state() {
        get_position(pos_x, pos_y);
        get_size(width, height);
    }

    using resize_func = std::function<bool(ui32, ui32)>;

    resize_func on_resize;

    void assign(input* callback) {
        input = callback;
    }

    void show_save_title(bool value = true) {
        save_title_active = value;
    }

    bool save_title() const {
        return save_title_active;
    }

    void update_title() {
        set_title(title);
    }

    VkSurfaceKHR create_surface();

    void set_icon(data_cptr data, uv2 size);

    index get_monitor() const;

    void center();

private:
    void handle_message();

    void handle_mouse_message();

    GLFWwindow* handle = nullptr;

    input_t* input = nullptr;

    string title = _lava_;

    string save_name = _default_;

    bool fullscreen_active = false;

    bool save_title_active = false;

    bool switch_mode_request_active = false;
```

```

    bool resize_request_active = false;

    ui32 framebuffer_width = 0;

    ui32 framebuffer_height = 0;

    i32 pos_x = 0;

    i32 pos_y = 0;

    ui32 width = 0;

    ui32 height = 0;
};

using window_t = window;

VkSurfaceKHR create_surface(GLFWwindow* window);

window* get_window(GLFWwindow* handle);

} // namespace lava
```

5.120 liblava/lava.hpp File Reference

All lava modules.

```
#include <liblava/app.hpp>
#include <liblava/asset.hpp>
#include <liblava/base.hpp>
#include <liblava/block.hpp>
#include <liblava/core.hpp>
#include <liblava/engine.hpp>
#include <liblava/file.hpp>
#include <liblava/frame.hpp>
#include <liblava/resource.hpp>
#include <liblava/util.hpp>
```

5.120.1 Detailed Description

All lava modules.

Author

Lava Block OÜ and contributors

Copyright

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5.121 lava.hpp

Go to the documentation of this file.

```

#pragma once

#include <liblava/app.hpp>
#include <liblava/asset.hpp>
#include <liblava/base.hpp>
#include <liblava/block.hpp>
#include <liblava/core.hpp>
#include <liblava/engine.hpp>
#include <liblava/file.hpp>
#include <liblava/frame.hpp>
#include <liblava/resource.hpp>
#include <liblava/util.hpp>
```

5.122 liblava/resource.hpp File Reference

Resource module.

```
#include <liblava/resource/buffer.hpp>
#include <liblava/resource/format.hpp>
#include <liblava/resource/image.hpp>
#include <liblava/resource/mesh.hpp>
#include <liblava/resource/texture.hpp>
```

5.122.1 Detailed Description

Resource module.

Author

Lava Block OÜ and contributors

Copyright

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5.123 resource.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/resource/buffer.hpp>
11 #include <liblava/resource/format.hpp>
12 #include <liblava/resource/image.hpp>
13 #include <liblava/resource/mesh.hpp>
14 #include <liblava/resource/texture.hpp>
```

5.124 liblava/resource/buffer.hpp File Reference

Vulkan buffer.

```
#include <liblava/base/device.hpp>
```

Classes

- struct lava::buffer

  Buffer.

Functions

- VkPipelineStageFlags lava::buffer_usage_to_possible_stages (VkBufferUsageFlags usage)

  Get possible stages by bufferusage flags.
- VkAccessFlags lava::buffer_usage_to_possible_access (VkBufferUsageFlags usage)

  Get possible access by buffer usage flags.

5.124.1 Detailed Description

Vulkan buffer.

Authors

Lava Block OÜ and contributors

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5.124.2 Function Documentation

5.124.2.1 buffer_usage_to_possible_access()

```
VkAccessFlags lava::buffer_usage_to_possible_access (
            VkBufferUsageFlags usage )
```

Get possible access by buffer usage flags.

Parameters

usage	Buffer usage flags

Returns

VkAccessFlags Access flags

5.124.2.2 buffer_usage_to_possible_stages()

```
VkPipelineStageFlags lava::buffer_usage_to_possible_stages (
            VkBufferUsageFlags usage )
```

Get possible stages by bufferusage flags.

Parameters

usage	Buffer usage flags

Returns

VkPipelineStageFlags Pipeline stage flags

5.125 buffer.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/base/device.hpp>
11
12 namespace lava {
13
17 struct buffer : entity {
19     using ptr = std::shared_ptr<buffer>;
20
22     using list = std::vector<ptr>;
23
28     static ptr make() {
29         return std::make_shared<buffer>();
30     }
31
35     ~buffer() {
36         destroy();
37     }
38
49     bool create(device_p device,
50                 void const* data,
51                 size_t size,
52                 VkBufferUsageFlags usage,
53                 bool mapped = false,
54                 VmaMemoryUsage memory_usage = VMA_MEMORY_USAGE_GPU_ONLY);
55
65     bool create_mapped(device_p device,
66                        void const* data,
67                        size_t size,
68                        VkBufferUsageFlags usage,
69                        VmaMemoryUsage memory_usage = VMA_MEMORY_USAGE_CPU_TO_GPU);
70
74     void destroy();
75
80     device_p get_device() {
81         return device;
82     }
83
88     bool valid() const {
89         return vk_buffer != VK_NULL_HANDLE;
90     }
91
96     VkBuffer get() const {
97         return vk_buffer;
98     }
99
104     VkDescriptorBufferInfo const* get_descriptor_info() const {
105         return &descriptor;
106     }
107
112     VkDeviceAddress get_address() const;
113
118     VkDeviceSize get_size() const {
119         return allocation_info.size;
120     }
121
126     void* get_mapped_data() const {
127         return allocation_info.pMappedData;
128     }
129
134     VkDeviceMemory get_device_memory() const {
135         return allocation_info.deviceMemory;
136     }
137
```

```
    void flush(VkDeviceSize offset = 0,
               VkDeviceSize size = VK_WHOLE_SIZE);

    VmaAllocation const& get_allocation() const {
        return allocation;
    }

    VmaAllocationInfo const& get_allocation_info() const {
        return allocation_info;
    }

private:
    device_p device = nullptr;

    VkBuffer vk_buffer = VK_NULL_HANDLE;

    VmaAllocation allocation = nullptr;

    VmaAllocationInfo allocation_info = {};

    VkDescriptorBufferInfo descriptor = {};
};

VkPipelineStageFlags buffer_usage_to_possible_stages(VkBufferUsageFlags usage);

VkAccessFlags buffer_usage_to_possible_access(VkBufferUsageFlags usage);

} // namespace lava
```

5.126 liblava/resource/format.hpp File Reference

Vulkan format.

```
#include <liblava/base/device.hpp>
#include <optional>
```

Classes

- struct lava::surface_format_request

  Surface format request.

Typedefs

- using **lava::VkFormat_optional** = std::optional< VkFormat >

  Optional format.

Functions

- bool lava::format_depth (VkFormat format)

  Check if format is depth compatible.
- bool lava::format_stencil (VkFormat format)

  Check if format is stencil compatible.
- bool lava::format_depth_stencil (VkFormat format)

  Check if format is depth or stencil compatible.
- bool lava::format_srgb (VkFormat format)

  Check if format is sRGB compatible.
- bool lava::format_bgr (VkFormat format)

  Check if format has BGR order.

- VkImageAspectFlags lava::format_aspect_mask (VkFormat format)

  Get image aspect mask of format.
- void lava::format_block_dim (VkFormat format, ui32 &width, ui32 &height)

  Get block dimension of format.
- void lava::format_align_dim (VkFormat format, ui32 &width, ui32 &height)

  Get align dimension of format.
- void lava::format_num_blocks (VkFormat format, ui32 &width, ui32 &height)

  Get format number of blocks.
- ui32 lava::format_block_size (VkFormat format)

  Get format block size.
- VkFormat_optional lava::get_supported_depth_format (VkPhysicalDevice physical_device)

  Get the supported depth format.
- VkFormat_optional lava::get_supported_format (VkPhysicalDevice physical_device, VkFormats const &possible_formats, VkImageUsageFlags usage)

  Get the supported format.
- VkImageMemoryBarrier lava::image_memory_barrier (VkImage image, VkImageLayout old_layout, VkImageLayout new_layout)

  Get image memory barrier.
- void lava::set_image_layout (device_p device, VkCommandBuffer cmd_buffer, VkImage image, VkImageLayout old_image_layout, VkImageLayout new_image_layout, VkImageSubresourceRange subresource_range, VkPipelineStageFlags src_stage_mask=VK_PIPELINE_STAGE_ALL_COMMANDS_BIT, VkPipelineStageFlags dst_stage_mask=VK_PIPELINE_STAGE_ALL_COMMANDS_BIT)

  Set the image layout.
- void lava::set_image_layout (device_p device, VkCommandBuffer cmd_buffer, VkImage image, VkImageAspectFlags aspect_mask, VkImageLayout old_image_layout, VkImageLayout new_image_layout, VkPipelineStageFlags src_stage_mask=VK_PIPELINE_STAGE_ALL_COMMANDS_BIT, VkPipelineStageFlags dst_stage_mask=VK_PIPELINE_STAGE_ALL_COMMANDS_BIT)

  Set the image layout.
- void lava::insert_image_memory_barrier (device_p device, VkCommandBuffer cmd_buffer, VkImage image, VkAccessFlags src_access_mask, VkAccessFlags dst_access_mask, VkImageLayout old_image_layout, VkImageLayout new_image_layout, VkPipelineStageFlags src_stage_mask, VkPipelineStageFlags dst_stage_mask, VkImageSubresourceRange subresource_range)

  Insert image memory barrier.
- VkSurfaceFormatKHR lava::get_surface_format (VkPhysicalDevice device, VkSurfaceKHR surface, surface_format_request request={})

  Get the surface format.
- bool lava::support_blit (VkPhysicalDevice device, VkFormat format)

  Check if format supports bltting.
- bool lava::support_vertex_buffer_format (VkPhysicalDevice device, VkFormat format)

  Check if vertex buffer format is supported.

5.126.1 Detailed Description

Vulkan format.

Authors

Lava Block OÜ and contributors

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5.126.2 Function Documentation

5.126.2.1 format_align_dim()

```
void lava::format_align_dim (
            VkFormat format,
            ui32 & width,
            ui32 & height )
```

Get align dimension of format.

Parameters

format	Target format
width	Align width
height	Align height

5.126.2.2 format_aspect_mask()

```
VkImageAspectFlags lava::format_aspect_mask (
            VkFormat format )
```

Get image aspect mask of format.

Parameters

format	Target format

Returns

VkImageAspectFlags Image aspect flags

5.126.2.3 format_bgr()

```
bool lava::format_bgr (
            VkFormat format )
```

Check if format has BGR order.

Parameters

format	Format to check

Returns

Format has BGR order or not

5.126.2.4 format_block_dim()

```
void lava::format_block_dim (
            VkFormat format,
            ui32 & width,
            ui32 & height )
```

Get block dimension of format.

Parameters

format	Target format
width	Block width
height	Block height

5.126.2.5 format_block_size()

```
ui32 lava::format_block_size (
            VkFormat format )
```

Get format block size.

Parameters

format	Target format

Returns

ui32 Size of block

5.126.2.6 format_depth()

```
bool lava::format_depth (
            VkFormat format )
```

Check if format is depth compatible.

Parameters

format	Format to check

Returns

Format is depth compatible or not

5.126.2.7 format_depth_stencil()

```
bool lava::format_depth_stencil (
            VkFormat format )
```

Check if format is depth or stencil compatible.

Parameters

format	Format to check

Returns

Format is depth or stencil compatible or not

5.126.2.8 format_num_blocks()

```
void lava::format_num_blocks (
            VkFormat format,
            ui32 & width,
            ui32 & height )
```

Get format number of blocks.

Parameters

format	Target format
width	Number blocks width
height	Number blocks height

5.126.2.9 format_srgb()

```
bool lava::format_srgb (
            VkFormat format )
```

Check if format is sRGB compatible.

Parameters

format	Format to check

Returns

Format is sRGB compatible or not

5.126.2.10 format_stencil()

```
bool lava::format_stencil (
            VkFormat format )
```

Check if format is stencil compatible.

Parameters

format	Format to check

Returns

Format is stencil compatible or not

5.126.2.11 get_supported_depth_format()

```
VkFormat_optional lava::get_supported_depth_format (
            VkPhysicalDevice physical_device )
```

Get the supported depth format.

Parameters

physical_device	Physical device

Returns

VkFormat_optional Optional format

5.126.2.12 get_supported_format()

```
VkFormat_optional lava::get_supported_format (
            VkPhysicalDevice physical_device,
            VkFormats const & possible_formats,
            VkImageUsageFlags usage )
```

Get the supported format.

Parameters

physical_device	Physical device
possible_formats	List of possible formats
usage	Image usage flags

Returns

VkFormat_optional Optional format

5.126.2.13 get_surface_format()

```
VkSurfaceFormatKHR lava::get_surface_format (
            VkPhysicalDevice device,
            VkSurfaceKHR surface,
            surface_format_request request = {} )
```

Get the surface format.

Parameters

device	Vulkan device
surface	Vulkan surface
request	Surface format request

Returns

VkSurfaceFormatKHR Chosen surface format

5.126.2.14 image_memory_barrier()

```
VkImageMemoryBarrier lava::image_memory_barrier (
            VkImage image,
            VkImageLayout old_layout,
            VkImageLayout new_layout )
```

Get image memory barrier.

Parameters

image	Target image
old_layout	Old image layout
new_layout	New image layout

Returns

VkImageMemoryBarrier Image memory barrier

5.126.2.15 insert_image_memory_barrier()

```
void lava::insert_image_memory_barrier (
            device_p device,
            VkCommandBuffer cmd_buffer,
            VkImage image,
            VkAccessFlags src_access_mask,
            VkAccessFlags dst_access_mask,
            VkImageLayout old_image_layout,
            VkImageLayout new_image_layout,
            VkPipelineStageFlags src_stage_mask,
            VkPipelineStageFlags dst_stage_mask,
            VkImageSubresourceRange subresource_range )
```

Insert image memory barrier.

Parameters

device	Vulkan device
cmd_buffer	Command buffer
image	Target image
src_access_mask	Source access mask
dst_access_mask	Destination access mask
old_image_layout	Old image layout
new_image_layout	New image layout
src_stage_mask	Source pipeline stage flags
dst_stage_mask	Destination pipeline stage flags
subresource_range	Image subresource range

5.126.2.16 set_image_layout() [1/2]

```
void lava::set_image_layout (
            device_p device,
            VkCommandBuffer cmd_buffer,
            VkImage image,
```

```
            VkImageAspectFlags aspect_mask,
            VkImageLayout old_image_layout,
            VkImageLayout new_image_layout,
            VkPipelineStageFlags src_stage_mask = VK_PIPELINE_STAGE_ALL_COMMANDS_BIT,
            VkPipelineStageFlags dst_stage_mask = VK_PIPELINE_STAGE_ALL_COMMANDS_BIT )
```

Set the image layout.

Parameters

device	Vulkan device
cmd_buffer	Command buffer
image	Target image
aspect_mask	Image aspect flags
old_image_layout	Old image layout
new_image_layout	New image layout
src_stage_mask	Source pipeline stage flags
dst_stage_mask	Destination pipeline stage flags

5.126.2.17 set_image_layout() [2/2]

```
void lava::set_image_layout (
            device_p device,
            VkCommandBuffer cmd_buffer,
            VkImage image,
            VkImageLayout old_image_layout,
            VkImageLayout new_image_layout,
            VkImageSubresourceRange subresource_range,
            VkPipelineStageFlags src_stage_mask = VK_PIPELINE_STAGE_ALL_COMMANDS_BIT,
            VkPipelineStageFlags dst_stage_mask = VK_PIPELINE_STAGE_ALL_COMMANDS_BIT )
```

Set the image layout.

Parameters

device	Vulkan device
cmd_buffer	Command buffer
image	Target image
old_image_layout	Old image layout
new_image_layout	New image layout
subresource_range	Image subresource range
src_stage_mask	Source pipeline stage flags
dst_stage_mask	Destination pipeline stage flags

5.126.2.18 support_blit()

```
bool lava::support_blit (
```

```
            VkPhysicalDevice device,
            VkFormat format )
```

Check if format supports bltting.

Parameters

device	Vulkan physical device
format	Format to check

Returns

Blitting is supported or not

5.126.2.19 support_vertex_buffer_format()

```
bool lava::support_vertex_buffer_format (
            VkPhysicalDevice device,
            VkFormat format )
```

Check if vertex buffer format is supported.

Parameters

device	Vulkan physical device
format	Format to check

Returns

Format is supported or not

5.127 format.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/base/device.hpp>
#include <optional>

namespace lava {

using VkFormat_optional = std::optional<VkFormat>;

bool format_depth(VkFormat format);

bool format_stencil(VkFormat format);

bool format_depth_stencil(VkFormat format);

bool format_srgb(VkFormat format);

bool format_bgr(VkFormat format);

VkImageAspectFlags format_aspect_mask(VkFormat format);
```

```

void format_block_dim(VkFormat format,
                      ui32& width,
                      ui32& height);

void format_align_dim(VkFormat format,
                      ui32& width,
                      ui32& height);

void format_num_blocks(VkFormat format,
                       ui32& width,
                       ui32& height);

ui32 format_block_size(VkFormat format);

VkFormat_optional get_supported_depth_format(VkPhysicalDevice physical_device);

VkFormat_optional get_supported_format(VkPhysicalDevice physical_device,
                                       VkFormats const& possible_formats,
                                       VkImageUsageFlags usage);

VkImageMemoryBarrier image_memory_barrier(VkImage image,
                                          VkImageLayout old_layout,
                                          VkImageLayout new_layout);

void set_image_layout(device_p device,
                      VkCommandBuffer cmd_buffer,
                      VkImage image,
                      VkImageLayout old_image_layout,
                      VkImageLayout new_image_layout,
                      VkImageSubresourceRange subresource_range,
                      VkPipelineStageFlags src_stage_mask = VK_PIPELINE_STAGE_ALL_COMMANDS_BIT,
                      VkPipelineStageFlags dst_stage_mask = VK_PIPELINE_STAGE_ALL_COMMANDS_BIT);

void set_image_layout(device_p device,
                      VkCommandBuffer cmd_buffer,
                      VkImage image,
                      VkImageAspectFlags aspect_mask,
                      VkImageLayout old_image_layout,
                      VkImageLayout new_image_layout,
                      VkPipelineStageFlags src_stage_mask = VK_PIPELINE_STAGE_ALL_COMMANDS_BIT,
                      VkPipelineStageFlags dst_stage_mask = VK_PIPELINE_STAGE_ALL_COMMANDS_BIT);

void insert_image_memory_barrier(device_p device,
                                 VkCommandBuffer cmd_buffer,
                                 VkImage image,
                                 VkAccessFlags src_access_mask,
                                 VkAccessFlags dst_access_mask,
                                 VkImageLayout old_image_layout,
                                 VkImageLayout new_image_layout,
                                 VkPipelineStageFlags src_stage_mask,
                                 VkPipelineStageFlags dst_stage_mask,
                                 VkImageSubresourceRange subresource_range);

struct surface_format_request {
    VkFormats formats = {
        VK_FORMAT_B8G8R8A8_UNORM,
        VK_FORMAT_R8G8B8A8_UNORM,
        VK_FORMAT_B8G8R8_UNORM,
        VK_FORMAT_R8G8B8_UNORM,
        VK_FORMAT_B8G8R8A8_SRGB,
        VK_FORMAT_R8G8B8A8_SRGB,
        VK_FORMAT_B8G8R8_SRGB,
        VK_FORMAT_R8G8B8_SRGB,
    };

    VkColorSpaceKHR color_space = VK_COLOR_SPACE_SRGB_NONLINEAR_KHR;
};

VkSurfaceFormatKHR get_surface_format(VkPhysicalDevice device,
                                      VkSurfaceKHR surface,
                                      surface_format_request request = {});

bool support_blit(VkPhysicalDevice device,
                  VkFormat format);

bool support_vertex_buffer_format(VkPhysicalDevice device,
                                  VkFormat format);

} // namespace lava
```

5.128 liblava/resource/image.hpp File Reference

Vulkan image.

```
#include <liblava/base/device.hpp>
```

Classes

- struct lava::image

  Image.

Functions

- image::ptr lava::create_image (device_p device, VkFormat format, uv2 size, VkImage vk_image=0)

  Create a new image.
- image::ptr lava::grab_image (image::ptr source)

  Grab an image (with blit/copy)

5.128.1 Detailed Description

Vulkan image.

Authors

Lava Block OÜ and contributors

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5.128.2 Function Documentation

5.128.2.1 create_image()

```
image::ptr lava::create_image (
            device_p device,
            VkFormat format,
            uv2 size,
            VkImage vk_image = 0 )
```

Create a new image.

Parameters

device	Vulkan device
format	Image format
size	Image size
vk_image	Vulkan image

Returns

image::ptr Shared pointer to image

5.128.2.2 grab_image()

```
image::ptr lava::grab_image (
            image::ptr source )
```

Grab an image (with blit/copy)

Parameters

source	Source image

Returns

image::ptr Grabbed image

5.129 image.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/base/device.hpp>
11
12 namespace lava {
13
17 struct image : entity {
19     using ptr = std::shared_ptr<image>;
20
22     using map = std::map<id, ptr>;
23
25     using list = std::vector<ptr>;
26
33     static ptr make(VkFormat format,
34                     VkImage vk_image = 0) {
35         return std::make_shared<image>(format, vk_image);
36     }
37
43     explicit image(VkFormat format,
44                    VkImage vk_image = 0);
45
54     bool create(device_p device,
55                 uv2 size,
56                 VmaMemoryUsage memory_usage = VMA_MEMORY_USAGE_GPU_ONLY,
57                 bool mip_levels_generation = false);
58
63     void destroy(bool view_only = false);
```

```
    void destroy_view() {
        destroy(true);
    }

    device_p get_device() {
        return device;
    }

    VkFormat get_format() const {
        return info.format;
    }

    uv2 get_size() const {
        return { info.extent.width, info.extent.height };
    }

    ui32 get_depth() const {
        return info.extent.depth;
    }

    VkImage get() const {
        return vk_image;
    }

    VkImageView get_view() const {
        return view;
    }

    VkImageCreateInfo const& get_info() const {
        return info;
    }

    VkImageViewCreateInfo const& get_view_info() const {
        return view_info;
    }

    VkImageSubresourceRange const& get_subresource_range() const {
        return subresource_range;
    }

    void set_flags(VkImageCreateFlagBits flags) {
        info.flags = flags;
    }

    void set_tiling(VkImageTiling tiling) {
        info.tiling = tiling;
    }

    void set_usage(VkImageUsageFlags usage) {
        info.usage = usage;
    }

    void set_layout(VkImageLayout initial) {
        info.initialLayout = initial;
    }

    void set_aspect_mask(VkImageAspectFlags aspectMask) {
        subresource_range.aspectMask = aspectMask;
    }

    void set_level_count(ui32 levels) {
        subresource_range.levelCount = levels;
        info.mipLevels = levels;
    }

    void set_layer_count(ui32 layers) {
        subresource_range.layerCount = layers;
        info.arrayLayers = layers;
    }

    void set_component(VkComponentMapping mapping = {}) {
        view_info.components = mapping;
    }

    void set_view_type(VkImageViewType type) {
        view_info.viewType = type;
    }

    VmaAllocation const& get_allocation() const {
        return allocation;
    }

private:
    device_p device = nullptr;

    VkImage vk_image = VK_NULL_HANDLE;
```

```
    VkImageCreateInfo info;

    VmaAllocation allocation = nullptr;

    VkImageView view = VK_NULL_HANDLE;

    VkImageViewCreateInfo view_info;

    VkImageSubresourceRange subresource_range;
};

image::ptr create_image(device_p device,
                        VkFormat format,
                        uv2 size,
                        VkImage vk_image = 0);

image::ptr grab_image(image::ptr source);

} // namespace lava
```

5.130 liblava/resource/mesh.hpp File Reference

Vulkan mesh.

```
#include <liblava/core/misc.hpp>
#include <liblava/resource/buffer.hpp>
#include <liblava/resource/primitive.hpp>
#include <liblava/util/hex.hpp>
#include <liblava/util/log.hpp>
```

Classes

- struct lava::mesh_template_data< T >

  Templated mesh data.
- struct lava::mesh_template< T >

  Temporary templated mesh.
- struct lava::mesh_meta

  Mesh meta.

Typedefs

- using **lava::mesh_data** = mesh_template_data< vertex >

  Mesh data with default vertex.
- using **lava::mesh** = mesh_template< vertex >

  Mesh with default vertex.
- using **lava::mesh_registry** = id_registry< mesh, mesh_meta >

  Mesh registry.

Functions

- template<typename T = vertex, bool generate_colors = true, bool generate_normals = true, bool generate_uvs = true, bool has_colors = true, bool has_normals = true, bool has_uvs = true>
  mesh_template_data< T > lava::create_mesh_data (mesh_type type)

  Create a new primitive mesh_data.
- template<typename T = vertex, bool generate_colors = true, bool generate_normals = true, bool generate_uvs = true, bool has_colors = true, bool has_normals = true, bool has_uvs = true>
  std::shared_ptr< mesh_template< T > > lava::create_mesh (device_p &device, mesh_type type)

  Create a new primitive mesh.
- template<typename PosType , size_t vert_count, bool is_complex>
  constexpr std::array< PosType, vert_count > lava::make_primitive_positions_cube ()

  Make primitive positions for cube.
- template<bool is_complex>
  std::vector< index > lava::make_primitive_indices_cube ()

  Make primitive indices for cube.
- template<typename NormType >
  constexpr std::array< NormType, 6 > lava::make_primitive_normals_cube ()

  Make primitive normals for cube.
- template<typename UVType >
  constexpr std::array< UVType, 24 > lava::make_primitive_uvs_cube ()

  Make primitive uvs for cube.

5.130.1 Detailed Description

Vulkan mesh.

Authors

Lava Block OÜ and contributors

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5.130.2 Function Documentation

5.130.2.1 create_mesh()

```
template<typename T = vertex, bool generate_colors = true, bool generate_normals = true, bool
generate_uvs = true, bool has_colors = true, bool has_normals = true, bool has_uvs = true>
std::shared_ptr< mesh_template< T > > lava::create_mesh (
            device_p & device,
            mesh_type type )
```

Create a new primitive mesh.

Template Parameters

T	Type of vertex struct
generate_colors	If color may be generated
generate_normals	If normals may be generated
generate_uvs	If UVs may be generated
has_colors	On MSVC, specifies if a `color` field exists
has_normals	On MSVC, specifies if a `normal` field exists
has_uvs	On MSVC, specifies if a `uv` field exists

Parameters

device	Vulkan device
type	Mesh type

Returns

std::shared_ptr<mesh_template<T>> Shared pointer to mesh

5.130.2.2 create_mesh_data()

```
template<typename T = vertex, bool generate_colors = true, bool generate_normals = true, bool
generate_uvs = true, bool has_colors = true, bool has_normals = true, bool has_uvs = true>
mesh_template_data< T > lava::create_mesh_data (
            mesh_type type )
```

Create a new primitive mesh_data.

Template Parameters

T	Type of vertex struct
generate_colors	If color may be generated
generate_normals	If normals may be generated
generate_uvs	If UVs may be generated
has_colors	On MSVC, specifies if a `color` field exists
has_normals	On MSVC, specifies if a `normal` field exists
has_uvs	On MSVC, specifies if a `uv` field exists

Parameters

type	Mesh type

Returns

mesh_template_data<T> Mesh data

5.130.2.3 make_primitive_indices_cube()

```
template<bool is_complex>
std::vector< index > lava::make_primitive_indices_cube ( )
```

Make primitive indices for cube.

Template Parameters

is_complex	Complex state

Returns

std::vector<index> Array for indices

5.130.2.4 make_primitive_normals_cube()

```
template<typename NormType >
constexpr std::array< NormType, 6 > lava::make_primitive_normals_cube ( )  [constexpr]
```

Make primitive normals for cube.

Template Parameters

NormType	Type of normal

Returns

constexpr std::array<NormType, 6> Array of normals

5.130.2.5 make_primitive_positions_cube()

```
template<typename PosType , size_t vert_count, bool is_complex>
constexpr std::array< PosType, vert_count > lava::make_primitive_positions_cube ( )  [constexpr]
```

Make primitive positions for cube.

Template Parameters

PosType	Type of position
vert_count	Number of vertices
is_complex	Complex state

Returns

constexpr std::array<PosType, vert_count> Array of positions

5.130.2.6 make_primitive_uvs_cube()

```
template<typename UVType >
constexpr std::array< UVType, 24 > lava::make_primitive_uvs_cube ( )  [constexpr]
```

Make primitive uvs for cube.

Template Parameters

UVType	Type of uv

Returns

constexpr std::array<UVType, 24> Array of uvs

5.131 mesh.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/core/misc.hpp>
11 #include <liblava/resource/buffer.hpp>
12 #include <liblava/resource/primitive.hpp>
13 #include <liblava/util/hex.hpp>
14 #include <liblava/util/log.hpp>
15
16 namespace lava {
17
22 template<typename T = vertex>
23 struct mesh_template_data {
25     std::vector<T> vertices;
26
28     index_list indices;
29
30 public:
36     template<typename PosType = float>
37     void move(std::array<PosType, 3> offset) {
38         for (T& vertex : vertices) {
39             for (auto i = 0u; i < 3; ++i) {
40                 vertex.position[i] += offset[i];
41             }
42         }
43     }
44
49     void scale(auto factor) {
50         for (T& vertex : vertices) {
51             for (auto i = 0u; i < 3; ++i) {
52                 vertex.position[i] *= factor;
53             }
54         }
55     }
56
62     template<typename PosType = float>
63     void scale_vector(std::array<PosType, 3> factors) {
64         for (T& vertex : vertices) {
65             for (auto i = 0u; i < 3; ++i) {
66                 vertex.position[i] *= factors[i];
67             }
68         }
```

```
    }
};

template<typename T = vertex>
struct mesh_template : entity {
    using ptr = std::shared_ptr<mesh_template<T>>;

    using map = std::map<id, ptr>;

    using list = std::vector<ptr>;

    using vertex_list = std::vector<T>;

    static ptr make() {
        return std::make_shared<mesh_template<T>>();
    }

    ~mesh_template() {
        destroy();
    }

    bool create(device_p device,
                bool mapped = false,
                VmaMemoryUsage memory_usage = VMA_MEMORY_USAGE_CPU_TO_GPU);

    void destroy();

    void bind(VkCommandBuffer cmd_buf) const;

    void draw(VkCommandBuffer cmd_buf) const;

    void bind_draw(VkCommandBuffer cmd_buf) const {
        bind(cmd_buf);
        draw(cmd_buf);
    }

    bool empty() const {
        return data.vertices.empty();
    }

    void set_data(mesh_template_data<T> const& value) {
        data = value;
    }

    mesh_template_data<T>& get_data() {
        return data;
    }

    void add_data(mesh_template_data<T> const& value) {
        data = value;
    }

    vertex_list& get_vertices() {
        return data.vertices;
    }

    vertex_list const& get_vertices() const {
        return data.vertices;
    }

    ui32 get_vertices_count() const {
        return to_ui32(data.vertices.size());
    }

    index_list& get_indices() {
        return data.indices;
    }

    index_list const& get_indices() const {
        return data.indices;
    }

    ui32 get_indices_count() const {
        return to_ui32(data.indices.size());
    }

    bool reload();

    buffer::ptr get_vertex_buffer() {
        return vertex_buffer;
    }

    buffer::ptr get_index_buffer() {
        return index_buffer;
    }

private:
```

```
    device_p device = nullptr;

    mesh_template_data<T> data;

    buffer::ptr vertex_buffer;

    buffer::ptr index_buffer;

    bool mapped = false;

    VmaMemoryUsage memory_usage = VMA_MEMORY_USAGE_CPU_TO_GPU;
};

//-----------------------------------------------------------------------------
template<typename T>
void mesh_template<T>::bind(VkCommandBuffer cmd_buf) const {
    if (vertex_buffer && vertex_buffer->valid()) {
        std::array<VkDeviceSize, 1> const buffer_offsets = { 0 };
        std::array<VkBuffer, 1> const buffers = { vertex_buffer->get() };

        vkCmdBindVertexBuffers(cmd_buf, 0,
                               to_ui32(buffers.size()), buffers.data(),
                               buffer_offsets.data());
    }

    if (index_buffer && index_buffer->valid())
        vkCmdBindIndexBuffer(cmd_buf,
                             index_buffer->get(),
                             0,
                             VK_INDEX_TYPE_UINT32);
}

//-----------------------------------------------------------------------------
template<typename T>
void mesh_template<T>::draw(VkCommandBuffer cmd_buf) const {
    if (!data.indices.empty())
        vkCmdDrawIndexed(cmd_buf,
                         to_ui32(data.indices.size()),
                         1, 0, 0, 0);
    else
        vkCmdDraw(cmd_buf,
                  to_ui32(data.vertices.size()),
                  1, 0, 0);
}

//-----------------------------------------------------------------------------
template<typename T>
void mesh_template<T>::destroy() {
    vertex_buffer = nullptr;
    index_buffer = nullptr;
    device = nullptr;
}

//-----------------------------------------------------------------------------
template<typename T>
bool mesh_template<T>::reload() {
    auto dev = device;
    destroy();

    return create(dev, mapped, memory_usage);
}

template<typename T = vertex,
         bool generate_colors = true,
         bool generate_normals = true,
         bool generate_uvs = true,
         bool has_colors = true,
         bool has_normals = true,
         bool has_uvs = true>
mesh_template_data<T> create_mesh_data(mesh_type type);

template<typename T = vertex,
         bool generate_colors = true,
         bool generate_normals = true,
         bool generate_uvs = true,
         bool has_colors = true,
         bool has_normals = true,
         bool has_uvs = true>
std::shared_ptr<mesh_template<T>> create_mesh(device_p& device,
                                              mesh_type type);

//-----------------------------------------------------------------------------
template<typename T>
bool mesh_template<T>::create(device_p d,
                              bool m,
                              VmaMemoryUsage mu) {
    device = d;
```

```
    mapped = m;
    memory_usage = mu;

    if (!data.vertices.empty()) {
        vertex_buffer = buffer::make();

        if (!vertex_buffer->create(device,
                                   data.vertices.data(),
                                   sizeof(T) * data.vertices.size(),
                                   VK_BUFFER_USAGE_VERTEX_BUFFER_BIT,
                                   mapped,
                                   memory_usage)) {
            log()->error("create mesh vertex buffer");
            return false;
        }
    }

    if (!data.indices.empty()) {
        index_buffer = buffer::make();

        if (!index_buffer->create(device,
                                  data.indices.data(),
                                  sizeof(ui32) * data.indices.size(),
                                  VK_BUFFER_USAGE_INDEX_BUFFER_BIT,
                                  mapped,
                                  memory_usage)) {
            log()->error("create mesh index buffer");
            return false;
        }
    }

    return true;
}

template<typename PosType, size_t vert_count, bool is_complex>
constexpr std::array<PosType, vert_count> make_primitive_positions_cube();

//-----------------------------------------------------------------------------
// NOTE: The C++20 spec allows std::vector<T> to be constexpr
// g++ does not currently implement this feature, however
//-----------------------------------------------------------------------------

template<bool is_complex>
std::vector<index> make_primitive_indices_cube();

template<typename NormType>
constexpr std::array<NormType, 6> make_primitive_normals_cube();

template<typename UVType>
constexpr std::array<UVType, 24> make_primitive_uvs_cube();

//-----------------------------------------------------------------------------
template<typename T,
         bool generate_colors,
         bool generate_normals,
         bool generate_uvs,
         bool has_colors,
         bool has_normals,
         bool has_uvs>
std::shared_ptr<mesh_template<T>> create_mesh(device_p& device,
                                              mesh_type type) {
    std::shared_ptr<mesh_template<T>> return_mesh =
        std::make_shared<mesh_template<T>>();

    return_mesh->add_data(create_mesh_data<T,
                                           generate_colors,
                                           generate_normals,
                                           generate_uvs,
                                           has_colors,
                                           has_normals,
                                           has_uvs>(type));
    return_mesh->create(device);
    return return_mesh;
}

//-----------------------------------------------------------------------------
template<typename T,
         bool generate_colors,
         bool generate_normals,
         bool generate_uvs,
         bool has_colors,
         bool has_normals,
         bool has_uvs>
mesh_template_data<T> create_mesh_data(mesh_type type) {
    mesh_template_data<T> return_mesh_data;

    constexpr bool auto_position = requires(const T t) {
```

```
        t.position;
    };
    static_assert(auto_position,
                  "Vertex struct `T` must contain field `position`");

    constexpr bool auto_colors = requires(const T t) {
        t.color;
    };
    constexpr bool auto_normals = requires(const T t) {
        t.normal;
    };
    constexpr bool auto_uvs = requires(const T t) {
        t.uv;
    };

    using PosType = decltype(T::position);

    switch (type) {
    case mesh_type::cube: {
        return_mesh_data.indices.reserve(36);
        return_mesh_data.indices = make_primitive_indices_cube<(generate_normals
                                                                && auto_normals)>();
        constexpr size_t vert_count = (generate_normals && auto_normals) ? 24 : 8;
        return_mesh_data.vertices.reserve(vert_count);
        auto positions = make_primitive_positions_cube<PosType, vert_count,
                                                       (generate_normals && auto_normals)>();
        for (size_t i = 0; i < vert_count; i++) {
            T vert;
            vert.position = positions[i];
            if constexpr (generate_normals && auto_normals) {
                // this array is generated inside of every loop because
                // that makes the scoping rules simplest to follow
                // my expectation is that a compiler should be able
                // to trivially optimize this
                using NormType = decltype(T::normal);
                auto normals = make_primitive_normals_cube<NormType>();
                vert.normal = normals[i / 4];
            }
            if constexpr (generate_uvs && auto_uvs) {
                using UVType = decltype(T::uv);
                auto uvs = make_primitive_uvs_cube<UVType>();
                vert.uv = uvs[i];
            }
            return_mesh_data.vertices.push_back(vert);
        }
        break;
    }

    case mesh_type::triangle: {
        return_mesh_data.vertices.reserve(3);
        T vert_one;
        vert_one.position = { 1, 1, 0 };
        T vert_two;
        vert_two.position = { -1, 1, 0 };
        T vert_three;
        vert_three.position = { 0, -1, 0 };
        if constexpr (generate_uvs && auto_uvs) {
            vert_one.uv = { 1, 1 };
            vert_two.uv = { 0, 1 };
            vert_three.uv = { 0.5, 0 };
        }
        if constexpr (generate_normals && auto_normals) {
            vert_one.normal = { 1, 1, 0 };
            vert_two.normal = { -1, 1, 0 };
            vert_three.normal = { 0, -1, 0 };
        }
        return_mesh_data.vertices.push_back(vert_one);
        return_mesh_data.vertices.push_back(vert_two);
        return_mesh_data.vertices.push_back(vert_three);
        break;
    }

    case mesh_type::quad: {
        return_mesh_data.vertices.reserve(4);
        return_mesh_data.indices.reserve(6);
        T vert_one;
        vert_one.position = { 1, 1, 0 };
        T vert_two;
        vert_two.position = { -1, 1, 0 };
        T vert_three;
        vert_three.position = { -1, -1, 0 };
        T vert_four;
        vert_four.position = { 1, -1, 0 };
        if constexpr (generate_uvs && auto_uvs) {
            vert_one.uv = { 1, 1 };
            vert_two.uv = { 0, 1 };
            vert_three.uv = { 0, 0 };
```

```
            vert_four.uv = { 1, 0 };
        }
        if constexpr (generate_normals && auto_normals) {
            vert_one.normal = { 0, 0, 1 };
            vert_two.normal = { 0, 0, 1 };
            vert_three.normal = { 0, 0, 1 };
            vert_four.normal = { 0, 0, 1 };
        }
        // clang-format off
        return_mesh_data.indices = {
            0, 1, 2,
            2, 3, 0,
        };
        // clang-format on
        return_mesh_data.vertices.push_back(vert_one);
        return_mesh_data.vertices.push_back(vert_two);
        return_mesh_data.vertices.push_back(vert_three);
        return_mesh_data.vertices.push_back(vert_four);
        break;
    }

    case mesh_type::hexagon: {
        return_mesh_data.vertices.reserve(7);
        return_mesh_data.indices.reserve(18);
        hex_layout layout;
        layout.orientation = hex_layout_point_y;
        layout.size = { 1, 1 };
        auto hex_corners = hex_polygon_corners(layout, {});
        T vert_center;
        vert_center.position = { 0, 0, 0 };
        T vert_sw;
        PosType temp;
        temp[0] = hex_corners.at(0).x;
        temp[1] = hex_corners.at(0).y;
        temp[2] = 0;
        vert_sw.position = temp;
        T vert_nw;
        temp[0] = hex_corners.at(1).x;
        temp[1] = hex_corners.at(1).y;
        vert_nw.position = temp;
        T vert_n;
        temp[0] = hex_corners.at(2).x;
        temp[1] = hex_corners.at(2).y;
        vert_n.position = temp;
        T vert_ne;
        temp[0] = hex_corners.at(3).x;
        temp[1] = hex_corners.at(3).y;
        vert_ne.position = temp;
        T vert_se;
        temp[0] = hex_corners.at(4).x;
        temp[1] = hex_corners.at(4).y;
        vert_se.position = temp;
        T vert_s;
        temp[0] = hex_corners.at(5).x;
        temp[1] = hex_corners.at(5).y;
        vert_s.position = temp;
        if constexpr (generate_uvs && auto_uvs) {
            vert_center.uv = { 0, 0 };
            vert_sw.uv = { 1, 0 };
            vert_nw.uv = { 0, 1 };
            vert_n.uv = { 1, 1 };
            vert_ne.uv = { 1, 0 };
            vert_se.uv = { 0, 1 };
            vert_s.uv = { 1, 1 };
        }
        if constexpr (generate_normals && auto_normals) {
            vert_center.normal = { 0, 0, 1 };
            vert_sw.normal = { 0, 0, 1 };
            vert_nw.normal = { 0, 0, 1 };
            vert_n.normal = { 0, 0, 1 };
            vert_ne.normal = { 0, 0, 1 };
            vert_se.normal = { 0, 0, 1 };
            vert_s.normal = { 0, 0, 1 };
        }
        // clang-format off
        return_mesh_data.indices = {
            0, 1, 6, 0, 6, 5, 0, 5, 4, 0, 4, 3, 0, 3, 2, 0, 2, 1
        };
        // clang-format on
        return_mesh_data.vertices.push_back(vert_center);
        return_mesh_data.vertices.push_back(vert_sw);
        return_mesh_data.vertices.push_back(vert_nw);
        return_mesh_data.vertices.push_back(vert_n);
        return_mesh_data.vertices.push_back(vert_ne);
        return_mesh_data.vertices.push_back(vert_se);
        return_mesh_data.vertices.push_back(vert_s);
        break;
```

```
    }

    case mesh_type::none:
    default:
        break;
    }

    if constexpr (generate_colors && auto_colors) {
        for (auto& vert : return_mesh_data.vertices) {
            // this does not work on glm vectors
            // for (auto& this_color : vert.color) {
            //     for (auto& color_component : this_color) {
            //         color_component = 1;
            //     }
            // }
            if constexpr (std::is_same<decltype(vert.color), glm::vec3>::value) {
                vert.color = { 1, 1, 1 };
            } else if constexpr (std::is_same<decltype(vert.color), glm::vec4>::value) {
                vert.color = { 1, 1, 1, 1 };
            } else {
                for (size_t i = 0; i < vert.color.size(); i++) {
                    vert.color[i] = 1;
                }
            }
        }
    }

    return return_mesh_data;
}

//-----------------------------------------------------------------------------
template<typename PosType,
         size_t vert_count,
         bool is_complex>
constexpr std::array<PosType, vert_count> make_primitive_positions_cube() {
    if constexpr (is_complex) {
        // clang-format off
        constexpr std::array<PosType, 24> positions = {{
            // front
            { 1, 1, 1 }, { -1, 1, 1}, { -1, -1, 1 }, { 1, -1, 1 },
            // back
            { 1, 1, -1 }, { -1, 1, -1 }, { -1, -1, -1 }, { 1, -1, -1 },
            // left
            { -1, 1, 1 }, { -1, 1, -1 }, { -1, -1, -1 }, { -1, -1, 1 },
            // right
            { 1, 1, 1 }, { 1, -1, 1 }, { 1, -1, -1 }, { 1, 1, -1 },
            // bottom
            { 1, 1, 1 }, { -1, 1, 1 }, { -1, 1, -1 }, { 1, 1, -1 },
            // top
            { 1, -1, 1 }, { -1, -1, 1 }, { -1, -1, -1 }, { 1, -1, -1 },
        }};
        return positions;
    } else {
        constexpr std::array<PosType, 8> positions = {{
            { -1, -1, -1 },
            { -1, -1, 1 },
            { -1, 1, -1 },
            { -1, 1, 1 },
            { 1, -1, -1 },
            { 1, -1, 1 },
            { 1, 1, -1 },
            { 1, 1, 1 },
        }};
        return positions;
    }
    // clang-format on
}

//-----------------------------------------------------------------------------
template<bool is_complex>
std::vector<index> make_primitive_indices_cube() {
    // clang-format off
    if constexpr (is_complex) {
        std::vector<index> indices = {
            0, 1, 2,
            2, 3, 0,
            4, 7, 6,
            6, 5, 4,
            8, 9, 10,
            10, 11, 8,
            12, 13, 14,
            14, 15, 12,
            16, 19, 18,
            18, 17, 16,
            20, 21, 22,
            22, 23, 20,
        };
```

```
        return indices;
    } else {
        // clockwise winding order
        std::vector<index> indices = {
            // left
            0, 1, 2,
            2, 1, 3,
            // right
            4, 5, 6,
            6, 5, 7,
            // top
            0, 1, 4,
            4, 1, 5,
            // bottom
            2, 3, 6,
            6, 3, 7,
            // back
            3, 1, 5,
            5, 7, 3,
            // front
            2, 0, 4,
            4, 6, 2,
        };
        return indices;
    }
    // clang-format on
}

//-----------------------------------------------------------------------------
template<typename NormType>
constexpr std::array<NormType, 6> make_primitive_normals_cube() {
    // clang-format off
    // front, back, left, right, bottom, and top normals, in that order
    constexpr std::array<NormType, 6> normals = {{ { 0, 0, 1 },  { 0, 0, -1 },
                                                   { -1, 0, 0 }, { 1, 0, 0 },
                                                   { 0, 1, 0 },  { 0, -1, 0 }, }};
    // clang-format on
    return normals;
}

//-----------------------------------------------------------------------------
template<typename UVType>
constexpr std::array<UVType, 24> make_primitive_uvs_cube() {
    // clang-format off
    constexpr std::array<UVType, 24> uvs = {{
        { 1, 1 }, { 0, 1 }, { 0, 0 }, { 1, 0 }, // front
        { 0, 1 }, { 1, 1 }, { 1, 0 }, { 0, 0 }, // back
        { 1, 1 }, { 0, 1 }, { 0, 0 }, { 1, 0 }, // left
        { 0, 1 }, { 0, 0 }, { 1, 0 }, { 1, 1 }, // right
        { 1, 0 }, { 0, 0 }, { 0, 1 }, { 1, 1 }, // bottom
        { 1, 1 }, { 0, 1 }, { 0, 0 }, { 1, 0 }, // top
    }};
    // clang-format on
    return uvs;
}

using mesh_data = mesh_template_data<vertex>;

using mesh = mesh_template<vertex>;

struct mesh_meta {
    string filename;

    mesh_type type = mesh_type::none;
};

using mesh_registry = id_registry<mesh, mesh_meta>;

} // namespace lava
```

5.132 liblava/resource/primitive.hpp File Reference

Vulkan primitives.

```
#include <liblava/util/math.hpp>
```

Classes

- struct lava::vertex

  Vertex.

Enumerations

- enum class lava::mesh_type : index { **none** = 0 , **cube** , **triangle** , **quad** , **hexagon** }

  Mesh types.

5.132.1 Detailed Description

Vulkan primitives.

Authors

Lava Block OÜ and contributors

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5.133 primitive.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/util/math.hpp>
11
12 namespace lava {
13
17 struct vertex {
19     using list = std::vector<vertex>;
20
22     v3 position;
23
25     v4 color;
26
28     v2 uv;
29
31     v3 normal;
32
38     bool operator==(vertex const& other) const {
39         return position == other.position
40                && color == other.color
41                && uv == other.uv
42                && normal == other.normal;
43     }
44 };
45
49 enum class mesh_type : index {
50     none = 0,
51     cube,
52     triangle,
53     quad,
54     hexagon,
55 };
56
57 } // namespace lava
```

5.134 liblava/resource/texture.hpp File Reference

Vulkan texture.

```
#include <liblava/resource/buffer.hpp>
#include <liblava/resource/image.hpp>
```

Classes

- struct lava::file_format

  File format.
- struct lava::texture

  Texture.
- struct lava::texture::mip_level

  Texture mip level.
- struct lava::texture::layer

  Texture layer.
- struct lava::staging

  Texture staging.

Typedefs

- using **lava::staging_t** = staging

  Staging type.
- using **lava::texture_registry** = id_registry< texture, file_format >

  Texture registry.

Enumerations

- enum class lava::texture_type : index { **none** = 0 , **tex_2d** , **array** , **cube_map** }

  Texture types.

5.134.1 Detailed Description

Vulkan texture.

Authors

Lava Block OÜ and contributors

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5.135 texture.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/resource/buffer.hpp>
#include <liblava/resource/image.hpp>

namespace lava {

enum class texture_type : index {
    none = 0,
    tex_2d,
    array,
    cube_map
};

struct file_format {
    using list = std::vector<file_format>;

    string path;

    VkFormat format = VK_FORMAT_UNDEFINED;
};

struct texture : entity {
    using ptr = std::shared_ptr<texture>;

    using map = std::map<id, ptr>;

    using list = std::vector<ptr>;

    struct mip_level {
        using list = std::vector<mip_level>;

        uv2 extent{};

        ui32 size = 0;
    };

    struct layer {
        using list = std::vector<layer>;

        mip_level::list levels;
    };

    static ptr make() {
        return std::make_shared<texture>();
    }

    ~texture() {
        destroy();
    }

    bool create(device_p device,
                uv2 size,
                VkFormat format,
                layer::list const& layers = {},
                texture_type type = texture_type::tex_2d);

    void destroy();

    bool upload(void const* data,
                size_t data_size);

    bool stage(VkCommandBuffer cmd_buffer);

    void destroy_upload_buffer();

    VkDescriptorImageInfo const* get_descriptor_info() const {
        return &descriptor;
    }

    image::ptr get_image() {
        return img;
    }

    uv2 get_size() const {
        return img ? img->get_size() : uv2();
    }

    texture_type get_type() const {
        return type;
    }
```

```
    VkFormat get_format() const {
        return img ? img->get_format() : VK_FORMAT_UNDEFINED;
    }

private:
    image::ptr img;

    texture_type type = texture_type::none;

    layer::list layers;

    VkSampler sampler = 0;

    VkDescriptorImageInfo descriptor = {};

    buffer::ptr upload_buffer;
};

struct staging {
    void add(texture::ptr texture) {
        todo.push_back(texture);
    }

    bool stage(VkCommandBuffer cmd_buf,
               index frame);

    void clear() {
        todo.clear();
        staged.clear();
    }

    bool busy() const {
        return !todo.empty() || !staged.empty();
    }

private:
    texture::list todo;

    using frame_stage_map = std::map<index, texture::list>;

    frame_stage_map staged;
};

using staging_t = staging;

using texture_registry = id_registry<texture, file_format>;

} // namespace lava
```

5.136 liblava/test.hpp File Reference

Unit tests.

```
#include <catch2/catch_test_macros.hpp>
#include <liblava/lava.hpp>
```

5.136.1 Detailed Description

Unit tests.

Authors

Lava Block OÜ and contributors

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5.137 test.hpp

Go to the documentation of this file.

```
#include <catch2/catch_test_macros.hpp>
#include <liblava/lava.hpp>

using namespace lava;
```

5.138 liblava/util.hpp File Reference

Util module.

```
#include <liblava/util/hex.hpp>
#include <liblava/util/log.hpp>
#include <liblava/util/math.hpp>
#include <liblava/util/random.hpp>
#include <liblava/util/telegram.hpp>
#include <liblava/util/thread.hpp>
```

5.138.1 Detailed Description

Util module.

Author

Lava Block OÜ and contributors

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5.139 util.hpp

Go to the documentation of this file.

```
#pragma once

#include <liblava/util/hex.hpp>
#include <liblava/util/log.hpp>
#include <liblava/util/math.hpp>
#include <liblava/util/random.hpp>
#include <liblava/util/telegram.hpp>
#include <liblava/util/thread.hpp>
```

5.140 liblava/util/hex.hpp File Reference

Hex point, cell and grid.

```
#include <cmath>
#include <liblava/core/types.hpp>
#include <numbers>
#include <unordered_map>
```

Classes

- struct lava::hex_point

  Hex point.
- struct lava::hex_cell

  Hex cell.
- struct lava::hex_fractional_cell

  Hex fractional cell.
- struct lava::hex_offset_coord

  Hex offset coordinates.
- struct lava::hex_orientation

  Hex orientation.
- struct lava::hex_layout

  Hex layout.
- struct lava::hex_grid

  Hex grid.

Typedefs

- using **lava::hex_frac_cell** = hex_fractional_cell

  Hex fractional cell.
- using **lava::hex_doubled_coord** = hex_offset_coord

  Hex doubled coordinates.

Enumerations

- enum class lava::hex_offset : i32 { **odd** = -1 , **even** = 1 }

  Hex offsets.
- enum class lava::hex_cardinal_direction : index {
  NE = 0 , **E** , **SE** , **SW** ,
  W , **NW** }

  Hex cardinal directions.

Functions

- i32 lava::hex_get_s (i32 q, i32 r)

  Get S axis from Q and R axes.
- hex_cell lava::hex_cell_from_pair (hex_cell::pair const &pair)

  Get hex cell from pair.
- bool lava::hex_is_valid (hex_cell const &cell)

  Check if hex cell is valid.
- hex_cell lava::hex_direction (index direction)

  Get the hex cell from direction.
- hex_cell lava::hex_neighbor (hex_cell const &cell, index direction)

  Get the neighbor of hex cell by direction.
- hex_cell lava::hex_diagonal (index direction)

  Get the diagonal from direction.
- hex_cell lava::hex_diagonal_neighbor (hex_cell const &cell, index direction)

  Get the diagonal neighbor of hex cell by direction.

- i32 lava::hex_length (hex_cell const &cell)

  Get the length of hex cell.
- i32 lava::hex_distance (hex_cell const &a, hex_cell const &b)

  Get the distance between 2 hex cells.
- hex_cell lava::hex_round (hex_frac_cell const &cell)

  Round a fractional cell to hex cell.
- hex_frac_cell lava::hex_lerp (hex_frac_cell const &a, hex_frac_cell const &b, r32 t)

  Get the linear interpolation between 2 hex cells.
- hex_cell::list lava::hex_line (hex_cell const &a, hex_cell const &b)

  Get the line between 2 hex cells.
- hex_offset_coord lava::hex_q_offset_from_cube (hex_offset offset, hex_cell const &cell)

  Get the Q offset from hex cube.
- hex_cell lava::hex_q_offset_to_cube (hex_offset offset, hex_offset_coord const &coord)

  Get the Q offset to hex cube.
- hex_offset_coord lava::hex_r_offset_from_cube (hex_offset offset, hex_cell const &cell)

  Get the R offset from hex cube.
- hex_cell lava::hex_r_offset_to_cube (hex_offset offset, hex_offset_coord const &coord)

  Get the R offset to hex cube.
- hex_doubled_coord lava::hex_q_doubled_from_cube (hex_cell const &cell)

  Get the Q doubled from hex cube.
- hex_cell lava::hex_q_doubled_to_cube (hex_doubled_coord const &coord)

  Get the Q doubled to hex cube.
- hex_doubled_coord lava::hex_r_doubled_from_cube (hex_cell const &cell)

  Get the R offset from hex cube.
- hex_cell lava::hex_r_doubled_to_cube (hex_doubled_coord const &coord)

  Get the R doubled to hex cube.
- hex_point lava::hex_to_pixel (hex_layout const &layout, hex_cell const &cell)

  Convert the hex cell to pixel.
- hex_frac_cell lava::hex_pixel_to_cell (hex_layout const &layout, hex_point const &p)

  Convert the hex point to cell.
- hex_point lava::hex_corner_offset (hex_layout const &layout, i32 corner)

  Get the hex corner offset.
- hex_point::list lava::hex_polygon_corners (hex_layout const &layout, hex_cell const &cell)

  Get the hex polygon corners.
- hex_point lava::hex_get_corner (hex_point const &center, r32 size, ui32 corner)

  Get the hex point by corner.
- string lava::to_string (hex_cardinal_direction direction)

  Convert hex cardinal direction to string.
- hex_cell lava::hex_get (hex_cardinal_direction direction)

  Get the hex cell from cardinal direction.
- hex_cardinal_direction lava::hex_opposite (hex_cardinal_direction direction)

  Get the opposite cardinal direction.
- r32 lava::hex_calculate_inner_radius (r32 outer_radius)

  Get the hex inner radius from outer radius.

Variables

- hex_cell::list const lava::hex_directions

  List of hex directions.
- hex_cell::list const lava::hex_diagonals

  List of hex diagonals.
- hex_orientation const lava::hex_layout_point_y

  Hex point Y layout.
- hex_orientation const lava::hex_layout_flat

  Hex flat layout.
- hex_cell::list const lava::hex_cardinal_directions

  List of hex cardinal directions.
- constexpr r32 const **lava::hex_inner_radius_factor** = 0.866025404f

  Hex inner radius factor = sqrt(3) / 2.
- constexpr r32 const **lava::hex_default_outer_radius** = 1.f

  Hex default outer radius.

5.140.1 Detailed Description

Hex point, cell and grid.

See also

https://www.redblobgames.com/grids/hexagons/

Authors

Lava Block OÜ and contributors

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5.140.2 Function Documentation

5.140.2.1 hex_calculate_inner_radius()

```
r32 lava::hex_calculate_inner_radius (
            r32 outer_radius )  [inline]
```

Get the hex inner radius from outer radius.

Parameters

outer_radius	Hex outer radius

Returns

r32 Hex inner radius

5.140.2.2 hex_cell_from_pair()

```
hex_cell lava::hex_cell_from_pair (
            hex_cell::pair const & pair )  [inline]
```

Get hex cell from pair.

Parameters

pair	Hex pair

Returns

hex_cell Hex cell

5.140.2.3 hex_corner_offset()

```
hex_point lava::hex_corner_offset (
            hex_layout const & layout,
            i32 corner )  [inline]
```

Get the hex corner offset.

Parameters

layout	Hex layout
corner	Corner

Returns

hex_point Hex point

5.140.2.4 hex_diagonal()

```
hex_cell lava::hex_diagonal (
            index direction )  [inline]
```

Get the diagonal from direction.

Parameters

direction	Direction index

Returns

hex_cell Hex cell

5.140.2.5 hex_diagonal_neighbor()

```
hex_cell lava::hex_diagonal_neighbor (
            hex_cell const & cell,
            index direction )  [inline]
```

Get the diagonal neighbor of hex cell by direction.

Parameters

cell	Target hex cell
direction	Direction index

Returns

hex_cell Diagonal neighbor hex cell

5.140.2.6 hex_direction()

```
hex_cell lava::hex_direction (
            index direction )  [inline]
```

Get the hex cell from direction.

Parameters

direction	Direction index

Returns

hex_cell Hex cell

5.140.2.7 hex_distance()

```
i32 lava::hex_distance (
            hex_cell const & a,
            hex_cell const & b )  [inline]
```

Get the distance between 2 hex cells.

Parameters

a	Source hex cell
b	Target hex cell

Returns

i32 Distance

5.140.2.8 hex_get()

```
hex_cell lava::hex_get (
            hex_cardinal_direction direction )  [inline]
```

Get the hex cell from cardinal direction.

Parameters

direction	Hex cardinal direction

Returns

hex_cell Hex cell

5.140.2.9 hex_get_corner()

```
hex_point lava::hex_get_corner (
            hex_point const & center,
            r32 size,
            ui32 corner )  [inline]
```

Get the hex point by corner.

Parameters

center	Center hex point
size	Size of hex cell
corner	Corner

Returns

hex_point Hex point

5.140.2.10 hex_get_s()

```
i32 lava::hex_get_s (
            i32 q,
            i32 r )  [inline]
```

Get S axis from Q and R axes.

Parameters

q	Q axis
r	R axis

Returns

i32 S axis

5.140.2.11 hex_is_valid()

```
bool lava::hex_is_valid (
            hex_cell const & cell )  [inline]
```

Check if hex cell is valid.

Parameters

cell	Hex cell to check

Returns

Hex cell is valid or not

5.140.2.12 hex_length()

```
i32 lava::hex_length (
            hex_cell const & cell )  [inline]
```

Get the length of hex cell.

Parameters

cell	Target hex cell

Returns

i32 Length of hex cell

5.140.2.13 hex_lerp()

```
hex_frac_cell lava::hex_lerp (
            hex_frac_cell const & a,
            hex_frac_cell const & b,
            r32 t )  [inline]
```

Get the linear interpolation between 2 hex cells.

Parameters

a	Source fractional hex cell
b	Target fractional hex cell
t	Factor

Returns

hex_frac_cell Fractional hex cell

5.140.2.14 hex_line()

```
hex_cell::list lava::hex_line (
            hex_cell const & a,
            hex_cell const & b )  [inline]
```

Get the line between 2 hex cells.

Parameters

a	Source hex cell
b	Target hex cell

Returns

hex_cell::list List of hex cells

5.140.2.15 hex_neighbor()

```
hex_cell lava::hex_neighbor (
            hex_cell const & cell,
            index direction )  [inline]
```

Get the neighbor of hex cell by direction.

Parameters

cell	Target hex cell
direction	Direction index

Returns

hex_cell Neighbor hex cell

5.140.2.16 hex_opposite()

```
hex_cardinal_direction lava::hex_opposite (
            hex_cardinal_direction direction )  [inline]
```

Get the opposite cardinal direction.

Parameters

direction	Hex cardinal direction

Returns

hex_cardinal_direction Hex cardinal direction

5.140.2.17 hex_pixel_to_cell()

```
hex_frac_cell lava::hex_pixel_to_cell (
            hex_layout const & layout,
            hex_point const & p )  [inline]
```

Convert the hex point to cell.

Parameters

layout	Hex layout
p	Hex point

Returns

hex_frac_cell Hex fractional cell

5.140.2.18 hex_polygon_corners()

```
hex_point::list lava::hex_polygon_corners (
            hex_layout const & layout,
            hex_cell const & cell )  [inline]
```

Get the hex polygon corners.

Parameters

layout	Hex layout
cell	Hex cell

Returns

hex_point::list List of hex points

5.140.2.19 hex_q_doubled_from_cube()

```
hex_doubled_coord lava::hex_q_doubled_from_cube (
            hex_cell const & cell )  [inline]
```

Get the Q doubled from hex cube.

Parameters

cell	Hex cell

Returns

hex_doubled_coord Hex doubled coordinates

5.140.2.20 hex_q_doubled_to_cube()

```
hex_cell lava::hex_q_doubled_to_cube (
            hex_doubled_coord const & coord )  [inline]
```

Get the Q doubled to hex cube.

Parameters

coord	Hex doubled coordinates

Returns

hex_cell Hex cell

5.140.2.21 hex_q_offset_from_cube()

```
hex_offset_coord lava::hex_q_offset_from_cube (
            hex_offset offset,
            hex_cell const & cell )  [inline]
```

Get the Q offset from hex cube.

Parameters

offset	Hex offset
cell	Hex cell

Returns

hex_offset_coord Hex offset coordinates

5.140.2.22 hex_q_offset_to_cube()

```
hex_cell lava::hex_q_offset_to_cube (
            hex_offset offset,
            hex_offset_coord const & coord )  [inline]
```

Get the Q offset to hex cube.

Parameters

offset	Hex offset
coord	Hex offset coordinates

Returns

hex_cell Hex cell

5.140.2.23 hex_r_doubled_from_cube()

```
hex_doubled_coord lava::hex_r_doubled_from_cube (
            hex_cell const & cell )  [inline]
```

Get the R offset from hex cube.

Parameters

cell	Hex cell

Returns

hex_doubled_coord Hex doubled coordinates

5.140.2.24 hex_r_doubled_to_cube()

```
hex_cell lava::hex_r_doubled_to_cube (
            hex_doubled_coord const & coord )  [inline]
```

Get the R doubled to hex cube.

Parameters

coord	Hex doubled coordinates

Returns

hex_cell Hex cell

5.140.2.25 hex_r_offset_from_cube()

```
hex_offset_coord lava::hex_r_offset_from_cube (
            hex_offset offset,
            hex_cell const & cell )  [inline]
```

Get the R offset from hex cube.

Parameters

offset	Hex offset
cell	Hex cell

Returns

hex_offset_coord Hex offset coordinates

5.140.2.26 hex_r_offset_to_cube()

```
hex_cell lava::hex_r_offset_to_cube (
            hex_offset offset,
            hex_offset_coord const & coord )  [inline]
```

Get the R offset to hex cube.

Parameters

offset	Hex offset
coord	Hex offset coordinates

Returns

hex_cell Hex cell

5.140.2.27 hex_round()

```
hex_cell lava::hex_round (
            hex_frac_cell const & cell )  [inline]
```

Round a fractional cell to hex cell.

Parameters

cell	Target fractional cell

Returns

hex_cell Rounded hex cell

5.140.2.28 hex_to_pixel()

```
hex_point lava::hex_to_pixel (
            hex_layout const & layout,
            hex_cell const & cell )  [inline]
```

Convert the hex cell to pixel.

Parameters

layout	Hex layout
cell	Hex cell

Returns

hex_point Hex point

5.140.2.29 to_string()

```
string lava::to_string (
            hex_cardinal_direction direction ) [inline]
```

Convert hex cardinal direction to string.

Parameters

direction	Hex cardinal direction

Returns

string String representation

5.140.3 Variable Documentation

5.140.3.1 hex_cardinal_directions

```
hex_cell::list const lava::hex_cardinal_directions
```

Initial value:

```
{
    { 1, 0, -1 },
    { 0, 1, -1 },
    { -1, 1, 0 },
    { -1, 0, 1 },
    { 0, -1, 1 },
    { 1, -1, 0 }
}
```

List of hex cardinal directions.

5.140.3.2 hex_diagonals

```
hex_cell::list const lava::hex_diagonals
```

Initial value:

```
{
    { 2, -1, -1 },
    { -1, -2, 1 },
    { -1, -1, 2 },
    { -2, 1, 1 },
    { -1, 2, -1 },
    { 1, 1, -2 }
}
```

List of hex diagonals.

5.140.3.3 hex_directions

```
hex_cell::list const lava::hex_directions
```

Initial value:

```
{
    { 1, 0, -1 },
    { 1, -1, 0 },
    { 0, -1, 1 },
    { -1, 0, 1 },
    { -1, 1, 0 },
    { 0, 1, -1 }
}
```

List of hex directions.

5.140.3.4 hex_layout_flat

```
hex_orientation const lava::hex_layout_flat
```

Initial value:

```
= {
    3.f / 2.f,
    0.f,
    std::sqrt(3.f) / 2.f,
    std::sqrt(3.f),
    2.f / 3.f,
    0.f,
    -1.f / 3.f,
    std::sqrt(3.f) / 3.f,
    0.f
}
```

Hex flat layout.

5.140.3.5 hex_layout_point_y

```
hex_orientation const lava::hex_layout_point_y
```

Initial value:

```
= {
    std::sqrt(3.f),
    std::sqrt(3.f) / 2.f,
    0.f,
    3.f / 2.f,
    std::sqrt(3.f) / 3.f,
    -1.f / 3.f,
    0.f,
    2.f / 3.f,
    0.5f
}
```

Hex point Y layout.

5.141 hex.hpp

Go to the documentation of this file.

```
1
9 #pragma once
10
11 #include <cmath>
12 #include <liblava/core/types.hpp>
13 #include <numbers>
14 #include <unordered_map>
15
16 namespace lava {
17
21 struct hex_point {
23     using list = std::vector<hex_point>;
24
26     r32 x{};
27
29     r32 y{};
30 };
31
35 struct hex_cell {
37     using list = std::vector<hex_cell>;
38
40     i32 q{};
41
43     i32 r{};
44
46     i32 s{};
47
51     auto operator<=>(hex_cell const&) const = default;
52
54     using pair = std::pair<i32, i32>;
55
57     using map = std::unordered_map<pair, index, pair_hash>;
58
63     inline pair to_pair() const {
64         return { q, r };
65     }
66
71     inline void add(hex_cell const& cell) {
72         *this = { q + cell.q,
73                   r + cell.r,
74                   s + cell.s };
75     }
76
81     inline void substract(hex_cell const& cell) {
82         *this = { q - cell.q,
83                   r - cell.r,
84                   s - cell.s };
85     }
86
91     inline void scale(i32 factor) {
92         *this = { q * factor,
93                   r * factor,
94                   s * factor };
95     }
96
```

```
    inline void rotate_left() {
        *this = { -s, -q, -r };
    }

    inline void rotate_right() {
        *this = { -r, -s, -q };
    }
};

inline i32 hex_get_s(i32 q, i32 r) {
    return -q - r;
}

inline hex_cell hex_cell_from_pair(hex_cell::pair const& pair) {
    return { pair.first, pair.second,
             hex_get_s(pair.first, pair.second) };
}

struct hex_fractional_cell {
    r32 q{};

    r32 r{};

    r32 s{};
};

using hex_frac_cell = hex_fractional_cell;

inline bool hex_is_valid(hex_cell const& cell) {
    return cell.q + cell.r + cell.s == 0;
}

struct hex_offset_coord {
    i32 col{};

    i32 row{};
};

using hex_doubled_coord = hex_offset_coord;

struct hex_orientation {
    r32 f0{};

    r32 f1{};

    r32 f2{};

    r32 f3{};

    r32 b0{};

    r32 b1{};

    r32 b2{};

    r32 b3{};

    r32 start_angle{};
};

struct hex_layout {
    hex_orientation orientation;

    hex_point origin;

    hex_point size;
};

hex_cell::list const hex_directions{
    { 1, 0, -1 },
    { 1, -1, 0 },
    { 0, -1, 1 },
    { -1, 0, 1 },
    { -1, 1, 0 },
    { 0, 1, -1 }
};

inline hex_cell hex_direction(index direction) {
    return hex_directions[direction];
}

inline hex_cell hex_neighbor(hex_cell const& cell,
                             index direction) {
    auto neighbor = cell;
    neighbor.add(hex_direction(direction));
    return neighbor;
}
```

```

hex_cell::list const hex_diagonals{
    { 2, -1, -1 },
    { -1, -2, 1 },
    { -1, -1, 2 },
    { -2, 1, 1 },
    { -1, 2, -1 },
    { 1, 1, -2 }
};

inline hex_cell hex_diagonal(index direction) {
    return hex_diagonals[direction];
}

inline hex_cell hex_diagonal_neighbor(hex_cell const& cell,
                                      index direction) {
    auto neighbor = cell;
    neighbor.add(hex_diagonal(direction));
    return neighbor;
}

inline i32 hex_length(hex_cell const& cell) {
    return to_i32(std::abs(cell.q)
                  + std::abs(cell.r)
                  + std::abs(cell.s) / 2);
}

inline i32 hex_distance(hex_cell const& a,
                        hex_cell const& b) {
    auto dist = a;
    dist.substract(b);
    return hex_length(dist);
}

inline hex_cell hex_round(hex_frac_cell const& cell) {
    auto qi = to_i32(std::round(cell.q));
    auto ri = to_i32(std::round(cell.r));
    auto si = to_i32(std::round(cell.s));

    auto const q_diff = std::abs(qi - cell.q);
    auto const r_diff = std::abs(ri - cell.r);
    auto const s_diff = std::abs(si - cell.s);

    if ((q_diff > r_diff) && (q_diff > s_diff))
        qi = -ri - si;
    else if (r_diff > s_diff)
        ri = -qi - si;
    else
        si = -qi - ri;

    return { qi, ri, si };
}

inline hex_frac_cell hex_lerp(hex_frac_cell const& a,
                              hex_frac_cell const& b,
                              r32 t) {
    return {
        a.q * (1.f - t) + b.q * t,
        a.r * (1.f - t) + b.r * t,
        a.s * (1.f - t) + b.s * t
    };
}

inline hex_cell::list hex_line(hex_cell const& a,
                               hex_cell const& b) {
    auto const a_nudge = hex_frac_cell{ a.q + 0.000001f,
                                        a.r + 0.000001f,
                                        a.s - 0.000002f };
    auto const b_nudge = hex_frac_cell{ b.q + 0.000001f,
                                        b.r + 0.000001f,
                                        b.s - 0.000002f };

    ui32 const n = hex_distance(a, b);
    auto const step = 1.0 / std::max(n, 1u);

    hex_cell::list results;
    for (auto i = 0u; i <= n; ++i)
        results.push_back(hex_round(hex_lerp(a_nudge,
                                             b_nudge,
                                             step * i)));

    return results;
}

enum class hex_offset : i32 {
    odd = -1,
    even = 1
```

```
};

inline hex_offset_coord hex_q_offset_from_cube(hex_offset offset,
                                               hex_cell const& cell) {
    auto const& col = cell.q;
    auto const row = cell.r
                     + to_i32((cell.q + (i32) offset * (cell.q & 1)) / 2);
    return { col, row };
}

inline hex_cell hex_q_offset_to_cube(hex_offset offset,
                                     hex_offset_coord const& coord) {
    auto const& q = coord.col;
    auto const r = coord.row
                    - to_i32((coord.col + (i32) offset * (coord.col & 1)) / 2);
    auto const s = -q - r;
    return { q, r, s };
}

inline hex_offset_coord hex_r_offset_from_cube(hex_offset offset,
                                               hex_cell const& cell) {
    auto const col = cell.q
                     + to_i32((cell.r + (i32) offset * (cell.r & 1)) / 2);
    auto const& row = cell.r;
    return { col, row };
}

inline hex_cell hex_r_offset_to_cube(hex_offset offset,
                                     hex_offset_coord const& coord) {
    auto const q = coord.col
                    - to_i32((coord.row + (i32) offset * (coord.row & 1)) / 2);
    auto const& r = coord.row;
    auto const s = -q - r;
    return { q, r, s };
}

inline hex_doubled_coord hex_q_doubled_from_cube(hex_cell const& cell) {
    auto const& col = cell.q;
    auto const row = 2 * cell.r + cell.q;
    return { col, row };
}

inline hex_cell hex_q_doubled_to_cube(hex_doubled_coord const& coord) {
    auto const& q = coord.col;
    auto const r = to_i32((coord.row - coord.col) / 2);
    auto const s = -q - r;
    return { q, r, s };
}

inline hex_doubled_coord hex_r_doubled_from_cube(hex_cell const& cell) {
    auto const col = 2 * cell.q + cell.r;
    auto const& row = cell.r;
    return { col, row };
}

inline hex_cell hex_r_doubled_to_cube(hex_doubled_coord const& coord) {
    auto const q = to_i32((coord.col - coord.row) / 2);
    auto const& r = coord.row;
    auto const s = -q - r;
    return { q, r, s };
}

hex_orientation const hex_layout_point_y = {
    std::sqrt(3.f),
    std::sqrt(3.f) / 2.f,
    0.f,
    3.f / 2.f,
    std::sqrt(3.f) / 3.f,
    -1.f / 3.f,
    0.f,
    2.f / 3.f,
    0.5f
};

hex_orientation const hex_layout_flat = {
    3.f / 2.f,
    0.f,
    std::sqrt(3.f) / 2.f,
    std::sqrt(3.f),
    2.f / 3.f,
    0.f,
    -1.f / 3.f,
    std::sqrt(3.f) / 3.f,
    0.f
};

inline hex_point hex_to_pixel(hex_layout const& layout,
```

```
                                hex_cell const& cell) {
    auto const& m = layout.orientation;
    auto const& size = layout.size;
    auto const& origin = layout.origin;
    auto const x = (m.f0 * cell.q + m.f1 * cell.r) * size.x;
    auto const y = (m.f2 * cell.q + m.f3 * cell.r) * size.y;
    return { x + origin.x, y + origin.y };
}

inline hex_frac_cell hex_pixel_to_cell(hex_layout const& layout,
                                       hex_point const& p) {
    auto const& m = layout.orientation;
    auto const& size = layout.size;
    auto const& origin = layout.origin;
    auto const pt = hex_point{ (p.x - origin.x) / size.x,
                               (p.y - origin.y) / size.y };
    auto const q = m.b0 * pt.x + m.b1 * pt.y;
    auto const r = m.b2 * pt.x + m.b3 * pt.y;
    return { q, r, -q - r };
}

inline hex_point hex_corner_offset(hex_layout const& layout,
                                   i32 corner) {
    auto const& m = layout.orientation;
    auto const& size = layout.size;
    auto const angle = 2.
                        * std::numbers::pi_v<r32> * (m.start_angle - corner)
                        / 6.f;
    return { size.x * (r32) std::cos(angle),
             size.y * (r32) std::sin(angle) };
}

inline hex_point::list hex_polygon_corners(hex_layout const& layout,
                                           hex_cell const& cell) {
    hex_point::list corners = {};
    auto const center = hex_to_pixel(layout, cell);
    for (int i = 0u; i < 6; ++i) {
        auto const offset = hex_corner_offset(layout, i);
        corners.push_back({ center.x + offset.x,
                            center.y + offset.y });
    }
    return corners;
}

inline hex_point hex_get_corner(hex_point const& center,
                                r32 size,
                                ui32 corner) {
    auto const angle_deg = 60 * corner - 30;
    auto const angle_rad = std::numbers::pi_v<r32> / 180 * angle_deg;

    return {
        center.x + size * std::cos(angle_rad),
        center.y + size * std::sin(angle_rad)
    };
}

enum class hex_cardinal_direction : index {
    NE = 0,
    E,
    SE,
    SW,
    W,
    NW
};

inline string to_string(hex_cardinal_direction direction) {
    switch (direction) {
    case hex_cardinal_direction::NE: {
        return "Northeast";
    }
    case hex_cardinal_direction::E: {
        return "East";
    }
    case hex_cardinal_direction::SE: {
        return "Southeast";
    }
    case hex_cardinal_direction::SW: {
        return "Southwest";
    }
    case hex_cardinal_direction::W: {
        return "West";
    }
    case hex_cardinal_direction::NW: {
        return "Northwest";
    }
    }
}
```

```
hex_cell::list const hex_cardinal_directions{
    { 1, 0, -1 },
    { 0, 1, -1 },
    { -1, 1, 0 },
    { -1, 0, 1 },
    { 0, -1, 1 },
    { 1, -1, 0 }
};

inline hex_cell hex_get(hex_cardinal_direction direction) {
    return hex_cardinal_directions[(index) direction];
}

inline hex_cardinal_direction hex_opposite(hex_cardinal_direction direction) {
    if ((index) direction < 3)
        return hex_cardinal_direction((i32) direction + 3);
    else
        return hex_cardinal_direction((i32) direction - 3);
}

constexpr r32 const hex_inner_radius_factor = 0.866025404f;

constexpr r32 const hex_default_outer_radius = 1.f;

inline r32 hex_calculate_inner_radius(r32 outer_radius) {
    return outer_radius * hex_inner_radius_factor;
}

struct hex_grid {
    r32 inner_radius = 0.f;

    r32 outer_radius = hex_default_outer_radius;

    hex_layout layout;

    hex_grid(r32 radius = hex_default_outer_radius)
    : outer_radius(radius) {
        update();
    }

    void update(hex_orientation orientation = hex_layout_point_y) {
        inner_radius = hex_calculate_inner_radius(outer_radius);
        layout = {
            orientation,
            {},
            { outer_radius, outer_radius }
        };
    }

    hex_cell find(r32 x, r32 y) const {
        return hex_round(hex_pixel_to_cell(layout,
                                           { x, y }));
    }

    hex_point to_pixel(hex_cell const& cell) const {
        return hex_to_pixel(layout, cell);
    }
};

} // namespace lava
```

5.142 liblava/util/log.hpp File Reference

Logging.

```
#include <spdlog/sinks/basic_file_sink.h>
#include <spdlog/sinks/stdout_color_sinks.h>
#include <spdlog/spdlog.h>
#include <liblava/core/version.hpp>
#include <memory>
```

Classes

- struct lava::log_config

Log configuration.

- struct lava::global_logger

  Global logger.

Typedefs

- using **lava::logger** = std::shared_ptr< spdlog::logger >

  Logger.

Functions

- string lava::to_string (string_ref id, string_ref name)

  Convert id and name to string.
- string lava::to_string (int_version const &version)

  Convert internal version to string.
- string lava::internal_version_string ()

  Convert global internal version to string.
- string lava::int_version_string ()
- name lava::to_string (version_stage stage)

  Convert version stage to string.
- string lava::to_string (version const &version)

  Convert version to string.
- string lava::version_string ()

  Convert global version to string.
- logger lava::setup_log (log_config config={})

  Set up logging.
- void lava::teardown_log (log_config config={})

  Tear down logging.
- logger lava::log ()

  Get global logger.

5.142.1 Detailed Description

Logging.

Authors

Lava Block OÜ and contributors

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5.142.2 Function Documentation

5.142.2.1 int_version_string()

```
string lava::int_version_string ( ) [inline]
```

See also

internal_version_string

5.142.2.2 internal_version_string()

```
string lava::internal_version_string ( ) [inline]
```

Convert global internal version to string.

Returns

string String representation

5.142.2.3 log()

```
logger lava::log ( ) [inline]
```

Get global logger.

Returns

logger Logger

5.142.2.4 setup_log()

```
logger lava::setup_log (
            log_config config = {} ) [inline]
```

Set up logging.

Parameters

config	Log configuration

5.142.2.5 teardown_log()

```
void lava::teardown_log (
            log_config config = {} )  [inline]
```

Tear down logging.

Parameters

config	Log configuration

5.142.2.6 to_string() [1/4]

```
string lava::to_string (
            int_version const & version )  [inline]
```

Convert internal version to string.

Parameters

version	Internal version to convert

Returns

string String representation

5.142.2.7 to_string() [2/4]

```
string lava::to_string (
            string_ref id,
            string_ref name )  [inline]
```

Convert id and name to string.

Parameters

id	Id to convert
name	Name to convert

Returns

string String representation

5.142.2.8 to_string() [3/4]

```
string lava::to_string (
            version const & version )  [inline]
```

Convert version to string.

Parameters

version	Version to convert

Returns

string String representation

5.142.2.9 to_string() [4/4]

```
name lava::to_string (
            version_stage stage )  [inline]
```

Convert version stage to string.

Parameters

stage	Version stage to convert

Returns

name Name representation

5.142.2.10 version_string()

```
string lava::version_string ( )  [inline]
```

Convert global version to string.

Returns

string String representation

5.143 log.hpp

Go to the documentation of this file.

```
#pragma once

#include <spdlog/sinks/basic_file_sink.h>
#include <spdlog/sinks/stdout_color_sinks.h>
#include <spdlog/spdlog.h>
#include <liblava/core/version.hpp>
#include <memory>

namespace lava {

using logger = std::shared_ptr<spdlog::logger>;

inline string to_string(string_ref id, string_ref name) {
    return fmt::format("{} | {}", id, name);
}

inline string to_string(int_version const& version) {
    return fmt::format("{}.{}.{}",
                       version.major,
                       version.minor,
                       version.patch);
}

inline string internal_version_string() {
    return to_string(int_version{});
}

inline string int_version_string() {
    return internal_version_string();
}

inline name to_string(version_stage stage) {
    switch (stage) {
    case version_stage::preview:
        return "preview";
    case version_stage::alpha:
        return "alpha";
    case version_stage::beta:
        return "beta";
    case version_stage::rc:
        return "rc";
    default:
        return "";
    }
}

inline string to_string(version const& version) {
    string stage_str = to_string(version.stage);
    if ((version.rev > 1) && (version.stage != version_stage::release))
        stage_str += fmt::format(" {}", version.rev);

    if (version.release == 0) {
        if (stage_str.empty())
            return fmt::format("{}", version.year);
        else
            return fmt::format("{} {}",
                               version.year, stage_str);
    } else
        return fmt::format("{}.{} {}",
                           version.year, version.release, stage_str);
}

inline string version_string() {
    return to_string(version{});
}

struct log_config {
    name logger = _lava_;

    name file = "lava.log";

    i32 level = -1;

    bool debug = false;
};

inline logger setup_log(log_config config = {}) {
    if (config.debug) {
        auto log = spdlog::stdout_color_mt(config.logger);
        log->set_level((config.level < 0)
                           ? spdlog::level::debug
```

```
                            : (spdlog::level::level_enum) config.level);
        return log;
    } else {
        auto log = spdlog::basic_logger_mt(config.logger, config.file);
        log->set_level((config.level < 0)
                            ? spdlog::level::warn
                            : (spdlog::level::level_enum) config.level);
        return log;
    }
}

inline void teardown_log(log_config config = {}) {
    spdlog::drop(config.logger);
}

struct global_logger {
    static global_logger& singleton() {
        static global_logger global_logger;
        return global_logger;
    }

    logger get() {
        return log;
    }

    void set(lava::logger l) {
        log = l;
    }

    void reset() {
        log = nullptr;
    }

private:
    logger log;
};

inline logger log() {
    return global_logger::singleton().get();
}

} // namespace lava
```

5.144 liblava/util/math.hpp File Reference

Math helpers.

```
#include <picosha2.h>
#include <liblava/core/types.hpp>
#include <glm/glm.hpp>
#include <glm/gtc/matrix_transform.hpp>
#include <glm/gtc/type_ptr.hpp>
```

Classes

- struct lava::rect

  Rectangle.

Typedefs

- using **lava::v2** = glm::vec2

  Vector 2D.
- using **lava::v3** = glm::vec3

  Vector 3D.
- using **lava::v4** = glm::vec4

Vector 4D.

- using **lava::uv2** = glm::uvec2

  UV pair.
- using **lava::mat3** = glm::mat3

  Matrix 3x3.
- using **lava::mat4** = glm::mat4

  Matrix 4x4.
- using **lava::iv2** = glm::ivec2

  Integer vector 2D.
- using **lava::iv3** = glm::ivec3

  Integer vector 3D.

Functions

- auto lava::ceil_div (auto x, auto y)

  Ceiling of division.
- mat4 lava::perspective_matrix (uv2 size, float fov=90.f, float far_plane=5.f)

  Calculate perspective matrix.
- string lava::hash256 (string_ref value)

  Get SHA-256 hash of string.

Variables

- constexpr v3 const lava::default_color

  Default color (Lava color: CF1020 : 207, 16, 32)

5.144.1 Detailed Description

Math helpers.

Authors

Lava Block OÜ and contributors

Copyright

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5.144.2 Function Documentation

5.144.2.1 ceil_div()

```
auto lava::ceil_div (
            auto x,
            auto y )  [inline]
```

Ceiling of division.

Parameters

x	X value
y	Y value

Returns

auto Result

5.144.2.2 hash256()

```
string lava::hash256 (
            string_ref value )  [inline]
```

Get SHA-256 hash of string.

Parameters

value	Value to hash

Returns

string Hash result

5.144.2.3 perspective_matrix()

```
mat4 lava::perspective_matrix (
            uv2 size,
            float fov = 90.f,
            float far_plane = 5.f )  [inline]
```

Calculate perspective matrix.

Parameters

size	Size for aspect ratio
fov	Field of view
far_plane	Far plane

Returns

mat4 Calculated matrix

5.144.3 Variable Documentation

5.144.3.1 default_color

```
constexpr v3 const lava::default_color  [constexpr]
```

Initial value:

```
= v3{ 0.8118f,
                                        0.0627f,
                                        0.1255f }
```

Default color (Lava color: CF1020 : 207, 16, 32)

5.145 math.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <picosha2.h>
11 #include <liblava/core/types.hpp>
12
13 #define GLM_FORCE_RADIANS
14 #define GLM_FORCE_DEPTH_ZERO_TO_ONE
15 #include <glm/glm.hpp>
16 #include <glm/gtc/matrix_transform.hpp>
17 #include <glm/gtc/type_ptr.hpp>
18
19 namespace lava {
20
22 using v2 = glm::vec2;
23
25 using v3 = glm::vec3;
26
28 using v4 = glm::vec4;
29
31 using uv2 = glm::uvec2;
32
34 using mat3 = glm::mat3;
35
37 using mat4 = glm::mat4;
38
40 using iv2 = glm::ivec2;
41
43 using iv3 = glm::ivec3;
44
48 struct rect {
50     using ref = rect const&;
51
55     rect() = default;
56
64     rect(i32 left, i32 top,
65          ui32 width, ui32 height)
66     : left_top({ left, top }) {
67         set_size({ width, height });
68     }
69
76     rect(iv2 const& left_top,
77          ui32 width, ui32 height)
78     : left_top(left_top) {
79         set_size({ width, height });
80     }
81
87     rect(iv2 const& left_top,
88          uv2 const& size)
89     : left_top(left_top) {
90         set_size(size);
91     }
92
97     iv2 const& get_origin() const {
98         return left_top;
```

```
    }

    iv2 const& get_end_point() const {
        return right_bottom;
    }

    uv2 get_size() const {
        LAVA_ASSERT(left_top.x <= right_bottom.x);
        LAVA_ASSERT(left_top.y <= right_bottom.y);
        return { right_bottom.x - left_top.x,
                 right_bottom.y - left_top.y };
    }

    void set_size(uv2 const& size) {
        right_bottom.x = left_top.x + size.x;
        right_bottom.y = left_top.y + size.y;
    }

    void move(iv2 const& offset) {
        left_top += offset;
        right_bottom += offset;
    }

    bool contains(iv2 point) const {
        return (left_top.x < point.x)
               && (left_top.y < point.y)
               && (right_bottom.x > point.x)
               && (right_bottom.y > point.y);
    }

private:
    iv2 left_top = iv2();

    iv2 right_bottom = iv2();
};

inline auto ceil_div(auto x, auto y) {
    return (x + y - 1) / y;
}

constexpr v3 const default_color = v3{ 0.8118f,
                                       0.0627f,
                                       0.1255f };

inline mat4 perspective_matrix(uv2 size,
                               float fov = 90.f,
                               float far_plane = 5.f) {
    // Vulkan NDC is right-handed with Y pointing down
    // we flip Y which makes it left-handed
    return glm::scale(glm::identity<glm::mat4>(),
                      { 1.f, -1.f, 1.f })
           * glm::perspectiveLH_ZO(
               glm::radians(fov),
               float(size.x) / size.y,
               0.1f,
               far_plane);
}

inline string hash256(string_ref value) {
    std::vector<uc8> hash(picosha2::k_digest_size);
    picosha2::hash256(value.begin(), value.end(),
                      hash.begin(), hash.end());

    return picosha2::bytes_to_hex_string(hash.begin(), hash.end());
}

} // namespace lava
```

5.146 liblava/util/random.hpp File Reference

Random generator.

```
#include <liblava/core/types.hpp>
#include <random>
```

Classes

- struct lava::random_generator

  Random generator.
- struct lava::pseudo_random_generator

  Pseudo random generator.

Functions

- auto lava::random (auto low, auto high)

  Get next random number.
- auto lava::random (auto high)

  Get next random number (lowest is 0)

5.146.1 Detailed Description

Random generator.

Authors

Lava Block OÜ and contributors

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5.146.2 Function Documentation

5.146.2.1 random() [1/2]

```
auto lava::random (
            auto high )  [inline]
```

Get next random number (lowest is 0)

Parameters

high	Highest number

Returns

auto Random number

5.146.2.2 random() [2/2]

```
auto lava::random (
            auto low,
            auto high )  [inline]
```

Get next random number.

Parameters

low	Lowest number
high	Highest number

Returns

auto Random number

5.147 random.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <liblava/core/types.hpp>
11 #include <random>
12
13 namespace lava {
14
18 struct random_generator {
22     random_generator() {
23         std::random_device rd;
24         mt = std::mt19937(rd());
25     }
26
33     i32 get(i32 low, i32 high) {
34         std::uniform_int_distribution<i32> dist(low, high);
35         return dist(mt);
36     }
37
45     template<typename T = real>
46     T get(T low, T high) {
47         std::uniform_real_distribution<T> dist(low, high);
48         return dist(mt);
49     }
50
51 private:
53     std::mt19937 mt;
54 };
55
62 inline auto random(auto low, auto high) {
63     return random_generator().get(low, high);
64 }
65
71 inline auto random(auto high) {
72     return random_generator().get({}, high);
73 }
74
78 struct pseudo_random_generator {
83     explicit pseudo_random_generator(ui32 seed)
84     : seed(seed) {}
85
90     void set_seed(ui32 value) {
91         seed = value;
92     }
93
98     ui32 get() {
99         return generate_fast() ^ (generate_fast() » 7);
100     }
101
102 private:
104     ui32 seed = 0;
```

```

    ui32 generate_fast() {
        return seed = (seed * 196314165 + 907633515);
    }
};

} // namespace lava
```

5.148 liblava/util/telegram.hpp File Reference

Telegram dispatcher.

```
#include <any>
#include <cmath>
#include <liblava/util/thread.hpp>
#include <set>
```

Classes

- struct lava::telegram

  Telegram.
- struct lava::dispatcher

  Telegram dispatcher.

Typedefs

- using **lava::any** = std::any

  Any type.

Variables

- constexpr ms const **lava::telegram_min_delay** { 250 }

  Minimal telegram delay in milliseconds.

5.148.1 Detailed Description

Telegram dispatcher.

Authors

Lava Block OÜ and contributors

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5.149 telegram.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
10 #include <any>
11 #include <cmath>
12 #include <liblava/util/thread.hpp>
13 #include <set>
14
15 namespace lava {
16
18 constexpr ms const telegram_min_delay{ 250 };
19
21 using any = std::any;
22
26 struct telegram {
28     using ref = telegram const&;
29
38     explicit telegram(id::ref sender,
39                       id::ref receiver,
40                       id::ref msg,
41                       ms dispatch_time = {},
42                       any info = {})
43     : sender(sender), receiver(receiver),
44       msg(msg), dispatch_time(dispatch_time),
45       info(std::move(info)) {}
46
52     bool operator==(ref rhs) const {
53         return ((dispatch_time - rhs.dispatch_time) < telegram_min_delay)
54                && (sender == rhs.sender)
55                && (receiver == rhs.receiver)
56                && (msg == rhs.msg);
57     }
58
64     bool operator<(ref rhs) const {
65         if (*this == rhs)
66             return false;
67
68         return (dispatch_time < rhs.dispatch_time);
69     }
70
72     id sender;
73
75     id receiver;
76
78     id msg;
79
81     ms dispatch_time;
82
84     any info;
85 };
86
90 struct dispatcher {
95     void setup(ui32 thread_count) {
96         pool.setup(thread_count);
97     }
98
102     void teardown() {
103         pool.teardown();
104     }
105
110     void update(ms current) {
111         current_time = current;
112         dispatch_delayed_messages(current_time);
113     }
114
123     void add_message(id::ref receiver,
124                      id::ref sender,
125                      id::ref message,
126                      ms delay = {},
127                      any const& info = {}) {
128         telegram msg(sender,
129                      receiver,
130                      message,
131                      current_time,
132                      info);
133
134         if (delay == ms{ 0 }) {
135             discharge(msg); // now
136             return;
137         }
138
139         msg.dispatch_time += delay;
```

```
        messages.insert(msg);
    }

    using message_func = std::function<void(telegram::ref, id::ref)>;

    message_func on_message;

private:
    void discharge(telegram::ref message) {
        pool.enqueue([&, message](id::ref thread_id) {
            if (on_message)
                on_message(message, thread_id);
        });
    }

    void dispatch_delayed_messages(ms time) {
        while (!messages.empty()
               && (messages.begin()->dispatch_time < time)
               && (messages.begin()->dispatch_time > ms{})) {
            discharge(*messages.begin());

            messages.erase(messages.begin());
        }
    }

    ms current_time;

    thread_pool pool;

    std::set<telegram> messages;
};

} // namespace lava
```

5.150 liblava/util/thread.hpp File Reference

Thread pool.

```
#include <condition_variable>
#include <deque>
#include <liblava/core/id.hpp>
#include <liblava/core/time.hpp>
#include <mutex>
#include <thread>
```

Classes

- struct lava::thread_pool

  Thread pool.

Functions

- void lava::sleep (ms time)

  Sleep for milliseconds.
- void lava::sleep (seconds time)

  Sleep for seconds.

5.150.1 Detailed Description

Thread pool.

Authors

Lava Block OÜ and contributors

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5.150.2 Function Documentation

5.150.2.1 sleep() [1/2]

```
void lava::sleep (
            ms time )  [inline]
```

Sleep for milliseconds.

Parameters

time	Milliseconds to sleep

5.150.2.2 sleep() [2/2]

```
void lava::sleep (
            seconds time )  [inline]
```

Sleep for seconds.

Parameters

time	Seconds to sleep

5.151 thread.hpp

Go to the documentation of this file.

```
1
8 #pragma once
9
```

```
#include <condition_variable>
#include <deque>
#include <liblava/core/id.hpp>
#include <liblava/core/time.hpp>
#include <mutex>
#include <thread>

namespace lava {

inline void sleep(ms time) {
    std::this_thread::sleep_for(time);
}

inline void sleep(seconds time) {
    std::this_thread::sleep_for(time);
}

struct thread_pool {
    using task = std::function<void(id::ref)>;

    void setup(ui32 count = 2) {
        for (auto i = 0u; i < count; ++i)
            workers.emplace_back(worker(*this));
    }

    void teardown() {
        stop = true;
        condition.notify_all();

        for (auto& worker : workers)
            worker.join();

        workers.clear();
    }

    void enqueue(auto f) {
        {
            std::unique_lock<std::mutex> lock(queue_mutex);
            tasks.push_back(task(f));
        }
        condition.notify_one();
    }

private:
    struct worker {
        explicit worker(thread_pool& pool)
        : pool(pool) {}

        void operator()() {
            auto thread_id = ids::instance().next();

            task task;
            while (true) {
                {
                    std::unique_lock<std::mutex> lock(pool.queue_mutex);

                    while (!pool.stop && pool.tasks.empty())
                         pool.condition.wait(lock);

                     if (pool.stop)
                         break;

                     task = pool.tasks.front();
                     pool.tasks.pop_front();
                }

                task(thread_id);
             }
         }

     private:
         thread_pool& pool;
     };

     std::vector<std::thread> workers;

     std::deque<task> tasks;

     std::mutex queue_mutex;

     std::condition_variable condition;

     bool stop = false;
};

} // namespace lava
```

Index

activated
- lava::block, 34
- lava::camera, 49
- lava::imgui, 163
- lava::pipeline, 210
- lava::subpass, 288

active
- lava::key_event, 180

add
- lava::descriptor, 71
- lava::gamepad_manager, 132
- lava::hex_cell, 135
- lava::id_listeners< T >, 144
- lava::id_registry< T, Meta >, 146
- lava::input, 168
- lava::input_events< T >, 171
- lava::json_file, 177
- lava::pipeline_layout, 215
- lava::property, 230
- lava::queue_family_info, 239
- lava::render_pass, 248
- lava::render_pipeline, 256
- lava::staging, 285
- lava::subpass, 289
- lava::tooltip_list, 318

add_binding
- lava::descriptor, 71

add_callback
- lava::render_target, 270
- lava::swapchain, 301

add_cmd
- lava::block, 35

add_color_blend_attachment
- lava::render_pipeline, 257

add_command
- lava::block, 35

add_data
- lava::mesh_template< T >, 187

add_dedicated_queues
- lava::device::create_param, 64
- queue.hpp, 443

add_descriptor
- lava::pipeline_layout, 215

add_dynamic_state
- lava::render_pipeline, 257

add_front
- lava::render_pass, 249
- lava::subpass, 289

add_id_map
- id.hpp, 478

add_mesh
- lava::producer, 225

add_message
- lava::dispatcher, 98

add_preserve_attachment
- lava::subpass, 289

add_push_constant_range
- lava::pipeline_layout, 216

add_queue
- lava::device::create_param, 64

add_queues
- lava::device::create_param, 64
- queue.hpp, 443

add_run
- lava::frame, 123

add_run_end
- lava::frame, 123

add_run_once
- lava::frame, 124

add_shader
- lava::render_pipeline, 257

add_shader_stage
- lava::render_pipeline, 258

add_specialization_entry
- lava::pipeline::shader_stage, 281

add_stage
- lava::driver, 100

add_texture
- lava::producer, 225

align
- data.hpp, 470, 472

align_up
- data.hpp, 472

alloc
- lava::device, 79
- lava::memory, 184

alloc_data
- data.hpp, 473

allocate
- lava::data, 67
- lava::descriptor, 73

allocate_set
- lava::descriptor, 73

allocate_sets
- lava::descriptor, 73

allocator
- lava::allocator, 16

app

lava::app, 19, 20
append
misc.hpp, 484
argh.hpp
log_command_line, 539
as_ptr
data.hpp, 473
assign
lava::window, 327
attachment
lava::attachment, 25
auto_bind
lava::pipeline, 210
auto_line_width
lava::render_pipeline, 258
auto_sizing
lava::render_pipeline, 258

base.hpp
check, 379
failed, 379
to_api_version, 380
to_string, 380
to_version, 380, 381
version_to_string, 381
begin
misc.hpp, 484
begin_frame
lava::renderer, 275
begin_label
debug_utils.hpp, 386
benchmark
benchmark.hpp, 343
benchmark.hpp
benchmark, 343
parse, 344
write_frames_json, 344
bind
lava::compute_pipeline, 58
lava::mesh_template< T >, 188
lava::pipeline, 210
lava::pipeline_layout, 216
lava::render_pipeline, 258
bind_descriptor_set
lava::pipeline_layout, 216
bind_draw
lava::mesh_template< T >, 188
block_cmd
lava::app, 20
buffer.hpp
buffer_usage_to_possible_access, 569
buffer_usage_to_possible_stages, 569
buffer_usage_to_possible_access
buffer.hpp, 569
buffer_usage_to_possible_stages
buffer.hpp, 569
busy
lava::staging, 285

c16
types.hpp, 505
c32
types.hpp, 505
c8
types.hpp, 505
call
lava::device, 79
capture_keyboard
lava::imgui, 163
capture_mouse
lava::imgui, 163
cdata
lava::cdata, 54
ceil_div
math.hpp, 629
check
base.hpp, 379
instance.hpp, 432
lava::property, 232
check_mod
input.hpp, 548
close_request
lava::window, 327
compile_shader
lava::producer, 225
compute_queue
lava::device, 79
compute_queues
lava::device, 79
config.hpp
set_window_icon, 348
contains
lava::rect, 244
misc.hpp, 484
copy_from
lava::compute_pipeline, 58
lava::render_pipeline, 259
copy_to
lava::compute_pipeline, 58
lava::render_pipeline, 259
count
lava::queue_family_info, 240
create
lava::allocator, 16
lava::block, 35
lava::buffer, 41
lava::camera, 49
lava::command, 56
lava::descriptor, 74
lava::descriptor::pool, 222
lava::device, 80
lava::forward_shading, 119
lava::id_registry< T, Meta >, 146
lava::image, 152
lava::imgui, 163, 164
lava::instance, 172
lava::mesh_template< T >, 188

lava::pipeline, 210
lava::pipeline::shader_stage, 282
lava::pipeline_layout, 217
lava::platform, 219
lava::render_pass, 249
lava::render_pipeline, 259
lava::render_target, 270
lava::renderer, 275
lava::swapchain, 303
lava::texture, 311
lava::window, 327
create_allocator
memory.hpp, 436
create_default_device_param
lava::physical_device, 203
create_default_texture
load_texture.hpp, 372
create_device
lava::platform, 220
create_folder
lava::file_system, 113
create_image
image.hpp, 582
create_mapped
lava::buffer, 42
create_mesh
lava::producer, 226
mesh.hpp, 586
create_mesh_data
mesh.hpp, 587
create_pipeline_color_blend_attachment
render_pipeline.hpp, 463
create_pipeline_shader_stage
pipeline.hpp, 456
create_shader_module
device.hpp, 423
create_surface
lava::window, 328
window.hpp, 563
create_target
render_target.hpp, 556
create_target_v_sync
render_target.hpp, 557
create_texture
lava::producer, 226

data
lava::data, 67
data.hpp
align, 470, 472
align_up, 472
alloc_data, 473
as_ptr, 473
free_data, 473
human_readable, 474
next_pow_2, 474
realloc_data, 474
debug_utils.hpp
begin_label, 386
end_label, 387
insert_label, 387
set_acceleration_structure_name, 388
set_acceleration_structure_nv_name, 388
set_acceleration_structure_nv_tag, 388
set_acceleration_structure_tag, 388
set_buffer_name, 389
set_buffer_tag, 389
set_buffer_view_name, 389
set_buffer_view_tag, 389
set_command_buffer_name, 390
set_command_buffer_tag, 390
set_command_pool_name, 390
set_command_pool_tag, 390
set_debug_report_callback_name, 391
set_debug_report_callback_tag, 391
set_debug_utils_messenger_name, 391
set_debug_utils_messenger_tag, 391
set_deferred_operation_name, 392
set_deferred_operation_tag, 392
set_descriptor_pool_name, 392
set_descriptor_pool_tag, 392
set_descriptor_set_layout_name, 393
set_descriptor_set_layout_tag, 393
set_descriptor_set_name, 393
set_descriptor_set_tag, 393
set_descriptor_update_template_name, 394
set_descriptor_update_template_tag, 394
set_device_memory_name, 394
set_device_memory_tag, 394
set_device_name, 395
set_device_tag, 395
set_display_mode_name, 395
set_display_mode_tag, 395
set_display_name, 396
set_display_tag, 396
set_event_name, 396
set_event_tag, 396
set_fence_name, 397
set_fence_tag, 397
set_framebuffer_name, 397
set_framebuffer_tag, 397
set_image_name, 398
set_image_tag, 398
set_image_view_name, 398
set_image_view_tag, 398
set_indirect_commands_layout_name, 399
set_indirect_commands_layout_tag, 399
set_instance_name, 399
set_instance_tag, 399
set_name, 400
set_object_name, 400
set_object_tag, 400
set_performance_configuration_name, 401
set_performance_configuration_tag, 401
set_physical_device_name, 401
set_physical_device_tag, 401
set_pipeline_cache_name, 402

set_pipeline_cache_tag, 402
set_pipeline_layout_name, 402
set_pipeline_layout_tag, 402
set_pipeline_name, 403
set_pipeline_tag, 403
set_private_data_slot_name, 403
set_private_data_slot_tag, 403
set_query_pool_name, 404
set_query_pool_tag, 404
set_queue_name, 404
set_queue_tag, 404
set_render_pass_name, 405
set_render_pass_tag, 405
set_sampler_name, 405
set_sampler_tag, 405
set_sampler_ycbcr_conversion_name, 406
set_sampler_ycbcr_conversion_tag, 406
set_semaphore_name, 406
set_semaphore_tag, 406
set_shader_module_name, 407
set_shader_module_tag, 407
set_surface_name, 407
set_surface_tag, 407
set_swapchain_name, 408
set_swapchain_tag, 408
set_tag, 408
set_validation_cache_name, 409
set_validation_cache_tag, 409
decorated
lava::window, 328
default_color
math.hpp, 631
default_queue_flags
queue.hpp, 444
destroy
lava::command, 56
lava::image, 153
device.hpp
create_shader_module, 423
one_time_submit, 423
one_time_submit_pool, 424
draw
lava::mesh_template< T >, 189
draw_about
lava::app, 20
driver.hpp
LAVA_STAGE, 541

elapsed
lava::timer, 316
empty
lava::mesh_template< T >, 189
lava::property, 232
end
lava::data, 67
misc.hpp, 485
end_frame
lava::renderer, 276
end_label
debug_utils.hpp, 387
enqueue
lava::thread_pool, 315
ENUM_FLAG_OPERATORS
types.hpp, 504
enumerate_extension_properties
instance.hpp, 432
enumerate_files
lava::file_system, 113
enumerate_layer_properties
instance.hpp, 434
exists
lava::file_system, 114
lava::id_registry< T, Meta >, 146
lava::property, 232
misc.hpp, 485
exists_subpass
lava::render_pass, 249
extension
file_utils.hpp, 528, 529

failed
base.hpp, 379
file
lava::file, 105
file.hpp
file_error, 524
file_data
lava::file_data, 110
file_delete
lava::file_delete, 111
file_error
file.hpp, 524
file_utils.hpp
extension, 528, 529
get_filename_from, 529
load_file_data, 529
read_file, 530
remove_existing_path, 530
write_file, 530
find
lava::hex_grid, 138
find_memory_type
memory.hpp, 437
find_memory_type_with_properties
memory.hpp, 437
floating
lava::window, 328
flush
lava::buffer, 42
focused
lava::window, 328
format.hpp
format_align_dim, 573
format_aspect_mask, 573
format_bgr, 573
format_block_dim, 574
format_block_size, 574
format_depth, 574

format_depth_stencil, 575
format_num_blocks, 575
format_srgb, 575
format_stencil, 576
get_supported_depth_format, 576
get_supported_format, 576
get_surface_format, 577
image_memory_barrier, 577
insert_image_memory_barrier, 578
set_image_layout, 578, 579
support_blit, 579
support_vertex_buffer_format, 580
format_align_dim
format.hpp, 573
format_aspect_mask
format.hpp, 573
format_bgr
format.hpp, 573
format_block_dim
format.hpp, 574
format_block_size
format.hpp, 574
format_depth
format.hpp, 574
format_depth_stencil
format.hpp, 575
format_num_blocks
format.hpp, 575
format_srgb
format.hpp, 575
format_stencil
format.hpp, 576
format_string
lava::tooltip_list, 318
formats
lava::surface_format_request, 300
frame
lava::frame, 122, 123
lava::renderer, 276
frame.hpp
handle_events, 535, 536
now, 536
frame_env
lava::frame_env, 128
free
lava::descriptor, 74
free_data
data.hpp, 473
free_set
lava::descriptor, 75
free_sets
lava::descriptor, 75
fullscreen
lava::window, 328
func
lava::input_callback, 170

gamepad
lava::gamepad, 129
gamepad.hpp
gamepads, 543
gamepads
gamepad.hpp, 543
get
lava::allocator, 16
lava::buffer, 42
lava::descriptor, 75
lava::descriptor::binding, 31
lava::descriptor::pool, 222
lava::device, 80
lava::global_logger, 133
lava::id_registry< T, Meta >, 147
lava::image, 153
lava::image_loader, 159
lava::instance, 173
lava::json_file, 178
lava::physical_device, 203
lava::pipeline, 211
lava::pipeline_layout, 217
lava::pseudo_random_generator, 237
lava::random_generator, 241
lava::render_pass, 250
lava::swapchain, 303
lava::window, 329
get_address
lava::buffer, 43
get_all
lava::id_registry< T, Meta >, 147
lava::property, 233
get_all_meta
lava::id_registry< T, Meta >, 147
get_allocation
lava::buffer, 43
lava::image, 153
get_allocation_info
lava::buffer, 43
get_allocator
lava::device, 80
get_app
lava::file_system, 114
get_aspect_ratio
lava::window, 329
get_backbuffer
lava::render_target, 270
get_backbuffer_count
lava::swapchain, 303
get_backbuffer_image
lava::render_target, 271
get_backbuffers
lava::render_target, 271
lava::swapchain, 303
get_base_dir
lava::file_system, 114
get_binding_count
lava::descriptor, 76
get_bindings
lava::descriptor, 76

get_buffers
lava::block, 36
get_channels
lava::image_loader, 160
get_clear_color
lava::render_pass, 250
get_clear_values
lava::render_pass, 250
get_cmd_line
lava::frame, 124
get_cmd_order
lava::block, 36
get_color_space
lava::swapchain, 304
get_command_buffer
lava::block, 36, 37
get_commands
lava::block, 37
get_compute_queue
lava::device, 80
get_compute_queues
lava::device, 81
get_config
lava::app_config, 23
lava::configurable, 61
lava::property, 233
get_create_info
lava::pipeline::shader_stage, 282
get_current_frame
lava::block, 37
get_current_time
time.hpp, 495
get_current_timestamp
time.hpp, 495
get_current_timestamp_ms
time.hpp, 496
get_data
lava::mesh_template< T >, 189
get_debug_config
lava::instance, 173
get_dependency
lava::subpass_dependency, 296
get_depth
lava::image, 153
get_depth_stencil
lava::forward_shading, 120
get_description
lava::attachment, 25
lava::subpass, 290
get_descriptor_info
lava::buffer, 43
lava::camera, 49
lava::texture, 311
get_descriptors
lava::pipeline_layout, 217
get_device
lava::block, 37
lava::buffer, 43
lava::descriptor, 76
lava::descriptor::pool, 222
lava::image, 153
lava::pipeline, 211
lava::pipeline_layout, 217
lava::render_pass, 250
lava::render_target, 271
lava::renderer, 276
lava::swapchain, 304
get_device_memory
lava::buffer, 44
get_device_name
lava::physical_device, 203
get_device_type_string
lava::physical_device, 204
get_devices
lava::platform, 220
get_dimensions
lava::image_loader, 160
get_driver_version
lava::physical_device, 204
get_end_point
lava::rect, 244
get_env
lava::frame, 124
get_ext
lava::file_system, 114
get_extension_properties
lava::physical_device, 204
get_features
lava::device, 81
lava::physical_device, 204
get_filename
lava::property, 233
get_filename_from
file_utils.hpp, 529
get_first_physical_device
lava::instance, 173
get_format
lava::image, 154
lava::render_target, 272
lava::swapchain, 304
lava::texture, 311
get_frame
lava::renderer, 277
get_frame_count
lava::block, 38
lava::render_target, 272
get_frame_counter
lava::app, 21
get_framebuffer_size
lava::window, 329
get_full_base_dir
lava::file_system, 115
get_graphics_queue
lava::device, 81
get_graphics_queues
lava::device, 82

get_id
lava::entity, 104
lava::gamepad, 129
get_image
lava::render_target, 272
lava::texture, 312
get_index_buffer
lava::mesh_template< T >, 189
get_indices
lava::mesh_template< T >, 190
get_indices_count
lava::mesh_template< T >, 190
get_info
lava::image, 154
lava::instance, 173
get_ini_file
lava::imgui, 165
get_input_callback
lava::imgui, 165
get_instance_version
instance.hpp, 434
get_layout
lava::pipeline, 211
get_line_width
lava::render_pipeline, 260
get_list
lava::id_listeners< T >, 144
lava::tooltip_list, 318
get_mapped_data
lava::buffer, 44
get_max
lava::descriptor::pool, 223
get_memory_properties
lava::physical_device, 205
get_mesh
lava::producer, 227
get_meta
lava::id_registry< T, Meta >, 148
get_monitor
lava::window, 330
get_mouse_position
lava::input, 168
lava::window, 330
get_name
lava::frame, 124
lava::gamepad, 129
get_org
lava::file_system, 115
get_origin
lava::rect, 244
get_pad_id
lava::gamepad, 130
get_pass
lava::forward_shading, 120
get_path
lava::file, 106
get_physical_device
lava::device, 82
get_physical_devices
lava::instance, 174
get_pipeline
lava::imgui, 165
get_position
lava::window, 330
get_pref_dir
lava::file_system, 115
get_projection
lava::camera, 49
get_properties
lava::device, 82
lava::physical_device, 205
get_push_constant_ranges
lava::pipeline_layout, 218
get_queue_family
lava::physical_device, 205
get_queue_family_properties
lava::physical_device, 206
get_queues
lava::device, 82
get_real_dir
lava::file_system, 115
get_render_pass
lava::render_pipeline, 260
get_res_dir
lava::file_system, 116
get_running_time
lava::frame, 125
get_running_time_sec
lava::frame, 125
get_save_name
lava::window, 331
get_scancode
input.hpp, 549
get_scissor
lava::render_pipeline, 260
get_shader
lava::producer, 227
get_shader_stage
lava::compute_pipeline, 59
get_shader_stages
lava::render_pipeline, 260
get_size
lava::buffer, 44
lava::file, 106
lava::image, 154
lava::rect, 245
lava::render_target, 272
lava::swapchain, 304
lava::texture, 312
lava::window, 331
get_sizes
lava::descriptor::pool, 223
get_sizing
lava::render_pipeline, 261
get_stages
lava::driver, 101

get_state
lava::window, 332
get_subpass
lava::render_pass, 250
lava::render_pipeline, 261
get_subpass_count
lava::render_pass, 251
get_subpasses
lava::render_pass, 251
get_subresource_range
lava::image, 154
get_supported_depth_format
format.hpp, 576
get_supported_format
format.hpp, 576
get_surface_format
format.hpp, 577
get_swapchain
lava::render_target, 273
get_target_callback
lava::render_pass, 251
get_texture
lava::producer, 227
get_title
lava::window, 332
get_transfer_queue
lava::device, 82
get_transfer_queues
lava::device, 83
get_type
lava::file, 106
lava::texture, 312
get_version
lava::file_system, 116
get_vertex_buffer
lava::mesh_template< T >, 190
get_vertices
lava::mesh_template< T >, 191
get_vertices_count
lava::mesh_template< T >, 191
get_view
lava::camera, 50
lava::image, 155
get_view_info
lava::image, 155
get_view_projection
lava::camera, 50
get_viewport
lava::render_pipeline, 261
get_vk_pass
lava::forward_shading, 120
get_vk_physical_device
lava::device, 83
get_window
window.hpp, 564
grab_image
image.hpp, 583
graphics_queue
lava::device, 83
graphics_queues
lava::device, 83

handle
lava::camera, 50, 51
handle_events
frame.hpp, 535, 536
handle_resize
lava::window, 332
hash256
math.hpp, 630
hash_combine
types.hpp, 508
hash_val
types.hpp, 508, 509
hex.hpp
hex_calculate_inner_radius, 604
hex_cardinal_directions, 615
hex_cell_from_pair, 605
hex_corner_offset, 605
hex_diagonal, 605
hex_diagonal_neighbor, 606
hex_diagonals, 615
hex_direction, 606
hex_directions, 616
hex_distance, 606
hex_get, 607
hex_get_corner, 607
hex_get_s, 608
hex_is_valid, 608
hex_layout_flat, 616
hex_layout_point_y, 616
hex_length, 608
hex_lerp, 609
hex_line, 609
hex_neighbor, 609
hex_opposite, 610
hex_pixel_to_cell, 610
hex_polygon_corners, 611
hex_q_doubled_from_cube, 611
hex_q_doubled_to_cube, 611
hex_q_offset_from_cube, 612
hex_q_offset_to_cube, 612
hex_r_doubled_from_cube, 612
hex_r_doubled_to_cube, 613
hex_r_offset_from_cube, 613
hex_r_offset_to_cube, 614
hex_round, 614
hex_to_pixel, 614
to_string, 615
hex_calculate_inner_radius
hex.hpp, 604
hex_cardinal_directions
hex.hpp, 615
hex_cell_from_pair
hex.hpp, 605
hex_corner_offset
hex.hpp, 605

hex_diagonal
hex.hpp, 605
hex_diagonal_neighbor
hex.hpp, 606
hex_diagonals
hex.hpp, 615
hex_direction
hex.hpp, 606
hex_directions
hex.hpp, 616
hex_distance
hex.hpp, 606
hex_get
hex.hpp, 607
hex_get_corner
hex.hpp, 607
hex_get_s
hex.hpp, 608
hex_grid
lava::hex_grid, 137
hex_is_valid
hex.hpp, 608
hex_layout_flat
hex.hpp, 616
hex_layout_point_y
hex.hpp, 616
hex_length
hex.hpp, 608
hex_lerp
hex.hpp, 609
hex_line
hex.hpp, 609
hex_neighbor
hex.hpp, 609
hex_opposite
hex.hpp, 610
hex_pixel_to_cell
hex.hpp, 610
hex_polygon_corners
hex.hpp, 611
hex_q_doubled_from_cube
hex.hpp, 611
hex_q_doubled_to_cube
hex.hpp, 611
hex_q_offset_from_cube
hex.hpp, 612
hex_q_offset_to_cube
hex.hpp, 612
hex_r_doubled_from_cube
hex.hpp, 612
hex_r_doubled_to_cube
hex.hpp, 613
hex_r_offset_from_cube
hex.hpp, 613
hex_r_offset_to_cube
hex.hpp, 614
hex_round
hex.hpp, 614
hex_to_pixel
hex.hpp, 614
hovered
lava::window, 332
human_readable
data.hpp, 474

i16
types.hpp, 505
i32
types.hpp, 505
i64
types.hpp, 506
i8
types.hpp, 506
iconified
lava::window, 333
id.hpp
add_id_map, 478
remove_id_map, 479
to_id, 479
image
lava::image, 152
image.hpp
create_image, 582
grab_image, 583
image_loader
lava::image_loader, 159
image_memory_barrier
format.hpp, 577
imgui
lava::imgui, 162
imgui.hpp
imgui_left_spacing, 365
setup_imgui_font, 365
setup_imgui_font_icons, 366
imgui_left_spacing
imgui.hpp, 365
initialize
lava::file_system, 116
lava::physical_device, 206
input.hpp
check_mod, 548
get_scancode, 549
to_string, 549
insert_image_memory_barrier
format.hpp, 578
insert_label
debug_utils.hpp, 387
instance
lava::driver, 101
lava::ids, 149
lava::memory, 184
instance.hpp
check, 432
enumerate_extension_properties, 432
enumerate_layer_properties, 434
get_instance_version, 434
int_version_string

log.hpp, 623
internal_version_string
log.hpp, 624

json_file
lava::json_file, 177

lava::allocator, 15
allocator, 16
create, 16
get, 16
make, 17
valid, 17
lava::app, 17
app, 19, 20
block_cmd, 20
draw_about, 20
get_frame_counter, 21
parse_config, 21
screenshot, 21
setup, 21
v_sync, 21
lava::app_config, 22
get_config, 23
set_config, 23
lava::attachment, 24
attachment, 25
get_description, 25
make, 25
set_final_layout, 26
set_format, 26
set_initial_layout, 26
set_layouts, 27
set_load_op, 27
set_op, 27
set_samples, 27
set_stencil_load_op, 28
set_stencil_op, 28
set_stencil_store_op, 28
set_store_op, 29
lava::benchmark_data, 29
lava::block, 33
activated, 34
add_cmd, 35
add_command, 35
create, 35
get_buffers, 36
get_cmd_order, 36
get_command_buffer, 36, 37
get_commands, 37
get_current_frame, 37
get_device, 37
get_frame_count, 38
make, 38
process, 38
remove_cmd, 39
remove_command, 39
set_active, 39
lava::buffer, 40
create, 41
create_mapped, 42
flush, 42
get, 42
get_address, 43
get_allocation, 43
get_allocation_info, 43
get_descriptor_info, 43
get_device, 43
get_device_memory, 44
get_mapped_data, 44
get_size, 44
make, 44
valid, 45
lava::camera, 47
activated, 49
create, 49
get_descriptor_info, 49
get_projection, 49
get_view, 50
get_view_projection, 50
handle, 50, 51
moving, 51
set_active, 51
set_movement_keys, 52
update_view, 52
valid, 53
lava::cdata, 53
cdata, 54
lava::command, 55
create, 56
destroy, 56
make, 56
lava::compute_pipeline, 57
bind, 58
copy_from, 58
copy_to, 58
get_shader_stage, 59
make, 59
set, 59
set_shader_stage, 60
lava::configurable, 61
get_config, 61
set_config, 61
lava::data, 66
allocate, 67
data, 67
end, 67
set, 68
lava::data_provider, 68
lava::descriptor, 70
add, 71
add_binding, 71
allocate, 73
allocate_set, 73
allocate_sets, 73
create, 74
free, 74

free_set, 75
free_sets, 75
get, 75
get_binding_count, 76
get_bindings, 76
get_device, 76
make, 76
lava::descriptor::binding, 30
get, 31
make, 31
set, 31
set_count, 31
set_samplers, 32
set_stage_flags, 32
set_type, 32
lava::descriptor::pool, 221
create, 222
get, 222
get_device, 222
get_max, 223
get_sizes, 223
make, 223
lava::device, 77
alloc, 79
call, 79
compute_queue, 79
compute_queues, 79
create, 80
get, 80
get_allocator, 80
get_compute_queue, 80
get_compute_queues, 81
get_features, 81
get_graphics_queue, 81
get_graphics_queues, 82
get_physical_device, 82
get_properties, 82
get_queues, 82
get_transfer_queue, 82
get_transfer_queues, 83
get_vk_physical_device, 83
graphics_queue, 83
graphics_queues, 83
make, 84
queues, 84
set_allocator, 84
surface_supported, 84
transfer_queue, 85
transfer_queues, 85
wait_for_idle, 85
lava::device::create_param, 63
add_dedicated_queues, 64
add_queue, 64
add_queues, 64
verify_queues, 65
lava::device_table, 86
vkAcquireNextImageKHR, 87
vkAllocateCommandBuffers, 88
vkCreateCommandPool, 88, 89
vkCreateFence, 89
vkCreateImageView, 89, 90
vkCreateSampler, 90
vkCreateSemaphore, 90, 91
vkCreateShaderModule, 91
vkCreateSwapchainKHR, 91, 92
vkDestroyCommandPool, 92
vkDestroyFence, 92
vkDestroyImageView, 92
vkDestroySampler, 93
vkDestroySemaphore, 93
vkDestroySwapchainKHR, 93
vkFreeCommandBuffers, 93
vkGetSwapchainImagesKHR, 94
vkQueuePresentKHR, 94
vkQueueSubmit, 94
vkResetFences, 94
vkUpdateDescriptorSets, 95, 96
vkWaitForFences, 96
lava::dispatcher, 97
add_message, 98
setup, 98
update, 98
lava::driver, 99
add_stage, 100
get_stages, 101
instance, 101
run, 101
lava::driver::result, 277
lava::engine, 102
setup, 102
lava::entity, 103
get_id, 104
lava::file, 104
file, 105
get_path, 106
get_size, 106
get_type, 106
open, 106
opened, 107
read, 107
seek, 108
tell, 108
writable, 108
write, 108
lava::file_data, 109
file_data, 110
lava::file_delete, 110
file_delete, 111
lava::file_format, 111
lava::file_system, 112
create_folder, 113
enumerate_files, 113
exists, 114
get_app, 114
get_base_dir, 114
get_ext, 114

get_full_base_dir, 115
get_org, 115
get_pref_dir, 115
get_real_dir, 115
get_res_dir, 116
get_version, 116
initialize, 116
mount, 117
mount_base, 117
mount_res, 117
ready, 118
lava::forward_shading, 119
create, 119
get_depth_stencil, 120
get_pass, 120
get_vk_pass, 120
lava::frame, 121
add_run, 123
add_run_end, 123
add_run_once, 124
frame, 122, 123
get_cmd_line, 124
get_env, 124
get_name, 124
get_running_time, 125
get_running_time_sec, 125
ready, 125
remove, 125
run, 126
set_wait_for_events, 126
shut_down, 126
waiting_for_events, 126
lava::frame_env, 127
frame_env, 128
lava::gamepad, 128
gamepad, 129
get_id, 129
get_name, 129
get_pad_id, 130
pressed, 130
ready, 130
update, 130
value, 131
lava::gamepad_manager, 131
add, 132
remove, 132
singleton, 132
lava::global_logger, 133
get, 133
set, 133
singleton, 134
lava::hex_cell, 134
add, 135
scale, 135
substract, 135
to_pair, 136
lava::hex_fractional_cell, 136
lava::hex_grid, 137
find, 138
hex_grid, 137
to_pixel, 138
update, 138
lava::hex_layout, 139
lava::hex_offset_coord, 139
lava::hex_orientation, 140
lava::hex_point, 140
lava::id, 142
to_string, 142
valid, 143
lava::id_listeners< T >, 143
add, 144
get_list, 144
remove, 144
lava::id_registry< T, Meta >, 145
add, 146
create, 146
exists, 146
get, 147
get_all, 147
get_all_meta, 147
get_meta, 148
remove, 148
update, 148
lava::ids, 149
instance, 149
next, 150
lava::image, 150
create, 152
destroy, 153
get, 153
get_allocation, 153
get_depth, 153
get_device, 153
get_format, 154
get_info, 154
get_size, 154
get_subresource_range, 154
get_view, 155
get_view_info, 155
image, 152
make, 155
set_aspect_mask, 156
set_component, 156
set_flags, 156
set_layer_count, 156
set_layout, 157
set_level_count, 157
set_tiling, 157
set_usage, 158
set_view_type, 158
lava::image_loader, 158
get, 159
get_channels, 160
get_dimensions, 160
image_loader, 159
ready, 160

size, 160
lava::imgui, 161
activated, 163
capture_keyboard, 163
capture_mouse, 163
create, 163, 164
get_ini_file, 165
get_input_callback, 165
get_pipeline, 165
imgui, 162
ready, 165
set_active, 165
set_ini_file, 166
setup, 166
upload_fonts, 166
lava::imgui::config, 60
lava::imgui::font, 118
lava::imgui::icon_font, 141
lava::input, 167
add, 168
get_mouse_position, 168
remove, 168
set_mouse_position, 168
lava::input_callback, 169
func, 170
lava::input_events< T >, 170
add, 171
lava::instance, 171
create, 172
get, 173
get_debug_config, 173
get_first_physical_device, 173
get_info, 173
get_physical_devices, 174
singleton, 174
lava::instance::create_param, 65
lava::instance::debug_config, 69
lava::instance_info, 174
lava::interface, 175
lava::internal_version, 176
lava::json_file, 176
add, 177
get, 178
json_file, 177
load, 178
remove, 178
save, 179
set, 179
lava::json_file::callback, 45
lava::key_event, 179
active, 180
pressed, 180, 181
released, 181
repeated, 181
lava::log_config, 183
lava::memory, 183
alloc, 184
instance, 184
set_callbacks, 184
set_use_custom_cpu_callbacks, 185
lava::mesh_meta, 185
lava::mesh_template< T >, 186
add_data, 187
bind, 188
bind_draw, 188
create, 188
draw, 189
empty, 189
get_data, 189
get_index_buffer, 189
get_indices, 190
get_indices_count, 190
get_vertex_buffer, 190
get_vertices, 191
get_vertices_count, 191
make, 191
reload, 192
set_data, 192
lava::mesh_template_data< T >, 192
move, 193
scale, 194
scale_vector, 194
lava::mouse_active_event, 195
lava::mouse_button_event, 196
pressed, 197
released, 197
lava::mouse_move_event, 197
lava::mouse_position, 198
lava::no_copy_no_move, 199
lava::pair_hash, 200
operator(), 200
lava::path_drop_event, 201
lava::physical_device, 201
create_default_device_param, 203
get, 203
get_device_name, 203
get_device_type_string, 204
get_driver_version, 204
get_extension_properties, 204
get_features, 204
get_memory_properties, 205
get_properties, 205
get_queue_family, 205
get_queue_family_properties, 206
initialize, 206
make, 206
physical_device, 203
supported, 206
surface_supported, 207
swapchain_supported, 207
lava::pipeline, 208
activated, 210
auto_bind, 210
bind, 210
create, 210
get, 211

get_device, 211
get_layout, 211
pipeline, 209
ready, 211
set_active, 212
set_auto_bind, 213
set_layout, 213
setup, 213
lava::pipeline::shader_stage, 281
add_specialization_entry, 281
create, 282
get_create_info, 282
make, 282
set_stage, 283
lava::pipeline_layout, 214
add, 215
add_descriptor, 215
add_push_constant_range, 216
bind, 216
bind_descriptor_set, 216
create, 217
get, 217
get_descriptors, 217
get_device, 217
get_push_constant_ranges, 218
make, 218
lava::platform, 218
create, 219
create_device, 220
get_devices, 220
remove, 220
lava::producer, 224
add_mesh, 225
add_texture, 225
compile_shader, 225
create_mesh, 226
create_texture, 226
get_mesh, 227
get_shader, 227
get_texture, 227
reload_shader, 228
lava::property, 229
add, 230
check, 232
empty, 232
exists, 232
get_all, 233
get_config, 233
get_filename, 233
load, 233
load_all, 234
operator(), 234
parse, 234
set_config, 235
set_filename, 235
unload, 235
lava::property::prop, 228
prop, 229
lava::pseudo_random_generator, 236
get, 237
pseudo_random_generator, 236
set_seed, 237
lava::queue, 237
operator<, 238
valid, 238
lava::queue_family_info, 239
add, 239
count, 240
lava::queue_info, 240
lava::random_generator, 241
get, 241
lava::rect, 242
contains, 244
get_end_point, 244
get_origin, 244
get_size, 245
move, 245
rect, 243
set_size, 245
lava::render_pass, 246
add, 248
add_front, 249
create, 249
exists_subpass, 249
get, 250
get_clear_color, 250
get_clear_values, 250
get_device, 250
get_subpass, 250
get_subpass_count, 251
get_subpasses, 251
get_target_callback, 251
make, 251
process, 252
remove, 252
render_pass, 247
set_clear_color, 252
set_clear_values, 253
lava::render_pipeline, 253
add, 256
add_color_blend_attachment, 257
add_dynamic_state, 257
add_shader, 257
add_shader_stage, 258
auto_line_width, 258
auto_sizing, 258
bind, 258
copy_from, 259
copy_to, 259
create, 259
get_line_width, 260
get_render_pass, 260
get_scissor, 260
get_shader_stages, 260
get_sizing, 261
get_subpass, 261

get_viewport, 261
make, 261
render_pipeline, 256
set, 262
set_auto_line_width, 262
set_auto_size, 262
set_depth_compare_op, 262
set_depth_test_and_write, 263
set_dynamic_states, 263
set_input_topology, 263
set_line_width, 264
set_rasterization_cull_mode, 264
set_rasterization_front_face, 264
set_rasterization_polygon_mode, 265
set_render_pass, 265
set_scissor, 265
set_sizing, 266
set_subpass, 266
set_vertex_input_attribute, 266
set_vertex_input_attributes, 266
set_vertex_input_binding, 267
set_vertex_input_bindings, 267
set_viewport, 267
set_viewport_and_scissor, 268
lava::render_pipeline::create_info, 62
lava::render_target, 268
add_callback, 270
create, 270
get_backbuffer, 270
get_backbuffer_image, 271
get_backbuffers, 271
get_device, 271
get_format, 272
get_frame_count, 272
get_image, 272
get_size, 272
get_swapchain, 273
make, 273
reload_request, 273
remove_callback, 273
resize, 274
lava::renderer, 274
begin_frame, 275
create, 275
end_frame, 276
frame, 276
get_device, 276
get_frame, 277
lava::reversion_wrapper< T >, 277
lava::run_time, 278
lava::scoped_label< T >, 278
scoped_label, 279
lava::scroll_event, 279
lava::scroll_offset, 280
lava::stage, 283
stage, 284
lava::staging, 284
add, 285
busy, 285
stage, 285
lava::subpass, 287
activated, 288
add, 289
add_front, 289
add_preserve_attachment, 289
get_description, 290
make, 290
process, 290
remove, 290
set, 291
set_active, 291
set_color_attachment, 291, 292
set_color_attachments, 292
set_depth_stencil_attachment, 292
set_input_attachment, 293
set_input_attachments, 293
set_preserve_attachments, 294
set_resolve_attachment, 294
set_resolve_attachments, 294
lava::subpass_dependency, 295
get_dependency, 296
make, 296
set_access_mask, 296
set_dependency_flags, 297
set_dst_access_mask, 297
set_dst_stage_mask, 297
set_dst_subpass, 298
set_src_access_mask, 298
set_src_stage_mask, 298
set_src_subpass, 298
set_stage_mask, 299
set_subpass, 299
lava::surface_format_request, 299
formats, 300
lava::swapchain, 300
add_callback, 301
create, 303
get, 303
get_backbuffer_count, 303
get_backbuffers, 303
get_color_space, 304
get_device, 304
get_format, 304
get_size, 304
reload_request, 305
remove_callback, 305
resize, 305
surface_supported, 306
v_sync, 306
lava::swapchain::callback, 46
lava::target_callback, 306
lava::telegram, 307
operator<, 308
operator==, 309
telegram, 308
lava::texture, 309

create, 311
get_descriptor_info, 311
get_format, 311
get_image, 312
get_size, 312
get_type, 312
make, 312
stage, 313
upload, 314
lava::texture::layer, 182
lava::texture::mip_level, 194
lava::thread_pool, 314
enqueue, 315
setup, 315
lava::timer, 315
elapsed, 316
lava::tooltip, 316
tooltip, 317
lava::tooltip_list, 317
add, 318
format_string, 318
get_list, 318
set, 319
lava::unique_data, 319
unique_data, 320
lava::version, 321
lava::vertex, 321
operator==, 322
lava::vk_result, 322
operator bool, 323
lava::window, 323
assign, 327
close_request, 327
create, 327
create_surface, 328
decorated, 328
floating, 328
focused, 328
fullscreen, 328
get, 329
get_aspect_ratio, 329
get_framebuffer_size, 329
get_monitor, 330
get_mouse_position, 330
get_position, 330
get_save_name, 331
get_size, 331
get_state, 332
get_title, 332
handle_resize, 332
hovered, 332
iconified, 333
maximized, 333
resizable, 333
resize_request, 333
save_title, 334
set_decorated, 334
set_floating, 334
set_fullscreen, 335
set_icon, 335
set_mouse_position, 335
set_position, 335
set_resizable, 336
set_save_name, 336
set_size, 336
set_title, 337
show_save_title, 337
switch_mode, 337
switch_mode_request, 337
visible, 338
window, 326
lava::window::state, 286
LAVA_STAGE
driver.hpp, 541
liblava/app.hpp, 339
liblava/app/app.hpp, 340, 341
liblava/app/benchmark.hpp, 342, 344
liblava/app/camera.hpp, 345, 346
liblava/app/config.hpp, 347, 348
liblava/app/forward_shading.hpp, 349, 350
liblava/app/icon.hpp, 350, 351
liblava/app/imgui.hpp, 364, 366
liblava/asset.hpp, 369
liblava/asset/image_loader.hpp, 369, 370
liblava/asset/load_mesh.hpp, 371, 372
liblava/asset/load_texture.hpp, 372, 374
liblava/asset/write_image.hpp, 374, 375
liblava/base.hpp, 375, 376
liblava/base/base.hpp, 376, 381
liblava/base/debug_utils.hpp, 383, 409
liblava/base/device.hpp, 422, 424
liblava/base/device_table.hpp, 427
liblava/base/instance.hpp, 431, 434
liblava/base/memory.hpp, 436, 438
liblava/base/physical_device.hpp, 439
liblava/base/platform.hpp, 440, 441
liblava/base/queue.hpp, 441, 445
liblava/block.hpp, 448
liblava/block/attachment.hpp, 446, 447
liblava/block/block.hpp, 449
liblava/block/compute_pipeline.hpp, 451, 452
liblava/block/descriptor.hpp, 452, 453
liblava/block/pipeline.hpp, 455, 456
liblava/block/pipeline_layout.hpp, 458, 459
liblava/block/render_pass.hpp, 460
liblava/block/render_pipeline.hpp, 462, 463
liblava/block/subpass.hpp, 466
liblava/core.hpp, 468, 469
liblava/core/data.hpp, 469, 475
liblava/core/id.hpp, 477, 480
liblava/core/misc.hpp, 482, 492
liblava/core/time.hpp, 494, 499
liblava/core/types.hpp, 501, 512
liblava/core/version.hpp, 515, 517
liblava/engine.hpp, 517, 518
liblava/engine/engine.hpp, 518, 519

liblava/engine/producer.hpp, 519, 520
liblava/engine/property.hpp, 521, 522
liblava/file.hpp, 523
liblava/file/file.hpp, 523, 525
liblava/file/file_system.hpp, 526
liblava/file/file_utils.hpp, 527, 531
liblava/file/json.hpp, 532
liblava/file/json_file.hpp, 532, 533
liblava/frame.hpp, 533, 534
liblava/frame/argh.hpp, 538, 539
liblava/frame/driver.hpp, 540, 541
liblava/frame/frame.hpp, 534, 536
liblava/frame/gamepad.hpp, 542, 544
liblava/frame/input.hpp, 545, 550
liblava/frame/render_target.hpp, 555, 557
liblava/frame/renderer.hpp, 559, 560
liblava/frame/swapchain.hpp, 560, 561
liblava/frame/window.hpp, 562, 564
liblava/lava.hpp, 567
liblava/resource.hpp, 568
liblava/resource/buffer.hpp, 568, 570
liblava/resource/format.hpp, 571, 580
liblava/resource/image.hpp, 582, 583
liblava/resource/mesh.hpp, 585, 589
liblava/resource/primitive.hpp, 596, 597
liblava/resource/texture.hpp, 598, 599
liblava/test.hpp, 600, 601
liblava/util.hpp, 601
liblava/util/hex.hpp, 601, 617
liblava/util/log.hpp, 622, 627
liblava/util/math.hpp, 628, 631
liblava/util/random.hpp, 632, 634
liblava/util/telegram.hpp, 635, 636
liblava/util/thread.hpp, 637, 638
load
    lava::json_file, 178
    lava::property, 233
load_all
    lava::property, 234
load_file_data
    file_utils.hpp, 529
load_mesh
    load_mesh.hpp, 371
load_mesh.hpp
    load_mesh, 371
load_texture
    load_texture.hpp, 373
load_texture.hpp
    create_default_texture, 372
    load_texture, 373
log
    log.hpp, 624
log.hpp
    int_version_string, 623
    internal_version_string, 624
    log, 624
    setup_log, 624
    teardown_log, 624
    to_string, 625, 626
    version_string, 626
log_command_line
    argh.hpp, 539
ltrim
    misc.hpp, 486
ltrim_copy
    misc.hpp, 486

make
    lava::allocator, 17
    lava::attachment, 25
    lava::block, 38
    lava::buffer, 44
    lava::command, 56
    lava::compute_pipeline, 59
    lava::descriptor, 76
    lava::descriptor::binding, 31
    lava::descriptor::pool, 223
    lava::device, 84
    lava::image, 155
    lava::mesh_template< T >, 191
    lava::physical_device, 206
    lava::pipeline::shader_stage, 282
    lava::pipeline_layout, 218
    lava::render_pass, 251
    lava::render_pipeline, 261
    lava::render_target, 273
    lava::subpass, 290
    lava::subpass_dependency, 296
    lava::texture, 312
make_primitive_indices_cube
    mesh.hpp, 587
make_primitive_normals_cube
    mesh.hpp, 588
make_primitive_positions_cube
    mesh.hpp, 588
make_primitive_uvs_cube
    mesh.hpp, 589
math.hpp
    ceil_div, 629
    default_color, 631
    hash256, 630
    perspective_matrix, 630
maximized
    lava::window, 333
memory.hpp
    create_allocator, 436
    find_memory_type, 437
    find_memory_type_with_properties, 437
mesh.hpp
    create_mesh, 586
    create_mesh_data, 587
    make_primitive_indices_cube, 587
    make_primitive_normals_cube, 588
    make_primitive_positions_cube, 588
    make_primitive_uvs_cube, 589
misc.hpp
    append, 484

begin, 484
contains, 484
end, 485
exists, 485
ltrim, 486
ltrim_copy, 486
remove, 486
remove_chars, 488
remove_chars_copy, 488
remove_chars_if_not, 488
remove_chars_if_not_copy, 489
remove_nondigit, 489
remove_nondigit_copy, 490
remove_punctuation_marks, 490
reverse, 490
rtrim, 491
rtrim_copy, 491
trim, 491
trim_copy, 492
mount
lava::file_system, 117
mount_base
lava::file_system, 117
mount_res
lava::file_system, 117
move
lava::mesh_template_data< T >, 193
lava::rect, 245
moving
lava::camera, 51

next
lava::ids, 150
next_pow_2
data.hpp, 474
now
frame.hpp, 536

one_time_submit
device.hpp, 423
one_time_submit_pool
device.hpp, 424
open
lava::file, 106
opened
lava::file, 107
operator bool
lava::vk_result, 323
operator<
lava::queue, 238
lava::telegram, 308
operator()
lava::pair_hash, 200
lava::property, 234
operator==
lava::telegram, 309
lava::vertex, 322

parse
benchmark.hpp, 344
lava::property, 234
parse_config
lava::app, 21
perspective_matrix
math.hpp, 630
physical_device
lava::physical_device, 203
pipeline
lava::pipeline, 209
pipeline.hpp
create_pipeline_shader_stage, 456
pressed
lava::gamepad, 130
lava::key_event, 180, 181
lava::mouse_button_event, 197
process
lava::block, 38
lava::render_pass, 252
lava::subpass, 290
prop
lava::property::prop, 229
pseudo_random_generator
lava::pseudo_random_generator, 236

queue.hpp
add_dedicated_queues, 443
add_queues, 443
default_queue_flags, 444
set_all_queues, 443
set_default_queues, 444
verify_queues, 444
queues
lava::device, 84

random
random.hpp, 633
random.hpp
random, 633
read
lava::file, 107
read_file
file_utils.hpp, 530
ready
lava::file_system, 118
lava::frame, 125
lava::gamepad, 130
lava::image_loader, 160
lava::imgui, 165
lava::pipeline, 211
realloc_data
data.hpp, 474
rect
lava::rect, 243
released
lava::key_event, 181
lava::mouse_button_event, 197
reload
lava::mesh_template< T >, 192

reload_request
lava::render_target, 273
lava::swapchain, 305
reload_shader
lava::producer, 228
remove
lava::frame, 125
lava::gamepad_manager, 132
lava::id_listeners< T >, 144
lava::id_registry< T, Meta >, 148
lava::input, 168
lava::json_file, 178
lava::platform, 220
lava::render_pass, 252
lava::subpass, 290
misc.hpp, 486
remove_callback
lava::render_target, 273
lava::swapchain, 305
remove_chars
misc.hpp, 488
remove_chars_copy
misc.hpp, 488
remove_chars_if_not
misc.hpp, 488
remove_chars_if_not_copy
misc.hpp, 489
remove_cmd
lava::block, 39
remove_command
lava::block, 39
remove_existing_path
file_utils.hpp, 530
remove_id_map
id.hpp, 479
remove_nondigit
misc.hpp, 489
remove_nondigit_copy
misc.hpp, 490
remove_punctuation_marks
misc.hpp, 490
render_pass
lava::render_pass, 247
render_pipeline
lava::render_pipeline, 256
render_pipeline.hpp
create_pipeline_color_blend_attachment, 463
render_target.hpp
create_target, 556
create_target_v_sync, 557
repeated
lava::key_event, 181
resizable
lava::window, 333
resize
lava::render_target, 274
lava::swapchain, 305
resize_request
lava::window, 333
reverse
misc.hpp, 490
rtrim
misc.hpp, 491
rtrim_copy
misc.hpp, 491
run
lava::driver, 101
lava::frame, 126

save
lava::json_file, 179
save_title
lava::window, 334
scale
lava::hex_cell, 135
lava::mesh_template_data< T >, 194
scale_vector
lava::mesh_template_data< T >, 194
scoped_label
lava::scoped_label< T >, 279
screenshot
lava::app, 21
seek
lava::file, 108
set
lava::compute_pipeline, 59
lava::data, 68
lava::descriptor::binding, 31
lava::global_logger, 133
lava::json_file, 179
lava::render_pipeline, 262
lava::subpass, 291
lava::tooltip_list, 319
set_acceleration_structure_name
debug_utils.hpp, 388
set_acceleration_structure_nv_name
debug_utils.hpp, 388
set_acceleration_structure_nv_tag
debug_utils.hpp, 388
set_acceleration_structure_tag
debug_utils.hpp, 388
set_access_mask
lava::subpass_dependency, 296
set_active
lava::block, 39
lava::camera, 51
lava::imgui, 165
lava::pipeline, 212
lava::subpass, 291
set_all_queues
queue.hpp, 443
set_allocator
lava::device, 84
set_aspect_mask
lava::image, 156
set_auto_bind
lava::pipeline, 213

set_auto_line_width
lava::render_pipeline, 262
set_auto_size
lava::render_pipeline, 262
set_buffer_name
debug_utils.hpp, 389
set_buffer_tag
debug_utils.hpp, 389
set_buffer_view_name
debug_utils.hpp, 389
set_buffer_view_tag
debug_utils.hpp, 389
set_callbacks
lava::memory, 184
set_clear_color
lava::render_pass, 252
set_clear_values
lava::render_pass, 253
set_color_attachment
lava::subpass, 291, 292
set_color_attachments
lava::subpass, 292
set_command_buffer_name
debug_utils.hpp, 390
set_command_buffer_tag
debug_utils.hpp, 390
set_command_pool_name
debug_utils.hpp, 390
set_command_pool_tag
debug_utils.hpp, 390
set_component
lava::image, 156
set_config
lava::app_config, 23
lava::configurable, 61
lava::property, 235
set_count
lava::descriptor::binding, 31
set_data
lava::mesh_template< T >, 192
set_debug_report_callback_name
debug_utils.hpp, 391
set_debug_report_callback_tag
debug_utils.hpp, 391
set_debug_utils_messenger_name
debug_utils.hpp, 391
set_debug_utils_messenger_tag
debug_utils.hpp, 391
set_decorated
lava::window, 334
set_default_queues
queue.hpp, 444
set_deferred_operation_name
debug_utils.hpp, 392
set_deferred_operation_tag
debug_utils.hpp, 392
set_dependency_flags
lava::subpass_dependency, 297
set_depth_compare_op
lava::render_pipeline, 262
set_depth_stencil_attachment
lava::subpass, 292
set_depth_test_and_write
lava::render_pipeline, 263
set_descriptor_pool_name
debug_utils.hpp, 392
set_descriptor_pool_tag
debug_utils.hpp, 392
set_descriptor_set_layout_name
debug_utils.hpp, 393
set_descriptor_set_layout_tag
debug_utils.hpp, 393
set_descriptor_set_name
debug_utils.hpp, 393
set_descriptor_set_tag
debug_utils.hpp, 393
set_descriptor_update_template_name
debug_utils.hpp, 394
set_descriptor_update_template_tag
debug_utils.hpp, 394
set_device_memory_name
debug_utils.hpp, 394
set_device_memory_tag
debug_utils.hpp, 394
set_device_name
debug_utils.hpp, 395
set_device_tag
debug_utils.hpp, 395
set_display_mode_name
debug_utils.hpp, 395
set_display_mode_tag
debug_utils.hpp, 395
set_display_name
debug_utils.hpp, 396
set_display_tag
debug_utils.hpp, 396
set_dst_access_mask
lava::subpass_dependency, 297
set_dst_stage_mask
lava::subpass_dependency, 297
set_dst_subpass
lava::subpass_dependency, 298
set_dynamic_states
lava::render_pipeline, 263
set_event_name
debug_utils.hpp, 396
set_event_tag
debug_utils.hpp, 396
set_fence_name
debug_utils.hpp, 397
set_fence_tag
debug_utils.hpp, 397
set_filename
lava::property, 235
set_final_layout
lava::attachment, 26

set_flags
lava::image, 156
set_floating
lava::window, 334
set_format
lava::attachment, 26
set_framebuffer_name
debug_utils.hpp, 397
set_framebuffer_tag
debug_utils.hpp, 397
set_fullscreen
lava::window, 335
set_icon
lava::window, 335
set_image_layout
format.hpp, 578, 579
set_image_name
debug_utils.hpp, 398
set_image_tag
debug_utils.hpp, 398
set_image_view_name
debug_utils.hpp, 398
set_image_view_tag
debug_utils.hpp, 398
set_indirect_commands_layout_name
debug_utils.hpp, 399
set_indirect_commands_layout_tag
debug_utils.hpp, 399
set_ini_file
lava::imgui, 166
set_initial_layout
lava::attachment, 26
set_input_attachment
lava::subpass, 293
set_input_attachments
lava::subpass, 293
set_input_topology
lava::render_pipeline, 263
set_instance_name
debug_utils.hpp, 399
set_instance_tag
debug_utils.hpp, 399
set_layer_count
lava::image, 156
set_layout
lava::image, 157
lava::pipeline, 213
set_layouts
lava::attachment, 27
set_level_count
lava::image, 157
set_line_width
lava::render_pipeline, 264
set_load_op
lava::attachment, 27
set_mouse_position
lava::input, 168
lava::window, 335
set_movement_keys
lava::camera, 52
set_name
debug_utils.hpp, 400
set_object_name
debug_utils.hpp, 400
set_object_tag
debug_utils.hpp, 400
set_op
lava::attachment, 27
set_performance_configuration_name
debug_utils.hpp, 401
set_performance_configuration_tag
debug_utils.hpp, 401
set_physical_device_name
debug_utils.hpp, 401
set_physical_device_tag
debug_utils.hpp, 401
set_pipeline_cache_name
debug_utils.hpp, 402
set_pipeline_cache_tag
debug_utils.hpp, 402
set_pipeline_layout_name
debug_utils.hpp, 402
set_pipeline_layout_tag
debug_utils.hpp, 402
set_pipeline_name
debug_utils.hpp, 403
set_pipeline_tag
debug_utils.hpp, 403
set_position
lava::window, 335
set_preserve_attachments
lava::subpass, 294
set_private_data_slot_name
debug_utils.hpp, 403
set_private_data_slot_tag
debug_utils.hpp, 403
set_query_pool_name
debug_utils.hpp, 404
set_query_pool_tag
debug_utils.hpp, 404
set_queue_name
debug_utils.hpp, 404
set_queue_tag
debug_utils.hpp, 404
set_rasterization_cull_mode
lava::render_pipeline, 264
set_rasterization_front_face
lava::render_pipeline, 264
set_rasterization_polygon_mode
lava::render_pipeline, 265
set_render_pass
lava::render_pipeline, 265
set_render_pass_name
debug_utils.hpp, 405
set_render_pass_tag
debug_utils.hpp, 405

set_resizable
lava::window, 336
set_resolve_attachment
lava::subpass, 294
set_resolve_attachments
lava::subpass, 294
set_sampler_name
debug_utils.hpp, 405
set_sampler_tag
debug_utils.hpp, 405
set_sampler_ycbcr_conversion_name
debug_utils.hpp, 406
set_sampler_ycbcr_conversion_tag
debug_utils.hpp, 406
set_samplers
lava::descriptor::binding, 32
set_samples
lava::attachment, 27
set_save_name
lava::window, 336
set_scissor
lava::render_pipeline, 265
set_seed
lava::pseudo_random_generator, 237
set_semaphore_name
debug_utils.hpp, 406
set_semaphore_tag
debug_utils.hpp, 406
set_shader_module_name
debug_utils.hpp, 407
set_shader_module_tag
debug_utils.hpp, 407
set_shader_stage
lava::compute_pipeline, 60
set_size
lava::rect, 245
lava::window, 336
set_sizing
lava::render_pipeline, 266
set_src_access_mask
lava::subpass_dependency, 298
set_src_stage_mask
lava::subpass_dependency, 298
set_src_subpass
lava::subpass_dependency, 298
set_stage
lava::pipeline::shader_stage, 283
set_stage_flags
lava::descriptor::binding, 32
set_stage_mask
lava::subpass_dependency, 299
set_stencil_load_op
lava::attachment, 28
set_stencil_op
lava::attachment, 28
set_stencil_store_op
lava::attachment, 28
set_store_op
lava::attachment, 29
set_subpass
lava::render_pipeline, 266
lava::subpass_dependency, 299
set_surface_name
debug_utils.hpp, 407
set_surface_tag
debug_utils.hpp, 407
set_swapchain_name
debug_utils.hpp, 408
set_swapchain_tag
debug_utils.hpp, 408
set_tag
debug_utils.hpp, 408
set_tiling
lava::image, 157
set_title
lava::window, 337
set_type
lava::descriptor::binding, 32
set_usage
lava::image, 158
set_use_custom_cpu_callbacks
lava::memory, 185
set_validation_cache_name
debug_utils.hpp, 409
set_validation_cache_tag
debug_utils.hpp, 409
set_vertex_input_attribute
lava::render_pipeline, 266
set_vertex_input_attributes
lava::render_pipeline, 266
set_vertex_input_binding
lava::render_pipeline, 267
set_vertex_input_bindings
lava::render_pipeline, 267
set_view_type
lava::image, 158
set_viewport
lava::render_pipeline, 267
set_viewport_and_scissor
lava::render_pipeline, 268
set_wait_for_events
lava::frame, 126
set_window_icon
config.hpp, 348
setup
lava::app, 21
lava::dispatcher, 98
lava::engine, 102
lava::imgui, 166
lava::pipeline, 213
lava::thread_pool, 315
setup_imgui_font
imgui.hpp, 365
setup_imgui_font_icons
imgui.hpp, 366
setup_log

log.hpp, 624
show_save_title
lava::window, 337
shut_down
lava::frame, 126
singleton
lava::gamepad_manager, 132
lava::global_logger, 134
lava::instance, 174
size
lava::image_loader, 160
sleep
thread.hpp, 638
stage
lava::stage, 284
lava::staging, 285
lava::texture, 313
str
types.hpp, 509
substract
lava::hex_cell, 135
support_blit
format.hpp, 579
support_vertex_buffer_format
format.hpp, 580
supported
lava::physical_device, 206
surface_supported
lava::device, 84
lava::physical_device, 207
lava::swapchain, 306
swapchain_supported
lava::physical_device, 207
switch_mode
lava::window, 337
switch_mode_request
lava::window, 337

teardown_log
log.hpp, 624
telegram
lava::telegram, 308
tell
lava::file, 108
thread.hpp
sleep, 638
time.hpp
get_current_time, 495
get_current_timestamp, 495
get_current_timestamp_ms, 496
timestamp, 496
to_delta, 496
to_dt, 497
to_ms, 497
to_sec, 499
timestamp
time.hpp, 496
to_api_version
base.hpp, 380
to_char
types.hpp, 509
to_delta
time.hpp, 496
to_dt
time.hpp, 497
to_i32
types.hpp, 510
to_i64
types.hpp, 510
to_id
id.hpp, 479
to_index
types.hpp, 510
to_ms
time.hpp, 497
to_pair
lava::hex_cell, 136
to_pixel
lava::hex_grid, 138
to_r32
types.hpp, 511
to_r64
types.hpp, 511
to_sec
time.hpp, 499
to_size_t
types.hpp, 511
to_string
base.hpp, 380
hex.hpp, 615
input.hpp, 549
lava::id, 142
log.hpp, 625, 626
to_ui32
types.hpp, 512
to_ui64
types.hpp, 512
to_version
base.hpp, 380, 381
version.hpp, 516
tooltip
lava::tooltip, 317
transfer_queue
lava::device, 85
transfer_queues
lava::device, 85
trim
misc.hpp, 491
trim_copy
misc.hpp, 492
types.hpp
c16, 505
c32, 505
c8, 505
ENUM_FLAG_OPERATORS, 504
hash_combine, 508
hash_val, 508, 509

i16, 505
i32, 505
i64, 506
i8, 506
str, 509
to_char, 509
to_i32, 510
to_i64, 510
to_index, 510
to_r32, 511
to_r64, 511
to_size_t, 511
to_ui32, 512
to_ui64, 512
uc16, 506
uc32, 506
uc8, 506
ui16, 507
ui32, 507
ui64, 507
ui8, 507

uc16
types.hpp, 506
uc32
types.hpp, 506
uc8
types.hpp, 506
ui16
types.hpp, 507
ui32
types.hpp, 507
ui64
types.hpp, 507
ui8
types.hpp, 507
unique_data
lava::unique_data, 320
unload
lava::property, 235
update
lava::dispatcher, 98
lava::gamepad, 130
lava::hex_grid, 138
lava::id_registry< T, Meta >, 148
update_view
lava::camera, 52
upload
lava::texture, 314
upload_fonts
lava::imgui, 166

v_sync
lava::app, 21
lava::swapchain, 306
valid
lava::allocator, 17
lava::buffer, 45
lava::camera, 53
lava::id, 143
lava::queue, 238
value
lava::gamepad, 131
verify_queues
lava::device::create_param, 65
queue.hpp, 444
version.hpp
to_version, 516
version_string
log.hpp, 626
version_to_string
base.hpp, 381
visible
lava::window, 338
vkAcquireNextImageKHR
lava::device_table, 87
vkAllocateCommandBuffers
lava::device_table, 88
vkCreateCommandPool
lava::device_table, 88, 89
vkCreateFence
lava::device_table, 89
vkCreateImageView
lava::device_table, 89, 90
vkCreateSampler
lava::device_table, 90
vkCreateSemaphore
lava::device_table, 90, 91
vkCreateShaderModule
lava::device_table, 91
vkCreateSwapchainKHR
lava::device_table, 91, 92
vkDestroyCommandPool
lava::device_table, 92
vkDestroyFence
lava::device_table, 92
vkDestroyImageView
lava::device_table, 92
vkDestroySampler
lava::device_table, 93
vkDestroySemaphore
lava::device_table, 93
vkDestroySwapchainKHR
lava::device_table, 93
vkFreeCommandBuffers
lava::device_table, 93
vkGetSwapchainImagesKHR
lava::device_table, 94
vkQueuePresentKHR
lava::device_table, 94
vkQueueSubmit
lava::device_table, 94
vkResetFences
lava::device_table, 94
vkUpdateDescriptorSets
lava::device_table, 95, 96
vkWaitForFences

lava::device_table, 96

wait_for_idle
lava::device, 85
waiting_for_events
lava::frame, 126
window
lava::window, 326
window.hpp
create_surface, 563
get_window, 564
writable
lava::file, 108
write
lava::file, 108
write_file
file_utils.hpp, 530
write_frames_json
benchmark.hpp, 344
write_image.hpp
write_image_png, 375
write_image_png
write_image.hpp, 375